M Y P

Mathematics

3 STUDENT COURSEBOOK

MYP 3 · YEAR 3 · TEN CHAPTERS IN FOUR PARTS

A concept-driven inquiry course in number, algebra, shape and space, trigonometry, data and probability. You investigate, justify and prove, then reason, communicate and apply.



THE REAL NUMBER SYSTEM · POWERS AND NOTATION · FUNCTIONS AND EXPRESSIONS · LINEAR MODELS · EQUATIONS AND SYSTEMS · TRANSFORMATIONS AND CONGRUENCE · SIMILARITY AND TRIGONOMETRY · THREE-DIMENSIONAL SOLIDS · BIVARIATE DATA · PROBABILITY AND EVIDENCE

HOW THIS BOOK WORKS

Every chapter opens with a task. You investigate, notice a pattern, state a mathematical conclusion and justify it before you meet the formal explanation.

MYP ASSESSMENT CRITERIA

CRITERION A	Knowing and understanding
CRITERION B	Investigating patterns
CRITERION C	Communicating
CRITERION D	Applying mathematics in real-life contexts

SADIKOU MOUKODDIM

MYP 3 · STUDENT COURSEBOOK

AUTHOR SADIKOU MOUKODDIM

About this edition

This is the second edition of the MYP 3 mathematics coursebook. It was rebuilt from the pre-rewrite MYP 3 audit, which recorded the chapter scope, the key and related concepts, the global contexts and the inquiry questions of the earlier edition, and identified where the teaching had to be strengthened.

The edition keeps that chapter scope and the visual identity of the collection, and rebuilds the teaching sequence around inquiry: engagement, inquiry questions, investigation, observation, mathematical conclusion, formal development, worked examples, extensive visual practice, and application, assessment and reflection.

Year 3 asks more of Criterion C than Year 2: reasoning must now be organised into a logical structure, and Criterion B moves from recognising and verifying a pattern to justifying a generalisation. The chapters train both explicitly.

All classroom data sets are simulated and are labelled as such. Real data would require provenance, so no claim in this book rests on unverified statistics.

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How to use this book

Each chapter follows the same sequence, so you always know what is coming next. The sequence is deliberate: you investigate before you are told the rule, because a rule you have discovered is a rule you can defend — and defending it is what Year 3 asks of you.

The chapter sequence

1. **Engagement.** A short situation that raises the question the chapter will answer.
2. **Inquiry questions.** Factual, conceptual and debatable questions that frame the chapter.
3. **Investigation.** A task you complete before the formal teaching: record evidence in a table and look for a pattern.
4. **Observation and discovery.** What you noticed, stated in your own words and compared with a partner.
5. **Mathematical conclusion.** The rule that the evidence supports, written as a general statement you can justify.
6. **Formal development.** The explanation, definitions and notation, with the reasoning made explicit.
7. **Worked examples.** Problem, visual representation, mathematical reasoning, step-by-step solution, final answer and interpretation.
8. **Practice.** Separated into Criterion A, B, C and D practice, with visual representations throughout.
9. **Application, assessment and reflection.** Real-context tasks, criterion-based assessment practice, and a reflection that returns to the statement of inquiry.

MYP 3 · STUDENT COURSEBOOK

How to use this book

The four MYP mathematics criteria

A · Knowing and understanding

Apply knowledge to familiar and unfamiliar problems, and check results.

B · Investigating patterns

Investigate relationships, describe patterns, formulate a conjecture, test it and justify the generalisation — at Year 3 level a justification is a logical argument, not a list of examples.

C · Communicating

Use correct mathematical language, notation and complete representations, and organise reasoning into a coherent structure that a reader can follow.

D · Applying mathematics in real-life contexts

Apply mathematics to authentic situations, identify relevant information, and comment on the accuracy and sense of the result.

Working with this book

- Practice is separated by criterion so that you can see which skill a task is training. Criterion B practice is additional genuine pattern investigation, not a repeat of the teaching.
- Visuals are part of the mathematics. Where a task supplies a diagram, grid, table, number line or model, use it; where it asks you to sketch, your drawing is part of the answer.
- Answers and student-facing reasoning are collected at the back of the book, with a formula bank, a glossary and references.
- Every chapter ends with project connections: a Service as Action opportunity, a Community Project connection and a Personal Project connection for MYP 5.

MYP 3 · STUDENT COURSEBOOK

MYP mathematics and this coursebook

The IB Middle Years Programme asks students to inquire into concepts through global contexts, to develop approaches to learning (ATL) skills, and to act on what they learn through service as action. In mathematics this means more than practising procedures: students investigate patterns, justify generalisations, communicate with precise language and apply mathematics to situations that matter to them.

MYP mathematics is organised into four branches — number, algebra, geometry and trigonometry, and statistics and probability — and the programme recommends at least 50 hours of teaching per subject each year. This coursebook covers the Year 3 content of those branches across ten chapters, grouped into four parts.

Year 3 is the level at which justification becomes central. A pattern spotted in three cases is a conjecture; a pattern shown to follow from something already established is a theorem. The chapters in this book keep that distinction visible, and the practice under each criterion says which kind of thinking it is training.

Assessment in MYP mathematics uses four equally weighted criteria, each with a maximum of 8: A knowing and understanding, B investigating patterns, C communicating, and D applying mathematics in real-life contexts. This book trains each criterion explicitly, and says so on the page where the training happens.

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MYP mathematics and this coursebook

Projects, service and action

Service as Action

Each chapter includes a service opportunity with a community need, a mathematical connection, a proposed action, an expected impact and a student reflection. Completing an exercise is not service: service requires action that benefits others and reflection on that action.

Community Project

MYP students completing the programme in Year 3 or Year 4 complete a community project. Each chapter offers a student-led project idea with a research question, the mathematics involved, an action or outcome, beneficiaries and evidence. Follow your school's requirements and supervision.

Personal Project (MYP 5)

Students completing MYP in Year 5 complete a personal project. Each chapter offers an adaptable MYP 5-oriented extension with a personal interest, a learning goal, a product, the mathematics involved, a process and evidence. It is deliberately different from the community project and from service as action.

What this book does not claim

- Programme requirements quoted here follow the official IB MYP curriculum, mathematics and projects pages and the MYP general regulations. Everything else — the chapter sequence, the investigations, the practice sets and the project ideas — is author-designed material built to serve those requirements, and is not an IB publication.
- No ISBN, publisher, imprint or print specification is claimed. The file is an A4 review copy with embedded fonts and continuous pagination, produced for proofing and classroom review rather than for commercial print production.

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Application and assessment · Review · Projects and service

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Application and assessment · Review · Projects and service

Every chapter follows the same sequence: engagement, inquiry questions, investigation, observation, mathematical conclusion, formal development, worked examples, Criterion A–D practice, application, assessment and reflection, and project connections.

CHAPTER 1 · PART I · NUMBER AND ABSTRACTION

01

The real number system

Every fraction is a terminating or a recurring decimal. The converse is false, and the numbers that escape the fractions are the ones that make the number line continuous.

ENGAGE

A surveyor measures a plot of land and records 12.7 m, then 12.73 m, then 12.732 m. Each measurement is better, and none of them is the true length. What kind of number is the true length?

Framing the inquiry

Key concept: Form
Related concepts: Quantity · Equivalence
Global context: Orientation in space and time
Global context exploration: Natural and human landscapes and resources

Statement of inquiry: The form of the real number system, and the equivalence between fractions, terminating decimals and recurring decimals, allow any position between two reference points to be described exactly, while the logic of proof shows that some positions on that line can never be written as a fraction.

Questions to guide us

- **Factual:** Which numbers can be written as fractions, and what are the only two decimal forms a fraction can take?
- **Conceptual:** Why can every fraction be written as a terminating or recurring decimal, and why is the converse false?
- **Debatable:** When a land boundary is recorded to the nearest centimetre, is the recorded number the same number as the true distance, and does the difference matter?

ATL · CRITICAL THINKING. In order for a student to evaluate a claim about all numbers, students must test the claim on several cases, search for a counterexample, and then decide whether a counterexample disproves the claim or widens the definition.

CHAPTER 1 · INVESTIGATION 1.1

Rational numbers and decimal form

INQUIRY QUESTION

How can every fraction be written as a decimal, and what are the only two forms that decimal can take?

YOUR TASK

Take the fractions 1/2, 1/3, 1/4, 1/6, 1/7, 1/8, 1/9, 1/11 and 1/12. Divide each one out to at least eight decimal places and record whether the decimal stops, repeats a single digit, or repeats a block of several digits. Then look at the denominators.

denominator	prime factors	decimal
2	2	0.5 stops
3	3	0.333… repeats
4	2 × 2	0.25 stops
6	2 × 3	0.1666… repeats

Terminating decimals come from denominators built only from 2s and 5s.

fraction	decimal	stops or repeats	denominator factors
1/2	0.5	stops	2
1/3	0.333…	repeats one digit	3
1/4	0.25	stops	2 × 2
1/6	0.1666…	repeats one digit	2 × 3
1/7	0.142857…	repeats six digits	7
1/8	0.125	stops	2 × 2 × 2
1/9	0.111…	repeats one digit	3 × 3
1/11	0.0909…	repeats two digits	11
1/12	0.08333…	repeats one digit	2 × 2 × 3

CHAPTER 1 · INVESTIGATION 1.1

Rational numbers and decimal form

What do you notice?

- Every fraction gave either a stopping decimal or a repeating one — never a decimal that wandered without repeating.
- The denominators that stop are built only from the primes 2 and 5.
- 1/7 repeats a block of six digits, and 7 − 1 = 6, which is not a coincidence.

Observation. Long division of a fraction can only leave a limited number of remainders, so the decimal must stop or repeat.

MATHEMATICAL CONCLUSION

Every fraction can be written as a decimal, and that decimal is either terminating or recurring — a terminating decimal exactly when the denominator in simplest form has no prime factor other than 2 or 5.

Your own words

CHAPTER 1 · 1.1

Rational numbers and decimal form

A rational number is any number that can be written as a/b where a and b are integers and b is not zero. Every integer is rational, because n = n/1.

Dividing the numerator by the denominator gives a decimal that either stops (terminating) or settles into a repeating block (recurring). Nothing else can happen: the remainders in a long division can only take b different values, so one of them must eventually repeat.

A terminating decimal is a fraction whose denominator has only the prime factors 2 and 5. A recurring decimal comes from any other denominator. This is why 3/8 terminates ($8 = 2^3$) but 1/3 recurs.

A recurring decimal is converted back to a fraction by multiplying by a power of 10 and subtracting, which cancels the repeating block.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 1 · 1.1 · WORKED EXAMPLE 1

Rational numbers and decimal form

WORKED EXAMPLE 1.1.1

Problem. Write 5/8 and 7/12 as decimals, and say which one terminates and why.

fraction	denominator	prime factors	decimal
5/8	8	2 × 2 × 2	0.625
7/12	12	2 × 2 × 3	0.5833...

Only the prime factors 2 and 5 give a terminating decimal.

Mathematical reasoning. A decimal terminates exactly when the denominator in simplest form has no prime factor other than 2 or 5.

1. 5/8: $8 = 2^3$, so $5 \div 8 = 0.625$, which stops.
2. 7/12: $12 = 2^2 \times 3$, and the factor 3 forces repetition.
3. $7 \div 12 = 0.58333\ldots = 0.58\dot{3}$, with the digit 3 repeating.

Final answer. 5/8 = 0.625 terminates; 7/12 = $0.58\dot{3}$ recurs

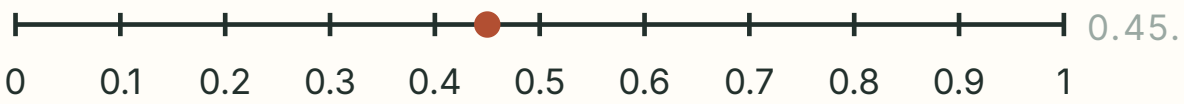
Interpretation. The shape of the denominator decides the shape of the decimal, so the fraction form carries information the decimal form hides.

CHAPTER 1 · 1.1 · WORKED EXAMPLE 2

Rational numbers and decimal form

WORKED EXAMPLE 1.1.2

Problem. Convert 0.4·5· back to a fraction.



Mathematical reasoning. Multiplying by a power of 10 shifts the repeating block, and subtracting removes it.

1. Let x = 0.4555....
2. 10x = 4.555..., and 100x = 45.555....
3. 100x − 10x = 41, so 90x = 41.
4. x = 41/90.

Final answer. 0.4·5· = 41/90

Interpretation. Every recurring decimal is rational, so the recurring bar is a claim about exactness, not about approximation.

CHAPTER 1 · INVESTIGATION 1.2

Irrational numbers and surds

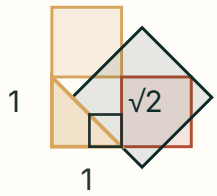
INQUIRY QUESTION

How can we show that a number exists on the number line that is not a fraction?

YOUR TASK

Draw a square of side 1 unit on grid paper and draw its diagonal. Use Pythagoras' theorem to find the length of the diagonal, then try to write that length as a fraction by testing denominators up to 20. Record how close each fraction gets.

a unit square and its diagonal



fraction a/b	value	square of the value	error against 2
7/5	1.4	1.96	0.04
17/12	1.41666…	2.00694…	0.00694…
41/29	1.41379…	1.99881…	0.00119…
99/70	1.41428…	2.00020…	0.00020…
239/169	1.41420…	1.99997…	0.00003…

CHAPTER 1 · INVESTIGATION 1.2

Irrational numbers and surds

What do you notice?

- Every fraction lands close to the diagonal, and then misses it.
- The errors get smaller, but they never reach zero.
- The fractions that get close have larger and larger denominators.

Observation. Fractions can approach the diagonal as closely as we like without ever landing on it, so the length of the diagonal is not a fraction.

MATHEMATICAL CONCLUSION

A proof by contradiction shows that no fraction can equal the diagonal of a unit square, so $\sqrt{2}$ exists on the number line and is irrational.

Your own words

CHAPTER 1 · 1.2

Irrational numbers and surds

An irrational number cannot be written as a/b for any integers a and b. Its decimal neither stops nor repeats, so it can never be written down in full — only named exactly (√2) or described (1.41421… to five decimal places).

The standard proof that √2 is irrational is a proof by contradiction: assume √2 = a/b in simplest form, square both sides, and show that a and b must both be even, which contradicts simplest form.

The parity argument is the engine of the proof. If $a^2 = 2b^2$ then a^2 is even, so a is even (because the square of an odd number is odd), so $a = 2c$, so $4c^2 = 2b^2$, so b^2 is even, so b is even too.

A surd is an exact irrational root left in root form. Keeping √2 as √2 is exact; writing 1.41 is a rounded claim that will not survive being squared.

> FORMAL DEVELOPMENT
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> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

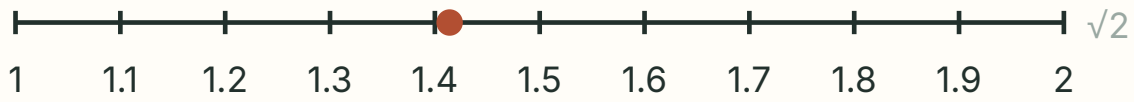
In your own words

CHAPTER 1 · 1.2 · WORKED EXAMPLE 1

Irrational numbers and surds

WORKED EXAMPLE 1.2.1

Problem. Prove that √2 is irrational.



Mathematical reasoning. Assume the opposite and derive a contradiction; a claim that leads to a contradiction must be false.

1. Suppose $\sqrt{2} = a/b$ with a and b integers, $b \neq 0$, and the fraction in simplest form.
2. Squaring gives $2 = a^2/b^2$, so $a^2 = 2b^2$.
3. Then a^2 is even, so a is even: write $a = 2c$.
4. Substituting: $4c^2 = 2b^2$, so $b^2 = 2c^2$, so b^2 is even and b is even.
5. Both a and b are even, contradicting simplest form.

Final answer. √2 is irrational

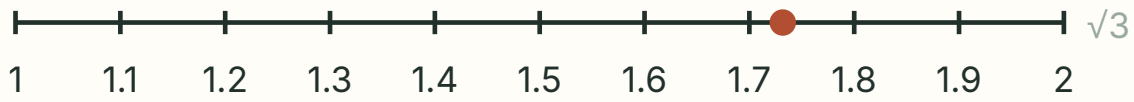
Interpretation. The contradiction came from the parity of a square, which is why no fraction can land exactly on √2.

CHAPTER 1 · 1.2 · WORKED EXAMPLE 2

Irrational numbers and surds

WORKED EXAMPLE 1.2.2

Problem. Show that √3 is irrational, using the same structure of argument.



Mathematical reasoning. The same contradiction argument works with any prime under the root, because a prime dividing a square must divide the number itself.

1. Suppose $\sqrt{3} = a/b$ in simplest form, so $a^2 = 3b^2$.
2. Then 3 divides a^2, and since 3 is prime, 3 divides a: write $a = 3c$.
3. Substituting gives $9c^2 = 3b^2$, so $b^2 = 3c^2$, so 3 divides b.
4. Both a and b are divisible by 3, contradicting simplest form.

Final answer. √3 is irrational

Interpretation. The argument transfers because the key fact — a prime dividing a square divides the root — holds for every prime.

CHAPTER 1 · INVESTIGATION 1.3

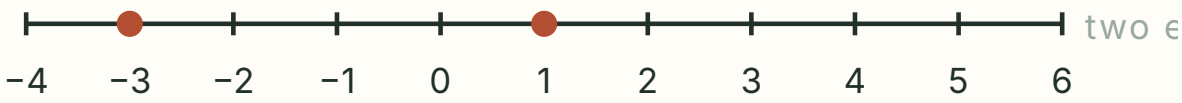
Order, absolute value and intervals

INQUIRY QUESTION

How can every position on a line be described, and how does an interval decide which positions are included?

YOUR TASK

On a number line from −4 to 6, mark the intervals [−3, 1], (−3, 1], [−3, 1) and (−3, 1). For each one, list three numbers that belong to it and one number that does not. Then find two intervals whose union is [−4, 6] and whose intersection is a single number.



interval	includes −3	includes 1	an example inside	an example outside
[−3, 1]	yes	yes	−3	1.5
(−3, 1]	no	yes	0	−3
[−3, 1)	yes	no	−2.5	1
(−3, 1)	no	no	0.5	1

CHAPTER 1 · INVESTIGATION 1.3

Order, absolute value and intervals

What do you notice?

- The four intervals differ only at the two endpoints.
- Every interval contains infinitely many numbers, including irrationals.
- The union of [−3, 1] and (1, 6] is [−3, 6], and their intersection is empty.

Observation. Square and round brackets record exactly which endpoints belong, and the union and intersection of intervals follow from that.

MATHEMATICAL CONCLUSION

An interval is the set of numbers between two endpoints, and the bracket type decides inclusion: [a, b] includes both ends, (a, b) excludes both, and the mixed forms include exactly one.

Your own words

CHAPTER 1 · 1.3

Order, absolute value and intervals

The real numbers fill the number line with no gaps: every point is a real number and every real number is a point. Quantity and order are therefore the same idea on a line: to compare two quantities is to compare their positions. Rational and irrational numbers are both dense — between any two there is another of each kind.

Absolute value measures distance from zero, so $|x| = x$ when $x \geq 0$ and $|x| = -x$ when $x < 0$. It turns an ordering question into a distance question.

An interval is the set of numbers between two endpoints. Square brackets include the endpoint, round brackets exclude it, and mixing them gives a half-open interval.

Interval and set notation are two languages for the same object: $[1, 4)$ is $\{x : 1 \leq x < 4\}$, and the complement of that set is everything outside it.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 1 · 1.3 · WORKED EXAMPLE 1

Order, absolute value and intervals

WORKED EXAMPLE 1.3.1

Problem. Solve $|x - 3| < 2$ and write the answer as an interval.



Mathematical reasoning. Absolute value measures distance, so $|x - 3| < 2$ means x is less than 2 units from 3.

1. $|x - 3| < 2$ means $-2 < x - 3 < 2$.
2. Add 3 to all three parts: $1 < x < 5$.
3. As an interval: $(1, 5)$.

Final answer. $x \in (1, 5)$

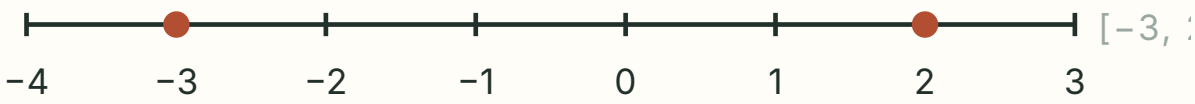
Interpretation. Reading absolute value as distance turns an inequality into a picture of a line segment.

CHAPTER 1 · 1.3 · WORKED EXAMPLE 2

Order, absolute value and intervals

WORKED EXAMPLE 1.3.2

Problem. Write $\{x : -3 \le x < 2\}$ in interval notation, and describe its complement.



Mathematical reasoning. A square bracket keeps the endpoint and a round bracket removes it, so the notation records inclusion directly.

1. -3 is included and 2 is excluded, so the interval is $[-3, 2)$.
2. The complement is everything outside: $x < -3$ or $x \ge 2$.
3. In interval notation: $(-\infty, -3) \cup [2, \infty)$.

Final answer. $[-3, 2)$, with complement $(-\infty, -3) \cup [2, \infty)$

Interpretation. The two notations disagree only about the endpoints, which is exactly the information a surveyor needs.

CHAPTER 1 · INVESTIGATION 1.4

Working with surds

INQUIRY QUESTION

How can an irrational quantity be written exactly and still be calculated with?

YOUR TASK

Take the surds $\sqrt{8}$, $\sqrt{18}$, $\sqrt{50}$ and $\sqrt{72}$. Simplify each one, then add them in pairs. Then try to simplify $\sqrt{2} + \sqrt{3}$ and record what happens. Finally, find a fraction with denominator $3 + \sqrt{2}$ that has an integer denominator.

surd	largest square factor	simplified
$\sqrt{8}$	4	$2\sqrt{2}$
$\sqrt{18}$	9	$3\sqrt{2}$
$\sqrt{50}$	25	$5\sqrt{2}$
$\sqrt{72}$	36	$6\sqrt{2}$
$\sqrt{2} + \sqrt{3}$	none	cannot be simplified

CHAPTER 1 · INVESTIGATION 1.4

Working with surds

What do you notice?

- All four surds simplify to a multiple of $\sqrt{2}$.
- $2\sqrt{2} + 3\sqrt{2} + 5\sqrt{2} + 6\sqrt{2} = 16\sqrt{2}$, but only because they are all like surds.
- $\sqrt{2} + \sqrt{3}$ resists simplification because the surds are different.
- Multiplying $1/(3 + \sqrt{2})$ by $(3 - \sqrt{2})/(3 - \sqrt{2})$ gives $(3 - \sqrt{2})/7$.

> **Observation.** Only like surds combine, and a surd denominator can always be cleared by multiplying by the conjugate.

MATHEMATICAL CONCLUSION

A surd can be written exactly and calculated with by taking out square factors, collecting like surds and rationalising with a conjugate — and the simplified form is unique.

Your own words

CHAPTER 1 · 1.4

Working with surds

Surds obey the same rules as powers with fractional indices: $\sqrt{a} \times \sqrt{b} = \sqrt{ab}$, and $\sqrt{a} \div \sqrt{b} = \sqrt{a/b}$ for non-negative a and b.

Simplifying a surd means taking out the largest square factor: $\sqrt{72} = \sqrt{36 \times 2} = 6\sqrt{2}$. The simplified form is unique, which makes it a good form for comparing.

Rationalising a denominator replaces a surd in a denominator by an integer, by multiplying top and bottom by a conjugate. $(1 + \sqrt{2})(1 - \sqrt{2}) = 1 - 2 = -1$, so the surd disappears.

Adding surds needs like surds: $3\sqrt{2} + 5\sqrt{2} = 8\sqrt{2}$, but $3\sqrt{2} + 5\sqrt{3}$ cannot be simplified. Treating $\sqrt{2}$ as if it were $\sqrt{3}$ is the standard error.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 1 · 1.4 · WORKED EXAMPLE 1

Working with surds

WORKED EXAMPLE 1.4.1

Problem. Simplify √200 and rationalise 6/√3.

expression	square factors	simplified
√200	200 = 100 × 2	10√2
6/√3	× √3/√3	2√3

Take out the largest square factor, then clear the surd denominator.

Mathematical reasoning. Simplifying removes square factors, and multiplying by a conjugate clears a surd from a denominator without changing the value.

1. √200 = √(100 × 2) = 10√2.
2. 6/√3 = (6 × √3)/(√3 × √3) = 6√3/3.
3. 6√3/3 = 2√3.

Final answer. √200 = 10√2 and 6/√3 = 2√3

Interpretation. Both answers are exact; writing 1.732… would lose the ability to check the work by squaring.

CHAPTER 1 · 1.4 · WORKED EXAMPLE 2

Working with surds

WORKED EXAMPLE 1.4.2

Problem. Show that $(2 + \sqrt{3})(2 - \sqrt{3}) = 1$, and use it to rationalise $5/(2 + \sqrt{3})$.

pair	product
$2 + \sqrt{3}$ and $2 - \sqrt{3}$	$4 - 3 = 1$
$5/(2 + \sqrt{3})$	$5(2 - \sqrt{3}) = 10 - 5\sqrt{3}$

Conjugate pairs differ only in the sign between the terms.

Mathematical reasoning. The product of a conjugate pair is rational, so multiplying by the conjugate is a legitimate way to clear a surd denominator.

1. $(2 + \sqrt{3})(2 - \sqrt{3}) = 4 - 2\sqrt{3} + 2\sqrt{3} - 3 = 1$.
2. $5/(2 + \sqrt{3}) = 5(2 - \sqrt{3})/((2 + \sqrt{3})(2 - \sqrt{3}))$.
3. $= 5(2 - \sqrt{3})/1 = 10 - 5\sqrt{3}$.

Final answer. $5/(2 + \sqrt{3}) = 10 - 5\sqrt{3}$

Interpretation. The conjugate works because the cross terms cancel, leaving only rational numbers.

CHAPTER 1 · INVESTIGATION 1.5

Sets, logic and proof

INQUIRY QUESTION

How can a claim about all numbers be proved rather than tested?

YOUR TASK

Test four claims about numbers. For each one, either find a counterexample or write down why you believe it is true for every case: (1) the sum of two rational numbers is rational; (2) the sum of two irrational numbers is irrational; (3) a non-zero rational number times an irrational number is irrational; (4) there is a rational number between any two rational numbers.

claim	counterexample or argument	verdict
rational + rational is rational	$a/b + c/d = (ad + bc)/bd$, a fraction	true for all
irrational + irrational is irrational	$\sqrt{2} + (-\sqrt{2}) = 0$	false
rational × irrational is irrational	if $qr = s$ then $r = s/q$ is rational	true for $q \neq 0$
a rational lies between any two rationals	$(p/q + r/s)/2$ is rational and between them	true for all

CHAPTER 1 · INVESTIGATION 1.5

Sets, logic and proof

What do you notice?

- Two claims were destroyed by a single counterexample each.
- Two claims survived because an argument covered every case.
- The average of two numbers always lies between them, which is why the rationals are dense.

> **Observation.** Examples can disprove a universal claim but never prove one; proving requires an argument that covers every case.

MATHEMATICAL CONCLUSION

A universal claim is proved by an argument that works for all cases and disproved by one counterexample, which is why proof and counterexample are complementary tools.

Your own words

CHAPTER 1 · 1.5

Sets, logic and proof

A set is a collection of objects, described by listing its elements or by a rule. The universal set is everything under discussion, and the complement of A is everything in the universal set that is not in A.

Union gathers what is in either set; intersection keeps only what is in both. These two operations behave like 'or' and 'and' in logic, and Venn diagrams make the difference visible.

A universal claim ('every fraction has a recurring or terminating decimal') cannot be proved by examples. It needs an argument that works for all cases, and a single counterexample destroys it.

Proof by contradiction assumes the claim is false and derives an impossible consequence. It is the tool that proves √2 is irrational, and it is used again in Chapter 7 to prove Pythagoras-related results.

> FORMAL DEVELOPMENT
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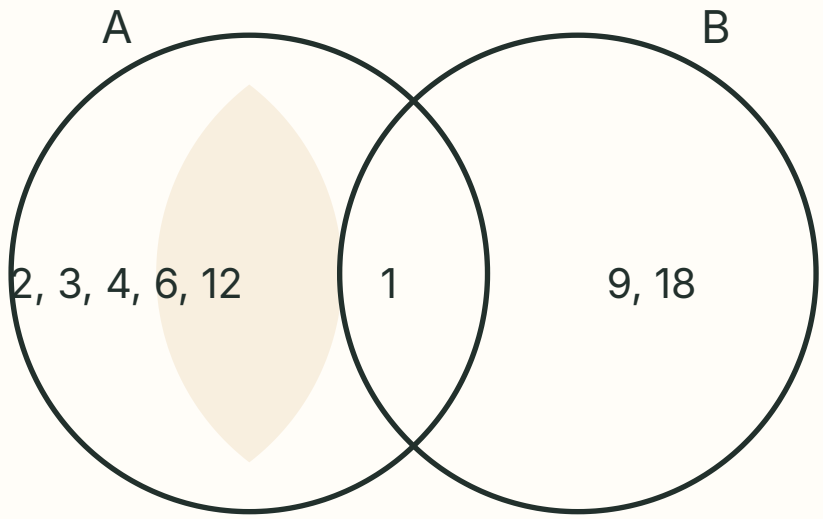
In your own words

CHAPTER 1 · 1.5 · WORKED EXAMPLE 1

Sets, logic and proof

WORKED EXAMPLE 1.5.1

Problem. A = {x : x is a factor of 12} and B = {x : x is a factor of 18}. List A ∪ B and A ∩ B.



Mathematical reasoning. Union collects everything in either set; intersection keeps only the common elements.

1. A = {1, 2, 3, 4, 6, 12} and B = {1, 2, 3, 6, 9, 18}.
2. A ∪ B = {1, 2, 3, 4, 6, 9, 12, 18}.
3. A ∩ B = {1, 2, 3, 6}.

Final answer. A ∪ B = {1, 2, 3, 4, 6, 9, 12, 18}; A ∩ B = {1, 2, 3, 6}

Interpretation. The intersection is the common structure the two sets share, which is why it is the useful part for comparing.

CHAPTER 1 · 1.5 · WORKED EXAMPLE 2

Sets, logic and proof

WORKED EXAMPLE 1.5.2

Problem. Show that the claim 'the product of two irrational numbers is irrational' is false.

first number	second number	product
√2	√2	2
√2	2√2	4

One counterexample is enough to destroy a universal claim.

Mathematical reasoning. A universal claim fails if a single case contradicts it, so a counterexample is a complete refutation.

1. The claim says: for all irrational a and b, ab is irrational.
2. Take a = √2 and b = √2. Both are irrational.
3. ab = √2 × √2 = 2, which is rational.
4. So the claim is false.

Final answer. False — √2 × √2 = 2

Interpretation. Finding one counterexample is faster than testing many cases, and it is decisive.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding

1.A1 FLUENCY
Write 7/8, 5/6 and 11/40 as decimals and say which terminate and why.

1.A2 FLUENCY
Convert $0.\dot{2}\dot{7}$ into a fraction.

1.A3 REASONING
Explain why 1/7 must produce a recurring decimal.

1.A4 FLUENCY
A land record gives a boundary as $0.3\dot{7}$ m. Write this as a fraction.

1.A5 REASONING
Is 0.10100100010000… rational? Justify your answer.

1.A6 APPLICATION
Write 22/7 as a decimal to four places and state whether it is exact.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding (continued)

1.A7 FLUENCY
Sort these into rational and irrational: 0.25, √5, 7/3, π, −4, 0.·1·.

1.A8 REASONING
Find a fraction between 1/3 and 1/2, and explain how you know it is between them.

1.A9 APPLICATION
A resource of 3/5 of a hectare is recorded as 0.6 ha. Is this exact?

1.A10 REASONING
Explain the difference between 0.·9· and 0.9999.

1.A11 CHALLENGE
Convert 0.0·4·5· into a fraction.

1.A12 FLUENCY
Why is every integer a rational number?

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns

1.B1 REASONING

Prove that √5 is irrational.

1.B2 REASONING

Investigate: is √(2/3) rational or irrational? Justify.

1.B3 REASONING

Find a counterexample to 'the square of an irrational number is irrational'.

1.B4 REASONING

Investigate the sequence of approximations 1, 1.4, 1.41, 1.414, 1.4142 to √2. What is happening to the error?

1.B5 REASONING

Test the claim: between any two irrational numbers there is a rational number.

1.B6 REASONING

Investigate: is 2√3 greater than, equal to or less than 3√2? Justify without a calculator.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns (continued)

1.B7 REASONING

Generalise: what can you say about √n when n is not a perfect square?

1.B8 FLUENCY

Investigate the pattern √4, √9, √16, √25, √36. Write the next two terms and a rule.

1.B9 REASONING

Find two irrational numbers whose sum is rational, and explain what this shows.

1.B10 REASONING

Test the claim: if x is irrational then 1/x is irrational.

1.B11 REASONING

Investigate: how many irrational numbers lie between 1 and 2?

CRITERION C Communicating

Criterion C practice · Communicating

1.C1 FLUENCY

Solve $|x + 2| \leq 5$ and write the answer as an interval.

1.C2 REASONING

Explain in words what $|a - b|$ measures, and give an example with two towns.

1.C3 FLUENCY

Write $(-\infty, 3) \cup (3, \infty)$ in set notation and say what is missing.

1.C4 FLUENCY

Simplify $\sqrt{75} + \sqrt{12}$ and explain why the two surds can be collected.

1.C5 FLUENCY

Rationalise $4/(\sqrt{5} - 1)$ and explain the role of the conjugate.

1.C6 REASONING

Explain why $\sqrt{2} + \sqrt{3}$ cannot be simplified, using the definition of like surds.

CRITERION C Communicating

Criterion C practice · Communicating (continued)

1.C7 APPLICATION
A survey gives a distance as 3.6 m to one decimal place. Write the interval of possible true distances.

1.C8 REASONING
Explain the difference between 3.6 m and 3.60 m as a record of measurement.

1.C9 FLUENCY
Draw a Venn diagram for A = {even numbers} and B = {multiples of 3} inside the universal set of the first 12 positive integers, and list $A \cap B$.

1.C10 REASONING
Explain why a proof by contradiction is not the same as finding a counterexample.

1.C11 FLUENCY
Write the complement of $[-2, 5]$ in interval notation and describe it in words.

1.C12 APPLICATION
A resource estimate is given as 4×10^3 kg to one significant figure. Write the interval of possible true values.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts

1.D1 APPLICATION

A plot of land measures √50 m by √18 m. Find its exact area and a decimal estimate to two places.

1.D2 APPLICATION

A boundary is recorded as 12.7 m to the nearest 10 cm. Write the interval of possible true lengths and say how wide it is.

1.D3 REASONING

Two farms measure $0.4\dot{3}$ km² and 0.43 km². Which is larger, and by how much? Give an exact answer.

1.D4 APPLICATION

A reservoir holds 3.5×10^6 litres. A second holds 3.5×10^5 litres. How many times larger is the first, and what does this say about reporting?

1.D5 REASONING

A path is measured as 25 m to the nearest metre. Explain why the true length cannot be known from this record, and state the best interval you can give.

1.D6 APPLICATION

A metal rod is √27 cm long. A slot is 5.1 cm wide to one decimal place. Will the rod fit? Justify with an exact comparison.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts (continued)

1.D7 APPLICATION

A field is $0.6\overline{1}$ hectares. Convert this to a fraction and find the area to the nearest square metre, given 1 ha = 10 000 m^2.

1.D8 REASONING

Explain, with a calculation, why a builder should not order material from a measurement rounded to the nearest metre.

1.D9 APPLICATION

A map scale is 1 : 25 000. A lake measures $\sqrt{8}$ cm on the map. Find the real width in exact form and to the nearest metre.

1.D10 REASONING

Two land records give areas $1.4\overline{9}$ km^2 and 1.49 km^2. Show which is larger and find the difference as a fraction.

1.D11 APPLICATION

A water tank holds $2\sqrt{3} \times 10^3$ litres. A second holds 3000 litres. Which holds more, and by how much, exactly?

1.D12 REASONING

Explain why an exact surd is preferred to a decimal when a measurement will be used in a further calculation.

CHAPTER 1 · APPLICATION AND ASSESSMENT

Applying chapter 1

These tasks use fraction-decimal conversion, ordering, absolute value, intervals, surds and set notation. State whether your answer is exact or a rounded claim.

A1 · The boundary of a plot

A triangular plot has sides √20 m, √45 m and √80 m. Find the exact perimeter, a decimal estimate to one decimal place, and the interval in which the true perimeter must lie if each side is measured to the nearest 0.1 m.

Criteria A and C · state which answers are exact and which are rounded.

A2 · The recurring record

An archive records a yield as $0.2\dot{7}$ tonnes per hectare. Convert this to a fraction, then find the yield of 4 hectares exactly and to the nearest tonne.

Criteria A and D · show the conversion, and comment on the precision of the original record.

CHAPTER 1 · REVIEW

What you should now be able to do

Key ideas

- A rational number is a fraction of integers; its decimal stops or recurs.
- A decimal terminates exactly when the denominator has no prime factor but 2 and 5.
- An irrational number cannot be written as a fraction; $\sqrt{2}$ is proved irrational by contradiction.
- Absolute value measures distance from zero, so it turns order into distance.
- An interval records which endpoints are included, using square and round brackets.
- Surds are simplified by taking out square factors and rationalised by multiplying by a conjugate.
- A universal claim is proved by an argument for all cases, or disproved by one counterexample.

Vocabulary

rational · irrational · terminating decimal · recurring decimal · surd · absolute value · interval · union · intersection · complement · proof by contradiction · counterexample · conjugate

CHAPTER 1 · REVIEW

Mixed practice · consolidate and extend

Do not use a calculator unless a question asks for one. Show a method where one is requested.

1.R1 Write 5/12 as a decimal and say whether it terminates.

1.R2 Convert $0.\dot{4}\dot{5}$ into a fraction.

1.R3 Simplify $\sqrt{98}$.

1.R4 Solve $|x - 1| < 3$ and write the answer as an interval.

1.R5 Is the sum of two irrational numbers always irrational? Justify.

1.R6 Write the complement of $(2, 7]$ in interval notation.

CHAPTER 1 · REVIEW

Returning to the statement of inquiry

STATEMENT OF INQUIRY

The form of the real number system, and the equivalence between fractions, terminating decimals and recurring decimals, allow any position between two reference points to be described exactly, while the logic of proof shows that some positions on that line can never be written as a fraction.

Reflect on the chapter

Answer in your own words, using evidence from this chapter.

- Write the statement of inquiry for this chapter in your own words.
- Which mathematical ideas from this chapter did you use to understand it? Name at least two.
- How did the key concept of form and the related concepts of quantity and equivalence appear in your work?
- How is the global context of orientation in space and time, and its exploration of natural and human landscapes and resources, visible in the investigations and examples you completed?
- Use evidence from an investigation or a worked example to justify your answer.
- Has your understanding of the statement of inquiry changed since the start of the chapter? Explain how.
- What are the limitations of a measurement that is rounded, and what would you still like to ask?

CHAPTER 1 · PROJECTS, SERVICE AND ACTION

Acting on what you have learned

Land, resource and boundary problems are real-number problems. Use exact forms where they exist, and say when a rounded value is the honest answer.

Service as Action

COMMUNITY NEED	Land and water records in the community are kept to different precisions, which makes sharing decisions hard to check.
MATHEMATICAL CONNECTION	Exact and rounded values, intervals and unit conversion.
PROPOSED ACTION	Re-record one shared resource to an agreed precision and publish the interval of uncertainty with it.
EXPECTED IMPACT	Decisions about the resource are made from numbers everyone can interpret the same way.
STUDENT REFLECTION	Record the precision you chose, why, and what changed when the record was made consistent.

Community Project

PROJECT IDEA	A precision audit of the school's measurements.
NEED	The field, the hall and the garden are all measured to different precisions.
RESEARCH QUESTION	What precision does each use of the school's spaces actually need?
MATHEMATICS	Rounding intervals, significant figures, unit conversion and error accumulation.
ACTION OR OUTCOME	A measurement register with an agreed precision and an uncertainty interval for each space.
BENEFICIARIES	Anyone planning events, seating or planting in those spaces.
EVIDENCE	The register, the method and a photograph of the measuring.

Personal Project connection · MYP 5

PERSONAL INTEREST	Mapping, hiking or gardening.
LEARNING GOAL	To record a real measurement honestly, with its uncertainty.
PRODUCT OR OUTCOME	A measured plan of a route or a bed, with intervals.
MATHEMATICS	Scale, surds, rounding and interval notation.
PROCESS	Measure, convert to a common unit, then state the interval.
EVIDENCE	The plan, the measurements and the stated uncertainty.

CHAPTER 2 · PART I · NUMBER AND ABSTRACTION

02

Powers, roots and scientific notation

Index form is a representation that carries the size of a quantity in its shape. The exponent laws are patterns in that shape, and they are what make large and small numbers calculable.

ENGAGE

A drop of seawater contains about 10^{22} atoms. A red blood cell is about 10^{-5} m wide. Writing these in full is impossible; writing them in index form is not, and the comparison becomes immediate.

Framing the inquiry

Key concept: Form
Related concepts: Quantity · Representation · Patterns
Global context: Scientific and technical innovation
Global context exploration: Precision, accuracy and measurement

Statement of inquiry: The form of a number written in index form, and the equivalence between index form and root form, allow any quantity to be represented compactly and compared across scales, while the pattern in the exponent laws makes those representations safe to manipulate.

Questions to guide us

- **Factual:** What do the exponent laws say, and how do you write a root as a power?
- **Conceptual:** Why does multiplying powers of the same base add the indices, and what does that reveal about the form of a number in index notation?
- **Debatable:** When a measurement is reported to a certain number of significant figures, does the extra precision of a longer decimal make the claim more scientific?

ATL · RESEARCH. In order for a student to use an index law correctly in an unfamiliar problem, students must test the law on small cases, record where it breaks, and state the conditions under which the law holds.

CHAPTER 2 · INVESTIGATION 2.1

Integer exponents and the exponent laws

INQUIRY QUESTION

How can the pattern in repeated multiplication be written once, as a rule that works for every case?

YOUR TASK

Compute $2^2 \times 2^3$, $2^3 \times 2^4$, $2^5 \times 2^5$ and $2^2 \times 2^5$ both as products of factors and as single powers of 2. Then do the same for division, and for $(2^2)^3$ and $(2^3)^2$. Record the pattern you see in the indices.

expression	as repeated factors	as one power
$2^2 \times 2^3$	$2 \times 2 \times 2 \times 2 \times 2$	2^5
$2^3 \times 2^4$	seven factors of 2	2^7
$2^5 \div 2^2$	five factors with two removed	2^3
$(2^2)^3$	three groups of two factors	2^6

The index law is a statement about counting factors.

calculation	written out	answer	what happened to the indices
$2^2 \times 2^3$	$2\times2\times2\times2\times2$	2^5	2 + 3 = 5
$2^3 \times 2^4$	seven factors	2^7	3 + 4 = 7
$2^5 \div 2^2$	five minus two factors	2^3	5 − 2 = 3
$(2^2)^3$	three groups of two	2^6	2 × 3 = 6
$(2^3)^2$	two groups of three	2^6	3 × 2 = 6
2^0	no factors at all	1	the pattern needs it to be 1

CHAPTER 2 · INVESTIGATION 2.1

Integer exponents and the exponent laws

What do you notice?

- Multiplying adds the indices, and the reason is that the factors pile up.
- Dividing subtracts them, because division removes factors.
- A power of a power multiplies the indices, because each factor is itself a repeated multiplication.
- $2^5 \div 2^5$ must equal 1, and the pattern says $2^{5-5} = 2^0$, so $2^0 = 1$.

Observation. Each law is the same statement about counting factors, written in index form.

MATHEMATICAL CONCLUSION

$a^m \times a^n = a^{m+n}$, $a^m \div a^n = a^{m-n}$ and $(a^m)^n = a^{mn}$, and these patterns are what force the definitions $a^0 = 1$ and $a^{-n} = 1/a^n$.

Your own words

CHAPTER 2 · 2.1

Integer exponents and the exponent laws

An index records repeated multiplication: $2^5 = 2 \times 2 \times 2 \times 2 \times 2$. The base is 2 and the index is 5, and the index is a count of factors, not a multiplier.

Multiplying powers of the same base adds the indices, because the factors simply accumulate: $2^3 \times 2^4 = 2^7$. Dividing subtracts them, because division removes factors.

A power of a power multiplies the indices, because each of the factors is itself a repeated multiplication: $(2^3)^4 = 2^{12}$.

The laws hold for every integer index once $a^0 = 1$ and $a^{-n} = 1/a^n$ are defined. Those definitions are chosen so the patterns in the exponent laws continue without a break, and the same patterns extend to roots.

> FORMAL DEVELOPMENT
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In your own words

CHAPTER 2 · 2.1 · WORKED EXAMPLE 1

Integer exponents and the exponent laws

WORKED EXAMPLE 2.1.1

Problem. Simplify $(3x^4)^3 \div (9x^2)$ and write the answer with positive indices.

expression	law used	result
$(3x^4)^3$	power of a power	$27x^{12}$
$27x^{12} \div 9x^2$	quotient law	$3x^{10}$

Each law changes only the form, never the value.

Mathematical reasoning. The power law multiplies the indices inside the bracket, and the quotient law subtracts them, with the coefficients handled separately.

1. $(3x^4)^3 = 3^3 \times (x^4)^3 = 27x^{12}$.
2. $27x^{12} \div 9x^2 = (27 \div 9) \times x^{12-2}$.
3. $= 3x^{10}$.

Final answer. $3x^{10}$

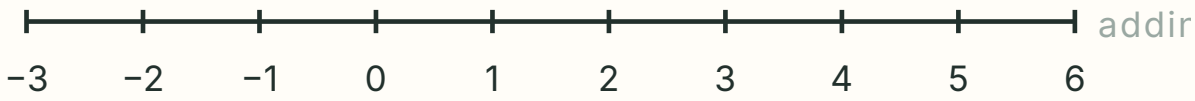
Interpretation. The coefficient and the variable part obey different laws, so keeping them separate avoids the commonest error.

CHAPTER 2 · 2.1 · WORKED EXAMPLE 2

Integer exponents and the exponent laws

WORKED EXAMPLE 2.1.2

Problem. Evaluate $5^{-2} \times 5^{5} \div 5^{0}$.



Mathematical reasoning. Negative and zero indices are defined so that the same laws continue to work, so they can be used in the same calculation.

1. $5^{-2} \times 5^{5} = 5^{-2+5} = 5^{3}$.
2. $5^{0} = 1$ by definition.
3. $5^{3} \div 1 = 125$.

Final answer. 125

Interpretation. The definitions of a^{0} and a^{-n} are not arbitrary: they are exactly what the pattern in the laws requires.

CHAPTER 2 · INVESTIGATION 2.2

Roots and rational exponents

INQUIRY QUESTION

How can a root be written as a power, and why are the two forms equivalent?

YOUR TASK

Take the numbers 4, 8, 27 and 32. Write each as a power of its smallest possible base, then write each root $\sqrt[n]{a}$ as a fractional index. Finally, test whether $\sqrt{(ab)} = \sqrt{a} \times \sqrt{b}$ holds for $a = 9$, $b = 16$ and for $a = -9$, $b = 16$, and record what you find.

number	as a power	root as an index
4	2^2	√4 = 4^(1/2) = 2
8	2^3	$\sqrt[3]{8}$ = 8^(1/3) = 2
27	3^3	$\sqrt[3]{27}$ = 27^(1/3) = 3
32	2^5	$\sqrt[5]{32}$ = 32^(1/5) = 2

CHAPTER 2 · INVESTIGATION 2.2

Roots and rational exponents

What do you notice?

- Every root could be written as a fractional index.
- $\sqrt{(9 \times 16)} = \sqrt{144} = 12$, and $\sqrt{9} \times \sqrt{16} = 3 \times 4 = 12$, so the law holds.
- $\sqrt{(-9 \times 16)}$ is not a real number, so the law cannot be tested there.
- Writing numbers as powers of the smallest base makes roots easy to see.

> **Observation.** A root and a fractional index are two notations for the same operation, so the exponent laws apply to roots too.

MATHEMATICAL CONCLUSION

$\sqrt[n]{a} = a^{(1/n)}$, so $a^{(m/n)} = (\sqrt[n]{a})^m$ and the exponent laws extend to roots — for non-negative a when n is even.

Your own words

CHAPTER 2 · 2.2

Roots and rational exponents

The nth root of a number is the value that, raised to the nth power, gives the number back: $^{n}\sqrt{a}$ = a^(1/n). The two notations are equivalent because (a^(1/n))^{n} = a.

Writing roots as fractional indices lets the exponent laws apply to roots as well: √a × √b = a^(1/2) × b^(1/2) = (ab)^(1/2) = √(ab).

A rational index a^(m/n) means the nth root raised to the mth power, and the order of the two operations does not matter for non-negative a.

Negative indices give reciprocals, so a^(−1/2) = 1/√a. Keeping the sign separate from the fraction is what prevents sign errors.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 2 · 2.2 · WORKED EXAMPLE 1

Roots and rational exponents

WORKED EXAMPLE 2.2.1

Problem. Write $\sqrt[3]{(8x^6)}$ using index notation, then simplify.

expression	index form	simplified
$\sqrt[3]{8}$	8^(1/3)	2
$\sqrt[3]{(x^6)}$	(x^6)^(1/3)	x^2

A root is a fractional index, so the power law applies.

Mathematical reasoning. The cube root is the power 1/3, and the power law multiplies indices, so 6 × 1/3 = 2.

1. $\sqrt[3]{(8x^6)} = (8x^6)^{(1/3)}$.
2. $= 8^{(1/3)} \times (x^6)^{(1/3)}$.
3. $8^{(1/3)} = 2$ and $(x^6)^{(1/3)} = x^{(6/3)} = x^2$.

Final answer. $2x^2$

Interpretation. Index form turns a root problem into an exponent problem, which is why it is the preferred form for manipulation.

CHAPTER 2 · 2.2 · WORKED EXAMPLE 2

Roots and rational exponents

WORKED EXAMPLE 2.2.2

Problem. Simplify (16/81)^(−3/4).

step	law	result
(16/81)^(−3/4)	negative index	(81/16)^(3/4)
(81/16)^(3/4)	root then power	$(3/2)^3$ = 27/8

Handle the sign, then the fraction, then the index.

Mathematical reasoning. A negative index inverts the fraction, and a fractional index takes a root before applying the power.

1. (16/81)^(−3/4) = (81/16)^(3/4).
2. (81/16)^(3/4) = ($\sqrt[4]{81}$ / $\sqrt[4]{16}$)^3 = $(3/2)^3$.
3. $(3/2)^3$ = 27/8.

Final answer. 27/8

Interpretation. Doing the sign first, then the root, then the power keeps each step checkable.

CHAPTER 2 · INVESTIGATION 2.3

Scientific notation

INQUIRY QUESTION

How can very large and very small quantities be written so that their size is visible at a glance?

YOUR TASK

Collect six quantities from science or geography at very different scales, write each in scientific notation, and order them from smallest to largest. Then multiply the smallest and the largest, and divide the largest by the smallest, without using a calculator for the exponents.

quantity	ordinary form	scientific notation
width of a hair	0.00008 m	8×10^{-5} m
mass of a grain of rice	0.000025 kg	2.5×10^{-5} kg
height of a room	3 m	3×10^{0} m
distance to the moon	384 000 000 m	3.84×10^{8} m
mass of the Earth	a 25-digit number	5.97×10^{24} kg
speed of light	300 000 000 m/s	3×10^{8} m/s

CHAPTER 2 · INVESTIGATION 2.3

Scientific notation

What do you notice?

- Ordering by the exponent gives the correct order immediately.
- The product of the smallest and largest had exponent −5 + 24 = 19.
- The quotient had exponent 24 − (−5) = 29, so the two differ by 29 orders of magnitude.
- Every coefficient stayed between 1 and 10, which is what makes the form readable.

> **Observation.** The exponent carries the size of the quantity, so comparing, multiplying and dividing become operations on the exponents.

MATHEMATICAL CONCLUSION

Scientific notation is $a \times 10^{n}$ with $1 \le a < 10$, and multiplying or dividing such numbers adds or subtracts the exponents — which is the exponent law applied to a representation.

Your own words

CHAPTER 2 · 2.3

Scientific notation

Scientific notation writes a number as $a \times 10^{n}$ with $1 \le a < 10$ and n an integer. The exponent counts how many places the decimal point has moved, so it carries the size of the quantity.

The form is a representation choice: 6 300 000 and 6.3×10^{6} are the same number, but only the second makes the order of magnitude visible.

Multiplying in scientific notation multiplies the coefficients and adds the exponents. Dividing does the reverse. This is the exponent law applied to a representation.

The form also prevents a common reporting error. A measurement written as 0.00042 m has hidden its scale; 4.2×10^{-4} m has not.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 2 · 2.3 · WORKED EXAMPLE 1

Scientific notation

WORKED EXAMPLE 2.3.1

Problem. Write 0.0000042 and 3 500 000 000 in scientific notation, then multiply them.

number	scientific notation
0.0000042	4.2×10^{-6}
3 500 000 000	3.5×10^{9}

The exponent records the position of the decimal point.

Mathematical reasoning. Move the decimal point until the coefficient lies between 1 and 10, then count the places moved to fix the exponent.

1. 0.0000042: move the point 6 places right $\longrightarrow 4.2 \times 10^{-6}$.
2. 3 500 000 000: move the point 9 places left $\longrightarrow 3.5 \times 10^{9}$.
3. Multiply: $(4.2 \times 3.5) \times 10^{-6+9} = 14.7 \times 10^{3}$.
4. Restore the form: 1.47×10^{4}.

Final answer. 4.2×10^{-6} and 3.5×10^{9}; their product is 1.47×10^{4}

Interpretation. The exponents add exactly as the index law says, and the coefficient is adjusted at the end to keep the form.

CHAPTER 2 · 2.3 · WORKED EXAMPLE 2

Scientific notation

WORKED EXAMPLE 2.3.2

Problem. A red blood cell is 7.5×10^{-6} m wide and a grain of sand is 2×10^{-4} m wide. How many times wider is the grain?

object	width
red blood cell	7.5×10^{-6} m
grain of sand	2×10^{-4} m

Dividing in scientific notation subtracts the exponents.

Mathematical reasoning. Dividing divides the coefficients and subtracts the exponents, which gives the comparison directly.

1. $(2 \times 10^{-4}) \div (7.5 \times 10^{-6}) = (2 \div 7.5) \times 10^{-4-(-6)}$.
2. $= 0.2666\ldots \times 10^{2}$.
3. $= 2.666\ldots \times 10^{1} \approx 27$.

Final answer. about 27 times wider

Interpretation. The subtraction of exponents gives the factor of 100 immediately, before any coefficient arithmetic.

CHAPTER 2 · INVESTIGATION 2.4

Orders of magnitude and estimation

INQUIRY QUESTION

How can an order of magnitude decide whether a calculation is worth doing in detail?

YOUR TASK

Estimate each of these without a calculator, then check with one: the number of heartbeats in a year, the number of seconds in a school day, the number of grains of rice in a 1 kg bag, and the number of litres of water a class uses in a week. Record the order of magnitude of each estimate.

quantity	estimate	order of magnitude	calculator value
heartbeats in a year	$70 \times 60 \times 24 \times 365$	10^7	3.68×10^7
seconds in a school day	$6 \times 60 \times 60 \times 6$	10^5	1.296×10^5
grains in 1 kg of rice	$1000\text{ g} \div 0.02\text{ g}$	10^5	5×10^4 to 1×10^5
litres used by a class in a week	$30 \times 200 \times 7$	10^5	4.2×10^4

CHAPTER 2 · INVESTIGATION 2.4

Orders of magnitude and estimation

What do you notice?

- Every estimate landed within a factor of 10 of the true value.
- Estimating first made the calculator check meaningful.
- The exponent of the estimate matched the exponent of the true value in three of the four cases.
- The rice estimate was the least reliable, because the mass of one grain is a guess.

Observation. An order-of-magnitude estimate fixes the scale of an answer, which is enough to catch most errors.

MATHEMATICAL CONCLUSION

The order of magnitude of a quantity is the power of 10 nearest to it, and estimating by orders of magnitude gives a value that is right in scale even when it is not right in detail.

Your own words

CHAPTER 2 · 2.4

Orders of magnitude and estimation

The order of magnitude of a quantity is the power of 10 nearest to it. Two quantities within a factor of 10 of each other share an order of magnitude.

Estimation by order of magnitude replaces every number by its leading digit times a power of 10, then works with the exponents. The answer is never exact, but it is always in the right scale.

An estimate is a check on a calculation. If a calculator answer is not within a factor of a few of the estimate, the calculation is wrong.

Rounding to one significant figure before calculating is the practical version of this idea, and it is the standard way to check a long computation.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 2 · 2.4 · WORKED EXAMPLE 1

Orders of magnitude and estimation

WORKED EXAMPLE 2.4.1

Problem. Estimate $4.9 \times 10^{7} \times 6.1 \times 10^{-3} \div (2.0 \times 10^{2})$ by order of magnitude, then compare with the calculator value.

quantity	rounded to one significant figure
4.9×10^{7}	5×10^{7}
6.1×10^{-3}	6×10^{-3}
2.0×10^{2}	2×10^{2}

Round the coefficients, then work with the exponents.

Mathematical reasoning. Rounding each quantity to one significant figure keeps the exponent exact and makes the arithmetic mental.

1. Rounded: $(5 \times 10^{7}) \times (6 \times 10^{-3}) \div (2 \times 10^{2})$.
2. Coefficients: $5 \times 6 \div 2 = 15$.
3. Exponents: $7 - 3 - 2 = 2$.
4. Estimate: $15 \times 10^{2} = 1.5 \times 10^{3}$.
5. Calculator: 1.4975×10^{3}, so the estimate is close.

Final answer. about 1.5×10^{3}, against a calculator value of 1.4975×10^{3}

Interpretation. The estimate caught no error this time, but it would have caught a decimal point slip of two places.

CHAPTER 2 · 2.4 · WORKED EXAMPLE 2

Orders of magnitude and estimation

WORKED EXAMPLE 2.4.2

Problem. A city of 2.4×10^5 people uses 1.8×10^{10} litres of water a year. Estimate the use per person.

quantity	value
population	2.4×10^5
water used	1.8×10^{10} litres

Division subtracts the exponents.

Mathematical reasoning. Dividing the total by the population gives the per-person value, and the exponents give the scale.

1. $(1.8 \times 10^{10}) \div (2.4 \times 10^5) = (1.8 \div 2.4) \times 10^5$.
2. $1.8 \div 2.4 = 0.75$, so the answer is 0.75×10^5.
3. $= 7.5 \times 10^4$ litres per person per year.

Final answer. about 7.5×10^4 litres per person per year

Interpretation. That is roughly 205 litres a day, which is a plausible order of magnitude for domestic use.

CHAPTER 2 · INVESTIGATION 2.5

Significant figures and precision

INQUIRY QUESTION

How many digits of a measurement actually carry meaning?

YOUR TASK

Take the measurements 0.0042 m, 4.20 m, 42 m, 4200 m and 0.042 m. For each one, state the number of significant figures and the interval of possible true values. Then multiply 4.2 by 3.65 and decide how many digits of the answer are meaningful.

measurement	significant figures	interval of true values
0.0042 m	2	[0.00415, 0.00425)
4.20 m	3	[4.195, 4.205)
42 m	2	[41.5, 42.5)
4200 m	2 or 4 — ambiguous without a note	[4150, 4250) if two
0.042 m	2	[0.0415, 0.0425)

CHAPTER 2 · INVESTIGATION 2.5

Significant figures and precision

What do you notice?

- Trailing zeros before a decimal point are ambiguous, which is why scientific notation is preferred.
- 0.0042 and 0.042 both have two significant figures, even though one is ten times larger.
- 4.20 has three significant figures and a narrower interval than 4.2.
- $4.2 \times 3.65 = 15.33$, but only two of those digits are meaningful.

> **Observation.** Significant figures count the digits that carry measured meaning, and the interval of true values gets narrower as the count rises.

MATHEMATICAL CONCLUSION

The number of significant figures records the precision of a measurement, and a calculated result cannot honestly be reported to more significant figures than the least precise value that went into it.

Your own words

CHAPTER 2 · 2.5

Significant figures and precision

Significant figures are the digits of a measurement that carry meaning, counted from the first non-zero digit. Leading zeros are placeholders and are never significant; trailing zeros after a decimal point always are.

The number of significant figures records the precision of the instrument. A length of 3.6 m claims the nearest 10 cm; 3.60 m claims the nearest centimetre.

Rounding to a given number of significant figures follows the usual rule, but the count starts at the first non-zero digit, which is why 0.00452 to three significant figures is 0.00452 and to two is 0.0045.

When measurements are combined, the result cannot be more precise than the least precise input. Reporting more digits than the data supports is a false claim of precision.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 2 · 2.5 · WORKED EXAMPLE 1

Significant figures and precision

WORKED EXAMPLE 2.5.1

Problem. Round 0.0040529 to three significant figures, and 604 512 to two.

number	significant figures	rounded
0.0040529	3	0.00405
604 512	2	$600\,000 = 6.0 \times 10^5$

Counting starts at the first non-zero digit.

Mathematical reasoning. The count begins at the first non-zero digit, so leading zeros are ignored and the decision digit is the one after the last significant figure.

1. 0.0040529: the first non-zero digit is 4, so the three significant figures are 4, 0, 5 and the next digit is 2.
2. $2 < 5$, so $0.0040529 \to 0.00405$ to three significant figures.
3. 604 512: the first two significant figures are 6 and 0, and the next digit is 4, which does not round up.
4. So $604\,512 \to 600\,000 = 6.0 \times 10^5$ to two significant figures.

Final answer. 0.00405 and 6.0×10^5

Interpretation. Writing 600 000 loses the claim about precision, so scientific notation is used to keep it.

CHAPTER 2 · 2.5 · WORKED EXAMPLE 2

Significant figures and precision

WORKED EXAMPLE 2.5.2

Problem. A rectangle measures 4.2 m by 3.65 m. Find the area, reporting it to a sensible precision.

side	measurement	significant figures
length	4.2 m	2
width	3.65 m	3

The area cannot be more precise than the least precise side.

Mathematical reasoning. Multiplying measurements combines their errors, so the result is reported to the precision of the least precise input.

1. Area = $4.2 \times 3.65 = 15.33$ m^2.
2. The least precise side has two significant figures, so report to two.
3. 15.33 m$^2 \to 15$ m^2 to two significant figures.

Final answer. 15 m² (to two significant figures)

Interpretation. Reporting 15.33 m^2 would claim a precision that neither measurement supports.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding

2.A1 FLUENCY

Simplify $4a^3 \times 5a^4$.

2.A2 FLUENCY

Evaluate 2^{-3}.

2.A3 REASONING

Write 3^0 and state why the definition is needed.

2.A4 FLUENCY

Simplify $(2x^2y)^3$.

2.A5 FLUENCY

Evaluate $(1/2)^{-2}$.

2.A6 FLUENCY

Simplify $12x^5y^3 \div 4x^2y$.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding (continued)

2.A7 FLUENCY

Write $10^{5} \div 10^{-3}$ as a single power of 10.

2.A8 REASONING

Explain why $(a + b)^{2}$ is not $a^{2} + b^{2}$, using index laws.

2.A9 FLUENCY

Simplify $(3^{2} \times 3^{5}) \div 3^{4}$.

2.A10 APPLICATION

A quantity doubles every hour, starting at 5 units. Write its size after n hours.

2.A11 FLUENCY

Simplify $x^{0} + x^{-1}$ for $x = 4$.

2.A12 REASONING

Explain the difference between $2x^{3}$ and $(2x)^{3}$.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns

2.B1 REASONING

Investigate: does $a^m \times b^m = (ab)^m$ hold for negative m? Test with a = 2, b = 3, m = −1.

2.B2 REASONING

Write $\sqrt{(x^8)}$ as a single power and state the condition on x.

2.B3 REASONING

Investigate the pattern 2^1, 2^0, 2^{-1}, 2^{-2}, 2^{-3} and generalise it.

2.B4 REASONING

Test the claim: a^(−1/2) = −√a. Find a counterexample or confirm it.

2.B5 REASONING

Investigate: for which values of n is $\sqrt{(x^n)}$ equal to x when x is negative?

2.B6 REASONING

Simplify 8^(2/3) + 27^(2/3) and describe the pattern you used.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns (continued)

2.B7 REASONING

Investigate: is (x^a)^b = (x^b)^a? Test with x = 2, a = 3, b = 5.

2.B8 REASONING

Generalise a rule for x^(1/n) × x^(1/m).

2.B9 REASONING

Investigate whether $x^0 = 1$ is true for x = 0, and report what you find.

2.B10 REASONING

Find a counterexample to 'a^m ÷ a^n = a^(m÷n)'.

2.B11 FLUENCY

Investigate the sequence 1, 4, 9, 16, 25 written as powers. State a rule.

CRITERION C Communicating

Criterion C practice · Communicating

2.C1 FLUENCY

Write 0.000000047 m in scientific notation.

2.C2 REASONING

Explain in words why the exponent in scientific notation can be negative.

2.C3 FLUENCY

Multiply (2.5×10^{6}) by (4×10^{-9}) and write the answer in scientific notation.

2.C4 REASONING

Explain the difference between 4×10^{3} and 4.0×10^{3} as a record of precision.

2.C5 FLUENCY

Divide (9×10^{12}) by (3×10^{-2}) and explain the step on the exponents.

2.C6 FLUENCY

A cell is 1.2×10^{-5} m long. Write this in ordinary form and explain the representation.

CRITERION C Communicating

Criterion C practice · Communicating (continued)

2.C7 REASONING

Explain why 6.3×10^6 is a better representation than 6 300 000 for a measurement.

2.C8 FLUENCY

Add (3×10^4) and (4×10^3) and write the answer in scientific notation.

2.C9 REASONING

Explain why 2.5×10^{-3} is not the same as 2.5×10^3, using the exponent.

2.C10 FLUENCY

Write the speed of light, 299 792 458 m/s, in scientific notation to three significant figures.

2.C11 REASONING

A mass is recorded as 5×10^{-7} kg. State its order of magnitude and explain what that means.

2.C12 REASONING

Explain how scientific notation makes it possible to compare the mass of an electron with the mass of the Earth.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts

2.D1 APPLICATION

Estimate the number of seconds in a school year of 180 days, 6 hours a day.

2.D2 APPLICATION

A city of 3×10^5 people each uses 200 litres of water a day. Estimate the daily total.

2.D3 APPLICATION

A factory makes 4.5×10^5 bottles a day. Estimate the yearly output.

2.D4 REASONING

Explain how an order-of-magnitude estimate can catch an error in a calculator answer.

2.D5 APPLICATION

A spacecraft is 2.4×10^8 km from Earth and travels at 4×10^4 km/h. Estimate the journey time.

2.D6 REASONING

A phone battery holds 3×10^3 mAh and a charger delivers 2×10^0 A. Estimate the charging time and comment on its sense.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts (continued)

2.D7 APPLICATION

A lake holds 8.2×10^9 litres and a town draws 1.5×10^7 litres a day. Estimate how many days the lake lasts.

2.D8 REASONING

Explain why reporting a measurement to more significant figures than the instrument allows is dishonest.

2.D9 APPLICATION

A field measures 42 m by 38 m, both to the nearest metre. Estimate the area and give the interval in which the true area must lie.

2.D10 APPLICATION

A vaccine dose is 5×10^{-4} litres. How many doses from 2×10^{-1} litres?

2.D11 REASONING

Two measurements are 3.6×10^4 and 2.4×10^5. Estimate their product and explain the order of magnitude of the answer.

2.D12 REASONING

Explain, with an example, when it is better to round before calculating rather than after.

CHAPTER 2 · APPLICATION AND ASSESSMENT

Applying chapter 2

These tasks use index laws, rational indices, scientific notation, orders of magnitude and significant figures. State the precision of every answer you report.

A1 · The water use audit

A school of 1.2×10^3 students uses 4.5×10^7 litres of water in a year. Estimate the use per student per day, report it to a sensible precision, and explain what the number of significant figures in your answer claims.

Criteria A and D · show the calculation in scientific notation and comment on precision.

A2 · The two scales

A virus is 1.2×10^{-7} m across and a football pitch is 1.05×10^2 m long. How many viruses fit across the pitch? Report your answer in scientific notation.

Criteria A and C · state the order of magnitude of your answer and what it means.

CHAPTER 2 · REVIEW

What you should now be able to do

Key ideas

- An index counts repeated multiplication, so multiplying powers adds indices.
- $a^0 = 1$ and $a^{-n} = 1/a^n$ are defined so that the exponent laws continue without a break.
- A root is a fractional index: $\sqrt[n]{a} = a^{(1/n)}$.
- Scientific notation $a \times 10^n$ with $1 \le a < 10$ makes the scale of a quantity visible.
- Multiplying or dividing in scientific notation adds or subtracts the exponents.
- The order of magnitude of a quantity is the nearest power of 10.
- Significant figures record precision, counted from the first non-zero digit.

Vocabulary

index · base · exponent law · zero index · negative index · fractional index · scientific notation · order of magnitude · significant figure · precision

CHAPTER 2 · REVIEW

Mixed practice · consolidate and extend

Do not use a calculator unless a question asks for one. Show a method where one is requested.

2.R1 Simplify $(2x^3)^2 \times x^{-1}$.

2.R2 Write 0.00000082 in scientific notation.

2.R3 Evaluate 27^(2/3).

2.R4 Round 0.007236 to three significant figures.

2.R5 Estimate $(6.1 \times 10^5) \times (2.9 \times 10^{-3})$.

2.R6 Explain why $2^0 = 1$ follows from the exponent laws.

CHAPTER 2 · REVIEW

Returning to the statement of inquiry

STATEMENT OF INQUIRY

The form of a number written in index form, and the equivalence between index form and root form, allow any quantity to be represented compactly and compared across scales, while the pattern in the exponent laws makes those representations safe to manipulate.

Reflect on the chapter

Answer in your own words, using evidence from this chapter.

- Write the statement of inquiry for this chapter in your own words.
- Which mathematical ideas from this chapter did you use to understand it? Name at least two.
- How did the key concept of form and the related concepts of quantity, representation and patterns appear in your work?
- How is the global context of scientific and technical innovation, and its exploration of precision, accuracy and measurement, visible in the investigations and examples you completed?
- Use evidence from an investigation or a worked example to justify your answer.
- Has your understanding of the statement of inquiry changed since the start of the chapter? Explain how.
- What are the limitations of a reported measurement, and what would you still like to ask?

CHAPTER 2 · PROJECTS, SERVICE AND ACTION

Acting on what you have learned

Innovation runs on measurement at very different scales. Use index form to keep the scales visible, and significant figures to keep the claims honest.

Service as Action

COMMUNITY NEED	Water and energy meters in the school are read and reported with inconsistent precision, so savings are hard to prove.
MATHEMATICAL CONNECTION	Scientific notation, orders of magnitude and significant figures.
PROPOSED ACTION	Agree a single reporting precision for the school's meter readings and publish a weekly estimate.
EXPECTED IMPACT	Savings become visible and checkable instead of disputed.
STUDENT REFLECTION	Record the precision you chose, why, and what the published figures changed.

Community Project

PROJECT IDEA	A one-week energy audit of the school.
NEED	The school does not know which activities use the most energy.
RESEARCH QUESTION	Where does the school's energy actually go, and what would save the most?
MATHEMATICS	Powers, scientific notation, estimation and proportional reasoning.
ACTION OR OUTCOME	A ranked list of savings with estimated daily impact.
BENEFICIARIES	The whole school community.
EVIDENCE	Meter readings, the estimates and the ranked list.

Personal Project connection · MYP 5

PERSONAL INTEREST	Photography, music or fitness tracking.
LEARNING GOAL	To use index form and precision language to describe a measurement you care about.
PRODUCT OR OUTCOME	A one-page data sheet for a personal measurement.
MATHEMATICS	Scientific notation, significant figures and estimation.
PROCESS	Measure, convert, write in scientific notation, then state the precision.
EVIDENCE	The data sheet and the stated uncertainty.

CHAPTER 3 · PART II · ALGEBRA AND MODELS

03

Functions, expressions and algebraic reasoning

A letter stands for a quantity that changes. A function is the relationship that gives each input one output, and equivalent forms of its rule reveal different things about that relationship.

ENGAGE

A phone plan costs 12 dollars a month plus 0.05 dollars a minute. For one month the bill depends entirely on the number of minutes used. What is the relationship, and how can it be written once for every month?

Framing the inquiry

Key concept: Relationships
Related concepts: Generalisation · Equivalence
Global context: Identities and relationships
Global context exploration: Relationships and communities

Statement of inquiry: The relationship between two quantities can be generalised as a function, and the equivalence of the different algebraic forms of that relationship allows it to be rearranged, compared and used to predict.

Questions to guide us

- **Factual:** What is a function, what are its domain and range, and how do you expand and factorise a quadratic expression?
- **Conceptual:** Why do equivalent forms of the same expression reveal different information about the relationship they describe?
- **Debatable:** Is a formula that has been rearranged a different formula, or the same formula in a different form — and does the difference matter?

ATL · THINKING. In order for a student to select an algebraic strategy for an unfamiliar expression, students must identify which form the problem is already in, decide which equivalent form the answer needs, and justify the choice.

CHAPTER 3 · INVESTIGATION 3.1

Variables, expressions and substitution

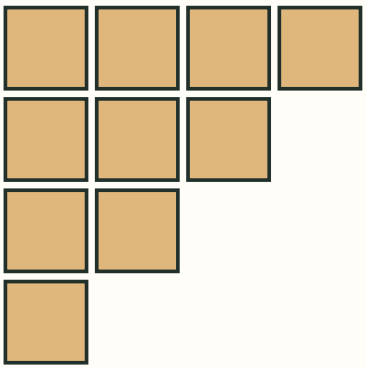
INQUIRY QUESTION

How can a changing quantity be represented by a letter, and what does substitution actually do?

YOUR TASK

A pattern of tiles has 1 tile in position 1, 4 in position 2, 7 in position 3 and 10 in position 4. Write the number of tiles as an expression in the position number n, then test it for n = 10 and n = 100. Then find a second, different expression that gives the same values.

position 1, 2, 3, 4



position n	tiles	3n − 2	check
1	1	1	3 × 1 − 2 = 1
2	4	4	3 × 2 − 2 = 4
3	7	7	3 × 3 − 2 = 7
4	10	10	3 × 4 − 2 = 10
10	28	28	3 × 10 − 2 = 28
100	298	298	3 × 100 − 2 = 298

CHAPTER 3 · INVESTIGATION 3.1

Variables, expressions and substitution

What do you notice?

- The number of tiles rises by 3 each time, so the expression contains 3n.
- Subtracting 2 makes the expression work from position 1 rather than position 0.
- The expression 1 + 3(n − 1) gives the same values, and it expands to 3n − 2.
- Both forms predict the same number of tiles at any position.

Observation. A pattern that grows by a constant amount is described by a linear expression, and several equivalent expressions can describe it.

MATHEMATICAL CONCLUSION

A generalisation of a growing pattern is an expression that works for every position, and equivalent expressions are different forms of the same generalisation.

Your own words

CHAPTER 3 · 3.1

Variables, expressions and substitution

A variable is a letter standing for a number that can change. It is not an unknown to be found: it is a quantity that takes different values in different situations.

An expression is a recipe built from variables, numbers and operations. 3n + 5 is a recipe that takes a number n, triples it and adds 5. The order of operations fixes what the recipe does.

Substitution replaces a variable by a value and evaluates. It is the point at which a general relationship becomes a specific number.

Two expressions that give the same value for every substitution are equivalent. Testing a few values suggests equivalence but never proves it; expansion or factorisation proves it.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

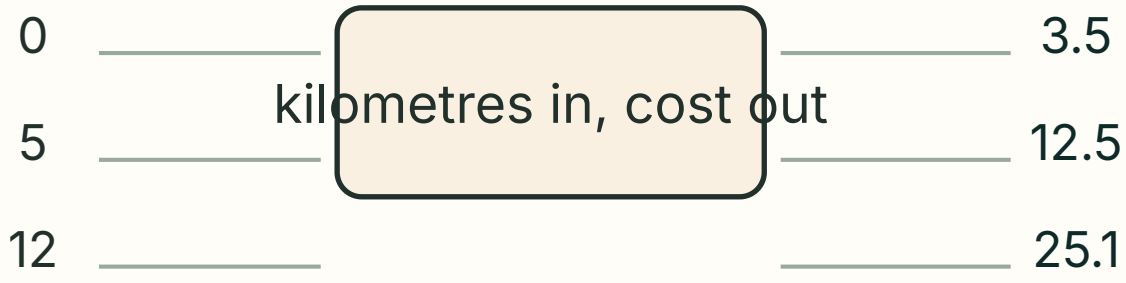
CHAPTER 3 · 3.1 · WORKED EXAMPLE 1

Variables, expressions and substitution

WORKED EXAMPLE 3.1.1

Problem. A taxi charges 3.50 dollars plus 1.80 dollars per kilometre. Write the cost as an expression and find the cost of a 12 km trip.

$$\text{cost} = 1.8k + 3.5$$



Mathematical reasoning. The cost has a fixed part and a variable part, so the expression is a linear recipe in the number of kilometres.

1. Let k be the number of kilometres.
2. Cost = 3.50 + 1.80k.
3. For k = 12: cost = 3.50 + 1.80 × 12 = 3.50 + 21.60.
4. = 25.10 dollars.

Final answer. C = 3.50 + 1.80k; a 12 km trip costs 25.10 dollars

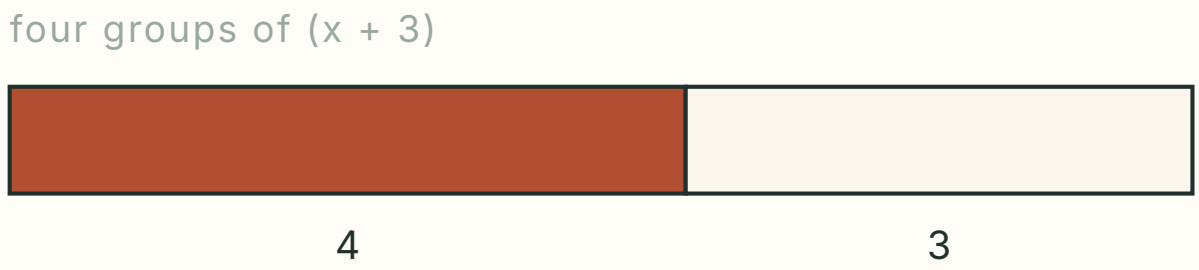
Interpretation. The expression works for every trip, not just this one, which is why it is worth writing.

CHAPTER 3 · 3.1 · WORKED EXAMPLE 2

Variables, expressions and substitution

WORKED EXAMPLE 3.1.2

Problem. Show that 4(x + 3) and 4x + 12 are equivalent.



Mathematical reasoning. Expanding the bracket shows the two forms have the same value for every x, which is what equivalence means.

1. 4(x + 3) means four copies of (x + 3).
2. Four copies of x give 4x, and four copies of 3 give 12.
3. So 4(x + 3) = 4x + 12.
4. Check with x = 2: 4 × 5 = 20 and 8 + 12 = 20.

Final answer. 4(x + 3) = 4x + 12 for every value of x

Interpretation. The expanded form is easier to evaluate; the factorised form is easier to read. Equivalence lets us choose.

CHAPTER 3 · INVESTIGATION 3.2

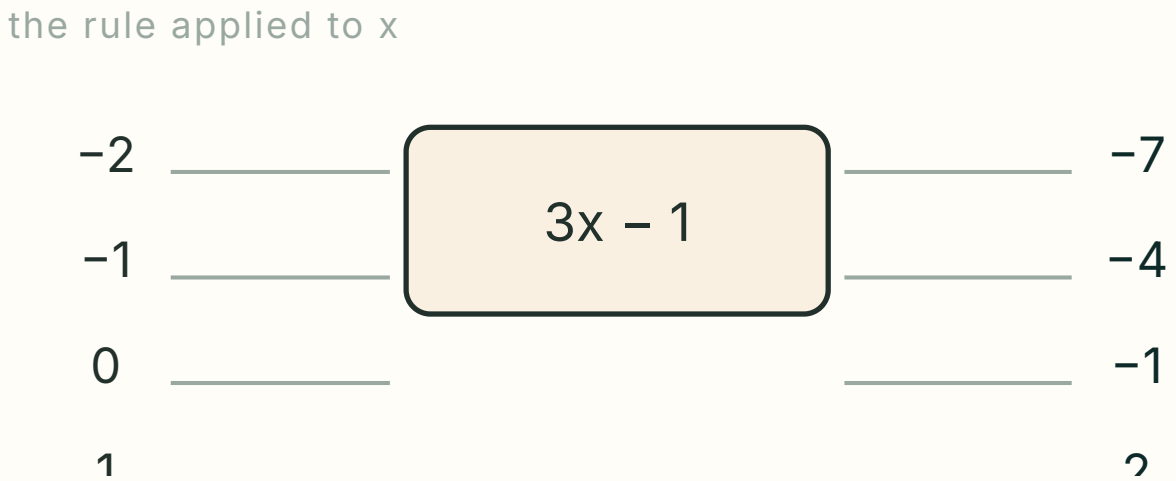
Functions, domain and range

INQUIRY QUESTION

How can a rule that gives each input exactly one output be represented, and what are its domain and range?

YOUR TASK

Take the rule 'multiply by 3, then subtract 1'. Build a table of values for inputs −2, −1, 0, 1, 2, 3 and 4, draw the mapping diagram, plot the points on a grid, and write the formula. Then change the rule to 'multiply by 3, then subtract a number of your choice' and repeat.



input x	3x	3x − 1	point on the graph
−2	−6	−7	(−2, −7)
−1	−3	−4	(−1, −4)
0	0	−1	(0, −1)
1	3	2	(1, 2)
2	6	5	(2, 5)
3	9	8	(3, 8)
4	12	11	(4, 11)

CHAPTER 3 · INVESTIGATION 3.2

Functions, domain and range

What do you notice?

- The outputs rise by 3 each time the input rises by 1.
- The points lie on a straight line, which is what a linear function looks like.
- Changing the subtracted number moves the line up or down but does not change its steepness.
- Every input gave exactly one output, so the rule is a function.

Observation. A rule that gives one output per input is a function, and a linear function gives outputs that change by a constant amount.

MATHEMATICAL CONCLUSION

A function assigns exactly one output to each permitted input; its domain is the set of permitted inputs and its range the set of outputs, and both are choices the modeller makes.

Your own words

CHAPTER 3 · 3.2

Functions, domain and range

A function is a rule that gives exactly one output for each permitted input. The input is the independent variable and the output is the dependent variable.

Function notation f(x) names the rule and the input together: f(x) = 2x + 1 means the rule is double and add one. f(3) = 7 is the value of the function at x = 3.

The domain is the set of permitted inputs and the range is the set of outputs the rule can produce. The domain is a modelling choice, not a mathematical fact: a rule can be defined for all real numbers but only make sense for positive ones.

A function can be represented as a mapping diagram, a table of values, a graph or a formula. The four representations describe the same relationship, and each one answers a different question about it.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

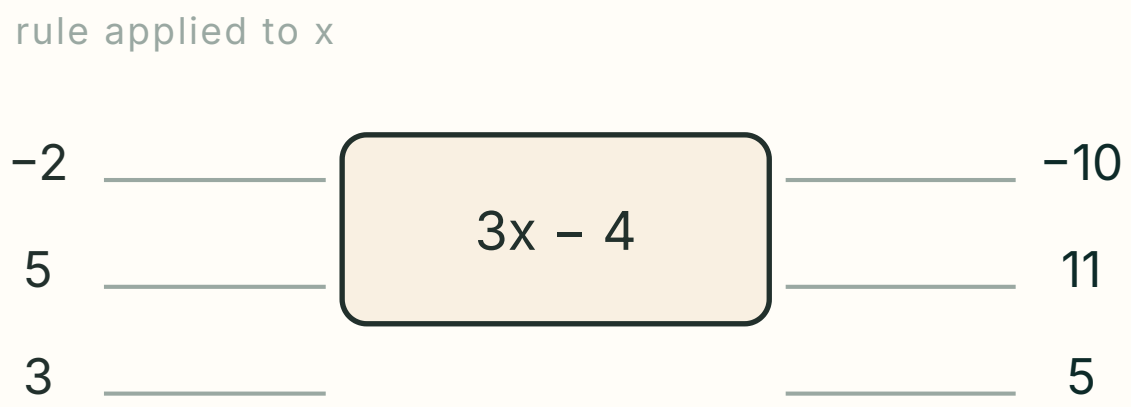
In your own words

CHAPTER 3 · 3.2 · WORKED EXAMPLE 1

Functions, domain and range

WORKED EXAMPLE 3.2.1

Problem. f(x) = 3x − 4. Find f(5), f(−2) and the value of x for which f(x) = 11.



Mathematical reasoning. Substitution evaluates the function, and solving a linear equation reverses it.

1. f(5) = 3 × 5 − 4 = 11.
2. f(−2) = 3 × (−2) − 4 = −10.
3. 3x − 4 = 11 gives 3x = 15, so x = 5.

Final answer. f(5) = 11, f(−2) = −10, and f(x) = 11 when x = 5

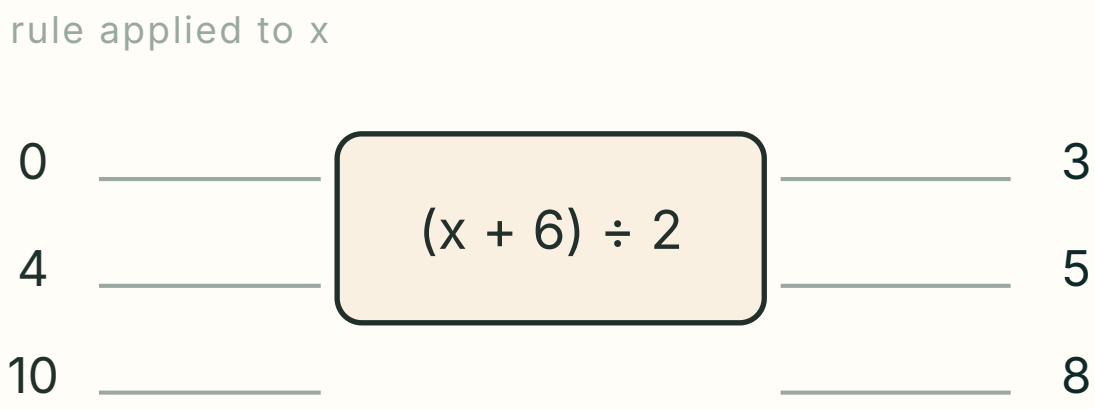
Interpretation. Evaluating and solving are opposite operations on the same relationship.

CHAPTER 3 · 3.2 · WORKED EXAMPLE 2

Functions, domain and range

WORKED EXAMPLE 3.2.2

Problem. A function is defined by the rule 'add 6 then divide by 2'. Write it as a formula, state a sensible domain, and find its range.



Mathematical reasoning. Translating a verbal rule into a formula makes the domain and range explicit.

1. $g(x) = (x + 6)/2$.
2. The formula is defined for every real x, so the domain is all real numbers.
3. Every real y can be produced, because $x = 2y - 6$ is real for any y.
4. So the range is also all real numbers.

Final answer. g(x) = (x + 6)/2, domain and range both all real numbers

Interpretation. A formula with no division by a variable and no root usually has the widest possible domain.

CHAPTER 3 · INVESTIGATION 3.3

Expanding and factorising

INQUIRY QUESTION

How can equivalent forms of the same expression reveal different information about it?

YOUR TASK

Expand $(x + 2)(x + 3)$, $(x + 2)(x + 5)$, $(x + 4)(x + 5)$ and $(x + 2)(x - 3)$. Record the coefficient of x and the constant term in each case, and look for a pattern that connects them to the numbers inside the brackets.

product	expanded	coefficient of x	constant
$(x + 2)(x + 3)$	$x^2 + 5x + 6$	5	6
$(x + 2)(x + 5)$	$x^2 + 7x + 10$	7	10
$(x + 4)(x + 5)$	$x^2 + 9x + 20$	9	20
$(x + 2)(x - 3)$	$x^2 - x - 6$	-1	-6

CHAPTER 3 · INVESTIGATION 3.3

Expanding and factorising

What do you notice?

- The coefficient of x is always the sum of the two numbers in the brackets.
- The constant term is always the product of those two numbers.
- A negative number in a bracket can give a negative coefficient or constant.
- The pattern works in reverse, which is exactly what factorising does.

> **Observation.** The expanded form of a product of two linear factors has coefficients that are the sum and product of the bracket numbers.

MATHEMATICAL CONCLUSION

$(x + a)(x + b) = x^2 + (a + b)x + ab$, so the expanded and factorised forms carry the same information in different arrangements, and factorising reads the pattern backwards.

Your own words

CHAPTER 3 · 3.3

Expanding and factorising

Expanding removes brackets by distributing: $a(b + c) = ab + ac$. It is the form that is easiest to evaluate and to collect like terms in.

Factorising is the reverse: it puts a common factor or a product of factors back in. The factorised form makes zeros and roots visible, which the expanded form hides.

Three products are worth knowing by sight: $(x + a)(x + b) = x^2 + (a + b)x + ab$, $(a + b)^2 = a^2 + 2ab + b^2$ and $a^2 - b^2 = (a - b)(a + b)$.

The difference of two squares is the most useful of the three, because it converts an addition into a product and is used again in Chapter 1 to rationalise a denominator.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 3 · 3.3 · WORKED EXAMPLE 1

Expanding and factorising

WORKED EXAMPLE 3.3.1

Problem. Expand and simplify $(x + 5)(x - 3)$, then factorise $x^2 + 7x + 10$.

form	expression	what it shows
factorised	$(x + 5)(x - 3)$	the two zeros
expanded	$x^2 + 2x - 15$	the coefficients
factorised	$(x + 2)(x + 5)$	the same zeros again

The same relationship, written in two forms.

Mathematical reasoning. Expanding distributes each term of one bracket over the other, and factorising looks for two numbers with the right sum and product.

1. $(x + 5)(x - 3) = x^2 - 3x + 5x - 15 = x^2 + 2x - 15$.
2. For $x^2 + 7x + 10$, find two numbers with product 10 and sum 7: 2 and 5.
3. $x^2 + 7x + 10 = (x + 2)(x + 5)$.

Final answer. $(x + 5)(x - 3) = x^2 + 2x - 15$ and $x^2 + 7x + 10 = (x + 2)(x + 5)$

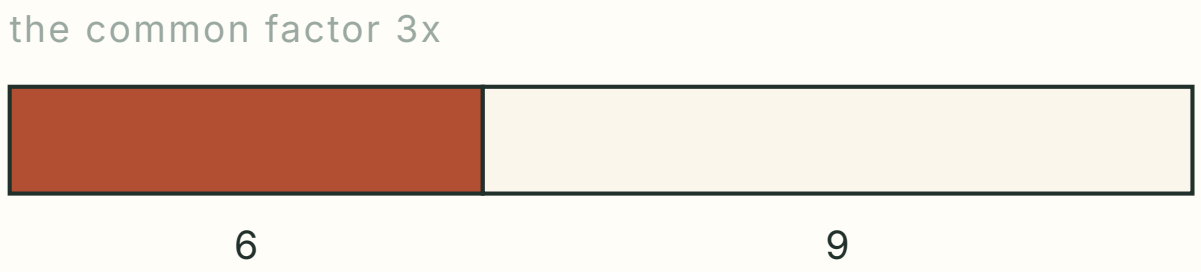
Interpretation. The expanded form shows the shape of the quadratic; the factorised form shows where it equals zero.

CHAPTER 3 · 3.3 · WORKED EXAMPLE 2

Expanding and factorising

WORKED EXAMPLE 3.3.2

Problem. Factorise $6x^2 + 9x$ completely, and use the result to solve $6x^2 + 9x = 0$.



Mathematical reasoning. Taking out the highest common factor leaves a product, and a product equals zero only when one of its factors does.

1. $6x^2 + 9x = 3x(2x + 3)$.
2. So $3x(2x + 3) = 0$.
3. Either $3x = 0$, giving $x = 0$, or $2x + 3 = 0$, giving $x = -3/2$.

Final answer. $6x^2 + 9x = 3x(2x + 3)$; the solutions are $x = 0$ and $x = -3/2$

Interpretation. The factorised form solved an equation that the expanded form did not make easy.

CHAPTER 3 · INVESTIGATION 3.4

Rearranging formulas

INQUIRY QUESTION

How can a formula be rewritten so that a different quantity is the subject?

YOUR TASK

Take the relationship $d = 5t + 20$, where d is a distance in metres and t is a time in seconds. Make t the subject, then find the time when $d = 100$. Then make a formula of your own with two letters and rearrange it the other way round.

form	expression	what it answers
original	$d = 5t + 20$	how far in a given time
rearranged	$t = (d - 20)/5$	how long for a given distance
evaluated	$t = (100 - 20)/5 = 16$	the time for 100 m

CHAPTER 3 · INVESTIGATION 3.4

Rearranging formulas

What do you notice?

- Rearranging did not change the relationship; it changed the question.
- Every step was the same operation on both sides.
- The constant 20 was removed by subtraction, which is the inverse of addition.
- The rearranged form is easier when the distance is known and the time is not.

Observation. Rearranging applies inverse operations to both sides, so the relationship is preserved while the subject changes.

MATHEMATICAL CONCLUSION

A formula can be rearranged into any subject by applying inverse operations to both sides in the reverse of the original order, and the rearranged formula is the same relationship answering a different question.

Your own words

CHAPTER 3 · 3.4

Rearranging formulas

Rearranging a formula means applying the same operation to both sides until the wanted letter is isolated. Every step must keep the statement true.

The operations are undone in the reverse of the order they were applied. To undo 'multiply by 3 then add 5', subtract 5 then divide by 3.

A formula rearranged into a different subject is not a different formula. It is the same relationship in a form that answers a different question.

Rearranging is the algebraic version of changing perspective: the same relationship, read from the other end.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 3 · 3.4 · WORKED EXAMPLE 1

Rearranging formulas

WORKED EXAMPLE 3.4.1

Problem. The formula for the area of a trapezium is $A = \frac{1}{2}(a + b)h$. Make h the subject.

step	operation	result
$A = \frac{1}{2}(a + b)h$	multiply by 2	$2A = (a + b)h$
$2A = (a + b)h$	divide by $(a + b)$	$h = 2A/(a + b)$

Undo the operations in the reverse order.

Mathematical reasoning. The height is multiplied by $(a + b)$ and by $\frac{1}{2}$, so both are undone by reversing the order.

1. $A = \frac{1}{2}(a + b)h$.
2. Multiply both sides by 2: $2A = (a + b)h$.
3. Divide both sides by $(a + b)$: $h = 2A/(a + b)$.

Final answer. $h = 2A/(a + b)$

Interpretation. The rearranged formula answers 'how high must it be' instead of 'what is the area', using the same relationship.

CHAPTER 3 · 3.4 · WORKED EXAMPLE 2

Rearranging formulas

WORKED EXAMPLE 3.4.2

Problem. Make x the subject of $y = (3x + 2)/(x - 1)$.

step	operation	result
$y = (3x + 2)/(x - 1)$	multiply by $(x - 1)$	$y(x - 1) = 3x + 2$
$yx - y = 3x + 2$	collect x terms	$yx - 3x = y + 2$
$x(y - 3) = y + 2$	divide by $(y - 3)$	$x = (y + 2)/(y - 3)$

When the wanted letter appears twice, collect it on one side.

Mathematical reasoning. Clearing the denominator first, then collecting every term containing x, isolates the wanted letter by factorising.

1. $y(x - 1) = 3x + 2$.
2. $yx - y = 3x + 2$.
3. $yx - 3x = y + 2$.
4. $x(y - 3) = y + 2$.
5. $x = (y + 2)/(y - 3)$, for $y \neq 3$.

Final answer. $x = (y + 2)/(y - 3)$

Interpretation. The restriction $y \neq 3$ appears because x would then be undefined, which the original formula also hides.

CHAPTER 3 · INVESTIGATION 3.5

Sequences, patterns and generalisation

INQUIRY QUESTION

How can a pattern in a sequence be written as a rule that works for any term?

YOUR TASK

Build three sequences from patterns of your own: one with a constant difference, one with a constant second difference, and one that is neither. For each, find the nth term and test it on two terms that were not in your original list.

sequence	type	nth term	test
3, 7, 11, 15, 19	linear	$4n - 1$	$n = 6$ gives 23
1, 4, 9, 16, 25	quadratic	n^2	$n = 7$ gives 49
1, 2, 4, 7, 11	neither	differences 1, 2, 3, 4	$n = 6$ gives 16 by the difference pattern

CHAPTER 3 · INVESTIGATION 3.5

Sequences, patterns and generalisation

What do you notice?

- The linear sequence had a constant difference of 4.
- The quadratic sequence had a constant second difference of 2.
- The third sequence had differences that grew by 1 each time, so it is not linear or quadratic.
- Each nth term predicted a term that was not in the original list.

Observation. The pattern of first and second differences identifies the form of the nth term.

MATHEMATICAL CONCLUSION

A generalisation is a rule that works for every position, and it is identified by the pattern of differences: a constant first difference gives a linear rule and a constant second difference gives a quadratic one.

Your own words

CHAPTER 3 · 3.5

Sequences, patterns and generalisation

A sequence is an ordered list of numbers. The term-to-term rule says how to get from one term to the next; the position-to-term rule gives the term in any position directly.

A linear sequence has a constant difference, so its nth term is $an + b$. The difference gives a and substituting one term gives b.

A quadratic sequence has a constant second difference, so its nth term contains n^2. The second difference is twice the coefficient of n^2.

A generalisation is a claim about every term, not just the ones listed. Checking the first five terms supports a generalisation; it does not prove it, and a counterexample at the sixth term destroys it.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

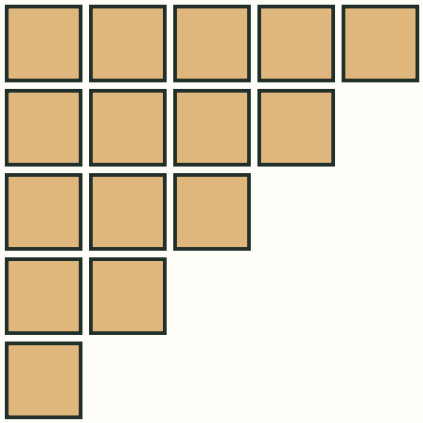
CHAPTER 3 · 3.5 · WORKED EXAMPLE 1

Sequences, patterns and generalisation

WORKED EXAMPLE 3.5.1

Problem. Find the nth term of the sequence 7, 11, 15, 19, 23, … and use it to find the 50th term.

7, 11, 15, 19, 23



Mathematical reasoning. A constant difference means the rule is linear, so the nth term is an + b.

1. The common difference is 4, so a = 4.
2. The nth term is 4n + b.
3. The first term is 7, so 4 × 1 + b = 7, giving b = 3.
4. The nth term is 4n + 3, and the 50th term is 4 × 50 + 3 = 203.

Final answer. the nth term is 4n + 3; the 50th term is 203

Interpretation. The position-to-term rule gives any term without listing all the previous ones.

CHAPTER 3 · 3.5 · WORKED EXAMPLE 2

Sequences, patterns and generalisation

WORKED EXAMPLE 3.5.2

Problem. The nth term of a sequence is $n^2 - 2n$. Find the first four terms and the second difference.

n	n^2	$-2n$	term
1	1	−2	−1
2	4	−4	0
3	9	−6	3
4	16	−8	8

A quadratic rule gives a constant second difference.

Mathematical reasoning. Substituting each position gives the terms, and the constant second difference confirms the quadratic form.

1. $n = 1$: $1 - 2 = -1$.
2. $n = 2$: $4 - 4 = 0$.
3. $n = 3$: $9 - 6 = 3$.
4. $n = 4$: $16 - 8 = 8$.
5. First differences: 1, 3, 5; second difference: 2.

Final answer. the first four terms are −1, 0, 3, 8 and the second difference is 2

Interpretation. The second difference of 2 matches the coefficient of n^2, which is the signature of a quadratic.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding

3.A1 FLUENCY

If f(x) = 4x − 7, find f(3) and f(−1).

3.A2 FLUENCY

Expand 5(2x − 3).

3.A3 FLUENCY

Factorise 12x + 18.

3.A4 FLUENCY

Expand and simplify (x + 4)(x − 2).

3.A5 FLUENCY

Find the nth term of 5, 8, 11, 14, ...

3.A6 REASONING

State the domain of f(x) = 1/(x − 2).

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding (continued)

3.A7 FLUENCY

Simplify $3(x + 2) + 2(x - 5)$.

3.A8 FLUENCY

Factorise $x^2 - 9$.

3.A9 FLUENCY

If $g(x) = x^2 - 3x$, find $g(4)$.

3.A10 FLUENCY

A sequence has nth term $2n^2 - 1$. Find its first three terms.

3.A11 FLUENCY

Make x the subject of $y = 4x + 9$.

3.A12 REASONING

Explain the difference between a term-to-term rule and a position-to-term rule.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns

3.B1 REASONING

Investigate: does the sequence 2, 4, 8, 16, 32 have a constant difference or second difference?

3.B2 REASONING

Test the claim: every sequence with a constant second difference is quadratic.

3.B3 FLUENCY

Investigate the sequence 1, 4, 9, 16, 25. Find the nth term and the next term.

3.B4 REASONING

Generalise: what is the nth term of 1, 8, 27, 64?

3.B5 REASONING

Investigate whether $3n + 1$ and $4n + 1$ ever give the same value.

3.B6 REASONING

Test the claim: the product of three consecutive integers is always divisible by 6.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns (continued)

3.B7 REASONING

Investigate the pattern of triangular numbers 1, 3, 6, 10, 15 and find the nth term.

3.B8 REASONING

Find a counterexample to 'a sequence with a constant difference has a positive nth term'.

3.B9 REASONING

Investigate: is $(x + y)^2$ ever equal to $x^2 + y^2$?

3.B10 REASONING

Generalise the pattern 1×1, 2×3, 3×5, 4×7 into an nth-term rule.

3.B11 REASONING

Investigate whether the nth term of 4, 9, 16, 25 is linear.

CRITERION C Communicating

Criterion C practice · Communicating

3.C1 REASONING

Explain in words why $2(x + 3)$ and $2x + 6$ are the same expression.

3.C2 REASONING

Explain the difference between an equation and an expression, with an example of each.

3.C3 REASONING

Write $f(x) = 2x + 5$ in words, and explain what $f(0)$ means.

3.C4 REASONING

Explain why a table of values and a formula can describe the same function.

3.C5 REASONING

Explain why $(x + 3)^2$ is not $x^2 + 9$.

3.C6 REASONING

A formula is written as $C = 2\pi r$. Explain what rearranging it to $r = C/(2\pi)$ changes.

CRITERION C Communicating

Criterion C practice · Communicating (continued)

3.C7 REASONING
Explain why the domain of 'height of a plant after t days' is not all real numbers.

3.C8 REASONING
Describe two representations of the function $f(x) = x^2$ and say what each one shows best.

3.C9 REASONING
Explain why factorising is useful for solving an equation.

3.C10 REASONING
Explain the difference between a conjecture and a generalisation.

3.C11 REASONING
Write the rule 'subtract 4, then square' as a formula and explain the order of operations.

3.C12 REASONING
Explain why the range of $f(x) = x^2$ is not all real numbers.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts

3.D1 APPLICATION

A plumber charges 40 dollars plus 55 dollars an hour. Write the cost expression and find the cost of a 4.5-hour job.

3.D2 APPLICATION

A phone plan costs 15 dollars a month plus 0.04 dollars a minute. How many minutes can be used for a 30-dollar bill?

3.D3 APPLICATION

A taxi fare is 3 dollars plus 1.5 dollars per kilometre. Two riders travel 9 km and 14 km. Find the difference in their fares.

3.D4 APPLICATION

A rectangular garden has length $(x + 3)$ m and width $(x - 1)$ m. Find the area in expanded form and its value when $x = 6$.

3.D5 APPLICATION

A community group pays 80 dollars to hire a hall and 6 dollars per person for food. Write the total cost for n people and find the cost per person.

3.D6 APPLICATION

A water tank fills at 12 litres a minute from an initial 50 litres. Write a formula for the volume after t minutes and find when it reaches 350 litres.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts (continued)

3.D7 APPLICATION

A square has side (2x + 1). Find its area and its perimeter, then find x if the perimeter is 30 cm.

3.D8 APPLICATION

A delivery charge is 5 dollars plus 0.80 dollars per kilogram. Find the weight that gives a total charge of 21 dollars.

3.D9 REASONING

Explain whether a service that charges 20 dollars plus 2 dollars per item is ever cheaper than one that charges 4 dollars per item.

3.D10 APPLICATION

A pattern of tiles has 2n + 3 tiles in row n. How many tiles are in rows 1 to 5 altogether?

3.D11 APPLICATION

A savings account pays 5% simple interest on P dollars for t years. Write the formula, make P the subject, and find the amount needed for 100 dollars of interest in 2 years.

3.D12 REASONING

Explain why a linear model of a phone bill is only sensible for a limited range of minutes.

CHAPTER 3 · APPLICATION AND ASSESSMENT

Applying chapter 3

These tasks use substitution, function notation, expansion, factorisation, rearranging formulas and sequences. State the form of your answer and why you chose it.

A1 · The community event budget

A community event has a fixed cost of 250 dollars and a cost of 8 dollars per attendee. Write the total cost as a formula, make the number of attendees the subject, and find how many can attend if the budget is 1 000 dollars.

Criteria A and C · show the rearrangement and state the form of your answer.

A2 · The tile pattern

A courtyard is tiled with a pattern in which row n has 3n + 2 tiles. Find the number of tiles in row 20, the total for the first 20 rows, and explain whether the pattern can continue indefinitely.

Criteria A and D · justify your total and comment on the limits of the model.

CHAPTER 3 · REVIEW

What you should now be able to do

Key ideas

- A variable stands for a changing quantity, and an expression is a recipe built from variables.
- A function gives exactly one output for each permitted input.
- The domain is the set of permitted inputs and the range is the set of outputs.
- Expanding and factorising produce equivalent forms of the same expression.
- A difference of two squares factorises: $a^2 - b^2 = (a - b)(a + b)$.
- Rearranging applies inverse operations to both sides, preserving the relationship.
- A constant first difference gives a linear rule; a constant second difference gives a quadratic rule.

Vocabulary

variable · expression · substitution · function · domain · range · function notation · expand · factorise · subject of a formula · sequence · nth term · generalisation

CHAPTER 3 · REVIEW

Mixed practice · consolidate and extend

Do not use a calculator unless a question asks for one. Show a method where one is requested.

3.R1 Expand and simplify $(x - 3)(x + 7)$.

3.R2 Factorise $x^2 - 16$.

3.R3 Find the nth term of 6, 10, 14, 18, ...

3.R4 Make t the subject of $v = u + at$.

3.R5 If $h(x) = x^2 - 4x$, find $h(6)$.

3.R6 Explain why a function cannot give two outputs for one input.

CHAPTER 3 · REVIEW

Returning to the statement of inquiry

> **STATEMENT OF INQUIRY**
>
> The relationship between two quantities can be generalised as a function, and the equivalence of the different algebraic forms of that relationship allows it to be rearranged, compared and used to predict.

Reflect on the chapter

Answer in your own words, using evidence from this chapter.

- Write the statement of inquiry for this chapter in your own words.
- Which mathematical ideas from this chapter did you use to understand it? Name at least two.
- How did the key concept of relationships and the related concepts of generalisation and equivalence appear in your work?
- How is the global context of identities and relationships, and its exploration of relationships and communities, visible in the investigations and examples you completed?
- Use evidence from an investigation or a worked example to justify your answer.
- Has your understanding of the statement of inquiry changed since the start of the chapter? Explain how.
- What are the limitations of a linear model of a real relationship, and what would you still like to ask?

CHAPTER 3 · PROJECTS, SERVICE AND ACTION

Acting on what you have learned

Relationships between quantities govern budgets, plans and schedules. Use algebra to make a relationship explicit enough to act on.

Service as Action

COMMUNITY NEED	Community groups plan events without a shared way to estimate costs, so budgets are guessed.
MATHEMATICAL CONNECTION	Linear functions, substitution and rearranging formulas.
PROPOSED ACTION	Build a simple cost model for one community event and publish it as a reusable formula.
EXPECTED IMPACT	The group can plan within a known budget instead of discovering the cost afterwards.
STUDENT REFLECTION	Record the formula, the assumptions it makes, and where they failed.

Community Project

PROJECT IDEA	A cost model for a school fair stall.
NEED	Stall holders set prices without knowing their break-even point.
RESEARCH QUESTION	What price covers the costs of a stall, and what does a profit of a given size require?
MATHEMATICS	Linear functions, substitution and rearranging formulas.
ACTION OR OUTCOME	A one-page cost model and a recommended price.
BENEFICIARIES	Stall holders and the groups they raise money for.
EVIDENCE	The model, the calculation and the recommended price.

Personal Project connection · MYP 5

PERSONAL INTEREST	Gaming, sport or music practice.
LEARNING GOAL	To model a personal routine with a formula and use it to make a decision.
PRODUCT OR OUTCOME	A training or practice plan with a cost or time model.
MATHEMATICS	Linear functions, sequences and substitution.
PROCESS	Record data, fit a linear rule, then use it to predict.
EVIDENCE	The data, the formula and the prediction.

CHAPTER 4 · PART II · ALGEBRA AND MODELS

04

Linear relationships and graphs

Two quantities that change together at a constant rate produce a straight line. The gradient is the rate of change and the intercept is where the line starts.

ENGAGE

A village water point is filled by two pumps. Pump A delivers 40 litres a minute and pump B delivers 55 litres a minute, starting 5 minutes later. Both graphs are straight lines. When does B overtake A, and what does the crossing point mean?

Framing the inquiry

Key concept: Relationships
Related concepts: Change · Representation
Global context: Fairness and development
Global context exploration: Distribution of resources

Statement of inquiry: The relationship between two quantities that change at a constant rate can be represented as a straight line, and the equivalence between a table, a graph and an equation allows that relationship to be used to predict — and the prediction to be judged.

Questions to guide us

- **Factual:** How do you find the gradient of a line from two points, and how do you write its equation?
- **Conceptual:** Why does a constant rate of change always produce a straight-line graph, and what does the y-intercept represent in a real situation?
- **Debatable:** Is a prediction made by extending a straight-line graph beyond the collected data a mathematical fact or an assumption?

ATL · COMMUNICATION. In order for a student to explain a linear relationship to an audience, students must choose a representation that suits that audience, label the axes and the units, and justify the choice of representation in the explanation.

CHAPTER 4 · INVESTIGATION 4.1

From tables to straight lines

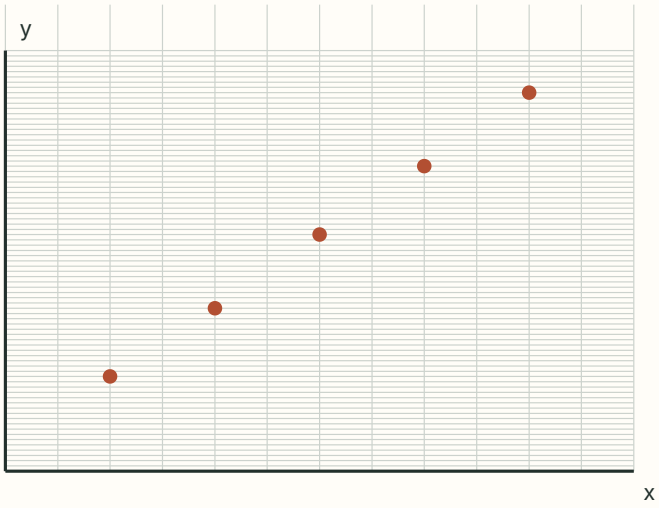
INQUIRY QUESTION

How can a table of paired values show that a relationship is linear, and what does the graph add?

YOUR TASK

Measure how far a toy car travels when it is released from ramps of different heights. Record five pairs of values, plot them, and decide whether the relationship is linear. Then use your graph to predict the distance for a ramp height you did not test.

ramp height against distance



ramp height (cm)	distance (cm)	first difference
2	18	—
4	31	13
6	45	14
8	58	13
10	72	14

CHAPTER 4 · INVESTIGATION 4.1

From tables to straight lines

What do you notice?

- The first differences are all close to 13 or 14 cm.
- The points nearly lie on a straight line, so the relationship is approximately linear.
- Small departures from the line come from measurement error, not from a different rule.
- Extending the line predicts about 86 cm for a 12 cm ramp.

> **Observation.** A roughly constant difference gives points that nearly lie on a straight line, which is what real measurement data looks like.

MATHEMATICAL CONCLUSION

A relationship is linear when equal changes in the input give equal changes in the output, and the graph of such a relationship is a straight line whose gradient is that equal change.

Your own words

CHAPTER 4 · 4.1

From tables to straight lines

A table of values lists matching input and output pairs. If the output changes by the same amount whenever the input changes by the same amount, the relationship is linear.

Plotting the pairs turns the table into a picture. Linear pairs give points that lie on a straight line, which is the visual signature of a constant rate of change, and this graph representation answers questions the table cannot.

The graph shows what the table hides: where the line crosses the axes, how steep it is, and where it crosses another line. Those features are answers to questions the table cannot answer directly.

Reading a graph back into a table is the reverse process, and it is the standard way to check that a plotted line has been drawn correctly.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 4 · 4.1 · WORKED EXAMPLE 1

From tables to straight lines

WORKED EXAMPLE 4.1.1

Problem. A car hire costs 30 dollars plus 12 dollars a day. Complete the table for 0 to 5 days and plot the points.

days d	cost C = 30 + 12d
0	30
1	42
2	54
3	66
4	78
5	90

A constant difference of 12 means the points lie on a straight line.

Mathematical reasoning. Each extra day adds the same 12 dollars, so the first differences are constant and the graph is linear.

1. C = 30 + 12d.
2. d = 0 gives 30, and each further day adds 12.
3. The values are 30, 42, 54, 66, 78, 90.
4. Plotted, they lie on a straight line with gradient 12.

Final answer. the costs are 30, 42, 54, 66, 78 and 90 dollars

Interpretation. The constant difference of 12 dollars is the rate of change, and it becomes the gradient of the line.

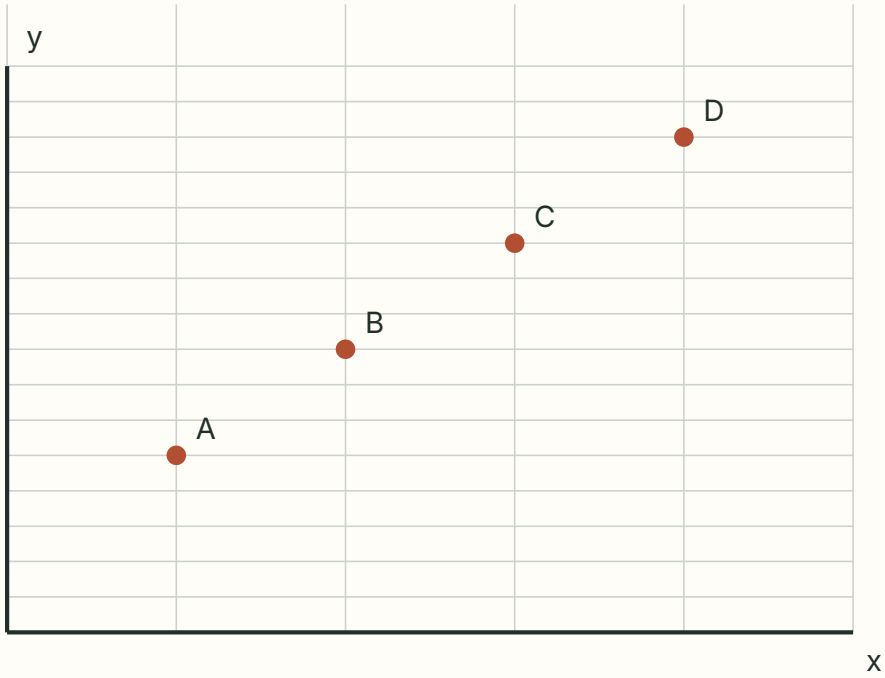
CHAPTER 4 · 4.1 · WORKED EXAMPLE 2

From tables to straight lines

WORKED EXAMPLE 4.1.2

Problem. A table gives (1, 5), (2, 8), (3, 11), (4, 14). Is the relationship linear, and what is its rule?

four points, one line



Mathematical reasoning. Constant first differences mean a linear rule, and the rule can be read from the pattern of the values.

1. The x-values rise by 1 and the y-values rise by 3 each time.
2. A constant difference of 3 means the relationship is linear.
3. The rule is y = 3x + b, and x = 1 gives y = 5, so b = 2.
4. The rule is y = 3x + 2.

Final answer. yes, the relationship is linear and the rule is y = 3x + 2

Interpretation. The graph confirms the rule: four points that lie on one line cannot disagree.

CHAPTER 4 · INVESTIGATION 4.2

Gradient as a rate of change

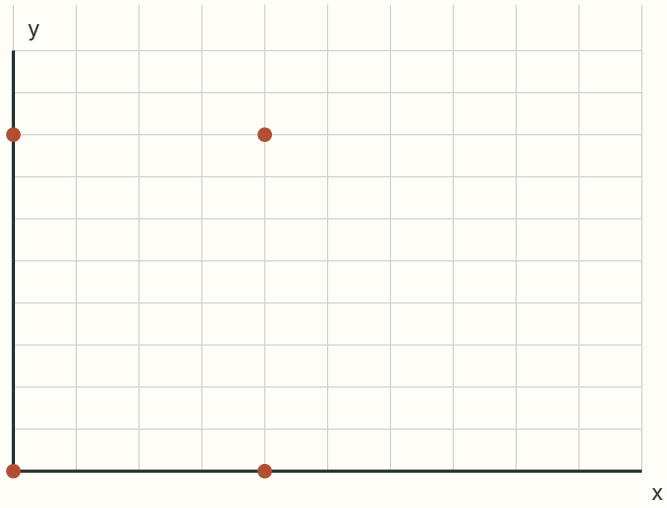
INQUIRY QUESTION

What does the gradient of a line measure, and how is it computed from two points?

YOUR TASK

Draw four lines on the same grid: one rising, one falling, one horizontal and one vertical. For each, choose two points and compute the gradient. Then draw a line parallel to the rising line and check whether the gradients agree.

four lines, four gradients



line	two points	change in y	change in x	gradient
rising	(0, 0) and (4, 8)	8	4	2
falling	(0, 8) and (4, 0)	−8	4	−2
horizontal	(0, 4) and (8, 4)	0	8	0
vertical	(4, 0) and (4, 8)	8	0	undefined

CHAPTER 4 · INVESTIGATION 4.2

Gradient as a rate of change

What do you notice?

- The rising line has a positive gradient and the falling line a negative one.
- The horizontal line has gradient 0 because y does not change.
- The vertical line has no gradient, because the change in x is zero.
- The parallel line through (0, 1) and (4, 9) also has gradient 2.

Observation. The gradient is the change in y per unit change in x, and it is the same for every pair of points on a line.

MATHEMATICAL CONCLUSION

The gradient of a line is $m = (y_2 - y_1)/(x_2 - x_1)$ for any two points on it, and it measures the rate of change of y with respect to x.

Your own words

CHAPTER 4 · 4.2

Gradient as a rate of change

The gradient of a line is the change in y divided by the change in x between any two points on it: $m = (y_2 - y_1)/(x_2 - x_1)$. It is a rate of change.

A positive gradient rises left to right, a negative gradient falls, and a zero gradient is horizontal. A vertical line has no gradient at all.

Because the gradient is the same between any two points on a line, it is enough to know two points to draw the line. This is what makes the equation of a line a complete description of it.

Parallel lines have equal gradients. Perpendicular lines have gradients whose product is −1, which is met formally in MYP 4.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

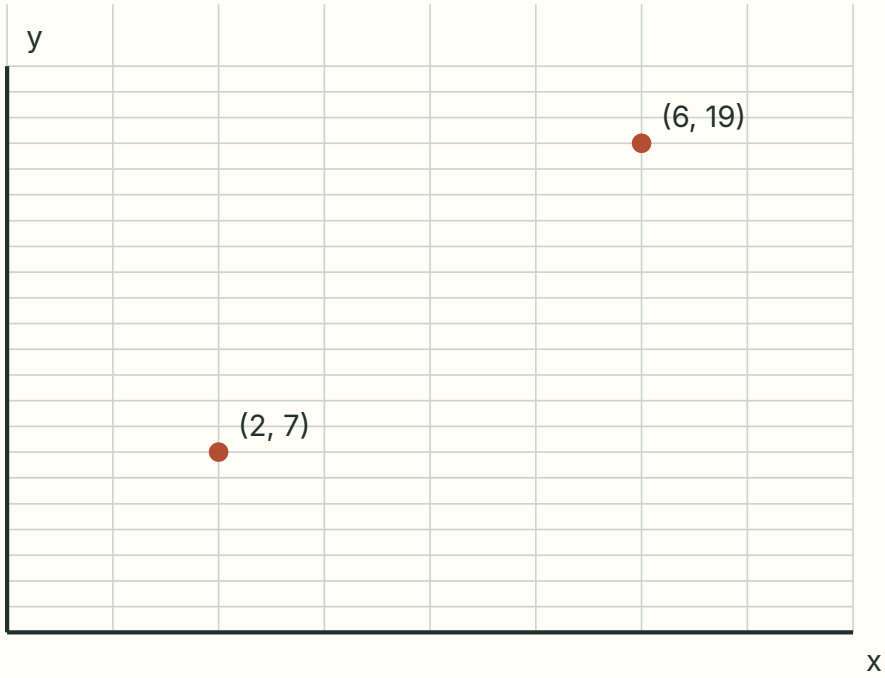
CHAPTER 4 · 4.2 · WORKED EXAMPLE 1

Gradient as a rate of change

WORKED EXAMPLE 4.2.1

Problem. Find the gradient of the line through (2, 7) and (6, 19), and interpret it.

gradient = 12 ÷ 4 = 3



Mathematical reasoning. The gradient is the change in y over the change in x, which is the rate of change of y with respect to x.

1. Change in y = 19 − 7 = 12.
2. Change in x = 6 − 2 = 4.
3. Gradient = 12/4 = 3.
4. So y rises by 3 units for every 1 unit rise in x.

Final answer. the gradient is 3

Interpretation. The gradient is the rate of change, so it answers 'how fast' rather than 'how much'.

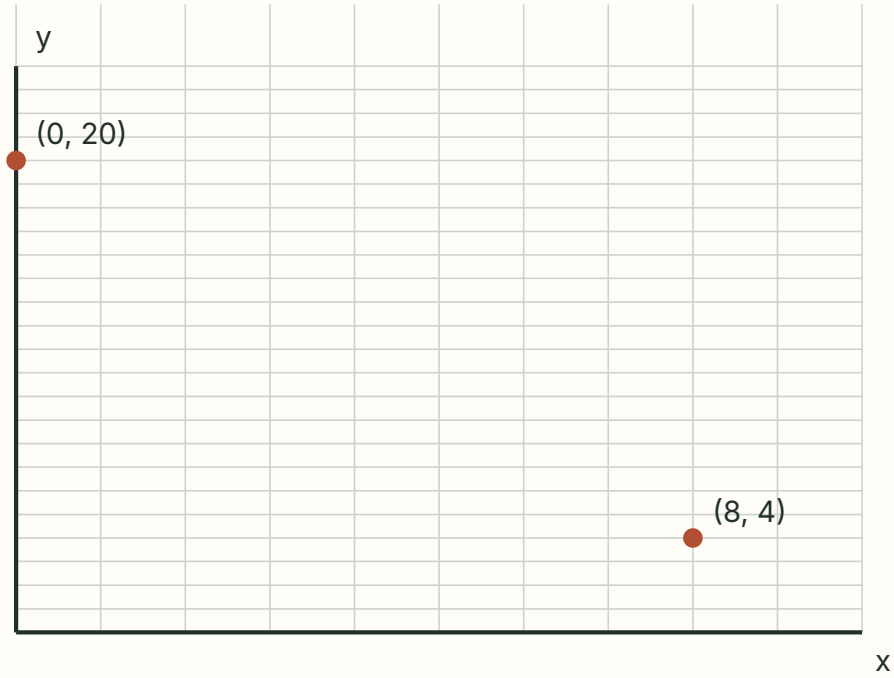
CHAPTER 4 · 4.2 · WORKED EXAMPLE 2

Gradient as a rate of change

WORKED EXAMPLE 4.2.2

Problem. A line falls from (0, 20) to (8, 4). Find its gradient and say what it means.

gradient = −16 ÷ 8 = −2



Mathematical reasoning. A falling line has a negative gradient, because the change in y is negative.

1. Change in y = 4 − 20 = −16.
2. Change in x = 8 − 0 = 8.
3. Gradient = −16/8 = −2.
4. So y falls by 2 units for every 1 unit rise in x.

Final answer. the gradient is −2

Interpretation. The negative sign is the direction of the change, and the 2 is its size.

CHAPTER 4 · INVESTIGATION 4.3

The equation of a straight line

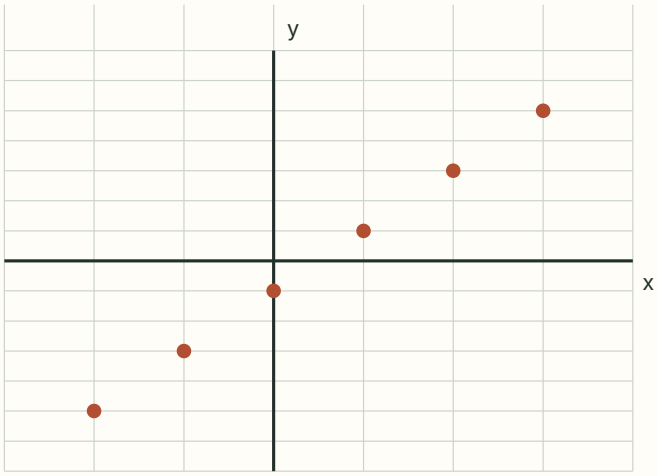
INQUIRY QUESTION

How can one equation describe every point on a line?

YOUR TASK

Take the equation y = 2x − 1. Build a table for x from −2 to 3, plot the points, then change the equation to y = 2x + 3 and repeat on the same grid. Then try y = −2x − 1. Record what changed and what stayed the same.

three lines from one gradient



equation	gradient	y-intercept	what it shows
y = 2x − 1	2	−1	rises, crossing y at −1
y = 2x + 3	2	3	rises the same way, crossing y at 3
y = −2x − 1	−2	−1	falls, crossing y at −1

CHAPTER 4 · INVESTIGATION 4.3

The equation of a straight line

What do you notice?

- Changing the constant moved the line up without changing its steepness.
- Changing the sign of the gradient turned the rising line into a falling one.
- The first two lines are parallel because their gradients are equal.
- All three lines pass through the y-axis at the value of c.

Observation. The gradient fixes the steepness and the constant fixes the position, and together they fix the line completely.

MATHEMATICAL CONCLUSION

Every straight line has an equation $y = mx + c$, where m is the gradient and c is the y-intercept, and two numbers describe the whole line.

Your own words

CHAPTER 4 · 4.3

The equation of a straight line

The equation $y = mx + c$ describes every point on the line with gradient m and y-intercept c. Substituting any x gives the matching y, and every such pair lies on the line.

The gradient-intercept form is the fastest to write when both are known. The point-gradient form $y - y_1 = m(x - x_1)$ is faster when a point and the gradient are known.

Two lines with the same gradient never meet. Two lines with different gradients meet exactly once, and the intersection is the point that satisfies both equations.

The equation is a complete description of the line: drawing it, reading a value from it, and finding where it crosses another line are all consequences of the same two numbers m and c.

> FORMAL DEVELOPMENT
>
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In your own words

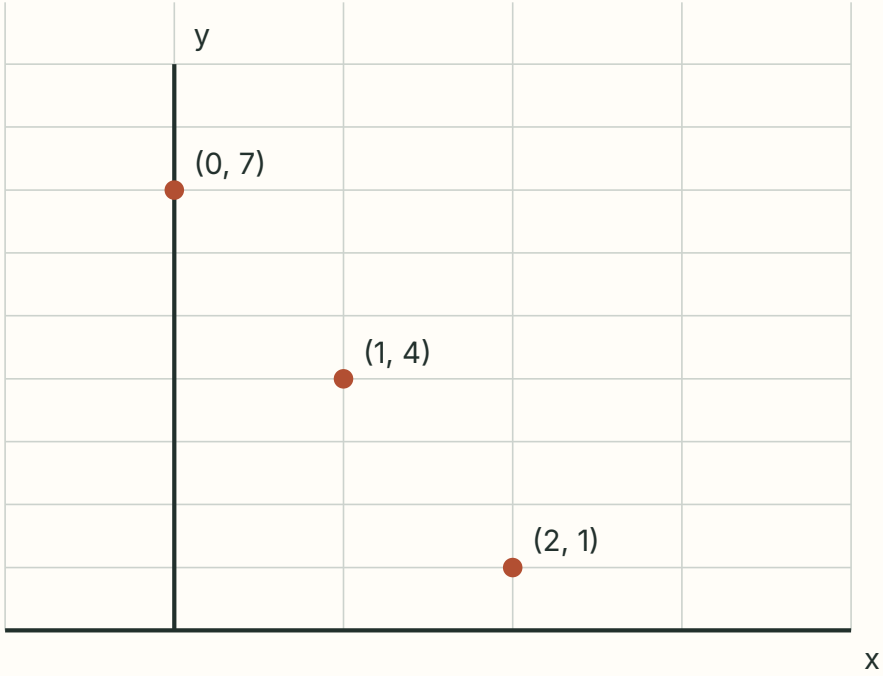
CHAPTER 4 · 4.3 · WORKED EXAMPLE 1

The equation of a straight line

WORKED EXAMPLE 4.3.1

Problem. Find the equation of the line through (1, 4) with gradient −3.

$$y = -3x + 7$$



Mathematical reasoning. One point and the gradient are enough to fix the line, and substituting the point into $y = mx + c$ gives c.

1. $m = -3$, so the equation is $y = -3x + c$.
2. The point (1, 4) lies on it: $4 = -3 \times 1 + c$.
3. So $c = 7$.
4. The equation is $y = -3x + 7$.

Final answer. $y = -3x + 7$

Interpretation. The intercept 7 is where the line meets the y-axis, and it is the value of y when $x = 0$.

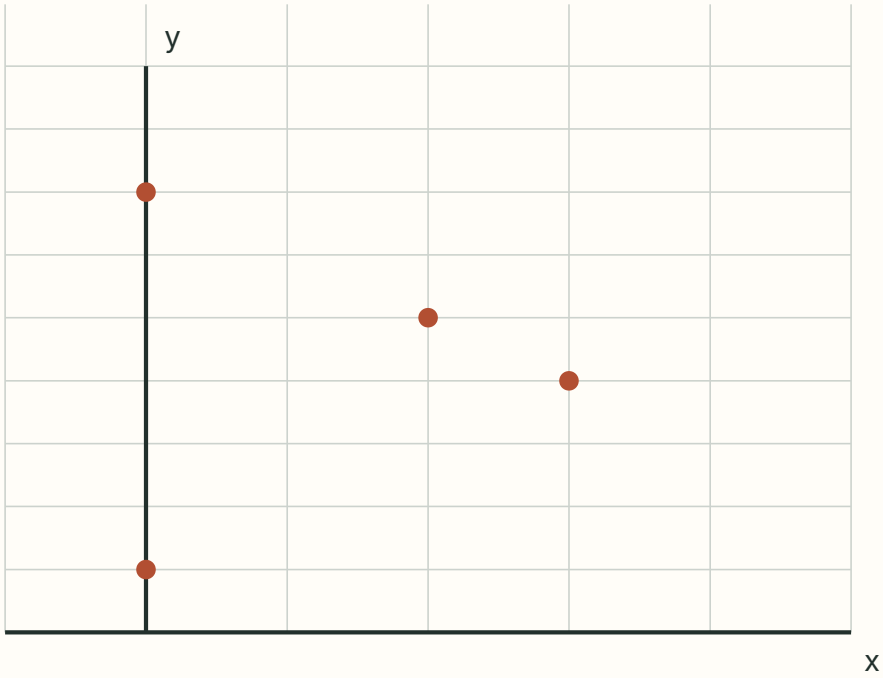
CHAPTER 4 · 4.3 · WORKED EXAMPLE 2

The equation of a straight line

WORKED EXAMPLE 4.3.2

Problem. Find where $y = 2x + 1$ meets $y = -x + 7$.

the crossing point (2, 5)



Mathematical reasoning. The intersection satisfies both equations, so solving them together gives the point.

1. $2x + 1 = -x + 7$.
2. $3x = 6$, so $x = 2$.
3. $y = 2 \times 2 + 1 = 5$.
4. The lines meet at (2, 5).

Final answer. the lines meet at (2, 5)

Interpretation. The crossing point is the only pair of values that makes both equations true at once.

CHAPTER 4 · INVESTIGATION 4.4

Direct proportion

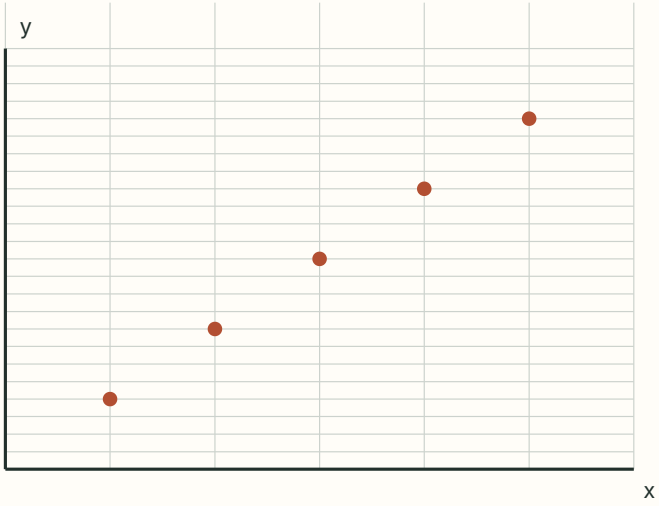
INQUIRY QUESTION

When does a linear relationship pass through the origin, and what does that mean?

YOUR TASK

Collect five pairs of values for a directly proportional relationship of your own (for example, cost against number of identical items). Compute y ÷ x for each pair, plot the points, and check whether the line passes through the origin. Then add a fixed delivery charge and repeat.

a line through the origin



x	y (no charge)	y ÷ x	y (with 5 charge)	y ÷ x
1	4	4	9	9
2	8	4	13	6.5
3	12	4	17	5.67
4	16	4	21	5.25
5	20	4	25	5

CHAPTER 4 · INVESTIGATION 4.4

Direct proportion

What do you notice?

- Without the charge, y ÷ x was 4 every time and the line passed through the origin.
- With the charge, the quotients changed and the line crossed the y-axis at 5.
- The fixed charge is a y-intercept, and it is what breaks direct proportion.
- Both graphs are straight lines, so both are linear — but only one is proportional.

Observation. Direct proportion is the special linear case where the quotient y/x is constant, which is why the line passes through the origin.

MATHEMATICAL CONCLUSION

y is directly proportional to x exactly when y = kx for a constant k, which means the graph passes through the origin and the quotient y/x is the same for every pair.

Your own words

CHAPTER 4 · 4.4

Direct proportion

Direct proportion is the relationship $y = kx$ for a constant k. Its graph is a straight line through the origin, because $y = 0$ when $x = 0$.

The constant k is the constant of proportionality. It is the gradient of the line and the value of y when $x = 1$.

A relationship is directly proportional exactly when y/x is the same for every pair. This is the test, and it is a division rather than a subtraction.

Direct proportion is the model behind unit prices, exchange rates and scale drawings. Recognising it saves work, because one pair of values gives the whole rule.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 4 · 4.4 · WORKED EXAMPLE 1

Direct proportion

WORKED EXAMPLE 4.4.1

Problem. 5 identical books weigh 11 kg. Find the weight of 8 books, and the constant of proportionality.



Mathematical reasoning. Identical items give a directly proportional relationship, so one unit value scales to any number of items.

1. Mass per book = 11 ÷ 5 = 2.2 kg.
2. So the constant of proportionality is 2.2.
3. For 8 books: 8 × 2.2 = 17.6 kg.

Final answer. 17.6 kg, with a constant of proportionality of 2.2 kg per book

Interpretation. The graph of mass against number of books is a straight line through the origin.

CHAPTER 4 · 4.4 · WORKED EXAMPLE 2

Direct proportion

WORKED EXAMPLE 4.4.2

Problem. Is y directly proportional to x for the pairs (3, 15), (5, 25), (7, 34)?

x	y	y ÷ x
3	15	5
5	25	5
7	34	4.857...

Direct proportion needs the same quotient every time.

Mathematical reasoning. The quotient y/x must be constant, and one different value is enough to rule the relationship out.

1. 15 ÷ 3 = 5.
2. 25 ÷ 5 = 5.
3. 34 ÷ 7 = 4.857..., which is not 5.
4. The quotients are not all equal, so y is not directly proportional to x.

Final answer. no, because the quotients are not constant

Interpretation. The third pair breaks the pattern, which is why the test is a check on every pair.

CHAPTER 4 · INVESTIGATION 4.5

Linear models, prediction and fairness

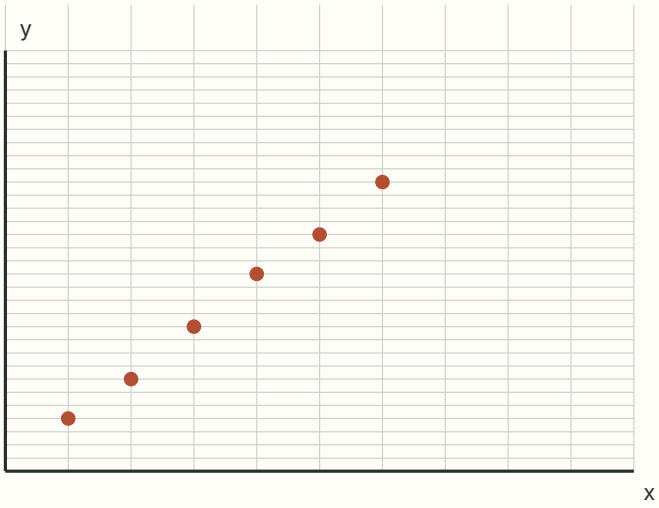
INQUIRY QUESTION

When is a prediction from a linear model justified, and when is it not?

YOUR TASK

Model the growth of a plant over six weeks by measuring its height each week. Fit a straight line by eye, use it to predict the height at week 10, and then measure the plant again in a later week if you can. Record how far the prediction was from the measurement.

height against week



week	height (cm)	predicted by the line	residual
1	4	3.8	0.2
2	7	7.4	−0.4
3	11	11.0	0.0
4	15	14.6	0.4
5	18	18.2	−0.2
6	22	21.8	0.2

CHAPTER 4 · INVESTIGATION 4.5

Linear models, prediction and fairness

What do you notice?

- The residuals are small and both positive and negative, so the line fits well.
- The predicted height at week 10 was about 37 cm.
- Real plants stop growing, so the week-10 prediction is an assumption rather than a fact.
- The residuals are the gaps between the model and the data.

> **Observation.** A linear model fits when its residuals are small and balanced, and its predictions are only as trustworthy as that fit.

MATHEMATICAL CONCLUSION

A linear model predicts reliably inside the range of the collected data, and outside that range it states an assumption about the rate of change continuing — which is why the range must be reported with the prediction.

Your own words

CHAPTER 4 · 4.5

Linear models, prediction and fairness

A linear model assumes that the rate of change stays constant. Inside the range of the collected data this is often reasonable; outside it, the model is an assumption.

Interpolation is prediction inside the range of the data, and it is usually safe. Extrapolation is prediction outside it, and it can fail badly.

A good model states its assumptions. 'Cost = 30 + 12 per day for the first 10 days' is honest; 'cost = 30 + 12 per day' is not, because eventually the rate changes.

Linear models are used to allocate resources, and a model that fits one group can be unfair to another. Checking the residuals — the gaps between the model and the data — is part of judging the model.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 4 · 4.5 · WORKED EXAMPLE 1

Linear models, prediction and fairness

WORKED EXAMPLE 4.5.1

Problem. A plumber's call-out cost follows $C = 45 + 38h$ for the first 4 hours. Predict the cost of a 3-hour job, and comment on using the model for 12 hours.

hours	model cost	comment
1	83	inside the range
3	159	inside the range
4	197	the last reliable value
12	501	outside the stated range

A model is reliable only where its assumptions hold.

Mathematical reasoning. Interpolation inside the stated range is reliable; extrapolation beyond it assumes the rate continues, which the model does not claim.

1. For 3 hours: $C = 45 + 38 \times 3 = 159$ dollars.
2. This is inside the stated range, so the prediction is reliable.
3. For 12 hours: $C = 45 + 38 \times 12 = 501$ dollars.
4. But the model was stated for the first 4 hours only, so this is extrapolation.

Final answer. 159 dollars for 3 hours; the 12-hour figure of 501 dollars is an extrapolation beyond the model's stated range

Interpretation. The mathematics is correct in both cases; only the justification differs.

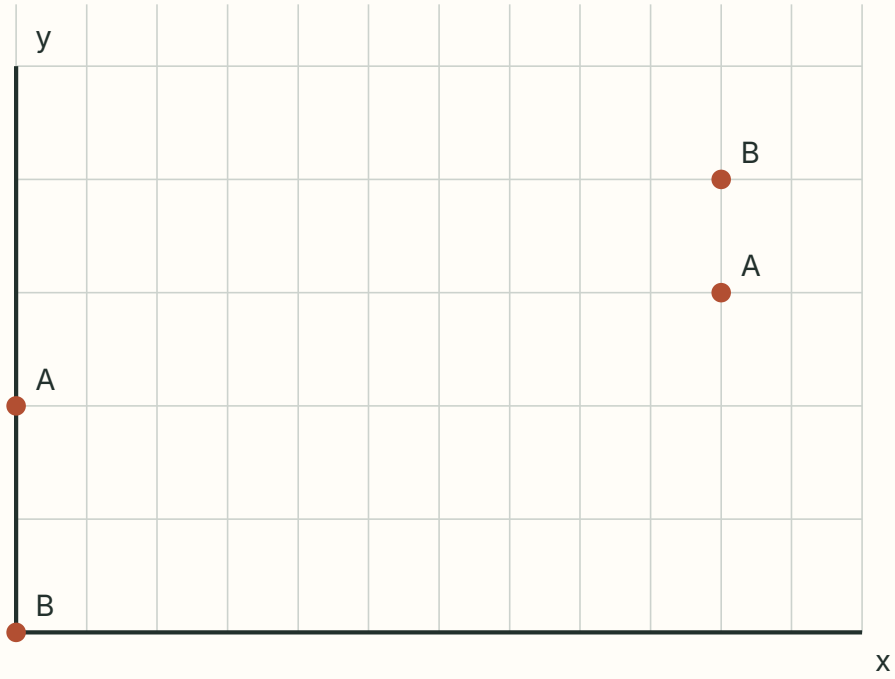
CHAPTER 4 · 4.5 · WORKED EXAMPLE 2

Linear models, prediction and fairness

WORKED EXAMPLE 4.5.2

Problem. Two shops charge for printing. Shop A: C = 2 + 0.10n. Shop B: C = 0.40n. For how many pages are they equal, and which is cheaper below that?

the crossing point



Mathematical reasoning. The two models are equal where their graphs cross, and the cheaper shop is the one with the lower value at that quantity.

1. 2 + 0.10n = 0.40n.
2. 2 = 0.30n, so n = 6.67.
3. For fewer than about 7 pages, shop A is cheaper.
4. For more than about 7 pages, shop B is cheaper.

Final answer. equal at about 6.67 pages; shop A is cheaper below that and shop B above

Interpretation. The crossing point is the decision boundary, and it comes from solving two linear models together.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding

4.A1 FLUENCY

Find the gradient of the line through (1, 3) and (5, 15).

4.A2 FLUENCY

Write the equation of the line with gradient 4 and y-intercept −2.

4.A3 FLUENCY

Find the gradient of the horizontal line through (2, 7) and (9, 7).

4.A4 FLUENCY

A line passes through (0, 5) and (4, 13). Find its equation.

4.A5 FLUENCY

Is y directly proportional to x for the pairs (2, 6), (4, 12), (6, 18)?

4.A6 FLUENCY

Find the y-intercept of $y = -3x + 8$.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding (continued)

4.A7 FLUENCY

Find the equation of the line through (2, 1) with gradient −5.

4.A8 FLUENCY

A table gives (0, 4), (2, 10), (4, 16). Find the gradient and the equation.

4.A9 FLUENCY

Where do $y = x + 1$ and $y = 5 - x$ meet?

4.A10 FLUENCY

A line has gradient 0.5 and passes through (6, 8). Find its equation.

4.A11 FLUENCY

Find the gradient of the line through (−2, 4) and (3, −6).

4.A12 REASONING

Explain why a vertical line has no gradient.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns

4.B1 REASONING

Investigate: do the lines $y = 2x + 1$ and $y = 2x - 7$ ever meet? Justify.

4.B2 REASONING

Test the claim: if two lines have the same y-intercept they must meet.

4.B3 REASONING

Investigate the gradients of the lines $y = 3x$, $y = 3x + 5$ and $y = -3x$. What do they have in common?

4.B4 REASONING

Generalise: what is the gradient of the line through (a, b) and (c, d)?

4.B5 REASONING

Investigate whether the points $(1, 2)$, $(3, 6)$ and $(5, 10)$ lie on one line.

4.B6 REASONING

Test the claim: a line with a negative gradient must have a negative y-intercept.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns (continued)

4.B7 REASONING

Investigate: for what value of k do y = kx and y = x + 4 meet at x = 8?

4.B8 REASONING

Generalise a rule for the gradient of a line perpendicular to a line of gradient m.

4.B9 REASONING

Investigate the pattern 2, 5, 8, 11, … as points (n, term). What is the gradient of the line through them?

4.B10 REASONING

Test the claim: two lines that meet must have different gradients.

4.B11 REASONING

Investigate whether the line through (0, 0) and (3, −9) has the same gradient as the line through (1, 1) and (4, −8).

CRITERION C Communicating

Criterion C practice · Communicating

4.C1 REASONING

Explain in words what the gradient of a line represents in a real situation.

4.C2 REASONING

Explain the difference between interpolation and extrapolation.

4.C3 REASONING

Explain why a straight-line graph through the origin indicates direct proportion.

4.C4 REASONING

Describe two ways to find the equation of a line, and say when each is quickest.

4.C5 REASONING

Explain what a residual is and what it tells you about a model.

4.C6 REASONING

Explain why the equation $y = mx + c$ is a complete description of a line.

CRITERION C Communicating

Criterion C practice · Communicating (continued)

4.C7 REASONING

A graph of cost against distance has a positive y-intercept. Explain what that means for a customer.

4.C8 REASONING

Explain why parallel lines have equal gradients, using the definition of gradient.

4.C9 REASONING

Describe how you would check whether three points lie on one straight line.

4.C10 REASONING

Explain the difference between a linear relationship and a directly proportional one.

4.C11 REASONING

Explain why a model that fits one community's data may not be fair to another.

4.C12 REASONING

Describe how you would use a graph to compare two service charges.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts

4.D1 APPLICATION

A school pays 120 dollars for a hall and 7 dollars per chair. Write the cost model and find the cost for 45 chairs.

4.D2 APPLICATION

A taxi charges 4 dollars plus 1.60 dollars per kilometre. Find the fare for 11 km and the distance that costs 20 dollars.

4.D3 APPLICATION

Two phone plans: A costs 10 dollars plus 0.05 dollars a minute, B costs 25 dollars plus 0.02 dollars a minute. Find the number of minutes at which they cost the same.

4.D4 APPLICATION

A water tank is filled at 15 litres a minute and already holds 60 litres. Find the time to reach 360 litres and sketch the graph.

4.D5 APPLICATION

A farmer sells maize at 0.35 dollars a kilogram and pays 50 dollars to transport it. Write a model for the profit and find the break-even mass.

4.D6 REASONING

Explain whether a model that charges a fixed fee plus a per-minute rate is directly proportional.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts (continued)

4.D7 APPLICATION

A cyclist covers 12 km in 40 minutes at a constant speed. Find the speed and the distance covered in 75 minutes.

4.D8 APPLICATION

A printing firm charges 15 dollars setup plus 0.08 dollars per page. Find the number of pages that makes the cost 47 dollars.

4.D9 REASONING

A village shares a water allowance in proportion to household size. Model the allocation and comment on whether the model is fair.

4.D10 REASONING

A linear model predicts a school roll of 620 in five years. The roll is growing by 20 a year. Find the current roll and comment on the assumption.

4.D11 APPLICATION

Two delivery services charge 8 dollars plus 0.50 dollars a kilogram and 3 dollars plus 0.75 dollars a kilogram. Find the mass at which they cost the same and say which is cheaper above it.

4.D12 REASONING

Explain why extrapolating a linear model of plant growth beyond the growing season gives an unreasonable answer.

CHAPTER 4 · APPLICATION AND ASSESSMENT

Applying chapter 4

These tasks use tables, gradients, equations of lines, direct proportion and linear models. State the representation you used and why it was the clearest one.

A1 · The two water pumps

Pump A delivers 40 litres a minute into an empty tank. Pump B delivers 55 litres a minute but starts 5 minutes later. Model both with linear equations, find when B overtakes A, and sketch both graphs on the same axes.

Criteria A and C · label the axes with units and state what the crossing point means.

A2 · The shared water allowance

A village of 240 households shares 60 000 litres of water a week in proportion to household size. Model the allocation, find the allowance for a household of 6, and discuss whether the model is fair to households of 1.

Criteria A and D · justify the model and comment on its limits.

CHAPTER 4 · REVIEW

What you should now be able to do

Key ideas

- A table with constant first differences describes a linear relationship.
- The gradient is the change in y per unit change in x: $m = (y_2 - y_1)/(x_2 - x_1)$.
- Every straight line has an equation $y = mx + c$.
- Parallel lines have equal gradients.
- Direct proportion is $y = kx$, a line through the origin.
- The intersection of two lines satisfies both equations.
- Interpolation is reliable; extrapolation states an assumption.

Vocabulary

gradient · y-intercept · rate of change · linear · direct proportion · constant of proportionality · parallel · intersection · interpolation · extrapolation · residual · model

CHAPTER 4 · REVIEW

Mixed practice · consolidate and extend

Do not use a calculator unless a question asks for one. Show a method where one is requested.

4.R1 Find the gradient of the line through (2, 5) and (7, 20).

4.R2 Write the equation of the line through (0, −3) with gradient 4.

4.R3 Where do $y = 3x - 2$ and $y = x + 4$ meet?

4.R4 Is y directly proportional to x for the pairs (2, 7), (4, 14), (5, 20)?

4.R5 A line has gradient −2 and passes through (3, 1). Find its equation.

4.R6 Explain the difference between a gradient and a y-intercept.

CHAPTER 4 · REVIEW

Returning to the statement of inquiry

STATEMENT OF INQUIRY

The relationship between two quantities that change at a constant rate can be represented as a straight line, and the equivalence between a table, a graph and an equation allows that relationship to be used to predict — and the prediction to be judged.

Reflect on the chapter

Answer in your own words, using evidence from this chapter.

- Write the statement of inquiry for this chapter in your own words.
- Which mathematical ideas from this chapter did you use to understand it? Name at least two.
- How did the key concept of relationships and the related concepts of change and representation appear in your work?
- How is the global context of fairness and development, and its exploration of the distribution of resources, visible in the investigations and examples you completed?
- Use evidence from an investigation or a worked example to justify your answer.
- Has your understanding of the statement of inquiry changed since the start of the chapter? Explain how.
- What are the limitations of a linear model for sharing resources, and what would you still like to ask?

CHAPTER 4 · PROJECTS, SERVICE AND ACTION

Acting on what you have learned

Resource distribution is a linear-modelling problem. Use the model to plan, and say plainly where the model stops being valid.

Service as Action

COMMUNITY NEED	Water and food distribution in the community is decided without a shared model, so some households are missed.
MATHEMATICAL CONNECTION	Linear models, gradients and direct proportion.
PROPOSED ACTION	Build a proportional allocation model for one shared resource and publish it with its assumptions.
EXPECTED IMPACT	Allocations become predictable and open to challenge.
STUDENT REFLECTION	Record the model, the assumptions, and what happened when it was used.

Community Project

PROJECT IDEA	A water-allowance model for the school.
NEED	The school's water allowance is shared by guesswork.
RESEARCH QUESTION	What allocation rule is both proportional to need and agreed by everyone?
MATHEMATICS	Linear functions, direct proportion and graph reading.
ACTION OR OUTCOME	A written allocation rule with a graph of the allocations.
BENEFICIARIES	Every household or class using the supply.
EVIDENCE	The model, the graph and the agreed rule.

Personal Project connection · MYP 5

PERSONAL INTEREST	Travel, sport or a savings goal.
LEARNING GOAL	To use a linear model to make a personal decision.
PRODUCT OR OUTCOME	A savings or travel plan with a graph and a prediction.
MATHEMATICS	Gradients, equations of lines and extrapolation.
PROCESS	Collect the data, fit a line, then predict and check.
EVIDENCE	The graph, the model and the prediction.

CHAPTER 5 · PART II · ALGEBRA AND MODELS

05

Equations, inequalities and systems

An equation is a balance, and every step of a solution must keep it balanced. When two conditions must hold at once, the answer is a system.

ENGAGE

A co-operative has 40 dollars a head to spend on seed and fertiliser. Seed costs 3 dollars a packet and fertiliser 5 dollars a bag, and at least 2 bags of fertiliser are needed. How many packets can they buy? The question has more than one answer, and inequalities describe them all.

Framing the inquiry

Key concept: Relationships
Related concepts: Equivalence · Systems
Global context: Globalisation and sustainability
Global context exploration: Resources and consumption

Statement of inquiry: The relationship between quantities that must satisfy several conditions at once can be represented as a system, and the equivalence of the different methods of solving such systems allows the feasible solutions to be identified, checked and justified.

Questions to guide us

- **Factual:** How do you solve a linear equation, an inequality, and a pair of simultaneous equations?
- **Conceptual:** Why must every step of a solution keep the equation equivalent to the original, and how does a system of inequalities produce a set of feasible answers rather than one?
- **Debatable:** When a resource is limited, is a solution that satisfies every constraint automatically the best one?

ATL · SELF-MANAGEMENT. In order for a student to organise the solution of a multi-step problem, students must plan the order of the steps before starting, record each step so it can be checked, and review the plan against the answer obtained.

CHAPTER 5 · INVESTIGATION 5.1

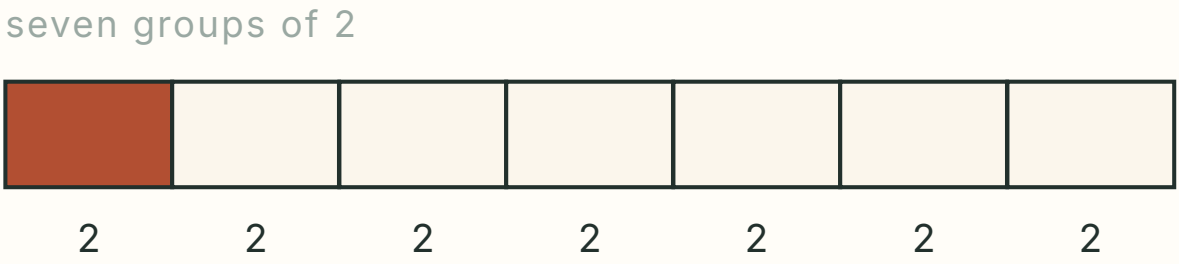
Solving linear equations

INQUIRY QUESTION

How can an equation be solved while keeping it equivalent to the original?

YOUR TASK

Solve $2(x + 3) = 14$ by three different routes: expanding first, dividing first, and using a bar model. Then try to solve $2(x + 3) = 15$ the same way and record what happens.



route	steps	answer
expand first	$2x + 6 = 14$, so $2x = 8$, so $x = 4$	$x = 4$
divide first	$x + 3 = 7$, so $x = 4$	$x = 4$
bar model	$14 - 6 = 8$ shared into two equal parts	$x = 4$

CHAPTER 5 · INVESTIGATION 5.1

Solving linear equations

What do you notice?

- All three routes gave the same answer, because each step kept the equation equivalent.
- Dividing first was the shortest route because it avoided expanding.
- For $2(x + 3) = 15$, dividing first gives $x + 3 = 7.5$, so $x = 4.5$.
- The answer need not be a whole number, and the bar model still works with half-units.

> **Observation.** Different routes through an equation give the same answer when every step preserves equivalence.

MATHEMATICAL CONCLUSION

A linear equation is solved by applying inverse operations to both sides, and any sequence of equivalent steps leads to the same solution.

Your own words

CHAPTER 5 · 5.1

Solving linear equations

An equation states that two expressions are equal. Solving it means finding the values that make the statement true, and every step of the solution must preserve that truth.

The permitted steps are: add or subtract the same quantity from both sides, multiply or divide both sides by the same non-zero quantity. Each step produces an equation equivalent to the last.

Multiplying or dividing both sides by a negative number is allowed when solving an equation, because the equality survives. This is the difference between an equation and an inequality.

Checking is not optional. Substituting the answer back into the original equation is the only way to be sure that a step did not break the equivalence.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 5 · 5.1 · WORKED EXAMPLE 1

Solving linear equations

WORKED EXAMPLE 5.1.1

Problem. Solve $3(x - 4) + 5 = 2x + 11$.

step	operation	result
$3(x - 4) + 5 = 2x + 11$	expand	$3x - 7 = 2x + 11$
$3x - 7 = 2x + 11$	subtract $2x$	$x - 7 = 11$
$x - 7 = 11$	add 7	$x = 18$

Each step keeps the equation balanced.

Mathematical reasoning. Expanding first, then collecting the variable terms on one side and the constants on the other, keeps each step equivalent.

1. Expand: $3x - 12 + 5 = 2x + 11$, so $3x - 7 = 2x + 11$.
2. Subtract $2x$ from both sides: $x - 7 = 11$.
3. Add 7 to both sides: $x = 18$.
4. Check: $3(14) + 5 = 47$ and $2(18) + 11 = 47$.

Final answer. $x = 18$

Interpretation. The check confirms that every step preserved the equality, not just the last one.

CHAPTER 5 · 5.1 · WORKED EXAMPLE 2

Solving linear equations

WORKED EXAMPLE 5.1.2

Problem. Solve $(2x - 1)/3 = (x + 2)/4$.

step	operation	result
$(2x - 1)/3 = (x + 2)/4$	multiply by 12	$4(2x - 1) = 3(x + 2)$
$8x - 4 = 3x + 6$	collect terms	$5x = 10$
$5x = 10$	divide by 5	$x = 2$

Clearing the denominators first keeps the arithmetic simple.

Mathematical reasoning. Multiplying both sides by the common denominator clears the fractions without changing the solution.

1. Multiply both sides by 12: $4(2x - 1) = 3(x + 2)$.
2. Expand: $8x - 4 = 3x + 6$.
3. Subtract $3x$ and add 4: $5x = 10$.
4. Divide by 5: $x = 2$.

Final answer. $x = 2$

Interpretation. Clearing fractions is a legitimate step because it multiplies both sides by the same non-zero number.

CHAPTER 5 · INVESTIGATION 5.2

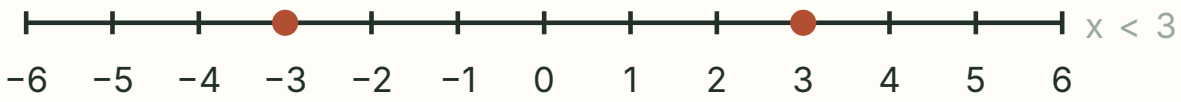
Inequalities and solution sets

INQUIRY QUESTION

How does solving an inequality differ from solving an equation, and why is the answer a set?

YOUR TASK

Solve $2x + 1 < 7$ and then $-2x + 1 < 7$. Draw both solution sets on the same number line, then solve $-2x + 1 < 2x + 5$. Record the effect of the negative coefficient in each case.



inequality	steps	solution set
$2x + 1 < 7$	$2x < 6$, so $x < 3$	$(-\infty, 3)$
$-2x + 1 < 7$	$-2x < 6$, so $x > -3$	$(-3, \infty)$
$-2x + 1 < 2x + 5$	$-4 < 4x$, so $x > -1$	$(-1, \infty)$

CHAPTER 5 · INVESTIGATION 5.2

Inequalities and solution sets

What do you notice?

- The first two inequalities differ only in the sign of the coefficient of x.
- That sign change reversed the direction of the solution set.
- Both solution sets are infinite, so an inequality rarely has one answer.
- The third inequality has the variable on both sides, which is solved by collecting it.

> **Observation.** A negative coefficient reverses the inequality sign when it is divided out, so the solution set can point the opposite way from an equation's solution.

MATHEMATICAL CONCLUSION

Solving an inequality follows the same steps as solving an equation, except that multiplying or dividing by a negative number reverses the sign — and the solution is a set, not a value.

Your own words

CHAPTER 5 · 5.2

Inequalities and solution sets

An inequality states that one expression is greater or less than another. Its solution is usually a set of values, not a single value, so it is written as an interval or with a number line.

Adding or subtracting the same quantity from both sides preserves the inequality. Multiplying or dividing by a positive number also preserves it.

Multiplying or dividing both sides by a negative number reverses the inequality sign. This is the one step that behaves differently from an equation, and it is the commonest source of error.

A compound inequality such as $1 < 2x + 1 \leq 7$ is solved by applying the same operation to all three parts, which keeps the structure intact.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

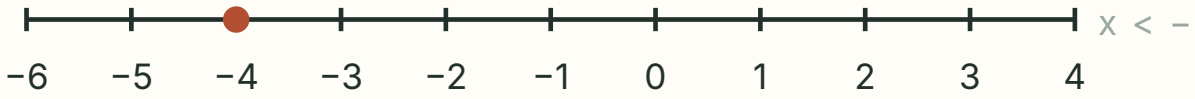
In your own words

CHAPTER 5 · 5.2 · WORKED EXAMPLE 1

Inequalities and solution sets

WORKED EXAMPLE 5.2.1

Problem. Solve $3 - 2x > 11$ and show the solution on a number line.



Mathematical reasoning. Dividing by a negative number reverses the inequality sign, which is the step that differs from solving an equation.

1. $3 - 2x > 11$.
2. Subtract 3: $-2x > 8$.
3. Divide by -2 and reverse the sign: $x < -4$.

Final answer. $x < -4$, or $x \in (-\infty, -4)$

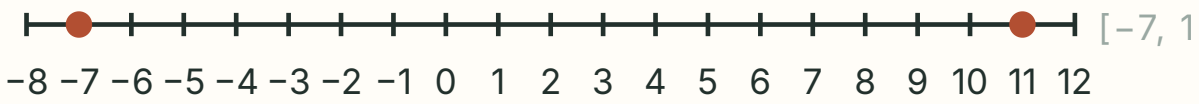
Interpretation. Reversing the sign is not a rule of convenience: it is what keeps the statement true.

CHAPTER 5 · 5.2 · WORKED EXAMPLE 2

Inequalities and solution sets

WORKED EXAMPLE 5.2.2

Problem. Solve $-2 \le (x + 1)/3 < 4$.



Mathematical reasoning. A compound inequality is solved by applying the same operations to all three parts.

1. Multiply all three parts by 3: $-6 \le x + 1 < 12$.
2. Subtract 1 from all three parts: $-7 \le x < 11$.
3. So $x \in [-7, 11)$.

Final answer. $-7 \le x < 11$, or $x \in [-7, 11)$

Interpretation. The square bracket at −7 and the round bracket at 11 record which endpoints are included.

CHAPTER 5 · INVESTIGATION 5.3

Simultaneous equations: graphs and substitution

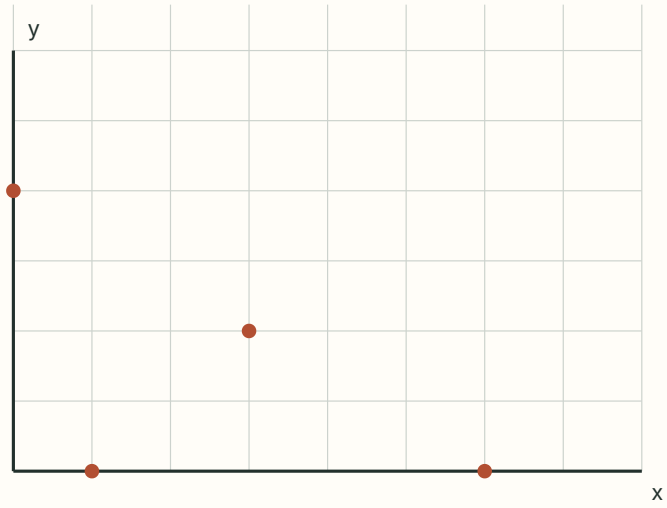
INQUIRY QUESTION

How can two conditions be satisfied at the same time?

YOUR TASK

Solve the pair $2x + 3y = 12$ and $x - y = 1$ three ways: by substitution, by elimination, and by drawing both lines. Record which method you found fastest and why.

the crossing point (3, 2)



method	steps	answer
substitution	$x = y + 1$, so $2(y + 1) + 3y = 12$, giving $5y = 10$	$x = 3$, $y = 2$
elimination	double the second equation and subtract	$x = 3$, $y = 2$
graph	draw both lines and read the crossing	$x = 3$, $y = 2$

CHAPTER 5 · INVESTIGATION 5.3

Simultaneous equations: graphs and substitution

What do you notice?

- All three methods gave the same pair, because they all enforce the same two conditions.
- Substitution was quickest here because x was easy to isolate.
- The graph showed the crossing but only to the accuracy of the drawing.
- The check in both equations confirmed the answer.

Observation. The solution of a system is the pair that satisfies every equation, and any exact method finds it.

MATHEMATICAL CONCLUSION

Simultaneous equations are solved by substitution or by elimination, and both methods produce the single pair of values that satisfies every equation in the system.

Your own words

CHAPTER 5 · 5.3

Simultaneous equations: graphs and substitution

A pair of simultaneous equations states two conditions that must both hold. The solution is the pair of values that satisfies both, which is the point where the two graphs cross.

The graphical method finds the intersection by drawing both lines. It is quick and it shows the structure, but it is limited by the accuracy of the drawing.

The substitution method solves one equation for one variable and substitutes the result into the other. It is exact and it works whenever one equation is easy to rearrange.

Both methods answer the same question. The choice between them is about accuracy and convenience, not about correctness.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

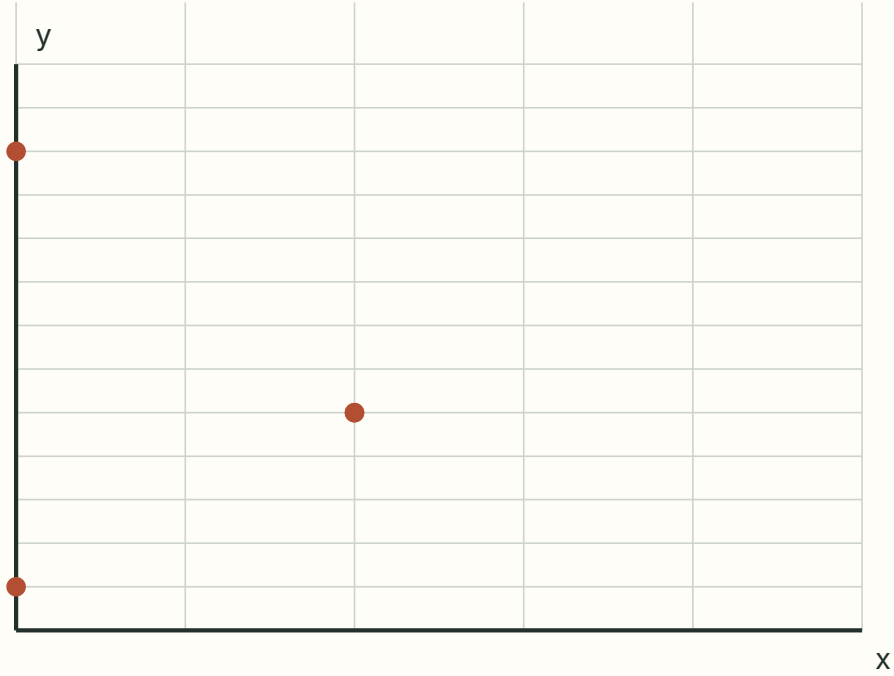
CHAPTER 5 · 5.3 · WORKED EXAMPLE 1

Simultaneous equations: graphs and substitution

WORKED EXAMPLE 5.3.1

Problem. Solve $y = 2x + 1$ and $3x + y = 11$ by substitution.

the crossing point (2, 5)



Mathematical reasoning. One equation already gives y in terms of x, so substituting that expression into the other removes y entirely.

1. Substitute $y = 2x + 1$ into $3x + y = 11$.
2. $3x + 2x + 1 = 11$.
3. $5x = 10$, so $x = 2$.
4. $y = 2 \times 2 + 1 = 5$.
5. Check: $3 \times 2 + 5 = 11$.

Final answer. x = 2 and y = 5

Interpretation. The check in the second equation confirms the pair satisfies both conditions.

CHAPTER 5 · 5.3 · WORKED EXAMPLE 2

Simultaneous equations: graphs and substitution

WORKED EXAMPLE 5.3.2

Problem. Solve $x + y = 10$ and $x - y = 4$ by elimination.

step	operation	result
$x + y = 10$	—	—
$x - y = 4$	add the equations	$2x = 14$
$2x = 14$	divide by 2	$x = 7$
$x = 7$	substitute back	$y = 3$

Adding the equations cancels y.

Mathematical reasoning. Adding the two equations removes y because its coefficients are equal and opposite, leaving one equation in x.

1. Add the equations: $(x + y) + (x - y) = 10 + 4$.
2. $2x = 14$, so $x = 7$.
3. Substitute into $x + y = 10$: $7 + y = 10$, so $y = 3$.
4. Check: $7 - 3 = 4$.

Final answer. $x = 7$ and $y = 3$

Interpretation. Elimination was faster here because the y terms were already set up to cancel.

CHAPTER 5 · INVESTIGATION 5.4

Elimination and special systems

INQUIRY QUESTION

How can a system be solved when the coefficients are arranged to cancel, and what happens when they are not?

YOUR TASK

Solve these four systems and classify each as having one solution, no solution, or infinitely many: (a) $x + y = 5$, $x - y = 1$; (b) $2x + 2y = 6$, $x + y = 3$; (c) $x + y = 5$, $2x + 2y = 8$; (d) $3x - y = 4$, $6x - 2y = 8$. Record how you decided.

system	what elimination gave	classification
(a) $x + y = 5$, $x - y = 1$	$2x = 6$, so $x = 3$ and $y = 2$	one solution
(b) $2x + 2y = 6$, $x + y = 3$	$0 = 0$ after elimination	infinitely many
(c) $x + y = 5$, $2x + 2y = 8$	$0 = -2$ after elimination	no solution
(d) $3x - y = 4$, $6x - 2y = 8$	$0 = 0$ after elimination	infinitely many

CHAPTER 5 · INVESTIGATION 5.4

Elimination and special systems

What do you notice?

- Elimination produced $0 = 0$ in two cases and $0 = -2$ in one.
- $0 = 0$ means the equations say the same thing, so there are infinitely many solutions.
- $0 = -2$ is impossible, so the lines are parallel and there is no solution.
- Only case (a) gave a single pair.

Observation. Elimination reveals the type of system: a true statement means infinitely many solutions, a false one means none, and a solvable equation means exactly one.

MATHEMATICAL CONCLUSION

A linear system has exactly one solution when its lines cross, no solution when they are parallel, and infinitely many when they are the same line, and elimination distinguishes the three cases directly.

Your own words

CHAPTER 5 · 5.4

Elimination and special systems

Elimination multiplies one or both equations by a constant so that one variable has equal and opposite coefficients, then adds or subtracts the equations.

The multiplier is chosen to make the coefficients match. If the coefficients of x are 2 and 3, multiplying by 3 and 2 makes them 6 and 6, and subtracting removes x.

A system with exactly one solution has lines that cross once. Parallel lines give no solution, and identical lines give infinitely many.

Recognising the special cases from the coefficients saves work: if the ratios of the x-coefficients and the y-coefficients are equal but the constants differ, the lines are parallel.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 5 · 5.4 · WORKED EXAMPLE 1

Elimination and special systems

WORKED EXAMPLE 5.4.1

Problem. Solve $3x + 2y = 16$ and $5x - 2y = 8$ by elimination.

equation	multiply by	result
$3x + 2y = 16$	1	$3x + 2y = 16$
$5x - 2y = 8$	1	$5x - 2y = 8$
add	—	$8x = 24$

The y coefficients already cancel.

Mathematical reasoning. The y coefficients are 2 and −2, so adding the equations removes y with no multiplier at all.

1. Add: $8x = 24$, so $x = 3$.
2. Substitute into $3x + 2y = 16$: $9 + 2y = 16$.
3. $2y = 7$, so $y = 3.5$.
4. Check in the second: $15 - 7 = 8$.

Final answer. $x = 3$ and $y = 3.5$

Interpretation. The answer need not be a pair of integers, and the check still works.

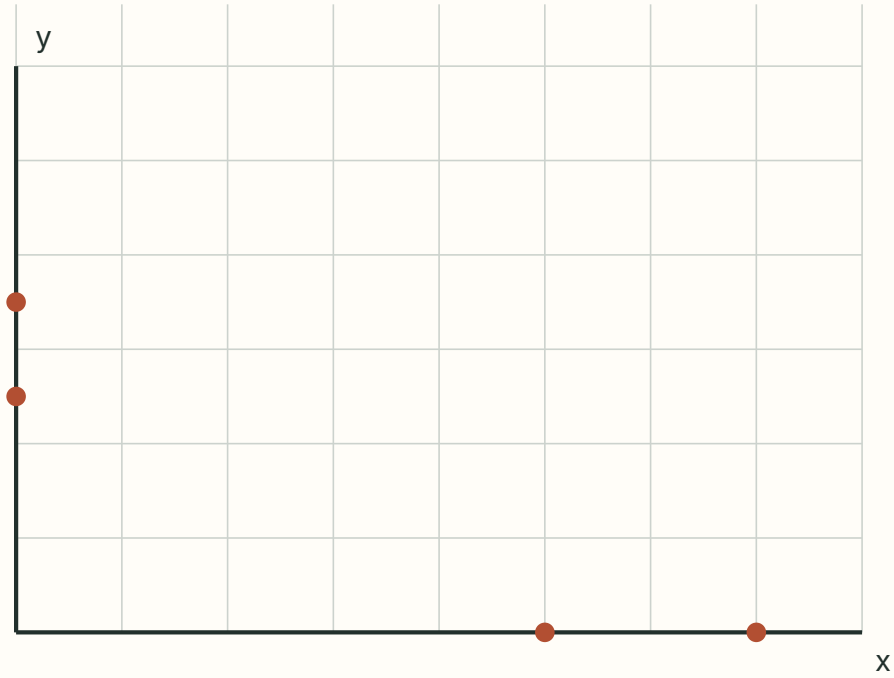
CHAPTER 5 · 5.4 · WORKED EXAMPLE 2

Elimination and special systems

WORKED EXAMPLE 5.4.2

Problem. Show that $2x + 4y = 10$ and $x + 2y = 7$ have no solution.

two parallel lines



Mathematical reasoning. If one equation is a multiple of the other in the variable terms but not in the constant, the lines are parallel and never meet.

1. Multiply the second equation by 2: $2x + 4y = 14$.
2. Compare with $2x + 4y = 10$.
3. The left sides are identical but the right sides differ by 4.
4. So no pair of values can satisfy both.

Final answer. there is no solution; the lines are parallel

Interpretation. Recognising the parallel case from the coefficients avoids a long search for an answer that does not exist.

CHAPTER 5 · INVESTIGATION 5.5

Systems of inequalities and resource allocation

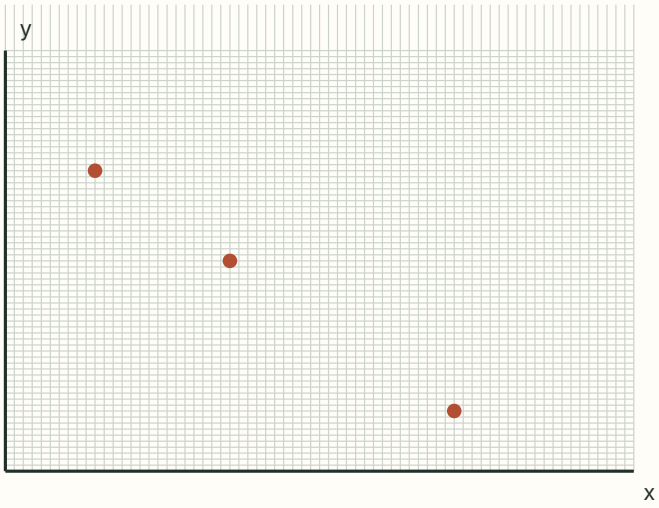
INQUIRY QUESTION

How can several limits at once define a set of feasible choices?

YOUR TASK

A school fair stall sells drinks at 2 dollars and snacks at 3 dollars. It has room for at most 60 items, at least 10 of each, and wants to raise at least 150 dollars. Write the system of inequalities, draw the feasible region, and find one combination that meets every condition.

the feasible region for the stall



condition	inequality
at most 60 items	$d + s \le 60$
at least 10 drinks	$d \ge 10$
at least 10 snacks	$s \ge 10$
at least 150 dollars	$2d + 3s \ge 150$

CHAPTER 5 · INVESTIGATION 5.5

Systems of inequalities and resource allocation

What do you notice?

- The first three inequalities bound a rectangle of possible stock.
- The fourth inequality removes the corner where the takings are too low.
- The feasible region is what remains after every condition is applied.
- $d = 25$ and $s = 35$ gives $2 \times 25 + 3 \times 35 = 155$ dollars with 60 items.

Observation. Each inequality removes part of the plane, and the feasible region is what survives all of them.

MATHEMATICAL CONCLUSION

A system of inequalities describes the set of choices that satisfy every condition at once, and that set is a region whose boundaries come from the individual inequalities.

Your own words

CHAPTER 5 · 5.5

Systems of inequalities and resource allocation

A system of inequalities describes a situation with several limits at once. The solution is the set of points that satisfy every inequality, and it is a region rather than a point.

Each inequality contributes a half-plane bounded by its line. The feasible region is the intersection of those half-planes.

The vertices of the feasible region are where two boundary lines cross, and they are usually where the best value of a quantity occurs.

Inequalities model scarcity honestly: they do not pick an answer, they describe which answers are possible. Choosing among them is a separate, human step.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

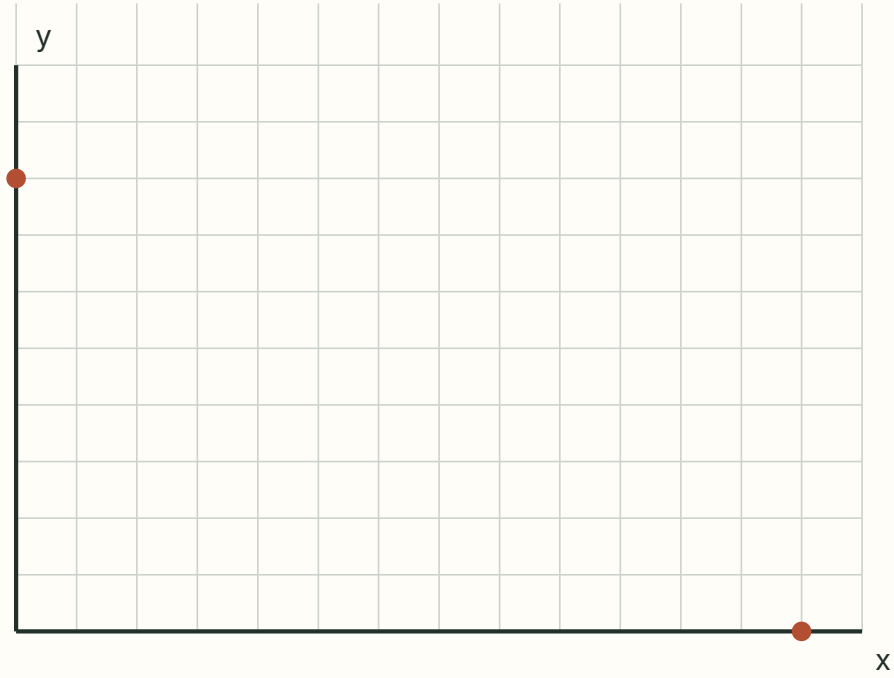
CHAPTER 5 · 5.5 · WORKED EXAMPLE 1

Systems of inequalities and resource allocation

WORKED EXAMPLE 5.5.1

Problem. A co-operative has 40 dollars, seed costs 3 dollars a packet and fertiliser 5 dollars a bag, and at least 2 bags of fertiliser are needed. Write the system and find the greatest number of seed packets.

the feasible region for $3x + 5y \le 40$



Mathematical reasoning. Each condition becomes an inequality, and the feasible region is where all of them hold at once.

1. Let x be seed packets and y be bags of fertiliser.
2. $3x + 5y \le 40$ (the budget) and $y \ge 2$ (the minimum fertiliser).
3. With $y = 2$: $3x + 10 \le 40$, so $3x \le 30$, giving $x \le 10$.
4. So the greatest number of seed packets is 10, with 2 bags.

Final answer. at most 10 seed packets, with exactly 2 bags of fertiliser

Interpretation. The budget limit binds at that corner, which is where the best value of x occurs.

CHAPTER 5 · 5.5 · WORKED EXAMPLE 2

Systems of inequalities and resource allocation

WORKED EXAMPLE 5.5.2

Problem. A farmer plants wheat and barley on at most 100 hectares, with at least 20 hectares of each. Write the system and state its feasible region.

$$20 \le x,\ 20 \le y,\ x + y \le 100$$



Mathematical reasoning. Each constraint is a half-plane, and the feasible region is their intersection.

1. $w + b \le 100$, $w \ge 20$ and $b \ge 20$.
2. The third condition gives $b \le 100 - w$.
3. Combining with $b \ge 20$ gives $20 \le b \le 100 - w$.
4. So w must satisfy $20 \le w \le 80$.

Final answer. $20 \le w \le 80$ and $20 \le b \le 100 - w$

Interpretation. The feasible region is a band of choices, and the farmer still has to choose.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding

5.A1 FLUENCY

Solve $5x - 3 = 22$.

5.A2 FLUENCY

Solve $3(x + 4) = 27$.

5.A3 FLUENCY

Solve $(2x + 1)/3 = 5$.

5.A4 FLUENCY

Solve $-4x = 20$.

5.A5 FLUENCY

Solve $7 - 2x = 1$.

5.A6 FLUENCY

Solve $x/4 - 2 = 3$.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding (continued)

5.A7 FLUENCY

Solve $3x + 2 = 2x + 9$.

5.A8 FLUENCY

Solve $2(3x - 1) = 4(x + 2)$.

5.A9 FLUENCY

Solve $5x - 8 = 3x + 12$.

5.A10 FLUENCY

Solve $(x - 2)/5 = (x + 4)/10$.

5.A11 FLUENCY

Solve $4x + 3 = 3x + 8$ and check your answer.

5.A12 REASONING

Explain why you cannot divide both sides of an equation by x.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns

5.B1 REASONING

Investigate: does the equation $2x + 3 = 2x + 5$ have a solution? Justify.

5.B2 REASONING

Test the claim: every linear equation has exactly one solution.

5.B3 REASONING

Investigate the system $x + y = 4$ and $2x + 2y = 9$. Classify it.

5.B4 REASONING

Generalise: when does $ax + by = c$ and $kax + kby = kc$ have infinitely many solutions?

5.B5 REASONING

Investigate whether the system $x + y = 3$ and $x + y = 3$ is one equation or two.

5.B6 REASONING

Test the claim: if two lines have different gradients they must cross.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns (continued)

5.B7 REASONING

Investigate: solve 2x + 3y = 12 for y in terms of x, and describe the set of solutions.

5.B8 REASONING

Generalise a rule for classifying a linear system from its coefficients.

5.B9 REASONING

Investigate whether the system 3x + 2y = 6 and 6x + 4y = 12 has one solution.

5.B10 REASONING

Test the claim: a system of three equations in two unknowns always has a solution.

5.B11 REASONING

Investigate the pattern of solutions to x + y = n for n = 1 to 5.

CRITERION C Communicating

Criterion C practice · Communicating

5.C1 REASONING

Explain in words what a solution of a pair of simultaneous equations means.

5.C2 REASONING

Explain the difference between an equation and an inequality in terms of the answer.

5.C3 REASONING

Explain why multiplying both sides of an inequality by a negative number reverses the sign.

5.C4 REASONING

Describe two methods of solving a simultaneous system and say when each is preferred.

5.C5 REASONING

Explain what it means for a system to have no solution.

5.C6 REASONING

Explain why the solution of a system of inequalities is drawn as a region.

CRITERION C Communicating

Criterion C practice · Communicating (continued)

5.C7 REASONING

Describe how you would check the solution of a simultaneous system.

5.C8 REASONING

Explain the difference between a feasible region and a solution.

5.C9 REASONING

Explain why an inequality is the right tool for a budget limit.

5.C10 REASONING

Describe the three possible outcomes of a linear system and how to recognise each.

5.C11 REASONING

Explain why $0 = 0$ after elimination means infinitely many solutions.

5.C12 REASONING

Describe how a vertex of a feasible region is found.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts

5.D1 APPLICATION

A co-operative has 50 dollars. Bags of seed cost 4 dollars and fertiliser 6 dollars, and at least 3 bags of fertiliser are needed. Find the greatest number of seed bags.

5.D2 APPLICATION

A school orders pencils at 0.30 dollars and notebooks at 1.20 dollars with a 60-dollar budget and at least 20 notebooks. Find the greatest number of pencils.

5.D3 APPLICATION

A delivery van carries at most 800 kg. Boxes weigh 12 kg and crates 25 kg, and at least 10 crates are needed. Find the greatest number of boxes.

5.D4 APPLICATION

A farmer has 60 hectares, plants wheat and barley, and needs at least 15 hectares of barley. Write the system and give one feasible plan.

5.D5 APPLICATION

A charity packs food parcels with a budget of 200 dollars. Rice costs 2 dollars a kilo and beans 3 dollars a kilo, and each parcel needs at least 1 kilo of beans. Write the system for 20 parcels.

5.D6 REASONING

Explain whether a plan that satisfies every constraint is automatically the best plan.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts (continued)

5.D7 APPLICATION

Two bus companies charge 3 dollars plus 0.50 dollars a kilometre and 5 dollars plus 0.30 dollars a kilometre. Find the distance at which they cost the same and say which is cheaper below it.

5.D8 APPLICATION

A water tank holds at most 500 litres and is filled by two pipes, one at 8 litres a minute for at most 30 minutes and one at 12 litres a minute. Find whether the tank can be filled in 30 minutes.

5.D9 REASONING

A community garden has 40 m of fencing and wants a rectangular plot of area at least 96 m^2. Find the feasible widths.

5.D10 APPLICATION

A shop sells two teas. Blend A costs 4 dollars a kilo and blend B 6 dollars a kilo. A 10-kilo mixture must cost at most 48 dollars. Find the greatest amount of blend B.

5.D11 APPLICATION

A school trip costs 8 dollars a student plus a 120-dollar bus. With a 500-dollar budget, find the greatest number of students.

5.D12 REASONING

Explain why a system of inequalities is a better model than an equation for a shared resource.

CHAPTER 5 · APPLICATION AND ASSESSMENT

Applying chapter 5

These tasks use equations, inequalities, simultaneous equations and systems of inequalities. Check every solution in the original conditions before reporting it.

A1 · The seed and fertiliser budget

A co-operative has 60 dollars to spend. Seed costs 4 dollars a packet, fertiliser 6 dollars a bag, and at least 3 bags of fertiliser are required. Write the system of inequalities, find the greatest number of seed packets, and state one other feasible plan.

Criteria A and C · show the system and justify the greatest value.

A2 · The two water carriers

Carrier A charges 5 dollars plus 0.40 dollars a litre and carrier B charges 2 dollars plus 0.60 dollars a litre. Find the volume at which the charges are equal, state which is cheaper below that volume, and find the volume that exactly uses a 30-dollar budget with each carrier.

Criteria A and D · interpret the crossing point and comment on the sense of your answers.

CHAPTER 5 · REVIEW

What you should now be able to do

Key ideas

- Solving an equation applies inverse operations to both sides.
- Multiplying or dividing an inequality by a negative number reverses the sign.
- The solution of an inequality is a set, written as an interval.
- Simultaneous equations are solved by substitution or elimination.
- A linear system has one solution, no solution, or infinitely many.
- A system of inequalities describes a feasible region.
- Every solution should be checked in the original conditions.

Vocabulary

equation · inequality · solution set · simultaneous equations · substitution · elimination · no solution · infinitely many solutions · feasible region · constraint · vertex

CHAPTER 5 · REVIEW

Mixed practice · consolidate and extend

Do not use a calculator unless a question asks for one. Show a method where one is requested.

5.R1 Solve $4x - 7 = 21$.

5.R2 Solve $-3x > 12$.

5.R3 Solve $x + y = 7$ and $x - y = 3$.

5.R4 Classify the system $2x + 4y = 8$ and $x + 2y = 5$.

5.R5 Solve $-2 \le 3x - 1 < 8$.

5.R6 Explain why dividing an inequality by a negative number reverses the sign.

CHAPTER 5 · REVIEW

Returning to the statement of inquiry

STATEMENT OF INQUIRY

The relationship between quantities that must satisfy several conditions at once can be represented as a system, and the equivalence of the different methods of solving such systems allows the feasible solutions to be identified, checked and justified.

Reflect on the chapter

Answer in your own words, using evidence from this chapter.

- Write the statement of inquiry for this chapter in your own words.
- Which mathematical ideas from this chapter did you use to understand it? Name at least two.
- How did the key concept of relationships and the related concepts of equivalence and systems appear in your work?
- How is the global context of globalisation and sustainability, and its exploration of resources and consumption, visible in the investigations and examples you completed?
- Use evidence from an investigation or a worked example to justify your answer.
- Has your understanding of the statement of inquiry changed since the start of the chapter? Explain how.
- What are the limitations of an inequality model for sharing a limited resource, and what would you still like to ask?

CHAPTER 5 · PROJECTS, SERVICE AND ACTION

Acting on what you have learned

Every shared resource is limited, and limits are inequalities. Use systems to see the feasible choices before choosing among them.

Service as Action

COMMUNITY NEED	Community groups share limited supplies without recording the limits, so some needs are missed.
MATHEMATICAL CONNECTION	Systems of inequalities and feasible regions.
PROPOSED ACTION	Write down the limits of one shared supply as inequalities and publish the feasible options.
EXPECTED IMPACT	Distribution decisions become transparent and checkable.
STUDENT REFLECTION	Record the constraints, the feasible region, and the choice you made.

Community Project

PROJECT IDEA	A resource-allocation plan for a school event.
NEED	Budgets and stock limits are decided informally.
RESEARCH QUESTION	What combinations of stock meet every limit and still raise enough money?
MATHEMATICS	Linear equations, inequalities and feasible regions.
ACTION OR OUTCOME	A written allocation plan with its constraints.
BENEFICIARIES	Stall holders and the groups they support.
EVIDENCE	The system of inequalities, the region and the chosen plan.

Personal Project connection · MYP 5

PERSONAL INTEREST	Budgeting, cooking or planning a trip.
LEARNING GOAL	To model a personal limit with inequalities and choose within it.
PRODUCT OR OUTCOME	A budget plan with stated constraints.
MATHEMATICS	Inequalities, feasible regions and linear models.
PROCESS	List the limits, write the inequalities, then choose a plan.
EVIDENCE	The system, the feasible region and the plan.

CHAPTER 6 · PART III · SHAPE, SPACE AND PROOF

06

Transformations, congruence and symmetry

A transformation moves a shape without changing it. Congruence is the claim that two shapes are related by such a movement, and a proof is what makes that claim secure.

ENGAGE

A tile maker cuts a triangular tile and needs a second tile that fits the gap exactly. Rather than measuring every side again, the maker places the first tile on the second and turns it over. What has been done mathematically, and how can it be justified?

Framing the inquiry

Key concept: Form
Related concepts: Space · Justification
Global context: Personal and cultural expression
Global context exploration: Artistry, craft, creation and beauty

Statement of inquiry: The form of a shape is preserved by a transformation, and the justification that a sequence of transformations maps one shape exactly onto another is what allows congruence and symmetry to be used in craft, design and pattern.

Questions to guide us

- **Factual:** What are the four transformations, and what are the four conditions for triangle congruence?
- **Conceptual:** Why does a transformation preserve form, and what exactly makes a justification of congruence complete?
- **Debatable:** Is a drawing that shows two triangles fitting together a proof that they are congruent?

ATL · THINKING. In order for a student to justify a claim about two shapes, students must decide what would count as sufficient evidence, test the claim against that standard, and state the justification in a form another person can check.

CHAPTER 6 · INVESTIGATION 6.1

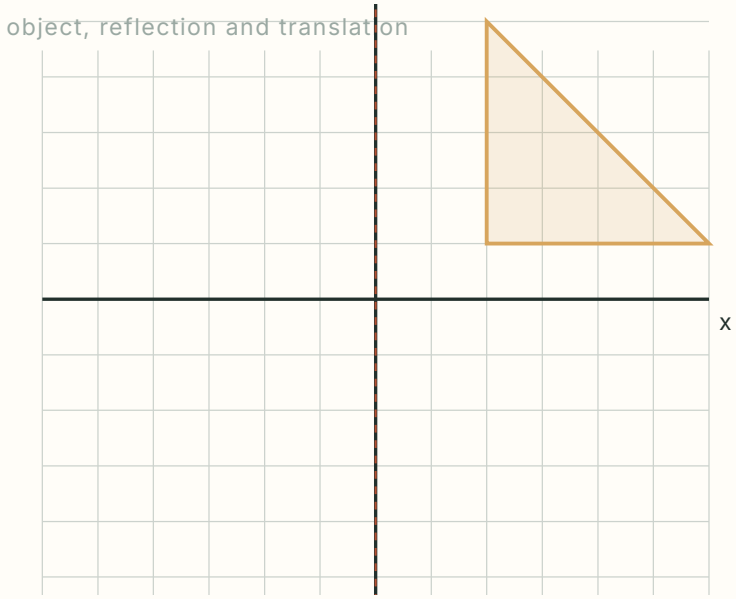
Moving shapes without changing them

INQUIRY QUESTION

How can a shape be moved without changing it, and what stays the same under every such movement?

YOUR TASK

Cut out a triangle from card. Perform a translation, a reflection and a rotation with it, and record the position of the vertices after each movement. Then check which measurements stayed the same: side lengths, angles and area.



transformation	what moved	side lengths	angles	area
translation	every point by the same vector	unchanged	unchanged	unchanged
reflection	every point mirrored in a line	unchanged	unchanged	unchanged
rotation	every point turned about a centre	unchanged	unchanged	unchanged
enlargement, factor 2	every point moved away from a centre	doubled	unchanged	quadrupled

CHAPTER 6 · INVESTIGATION 6.1

Moving shapes without changing them

What do you notice?

- The three isometries left every measurement unchanged.
- The enlargement doubled the lengths and quadrupled the area, but kept the angles.
- Each transformation could be described completely by a small amount of information.
- The transformed triangle fitted exactly into the gap it was moved to.

Observation. Translations, reflections and rotations preserve lengths, angles and area, so the form of a shape is unchanged.

MATHEMATICAL CONCLUSION

A transformation maps a shape to its image, and the three isometries preserve form exactly, which is why a shape and its image can be used interchangeably.

Your own words

CHAPTER 6 · 6.1

Moving shapes without changing them

A transformation maps every point of a shape to a new point. The original is the object and the result is the image.

The four transformations are translation, reflection, rotation and enlargement. The first three preserve form exactly; an enlargement changes size but preserves shape.

What is preserved under all four is called invariance. Lengths, angles and area are invariant under translation, reflection and rotation, which is why they are called isometries and why the position of a shape in space can change without the shape itself changing.

Because form is preserved, a transformed shape can be used in place of the original: a tile cut once can be placed many times.

> FORMAL DEVELOPMENT
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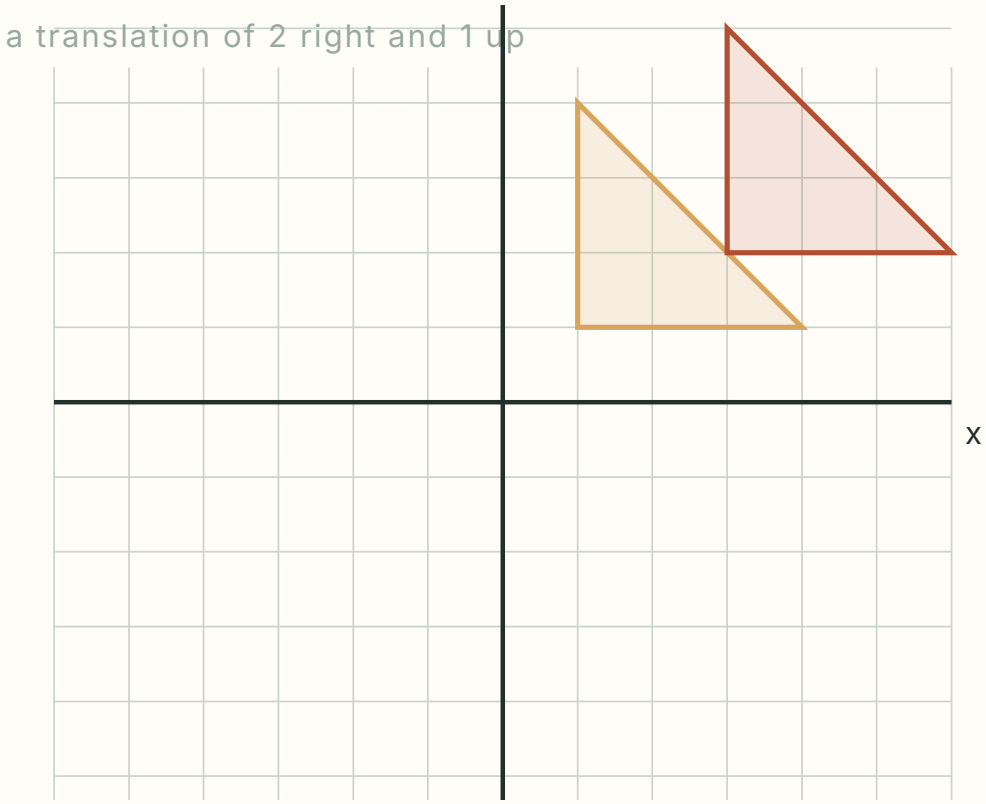
In your own words

CHAPTER 6 · 6.1 · WORKED EXAMPLE 1

Moving shapes without changing them

WORKED EXAMPLE 6.1.1

Problem. Describe the transformation that maps triangle A with vertices (1, 1), (4, 1), (1, 4) onto triangle B with vertices (3, 2), (6, 2), (3, 5).



Mathematical reasoning. Every vertex moved by the same amount, which is the signature of a translation.

1. Compare corresponding vertices: (1, 1) → (3, 2).
2. The change is +2 in x and +1 in y.
3. Check the other vertices: (4, 1) → (6, 2) and (1, 4) → (3, 5).
4. Every point moved by the same vector, so the transformation is a translation.

Final answer. a translation by the vector (2, 1)

Interpretation. A single vector describes the whole movement, which is why translations are easy to specify.

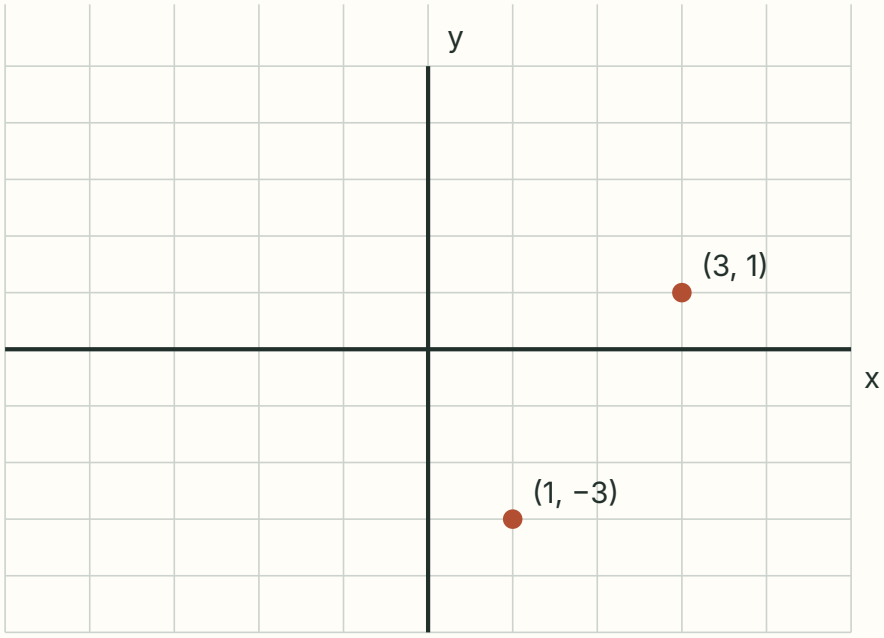
CHAPTER 6 · 6.1 · WORKED EXAMPLE 2

Moving shapes without changing them

WORKED EXAMPLE 6.1.2

Problem. A shape is rotated 90° clockwise about the origin. Where does the point (3, 1) go?

90° clockwise about the origin



Mathematical reasoning. A rotation preserves distance from the centre and turns every direction through the same angle.

1. The rule for a 90° clockwise rotation about the origin is $(x, y) \to (y, -x)$.
2. Apply it to (3, 1): (1, −3).
3. Check the distance from the origin: $\sqrt{10}$ before and after.

Final answer. (3, 1) maps to (1, −3)

Interpretation. The distance from the centre is unchanged, which is what makes a rotation an isometry.

CHAPTER 6 · INVESTIGATION 6.2

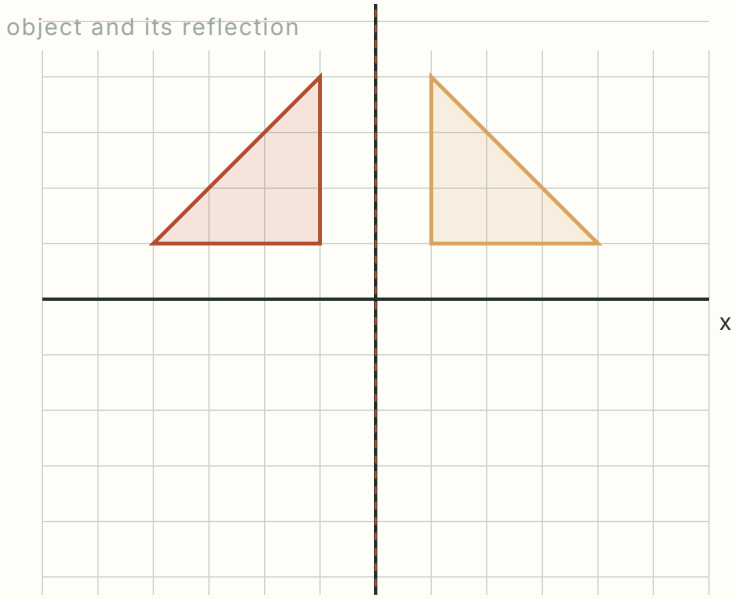
Reflecting across a line

INQUIRY QUESTION

What is preserved when a shape is reflected, and how is the mirror line described?

YOUR TASK

Draw a shape on grid paper and reflect it in the y-axis, then in the x-axis, then in the line $y = x$. Record the coordinates of each image. Then reflect the shape twice in the same line and describe what happens.



reflection	image of (x, y)	image of (3, 2)
in the y-axis	(−x, y)	(−3, 2)
in the x-axis	(x, −y)	(3, −2)
in the line y = x	(y, x)	(2, 3)
twice in the y-axis	(x, y) again	(3, 2) — back to the start

CHAPTER 6 · INVESTIGATION 6.2

Reflecting across a line

What do you notice?

- Each reflection changed only the coordinate that measures distance from the mirror line.
- Reflecting twice in the same line returned the original point.
- The reflected shape is congruent to the original, so every length and angle was preserved.
- The orientation of the vertices reversed, which reflection always does.

Observation. A reflection preserves lengths and angles and reverses orientation, and reflecting twice in the same line is the identity.

MATHEMATICAL CONCLUSION

A reflection maps each point to the point the same distance from the mirror line on the opposite side, preserving form and reversing orientation.

Your own words

CHAPTER 6 · 6.2

Reflecting across a line

A reflection maps every point to the point the same distance from the mirror line, on the opposite side. The mirror line is the perpendicular bisector of every segment joining a point to its image.

Reflection reverses orientation: a shape and its image are mirror images, and a clockwise order of vertices becomes anticlockwise.

The mirror line can be described by an equation. Reflection in the y-axis maps $(x, y) \to (-x, y)$; reflection in the x-axis maps $(x, y) \to (x, -y)$; reflection in $y = x$ maps $(x, y) \to (y, x)$.

Because a reflection preserves lengths and angles, a reflected shape is congruent to the original. The justification is the invariance, not the appearance.

> FORMAL DEVELOPMENT
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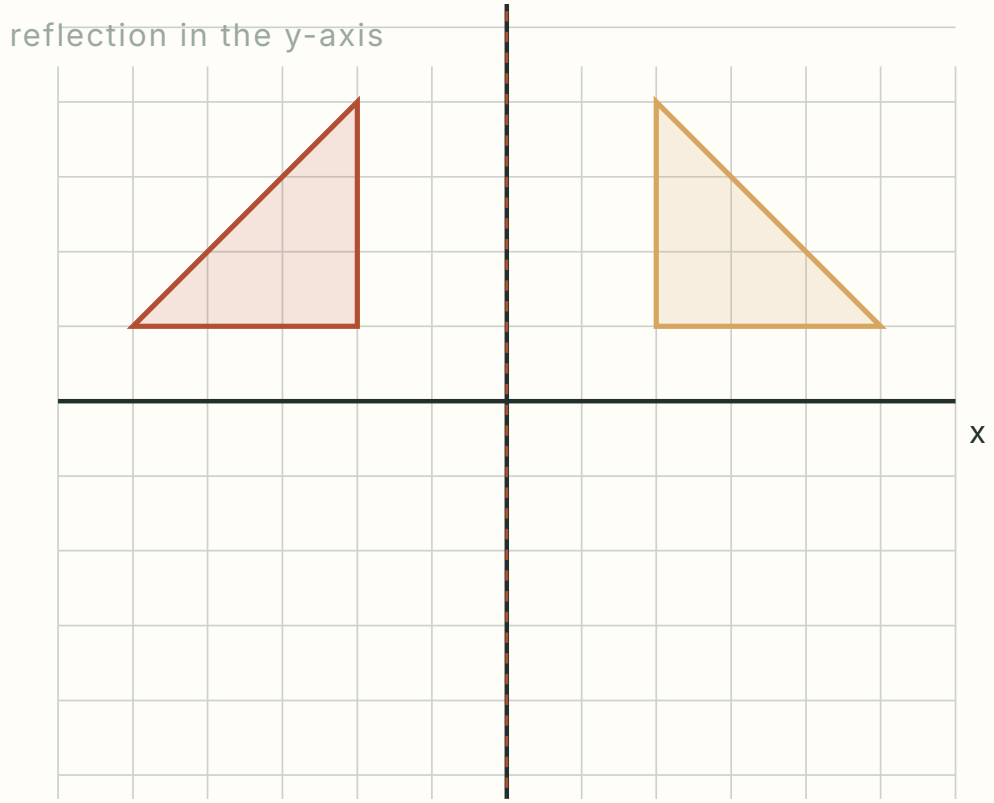
In your own words

CHAPTER 6 · 6.2 · WORKED EXAMPLE 1

Reflecting across a line

WORKED EXAMPLE 6.2.1

Problem. Reflect the triangle with vertices (2, 1), (5, 1), (2, 4) in the y-axis.



Mathematical reasoning. Reflection in the y-axis changes the sign of every x-coordinate and leaves y unchanged.

1. The rule is $(x, y) \to (-x, y)$.
2. $(2, 1) \to (-2, 1)$, $(5, 1) \to (-5, 1)$ and $(2, 4) \to (-2, 4)$.
3. Each image point is the same distance from the y-axis as its object point.

Final answer. the image has vertices (−2, 1), (−5, 1) and (−2, 4)

Interpretation. The y-axis is the perpendicular bisector of every segment joining a vertex to its image.

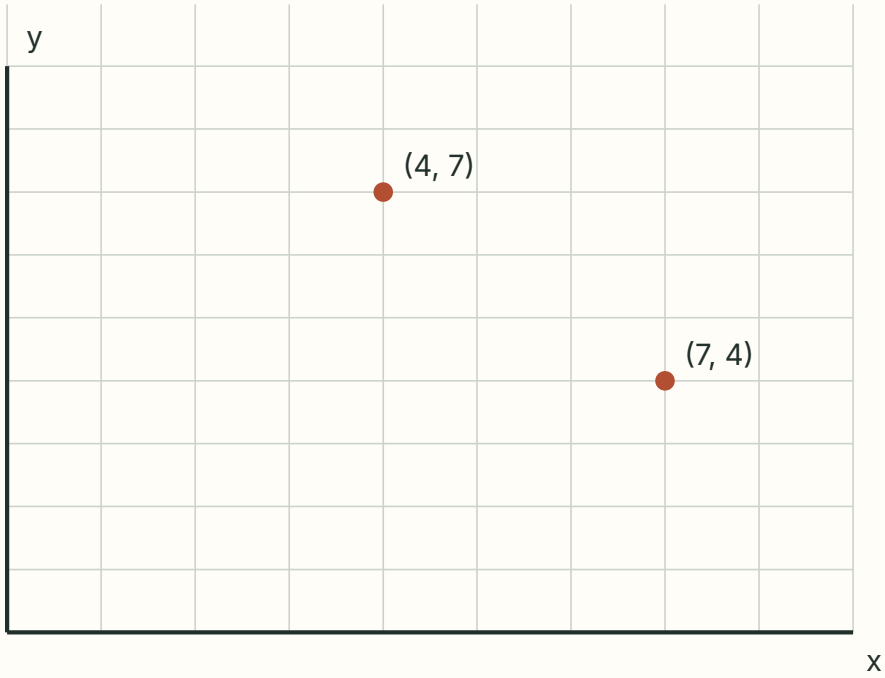
CHAPTER 6 · 6.2 · WORKED EXAMPLE 2

Reflecting across a line

WORKED EXAMPLE 6.2.2

Problem. Reflect the point (4, 7) in the line y = x.

reflection in y = x



Mathematical reasoning. Reflection in y = x interchanges the coordinates, because the line is the set of points where they are equal.

1. The rule for reflection in y = x is (x, y) ⟶ (y, x).
2. So (4, 7) ⟶ (7, 4).
3. Both points are the same distance from the line y = x.

Final answer. (7, 4)

Interpretation. Interchanging the coordinates is the algebraic version of swapping the two sides of the line.

CHAPTER 6 · INVESTIGATION 6.3

Rotating about a point

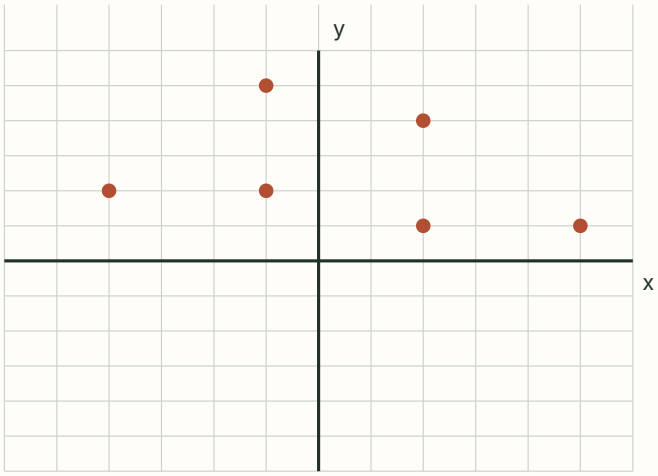
INQUIRY QUESTION

How does a rotation preserve form, and what information is needed to describe it?

YOUR TASK

Take a shape and rotate it 90°, 180°, 270° and 360° anticlockwise about the same centre. Record the vertices after each rotation, then describe what a rotation of 360° does and what a rotation of 270° anticlockwise is equivalent to.

four rotations of one triangle



rotation	image of (x, y)	image of (2, 1)
90° anticlockwise	(−y, x)	(−1, 2)
180°	(−x, −y)	(−2, −1)
270° anticlockwise	(y, −x)	(1, −2)
360°	(x, y) again	(2, 1) — back to the start

CHAPTER 6 · INVESTIGATION 6.3

Rotating about a point

What do you notice?

- Each rotation preserved the distances from the centre.
- The side lengths and angles of the triangle never changed.
- A rotation of 360° returned the shape to its starting position.
- A rotation of 270° anticlockwise is the same as 90° clockwise.

Observation. A rotation preserves every length and angle and moves every point along a circle about the centre.

MATHEMATICAL CONCLUSION

A rotation turns every point through the same angle about a fixed centre, preserving form, and it is fully described by the centre, the angle and the direction.

Your own words

CHAPTER 6 · 6.3

Rotating about a point

A rotation turns every point through the same angle about a fixed centre. It needs three pieces of information: the centre, the angle and the direction.

Distances from the centre are preserved, and so are all lengths and angles, which makes a rotation an isometry.

The rules for rotations about the origin are: 90° clockwise $(x, y) \to (y, -x)$; 90° anticlockwise $(x, y) \to (-y, x)$; 180° $(x, y) \to (-x, -y)$.

A rotation of 180° about the origin is the same as a reflection in the origin, and it maps every line through the origin to itself.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

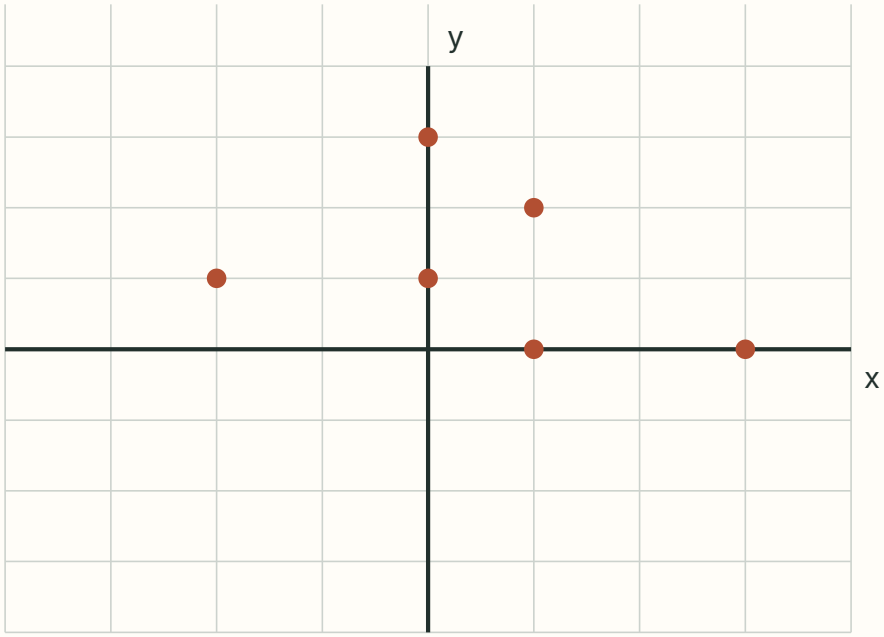
CHAPTER 6 · 6.3 · WORKED EXAMPLE 1

Rotating about a point

WORKED EXAMPLE 6.3.1

Problem. Rotate the triangle with vertices (1, 0), (3, 0), (1, 2) through 90° anticlockwise about the origin.

90° anticlockwise about the origin



Mathematical reasoning. The rule for a 90° anticlockwise rotation about the origin interchanges the coordinates with a sign change.

1. The rule is $(x, y) \to (-y, x)$.
2. $(1, 0) \to (0, 1)$.
3. $(3, 0) \to (0, 3)$.
4. $(1, 2) \to (-2, 1)$.

Final answer. the image has vertices (0, 1), (0, 3) and (−2, 1)

Interpretation. Each image point is the same distance from the origin as its object point, which is the invariance that makes a rotation an isometry.

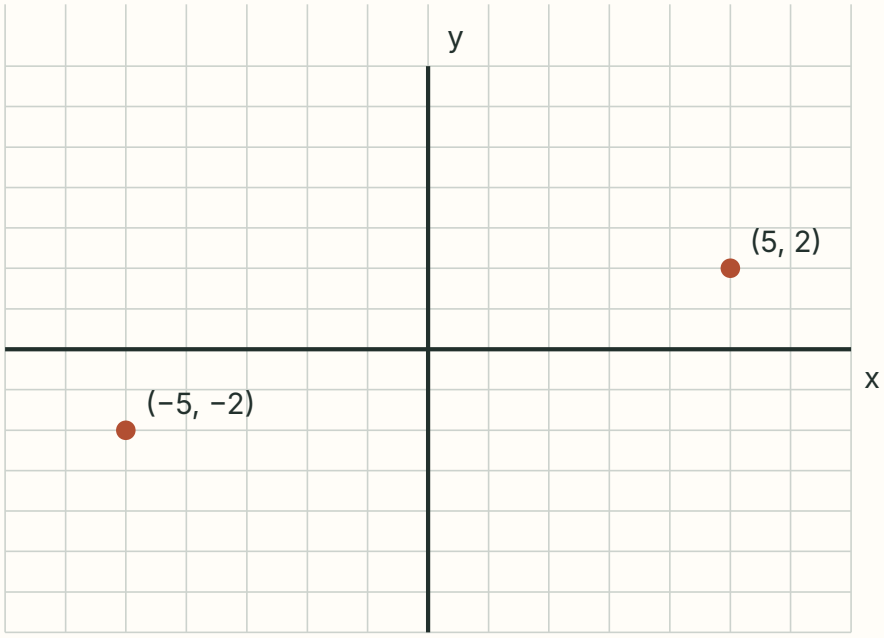
CHAPTER 6 · 6.3 · WORKED EXAMPLE 2

Rotating about a point

WORKED EXAMPLE 6.3.2

Problem. Rotate the point (5, 2) through 180° about the origin and describe the result.

180° about the origin



Mathematical reasoning. A half turn reverses both coordinates, which is the same as reflecting through the origin.

1. The rule for 180° is (x, y) → (−x, −y).
2. So (5, 2) → (−5, −2).
3. The distance from the origin is √29 before and after.

Final answer. (−5, −2)

Interpretation. A half turn about the origin is a point reflection, and it is its own inverse.

CHAPTER 6 · INVESTIGATION 6.4

Congruence and its criteria

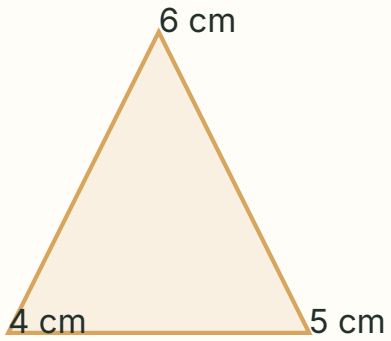
INQUIRY QUESTION

How can you justify that two triangles are congruent without measuring every side?

YOUR TASK

Build triangles from the following measurements and record how many different triangles are possible each time: (a) 4 cm, 5 cm, 6 cm; (b) 4 cm, 5 cm, 40° with the angle included; (c) 4 cm, 5 cm, 40° with the angle not included; (d) 40°, 60°, 5 cm with the side included.

one triangle, three sides



measurements	condition	how many triangles
4, 5, 6 cm	SSS	exactly one
4 cm, 5 cm, 40° included	SAS	exactly one
4 cm, 5 cm, 40° not included	SSA — ambiguous	two
40°, 60°, 5 cm included	ASA	exactly one

CHAPTER 6 · INVESTIGATION 6.4

Congruence and its criteria

What do you notice?

- Three cases produced exactly one triangle, and one produced two.
- The ambiguous case is two sides and a non-included angle.
- Where exactly one triangle is possible, the shape is fixed and congruence is justified.
- Where two are possible, the measurements do not determine the shape.

Observation. Measurements that produce exactly one triangle justify a congruence claim, and measurements that produce more than one do not.

MATHEMATICAL CONCLUSION

SSS, SAS, ASA and RHS are sufficient conditions for triangle congruence because each determines a unique triangle, and a justification of congruence must cite one of them.

Your own words

CHAPTER 6 · 6.4

Congruence and its criteria

Two shapes are congruent when one can be mapped onto the other by an isometry. Congruent shapes have the same form and the same size.

For triangles, four conditions are sufficient: SSS (three sides), SAS (two sides and the included angle), ASA (two angles and the included side) and RHS (right angle, hypotenuse and one side).

These conditions are sufficient because they fix the triangle: no other triangle can be built with the same three measurements, so the shape is determined.

A drawing is not a justification. Two triangles that look congruent may differ by a small amount that the eye cannot see, which is why the conditions are stated in measurements.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

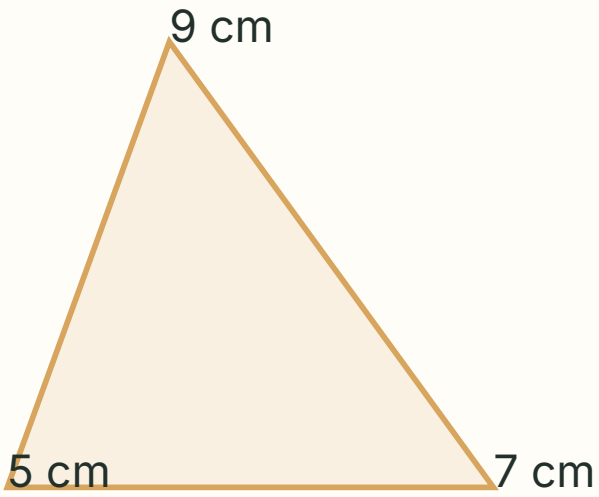
CHAPTER 6 · 6.4 · WORKED EXAMPLE 1

Congruence and its criteria

WORKED EXAMPLE 6.4.1

Problem. Triangle P has sides 5 cm, 7 cm and 9 cm. Triangle Q has sides 9 cm, 5 cm and 7 cm. Are they congruent, and which condition applies?

three sides fix the triangle



Mathematical reasoning. Three matching sides are enough to fix a triangle, and the order in which they are listed does not matter.

1. The three sides of P are 5, 7 and 9 cm.
2. The three sides of Q are 9, 5 and 7 cm, which are the same three lengths.
3. So the SSS condition is satisfied.
4. The triangles are congruent.

Final answer. yes, congruent by SSS

Interpretation. Only three measurements are needed, not six, which is why the criteria save work.

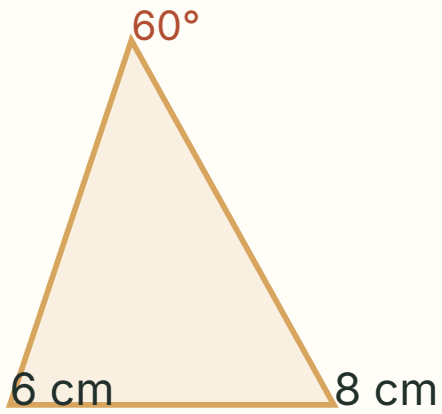
CHAPTER 6 · 6.4 · WORKED EXAMPLE 2

Congruence and its criteria

WORKED EXAMPLE 6.4.2

Problem. Two triangles have sides 6 cm and 8 cm with an included angle of 60°, and sides 6 cm and 8 cm with an angle of 60° that is not included. Are they congruent?

the angle must be included



Mathematical reasoning. SAS requires the angle between the two given sides; an angle elsewhere does not fix the triangle.

1. The first triangle satisfies SAS, because 60° is between the 6 cm and 8 cm sides.
2. The second gives two sides and a non-included angle, which is the ambiguous case.
3. Two different triangles can be built from those measurements.
4. So SAS does not apply and congruence cannot be claimed.

Final answer. only the first is determined; congruence cannot be claimed for the second

Interpretation. The word 'included' is doing real work, and leaving it out changes the claim.

CHAPTER 6 · INVESTIGATION 6.5

Symmetry and tessellation

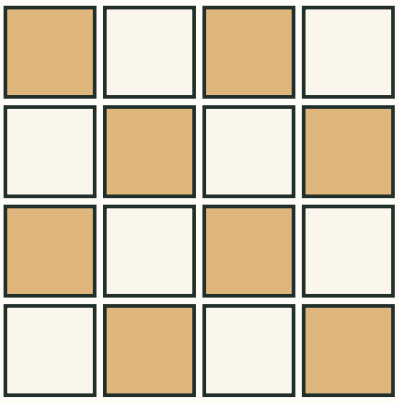
INQUIRY QUESTION

How does symmetry generate a pattern that covers a surface?

YOUR TASK

Test which regular polygons tessellate by computing 360° divided by each interior angle. Then take an irregular quadrilateral and check whether it tessellates by tracing copies. Record the angle sum at each meeting point.

a quadrilateral tessellation



polygon	interior angle	360° ÷ angle	tessellates?
triangle	60°	6	yes
square	90°	4	yes
pentagon	108°	3.33…	no
hexagon	120°	3	yes
octagon	135°	2.67…	no
any quadrilateral	sums to 360°	4 angles at a point	yes

CHAPTER 6 · INVESTIGATION 6.5

Symmetry and tessellation

What do you notice?

- Only the triangle, square and hexagon tessellate among the regular polygons.
- Each works because its interior angle divides 360° exactly.
- The quadrilateral tessellates because its four angles sum to 360°, so one of each fits around a point.
- The pattern can be extended indefinitely in every direction.

> **Observation.** A shape tessellates when its copies meet edge to edge with the angles at every meeting point summing to exactly 360°.

MATHEMATICAL CONCLUSION

A regular polygon tessellates exactly when its interior angle divides 360°, and any quadrilateral tessellates because its four angles sum to 360°.

Your own words

CHAPTER 6 · 6.5

Symmetry and tessellation

A shape has line symmetry if a reflection maps it onto itself, and rotational symmetry if a rotation does. The order of rotational symmetry is the number of times the shape fits onto itself in a full turn.

A tessellation covers a surface with shapes that meet edge to edge and leave no gaps. Regular tessellations use one regular polygon, and only three exist: triangles, squares and hexagons.

A shape tessellates if its copies can be arranged so that the angles at every meeting point sum to 360°. This is why the interior angle must divide 360°.

Symmetry and tessellation are used in craft, tiling and design. The mathematics is the constraint that makes the pattern possible, and the design is the choice inside that constraint.

> FORMAL DEVELOPMENT
>
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In your own words

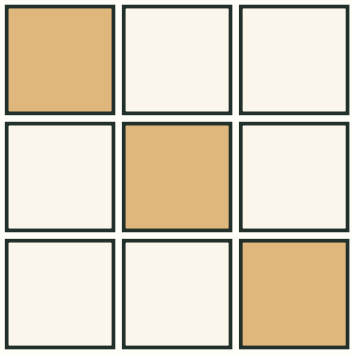
CHAPTER 6 · 6.5 · WORKED EXAMPLE 1

Symmetry and tessellation

WORKED EXAMPLE 6.5.1

Problem. A regular hexagon has interior angle 120°. Does it tessellate, and why?

a hexagonal tessellation



Mathematical reasoning. A shape tessellates if copies meet edge to edge with the angles summing to 360° at every meeting point.

1. The interior angle of a regular hexagon is 120°.
2. 360° ÷ 120° = 3, so three hexagons meet at each point.
3. The angles sum to 360° with no gap and no overlap.
4. So the hexagon tessellates.

Final answer. yes, because 3 × 120° = 360°

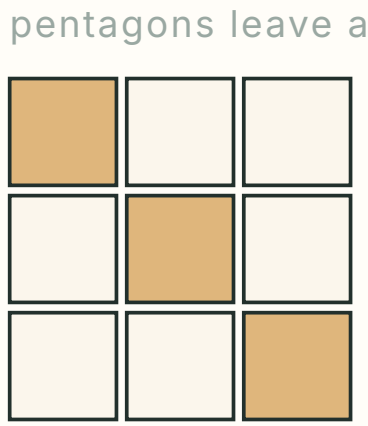
Interpretation. The division of 360° by the interior angle is the whole test for a regular polygon.

CHAPTER 6 · 6.5 · WORKED EXAMPLE 2

Symmetry and tessellation

WORKED EXAMPLE 6.5.2

Problem. Does a regular pentagon tessellate? Justify your answer.



Mathematical reasoning. If the interior angle does not divide 360°, copies cannot meet without a gap or an overlap.

1. The interior angle of a regular pentagon is 108°.
2. 360° ÷ 108° = 3.33..., which is not a whole number.
3. Three pentagons leave a gap of 36°, and four overlap by 72°.
4. So a regular pentagon does not tessellate.

Final answer. no, because 108° does not divide 360°

Interpretation. The failure of the division is a proof, not an observation about one attempt.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding

6.A1 FLUENCY
Translate the point (3, 5) by the vector (−2, 4).

6.A2 FLUENCY
Reflect the point (4, 3) in the y-axis.

6.A3 FLUENCY
Reflect the point (4, 3) in the x-axis.

6.A4 FLUENCY
Rotate the point (2, 5) through 180° about the origin.

6.A5 FLUENCY
Rotate the point (3, 1) through 90° clockwise about the origin.

6.A6 REASONING
State which measurements are preserved by a translation.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding (continued)

6.A7 FLUENCY

Reflect the point (6, 2) in the line y = x.

6.A8 FLUENCY

An enlargement of scale factor 3 maps a side of 4 cm. How long is the image?

6.A9 FLUENCY

State the order of rotational symmetry of a square.

6.A10 FLUENCY

Reflect the triangle with vertices (1, 1), (4, 1), (1, 5) in the y-axis.

6.A11 FLUENCY

Rotate the point (−3, 2) through 90° anticlockwise about the origin.

6.A12 REASONING

Explain the difference between a translation and a rotation.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns

6.B1 REASONING

Investigate: does the order of a reflection and a translation matter? Test with the point (2, 3), a reflection in the y-axis and a translation of (1, 0).

6.B2 REASONING

Test the claim: two reflections in perpendicular lines give a rotation.

6.B3 REASONING

Investigate whether a rotation of 90° followed by the same rotation again gives a rotation of 180°.

6.B4 REASONING

Generalise: what is the result of reflecting twice in the same line?

6.B5 REASONING

Investigate whether every triangle has at least one line of symmetry.

6.B6 REASONING

Test the claim: if two triangles have two equal angles and one equal side they are congruent.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns (continued)

6.B7 REASONING

Investigate the minimum information needed to fix a triangle, and justify your answer.

6.B8 REASONING

Generalise: for what values of n does a regular n-gon tessellate?

6.B9 REASONING

Investigate whether an irregular triangle always tessellates.

6.B10 REASONING

Test the claim: a shape with rotational symmetry must have line symmetry.

6.B11 FLUENCY

Investigate the order of rotational symmetry of a regular hexagon.

CRITERION C Communicating

Criterion C practice · Communicating

6.C1 REASONING

Explain in words what it means for two shapes to be congruent.

6.C2 REASONING

Explain why a reflection reverses orientation.

6.C3 REASONING

Describe the information needed to specify a rotation completely.

6.C4 REASONING

Explain why SSS is a sufficient condition for triangle congruence.

6.C5 REASONING

Describe the difference between SAS and the ambiguous SSA case.

6.C6 REASONING

Explain why a drawing alone is not a proof of congruence.

CRITERION C Communicating

Criterion C practice · Communicating (continued)

6.C7 REASONING
Describe two representations of the same reflection and say what each shows.

6.C8 REASONING
Explain the difference between line symmetry and rotational symmetry.

6.C9 REASONING
Describe why only three regular polygons tessellate.

6.C10 REASONING
Explain what is preserved by an enlargement and what is not.

6.C11 REASONING
Describe how to check whether two triangles are congruent using ASA.

6.C12 REASONING
Explain why a tessellation of a craft surface requires edge-to-edge contact.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts

6.D1 APPLICATION

A tile pattern uses a triangle rotated 120° about a vertex. Describe the transformation and the angle sum at that vertex.

6.D2 APPLICATION

A floor is tiled with squares of side 40 cm. A room is 4.8 m by 3.6 m. How many tiles are needed?

6.D3 APPLICATION

A mirror tile of side 15 cm is reflected to make a pair. Find the total perimeter of the pair when they share an edge.

6.D4 APPLICATION

A craft pattern uses a regular hexagon of side 6 cm. Find its perimeter and explain why three fit around a point.

6.D5 REASONING

A triangular tile has angles 50°, 60° and 70°. Explain how many fit around a point.

6.D6 REASONING

A logo is designed as a shape and its reflection. Explain how congruence is justified.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts (continued)

6.D7 APPLICATION
A wall panel is made by rotating a motif 90° four times about one corner. Find the angle at that corner and the number of copies.

6.D8 REASONING
Explain whether a photograph of a building and its reflection in water shows congruent shapes.

6.D9 REASONING
A tiled floor uses two different regular polygons at each vertex. Explain how to find which pairs work.

6.D10 REASONING
A triangular garden bed has sides 3 m, 4 m and 5 m. Explain how to check that a second bed of the same design is congruent without measuring every side.

6.D11 REASONING
A kite design has two lines of symmetry. Explain what each line does.

6.D12 REASONING
Explain why an enlargement of a tile pattern changes the area but not the angles.

CHAPTER 6 · APPLICATION AND ASSESSMENT

Applying chapter 6

These tasks use transformations, congruence, symmetry and tessellation. Justify every claim with a transformation or a congruence condition, not with an impression.

A1 · The tiled panel

A craft panel is made from copies of one triangular tile rotated about a shared vertex. The tile has angles 40°, 60° and 80°. Explain how many copies can meet at that vertex, whether the panel tiles the plane, and justify your answer with an angle calculation.

Criteria A and C · justify with an angle sum, not with a drawing.

A2 · The congruent frame

A picture frame is a parallelogram with sides 12 cm and 7 cm and an included angle of 65°. A second frame is claimed to be congruent. State the minimum measurements needed to check the claim, and justify the congruence condition you used.

Criteria A and B · identify the condition and explain why it is sufficient.

CHAPTER 6 · REVIEW

What you should now be able to do

Key ideas

- A transformation maps an object to an image.
- Translation, reflection and rotation are isometries: they preserve lengths, angles and area.
- A reflection reverses orientation; a rotation preserves it.
- Congruent shapes are related by an isometry.
- SSS, SAS, ASA and RHS are sufficient conditions for triangle congruence.
- A shape tessellates if the angles at each meeting point sum to 360°.
- Only three regular polygons tessellate: the triangle, square and hexagon.

Vocabulary

transformation · object · image · translation · reflection · rotation · enlargement · isometry · invariant · congruent · SSS · SAS · ASA · RHS · line symmetry · rotational symmetry · tessellation

CHAPTER 6 · REVIEW

Mixed practice · consolidate and extend

Do not use a calculator unless a question asks for one. Show a method where one is requested.

6.R1 Translate (5, −2) by the vector (−3, −4).

6.R2 Reflect (7, 1) in the line y = x.

6.R3 Rotate (4, 3) through 90° anticlockwise about the origin.

6.R4 State the order of rotational symmetry of a regular pentagon.

6.R5 Which congruence condition applies to two triangles with sides 5, 6, 7?

6.R6 Explain why a regular octagon does not tessellate.

CHAPTER 6 · REVIEW

Returning to the statement of inquiry

> **STATEMENT OF INQUIRY**
>
> The form of a shape is preserved by a transformation, and the justification that a sequence of transformations maps one shape exactly onto another is what allows congruence and symmetry to be used in craft, design and pattern.

Reflect on the chapter

Answer in your own words, using evidence from this chapter.

- Write the statement of inquiry for this chapter in your own words.
- Which mathematical ideas from this chapter did you use to understand it? Name at least two.
- How did the key concept of form and the related concepts of space and justification appear in your work?
- How is the global context of personal and cultural expression, and its exploration of artistry, craft, creation and beauty, visible in the investigations and examples you completed?
- Use evidence from an investigation or a worked example to justify your answer.
- Has your understanding of the statement of inquiry changed since the start of the chapter? Explain how.
- What are the limitations of a drawing as a justification of congruence, and what would you still like to ask?

CHAPTER 6 · PROJECTS, SERVICE AND ACTION

Acting on what you have learned

Craft and design depend on form being preserved. Use transformations and congruence to make something that fits exactly.

Service as Action

COMMUNITY NEED	A community space needs a floor or wall treatment that can be made and fitted by volunteers.
MATHEMATICAL CONNECTION	Transformations, congruence and tessellation.
PROPOSED ACTION	Design and cut a tiling pattern from one template and fit it in a shared space.
EXPECTED IMPACT	A durable, locally made surface treatment that the community helped to produce.
STUDENT REFLECTION	Record the template, the transformation used, and how the fitting went.

Community Project

PROJECT IDEA	A tiling or mural design for a community wall.
NEED	A shared wall is bare and there is no budget for a commercial treatment.
RESEARCH QUESTION	What pattern can be cut from one template and fitted without gaps?
MATHEMATICS	Reflection, rotation, congruence and tessellation.
ACTION OR OUTCOME	A fitted pattern with a reusable template.
BENEFICIARIES	Everyone who uses the space.
EVIDENCE	The template, the plan and a photograph of the fitted work.

Personal Project connection · MYP 5

PERSONAL INTEREST	Drawing, textiles or model making.
LEARNING GOAL	To use a transformation to make a design that repeats exactly.
PRODUCT OR OUTCOME	A repeating pattern made from one template.
MATHEMATICS	Isometries, congruence and tessellation.
PROCESS	Design the template, apply the transformation, then check the fit.
EVIDENCE	The template, the pattern and a photograph.

CHAPTER 7 · PART III · SHAPE, SPACE AND PROOF

07

Similarity, proof and trigonometry

Similar figures have the same shape at different sizes. The scale factor is the constant ratio between corresponding lengths, and it is what makes indirect measurement possible.

ENGAGE

A surveyor needs the height of a cliff. No one can climb it with a tape measure. The surveyor measures a shadow, paces a distance and records two angles. What has been measured, and what has been inferred?

Framing the inquiry

Key concept: Relationships
Related concepts: Justification · Measurement
Global context: Orientation in space and time
Global context exploration: Natural and human landscapes and resources

Statement of inquiry: The relationship between corresponding lengths in similar figures can be justified as a constant ratio, and that justification, combined with the measurement of angles, allows indirect measurement of distances and heights that cannot be measured directly.

Questions to guide us

- **Factual:** What are the conditions for two triangles to be similar, and what are the three trigonometric ratios?
- **Conceptual:** Why does a constant scale factor justify a claim about every corresponding pair of lengths, and how does Pythagoras' theorem follow from the properties of similar triangles?
- **Debatable:** Is a measurement taken indirectly with trigonometry as reliable as one taken directly with a tape?

ATL · RESEARCH. In order for a student to justify a general claim about shapes, students must gather evidence from several cases, identify what the cases have in common, and express the common structure as a statement that can be proved rather than tested.

CHAPTER 7 · INVESTIGATION 7.1

Similar figures and scale factors

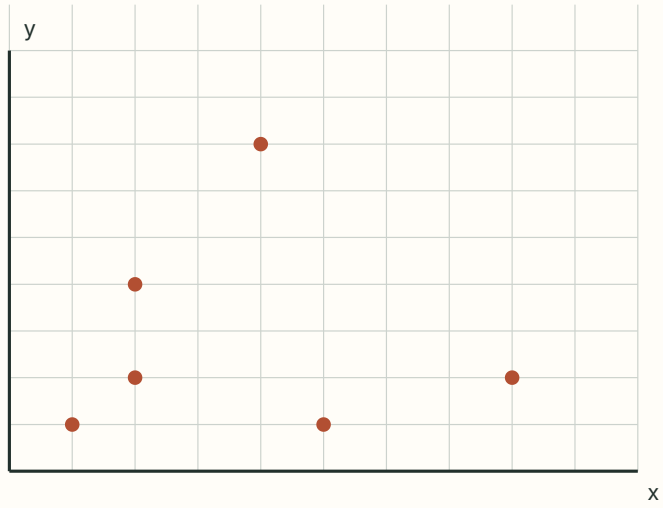
INQUIRY QUESTION

How can you justify that two figures are similar without measuring every angle?

YOUR TASK

Draw a triangle, then enlarge it by scale factor 2 about a chosen centre using a ruler. Measure all six sides and all six angles, and record the ratios of corresponding sides and the ratio of the two areas.

a triangle and its enlargement



measurement	original	enlarged	ratio
side 1	4 cm	8 cm	2
side 2	3.6 cm	7.2 cm	2
side 3	3 cm	6 cm	2
angle A	37°	37°	1
angle B	90°	90°	1
area	5.4 cm^2	21.6 cm^2	4

CHAPTER 7 · INVESTIGATION 7.1

Similar figures and scale factors

What do you notice?

- Every length ratio was exactly 2, the scale factor.
- Every angle was unchanged, so the shape was preserved.
- The area ratio was 4, which is 2^2.
- The enlargement could be described by a single number and a centre.

Observation. Corresponding lengths are in a constant ratio, corresponding angles are equal, and areas are in the square of that ratio.

MATHEMATICAL CONCLUSION

Two figures are similar when corresponding angles are equal and corresponding sides are in a constant ratio k, with areas in the ratio k^2.

Your own words

CHAPTER 7 · 7.1

Similar figures and scale factors

Two figures are similar when their corresponding angles are equal and their corresponding sides are in the same ratio. The ratio is the scale factor.

For triangles, two conditions are sufficient: AA (two equal angles, which forces the third), and SSS (all three pairs of sides in the same ratio). SAS with the included angle also works.

The scale factor from figure A to figure B is k, and the scale factor from B to A is 1/k. Lengths multiply by k and areas by k^2, which is why area scales differently from length.

Similarity is a claim about every pair of corresponding lengths at once, so it needs a justification that covers all of them. Measuring three pairs is evidence; the angle conditions are the justification.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

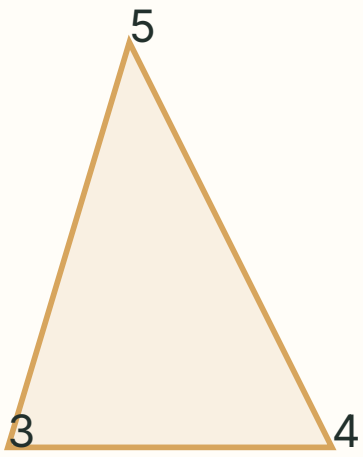
CHAPTER 7 · 7.1 · WORKED EXAMPLE 1

Similar figures and scale factors

WORKED EXAMPLE 7.1.1

Problem. Two triangles have sides 3 cm, 4 cm, 5 cm and 6 cm, 8 cm, 10 cm. Are they similar, and what is the scale factor?

a 3-4-5 triangle



Mathematical reasoning. If all three pairs of sides are in the same ratio, the triangles are similar by the SSS condition.

1. Compare corresponding sides: $6/3 = 2$, $8/4 = 2$ and $10/5 = 2$.
2. All three ratios are equal to 2.
3. So the triangles are similar, with scale factor 2.
4. The angles must also be equal, and indeed both are right-angled.

Final answer. yes, similar with scale factor 2

Interpretation. The scale factor applies to every length, so one number describes the whole relationship.

CHAPTER 7 · 7.1 · WORKED EXAMPLE 2

Similar figures and scale factors

WORKED EXAMPLE 7.1.2

Problem. A photograph 8 cm wide is enlarged to 30 cm wide. By what factor does the area increase?

measurement	original	enlarged	factor
width	8 cm	30 cm	3.75
area	A	$3.75^2 \times A$	14.0625

Area scales as the square of the length factor.

Mathematical reasoning. Lengths multiply by the scale factor and areas by its square, because area is a product of two lengths.

1. Scale factor = $30 \div 8 = 3.75$.
2. Area factor = $3.75^2 = 14.0625$.
3. So the area increases by a factor of about 14.

Final answer. the area increases by a factor of 14.0625

Interpretation. A modest change in width produces a large change in area, which matters for printing and for material cost.

CHAPTER 7 · INVESTIGATION 7.2

Similar triangles and indirect measurement

INQUIRY QUESTION

How can a height that cannot be measured directly be found?

YOUR TASK

On a sunny day, measure the shadow of a partner and the shadow of a tall object such as a tree, a building or a floodlight. Use similarity to estimate the height, then compare with any direct measurement or published figure you can find.

object, shadow and the similar triangle



object	shadow (m)	estimated height (m)	method
partner	1.2	1.55	measured directly
tree	8.4	1.55 × (8.4/1.2) = 10.85	similarity ratio
building	22.5	1.55 × (22.5/1.2) = 29.06	similarity ratio

CHAPTER 7 · INVESTIGATION 7.2

Similar triangles and indirect measurement

What do you notice?

- The ratio height/shadow was the same for every object.
- The two tall estimates were consistent with the number of storeys.
- The method needs level ground and vertical objects.
- The precision is limited by the shadow measurement, not by the mathematics.

Observation. Similar triangles convert a measurable shadow into an unmeasurable height, using one ratio for every object.

MATHEMATICAL CONCLUSION

Two objects measured at the same time form similar triangles with their shadows, so the ratio of height to shadow length is constant and one measured pair determines every height.

Your own words

CHAPTER 7 · 7.2

Similar triangles and indirect measurement

Two triangles formed by the sun's rays and vertical objects are similar, because the rays are parallel and the objects are both perpendicular to the ground. The angle at each object is equal.

This gives the shadow method: the ratio of height to shadow length is the same for every object at the same time, so one measurable pair fixes the ratio for all of them.

Indirect measurement turns an inaccessible quantity into a ratio problem. What is measured is a distance and an angle; what is inferred is a height.

The reliability of the result depends on the assumptions: that the ground is level, that the object is vertical, and that the light rays are effectively parallel.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

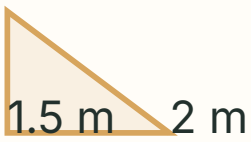
CHAPTER 7 · 7.2 · WORKED EXAMPLE 1

Similar triangles and indirect measurement

WORKED EXAMPLE 7.2.1

Problem. A 1.5 m pole casts a 2 m shadow. At the same time a tree casts an 18 m shadow. Find the height of the tree.

two similar right-angled triangles



Mathematical reasoning. The triangles are similar because the sun's rays are parallel, so the ratio of height to shadow is the same for both objects.

1. Pole: height/shadow = 1.5/2 = 0.75.
2. Tree: height/18 = 0.75.
3. Height = 18 × 0.75 = 13.5 m.

Final answer. 13.5 m

Interpretation. The tree's height was never measured; it was inferred from the ratio.

CHAPTER 7 · 7.2 · WORKED EXAMPLE 2

Similar triangles and indirect measurement

WORKED EXAMPLE 7.2.2

Problem. A student 1.6 m tall stands so that her shadow just reaches the base of a flagpole. She is 4 m from the pole and her shadow is 2.5 m long. Find the height of the pole.



Mathematical reasoning. The two triangles are similar, so the ratio of height to horizontal distance is the same for the student and the pole.

1. Student: 1.6/2.5.
2. Pole: h/(4 + 2.5) = h/6.5.
3. 1.6/2.5 = h/6.5, so h = 6.5 × 0.64.
4. h = 4.16 m.

Final answer. about 4.16 m

Interpretation. The distance from the student to the pole is added to her shadow because both lie on the same horizontal line.

CHAPTER 7 · INVESTIGATION 7.3

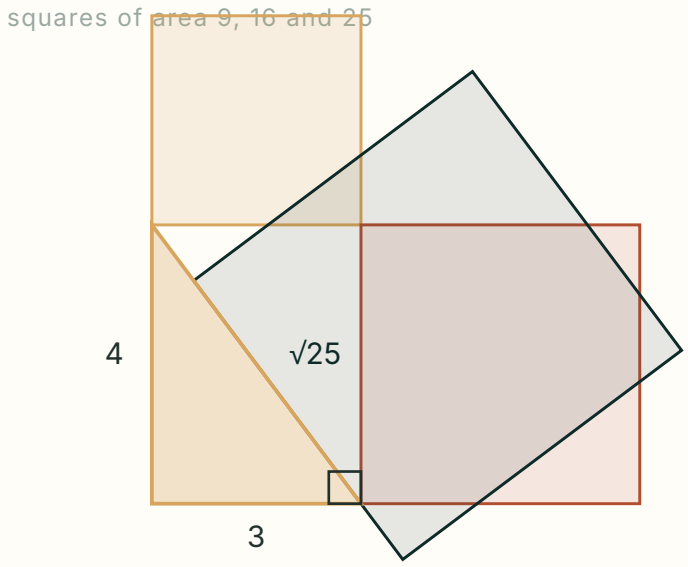
Pythagoras' theorem

INQUIRY QUESTION

How can the relationship between the sides of a right-angled triangle be proved, and why does the proof work?

YOUR TASK

Draw a right-angled triangle with legs a and b, then draw squares on all three sides. Count grid squares to estimate each area, and test whether $a^2 + b^2 = c^2$. Then repeat with two other triangles of your own.



triangle	a	b	$a^2 + b^2$	c^2	c
1	3	4	25	25	5
2	6	8	100	100	10
3	5	12	169	169	13
4	1.5	2	6.25	6.25	2.5

CHAPTER 7 · INVESTIGATION 7.3

Pythagoras' theorem

What do you notice?

- In every case $a^2 + b^2$ equalled c^2 exactly.
- The pattern held for integer sides and for decimal sides.
- Triangles 1 and 2 are similar, and their sides are in the ratio 1:2.
- The largest square always had the same area as the two smaller squares combined.

Observation. The sum of the squares on the two shorter sides always equals the square on the hypotenuse.

MATHEMATICAL CONCLUSION

In a right-angled triangle $a^2 + b^2 = c^2$, where c is the hypotenuse, and the converse holds: equal sums of squares force a right angle.

Your own words

CHAPTER 7 · 7.3

Pythagoras' theorem

Pythagoras' theorem states that in a right-angled triangle, $a^2 + b^2 = c^2$, where c is the hypotenuse. It applies only to right-angled triangles.

The theorem can be proved by arranging four copies of the triangle inside a square of side a + b in two ways, and comparing the areas that are left over.

Both arrangements leave the same total area, but one leaves two squares of sides a and b and the other leaves one square of side c. So $a^2 + b^2 = c^2$.

The converse is also true: if $a^2 + b^2 = c^2$ then the triangle is right-angled. This is how the theorem is used to test whether an angle is a right angle.

> FORMAL DEVELOPMENT
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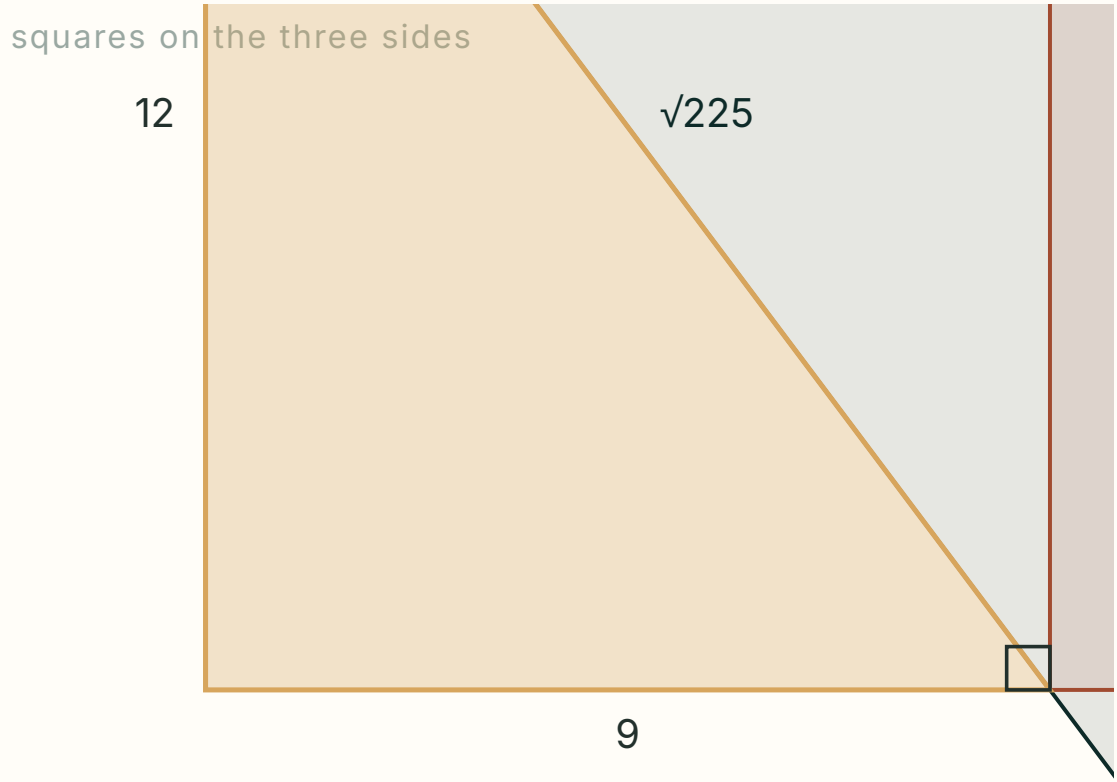
In your own words

CHAPTER 7 · 7.3 · WORKED EXAMPLE 1

Pythagoras' theorem

WORKED EXAMPLE 7.3.1

Problem. A right-angled triangle has legs 9 cm and 12 cm. Find the hypotenuse.



Mathematical reasoning. The theorem relates the squares of the three sides, and the hypotenuse is the longest side.

1. $c^2 = 9^2 + 12^2 = 81 + 144 = 225$.
2. $c = \sqrt{225} = 15$ cm.

Final answer. 15 cm

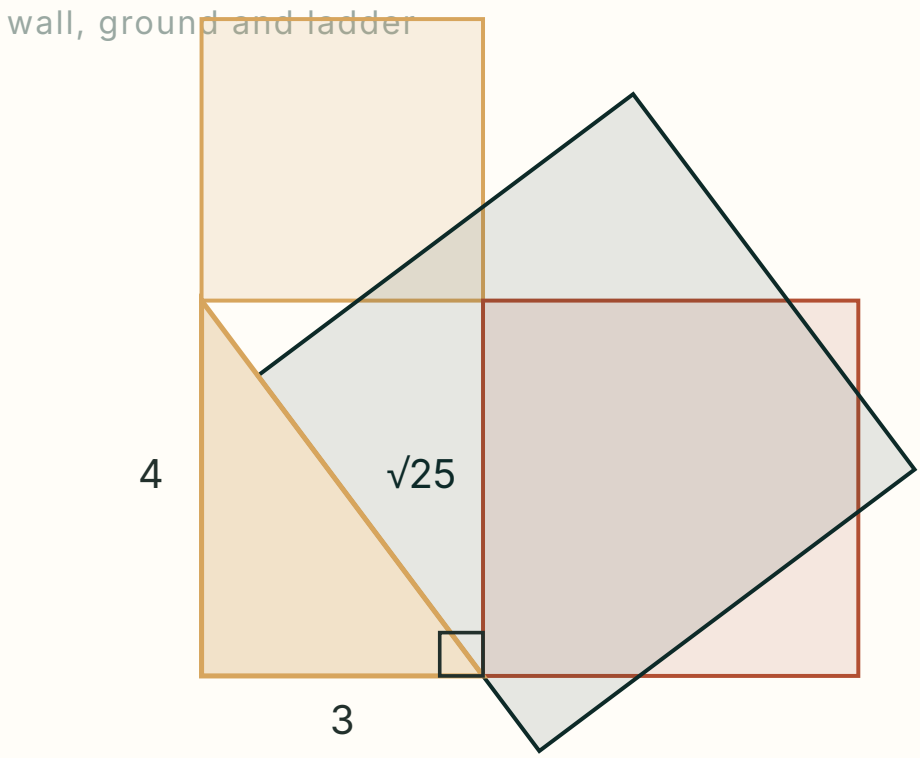
Interpretation. The answer is exact, and the diagram shows why: the two smaller squares exactly fill the largest one.

CHAPTER 7 · 7.3 · WORKED EXAMPLE 2

Pythagoras' theorem

WORKED EXAMPLE 7.3.2

Problem. A ladder 5 m long rests against a wall with its foot 3 m from the wall. How high up the wall does it reach?



Mathematical reasoning. The ladder is the hypotenuse, so the height is found by subtracting the square of the known leg.

1. $5^2 = 3^2 + h^2$.
2. $25 = 9 + h^2$, so $h^2 = 16$.
3. $h = 4$ m.

Final answer. 4 m

Interpretation. The theorem works in both directions: given the hypotenuse and one leg, the other leg follows.

CHAPTER 7 · INVESTIGATION 7.4

Trigonometric ratios

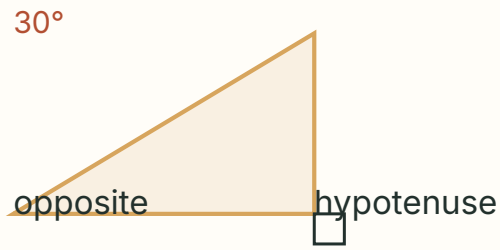
INQUIRY QUESTION

How can an angle determine a ratio between two sides of a right-angled triangle?

YOUR TASK

Draw three right-angled triangles with a 30° angle but different sizes. For each, measure the opposite and hypotenuse, compute opposite ÷ hypotenuse, and record the value. Then repeat with a 45° angle and with a 60° angle.

three triangles, one angle



angle	triangle	opposite	hypotenuse	opposite ÷ hypotenuse
30°	small	2.0	4.0	0.500
30°	medium	3.5	7.0	0.500
30°	large	5.2	10.4	0.500
45°	any	4.2	5.9	0.712
60°	any	5.2	6.0	0.867

CHAPTER 7 · INVESTIGATION 7.4

Trigonometric ratios

What do you notice?

- For a 30° angle the ratio was 0.5 in all three triangles, despite their different sizes.
- For 45° the ratio was about 0.707, and for 60° about 0.866.
- The ratios depend only on the angle, which is what makes trigonometry possible.
- The same values appear for every 30° triangle in the world.

Observation. The ratio of two sides of a right-angled triangle depends only on the angles, not on the size of the triangle.

MATHEMATICAL CONCLUSION

In a right-angled triangle the ratios sin θ = opposite/hypotenuse, cos θ = adjacent/hypotenuse and tan θ = opposite/adjacent depend only on the angle, because similar triangles have proportional sides.

Your own words

CHAPTER 7 · 7.4

Trigonometric ratios

In a right-angled triangle, the ratio of two sides depends only on the angles, not on the size of the triangle. This is why trigonometry works for triangles of every scale.

The three ratios are sine, cosine and tangent, defined for an acute angle θ: sin θ = opposite/hypotenuse, cos θ = adjacent/hypotenuse and tan θ = opposite/adjacent.

Because similar triangles have the same angles and proportional sides, these ratios are the same for every triangle with the same angles. That is the justification for the whole subject.

The ratios are used in two directions: given an angle and one side, find another side; given two sides, find the angle with an inverse ratio.

> FORMAL DEVELOPMENT
>
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In your own words

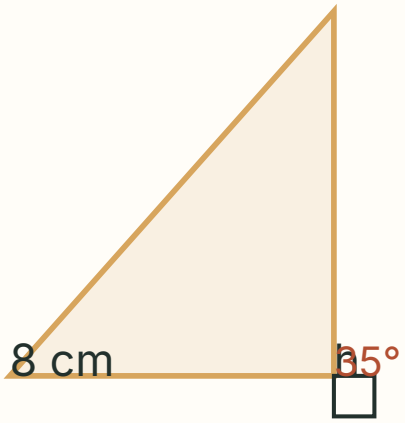
CHAPTER 7 · 7.4 · WORKED EXAMPLE 1

Trigonometric ratios

WORKED EXAMPLE 7.4.1

Problem. In a right-angled triangle, the side opposite a 35° angle is 8 cm and the hypotenuse is h. Find h.

sine relates the opposite side to the hypotenuse



Mathematical reasoning. The sine ratio links the opposite side to the hypotenuse, so it can be solved for the unknown hypotenuse.

1. sin 35° = opposite/hypotenuse = 8/h.
2. 0.5736 = 8/h.
3. h = 8 ÷ 0.5736.
4. h ≈ 13.95 cm.

Final answer. about 13.95 cm

Interpretation. The angle fixes the ratio, so the size of the triangle does not affect the method.

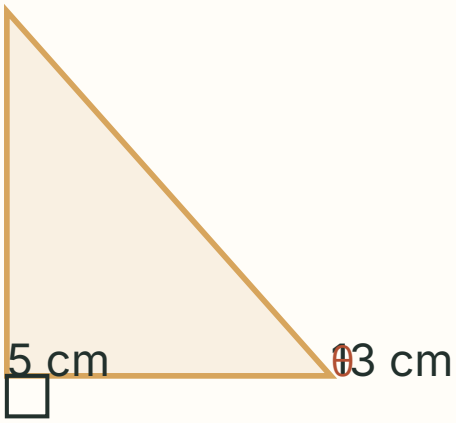
CHAPTER 7 · 7.4 · WORKED EXAMPLE 2

Trigonometric ratios

WORKED EXAMPLE 7.4.2

Problem. A right-angled triangle has adjacent side 5 cm and hypotenuse 13 cm. Find the angle.

cosine relates the adjacent side to the hypotenuse



Mathematical reasoning. The cosine ratio links the adjacent side to the hypotenuse, and the inverse cosine recovers the angle.

1. $\cos\theta = 5/13 = 0.3846$.
2. $\theta = \cos^{-1}(0.3846)$.
3. $\theta \approx 67.4°$.

Final answer. about 67.4°

Interpretation. The inverse ratio turns a pair of lengths into an angle, which is how slopes are measured.

CHAPTER 7 · INVESTIGATION 7.5

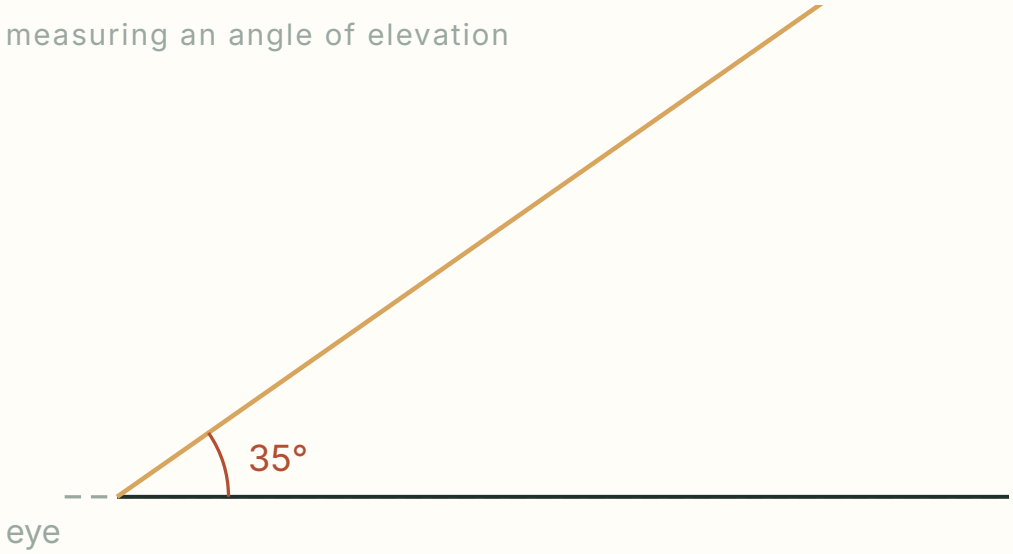
Angles of elevation and depression

INQUIRY QUESTION

How can an angle measured from the ground give a height?

YOUR TASK

Measure the angle of elevation to the top of a tall object using a clinometer made from a protractor, a straw and a plumb line. Record the angle and the horizontal distance, then compute the height. Add your eye height and compare with a direct measurement if one is possible.



measurement	value	how it was obtained
angle of elevation	35°	clinometer
horizontal distance	20 m	tape or pacing
height above eye level	20 × tan 35° = 14.0 m	tangent ratio
eye height	1.55 m	measured
total height	15.55 m	sum of the two

CHAPTER 7 · INVESTIGATION 7.5

Angles of elevation and depression

What do you notice?

- The angle and the distance gave the height above eye level directly.
- Adding the eye height was essential, and omitting it understates the result by about 1.5 m.
- The precision is limited by the angle, which was read to the nearest degree.
- A one-degree error at this distance changes the height by about 0.4 m.

Observation. An angle of elevation and a horizontal distance form a right-angled triangle whose height is given by the tangent ratio.

MATHEMATICAL CONCLUSION

The height of an object is found from the angle of elevation and the horizontal distance using tan θ = height/distance, and the observer's eye height must be added to the result.

Your own words

CHAPTER 7 · 7.5

Angles of elevation and depression

The angle of elevation is measured upward from the horizontal to the line of sight; the angle of depression is measured downward from the horizontal.

The two angles are equal by alternate angles, which is why a measurement taken from the top of a building gives the same information as one taken from the ground.

The measurement gives a right-angled triangle: the horizontal distance and the angle determine the height through a trigonometric ratio.

The height found is the height above the observer's eye, so the observer's own height must be added. This is the step that is most often forgotten.

> FORMAL DEVELOPMENT
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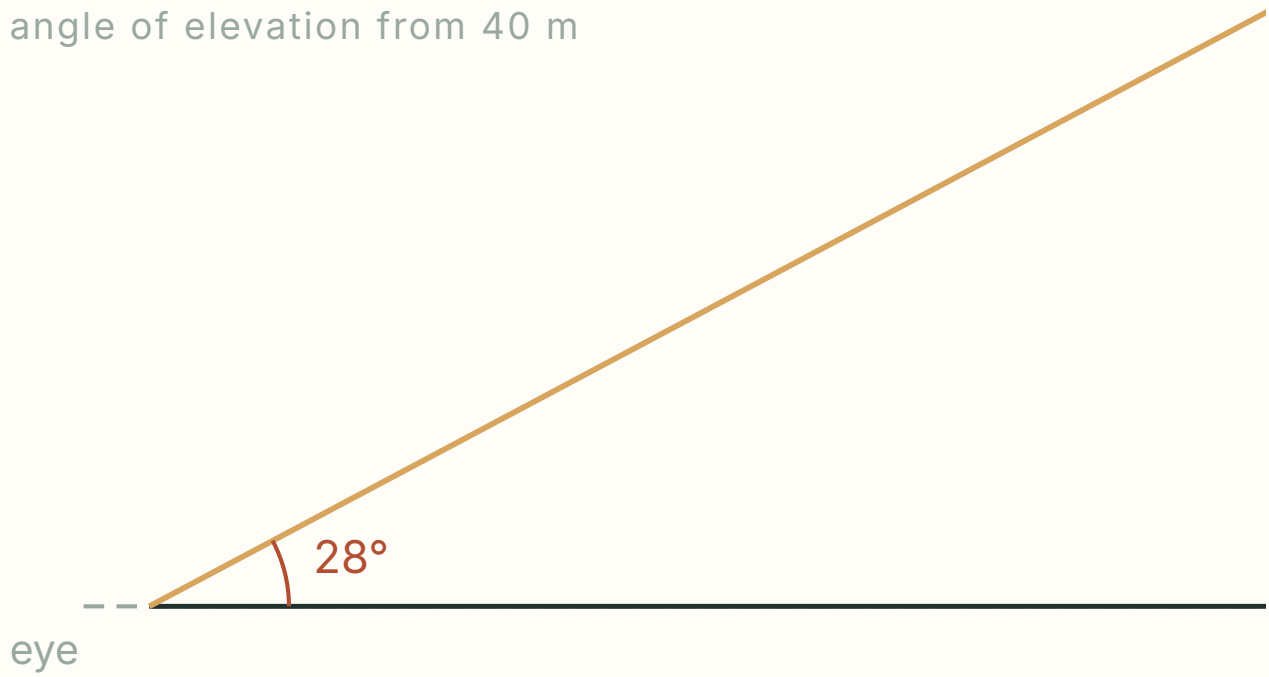
In your own words

CHAPTER 7 · 7.5 · WORKED EXAMPLE 1

Angles of elevation and depression

WORKED EXAMPLE 7.5.1

Problem. From 40 m away, the angle of elevation to the top of a tree is 28°. Find the height of the tree above eye level.



Mathematical reasoning. The horizontal distance and the angle of elevation form a right-angled triangle whose height is found with the tangent ratio.

1. $\tan 28° = \text{height}/40$.
2. $0.5317 = \text{height}/40$.
3. $\text{height} = 40 \times 0.5317$.
4. $\text{height} \approx 21.27$ m above eye level.

Final answer. about 21.3 m above eye level

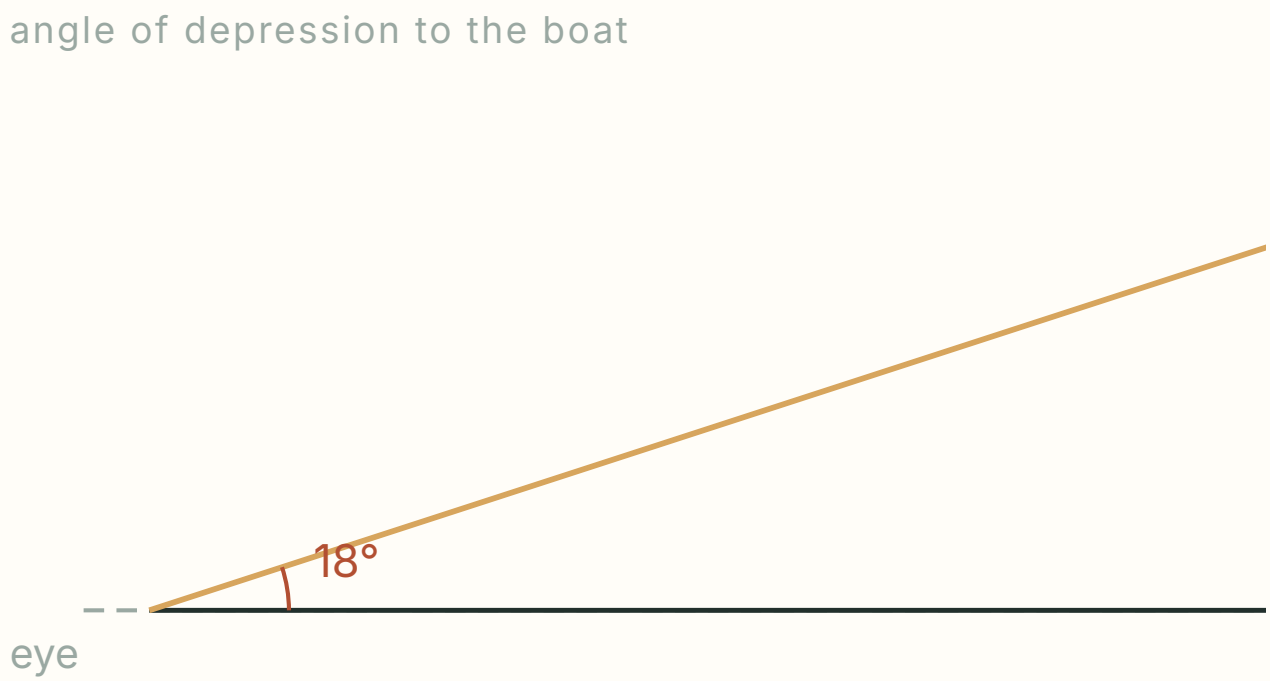
Interpretation. The eye height of the observer must be added to obtain the total height.

CHAPTER 7 · 7.5 · WORKED EXAMPLE 2

Angles of elevation and depression

WORKED EXAMPLE 7.5.2

Problem. From the top of a 30 m cliff, the angle of depression to a boat is 18°. How far is the boat from the base of the cliff?



Mathematical reasoning. The angle of depression equals the angle of elevation from the boat, so the same right-angled triangle applies.

1. The angle at the boat is also 18° by alternate angles.
2. tan 18° = 30/distance.
3. distance = 30 ÷ 0.3249.
4. distance ≈ 92.3 m.

Final answer. about 92.3 m

Interpretation. The equality of the two angles means one measurement answers the question from either end.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding

7.A1 FLUENCY

Two similar triangles have a scale factor of 3. A side of the smaller is 4 cm. Find the corresponding side of the larger.

7.A2 FLUENCY

Find the hypotenuse of a right-angled triangle with legs 6 cm and 8 cm.

7.A3 FLUENCY

In a right-angled triangle, $\sin\theta = 0.5$. What is θ?

7.A4 FLUENCY

Find the leg of a right-angled triangle with hypotenuse 13 cm and one leg 5 cm.

7.A5 FLUENCY

A rectangle 3 cm by 4 cm is enlarged by scale factor 5. Find the area of the image.

7.A6 FLUENCY

Write $\tan\theta$ in terms of the sides of a right-angled triangle.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding (continued)

7.A7 FLUENCY

Find cos θ when the adjacent side is 7 cm and the hypotenuse 25 cm.

7.A8 REASONING

Two triangles have angles 40°, 60°, 80° and 40°, 60°, 80°. Are they similar?

7.A9 FLUENCY

A map has scale 1 : 50 000. A path measures 3.4 cm. Find the real distance.

7.A10 FLUENCY

Find the hypotenuse when the legs are 1 cm and 2 cm.

7.A11 FLUENCY

An angle of elevation to a tower is 45° from 60 m away. Find the height above eye level.

7.A12 REASONING

Explain what the scale factor does to lengths and to areas.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns

7.B1 REASONING

Investigate: do all right-angled triangles with a 30° angle have the same side ratios?

7.B2 REASONING

Test the claim: two triangles with all three pairs of sides in the same ratio are similar.

7.B3 REASONING

Investigate the pattern of Pythagorean triples 3-4-5, 5-12-13, 8-15-17 and find the next one.

7.B4 REASONING

Generalise: if a triangle has sides in the ratio 3:4:5, what kind of triangle is it?

7.B5 REASONING

Investigate whether a triangle with sides 5, 6 and 10 is right-angled.

7.B6 REASONING

Test the claim: the altitude to the hypotenuse of a right-angled triangle creates two triangles similar to the original.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns (continued)

7.B7 REASONING

Investigate how the area ratio of two similar figures relates to their length ratio.

7.B8 REASONING

Generalise a rule for the height of an object given the angle of elevation and the distance.

7.B9 REASONING

Investigate whether the angle of elevation equals the angle of depression in the same situation.

7.B10 REASONING

Test the claim: a larger similar triangle has larger trigonometric ratios.

7.B11 REASONING

Investigate the values of sin θ for θ = 0°, 30°, 45°, 60° and 90°.

CRITERION C Communicating

Criterion C practice · Communicating

7.C1 REASONING

Explain in words what it means for two figures to be similar.

7.C2 REASONING

Explain why the trigonometric ratios are the same for every triangle with the same angle.

7.C3 REASONING

Describe the difference between the angle of elevation and the angle of depression.

7.C4 REASONING

Explain why the eye height of the observer must be added to a height found by trigonometry.

7.C5 REASONING

Describe two ways to prove Pythagoras' theorem.

7.C6 REASONING

Explain the difference between SSS and AA as similarity conditions.

CRITERION C Communicating

Criterion C practice · Communicating (continued)

7.C7 REASONING
Describe how you would use a clinometer and a tape to find the height of a building.

7.C8 REASONING
Explain why indirect measurement can be less reliable than direct measurement.

7.C9 REASONING
Describe what a scale drawing shows and what it does not.

7.C10 REASONING
Explain why the converse of Pythagoras' theorem is useful.

7.C11 REASONING
Describe how you would check that a corner of a room is square.

7.C12 REASONING
Explain why a one-degree error in an angle of elevation matters more for a distant object.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts

7.D1 APPLICATION

A tree casts a 15 m shadow while a 2 m stick casts a 2.5 m shadow. Find the height of the tree.

7.D2 APPLICATION

From 25 m away the angle of elevation to a roof is 32°. Find the height above eye level, to one decimal place.

7.D3 APPLICATION

A 6 m ladder leans against a wall with its foot 2 m out. How high does it reach?

7.D4 APPLICATION

A map scale is 1 : 20 000. Two towns are 7.5 cm apart on the map. Find the real distance and the walking time at 5 km/h.

7.D5 APPLICATION

A photograph 10 cm by 15 cm is enlarged so that the longer side is 45 cm. Find the shorter side and the area factor.

7.D6 REASONING

Explain whether measuring a height by shadow is reliable on a cloudy day.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts (continued)

7.D7 APPLICATION

A boat is 120 m from the base of a 40 m cliff. Find the angle of depression from the top of the cliff.

7.D8 APPLICATION

A roof rafter is 4.2 m long and spans 3.6 m. Find the angle the rafter makes with the horizontal.

7.D9 REASONING

A reservoir is surveyed from two points 60 m apart, with angles of elevation 45° and 30°. Explain whether the height can be found.

7.D10 APPLICATION

A scale model of a bridge is 1 : 200. A cable on the model is 12 cm. Find the real length and the real length if the model were 1 : 50.

7.D11 APPLICATION

A flagpole is held by a 10 m wire fixed 6 m up the pole. Find the angle the wire makes with the ground.

7.D12 REASONING

Explain why a surveyor reports the precision of an indirectly measured height.

CHAPTER 7 · APPLICATION AND ASSESSMENT

Applying chapter 7

These tasks use similarity, scale factors, Pythagoras' theorem, trigonometric ratios and angles of elevation. State the precision of every measurement you report.

A1 · The height of the water tower

From a point 35 m from the base of a water tower, the angle of elevation to the top is 41°. Find the height above eye level, then the total height given an eye height of 1.6 m. State the precision of your answer and what limits it.

Criteria A and D · justify the method and comment on the precision.

A2 · The similar frame

Two triangular supports have sides in the ratio 2:5. The smaller has area 14 cm². Find the area of the larger, and explain why the area ratio is not 2:5.

Criteria A and C · justify with the square of the scale factor.

CHAPTER 7 · REVIEW

What you should now be able to do

Key ideas

- Similar figures have equal corresponding angles and proportional corresponding sides.
- For triangles, AA and SSS are sufficient similarity conditions.
- Lengths scale by k and areas by k^2.
- Pythagoras' theorem: $a^2 + b^2 = c^2$ for a right-angled triangle.
- The converse tests whether a triangle is right-angled.
- $\sin\theta$ = opposite/hypotenuse, $\cos\theta$ = adjacent/hypotenuse, $\tan\theta$ = opposite/adjacent.
- The angle of elevation equals the angle of depression by alternate angles.
- Indirect measurement gives the height above eye level, so the eye height must be added.

Vocabulary

similar · scale factor · AA · SSS similarity · hypotenuse · leg · Pythagoras' theorem · sine · cosine · tangent · angle of elevation · angle of depression · indirect measurement · clinometer

CHAPTER 7 · REVIEW

Mixed practice · consolidate and extend

Do not use a calculator unless a question asks for one. Show a method where one is requested.

7.R1 Find the hypotenuse when the legs are 7 cm and 24 cm.

7.R2 Find tan θ when the opposite side is 5 and the adjacent 12.

7.R3 Two similar figures have a length ratio of 1:4. Find the area ratio.

7.R4 From 30 m the angle of elevation to a roof is 25°. Find the height above eye level.

7.R5 Is a triangle with sides 8, 15 and 17 right-angled? Justify.

7.R6 Explain why the trigonometric ratios depend only on the angle.

CHAPTER 7 · REVIEW

Returning to the statement of inquiry

STATEMENT OF INQUIRY

The relationship between corresponding lengths in similar figures can be justified as a constant ratio, and that justification, combined with the measurement of angles, allows indirect measurement of distances and heights that cannot be measured directly.

Reflect on the chapter

Answer in your own words, using evidence from this chapter.

- Write the statement of inquiry for this chapter in your own words.
- Which mathematical ideas from this chapter did you use to understand it? Name at least two.
- How did the key concept of relationships and the related concepts of justification and measurement appear in your work?
- How is the global context of orientation in space and time, and its exploration of natural and human landscapes and resources, visible in the investigations and examples you completed?
- Use evidence from an investigation or a worked example to justify your answer.
- Has your understanding of the statement of inquiry changed since the start of the chapter? Explain how.
- What are the limitations of indirect measurement, and what would you still like to ask?

CHAPTER 7 · PROJECTS, SERVICE AND ACTION

Acting on what you have learned

Indirect measurement is how landscapes, buildings and resources are measured. Use it to plan, and report the precision of the result.

Service as Action

COMMUNITY NEED	The community needs accurate heights and distances for a water tank, a ramp or a shade structure, but has no surveying equipment.
MATHEMATICAL CONNECTION	Similarity, Pythagoras' theorem and trigonometry.
PROPOSED ACTION	Measure one height or distance indirectly with a clinometer and publish the method with the result.
EXPECTED IMPACT	A structure can be planned from a measurement that was previously guessed.
STUDENT REFLECTION	Record the method, the assumptions and the uncertainty.

Community Project

PROJECT IDEA	A measured plan for a community shade structure.
NEED	A proposed shade structure has no measured dimensions.
RESEARCH QUESTION	What height and span will give useful shade at this site?
MATHEMATICS	Angles of elevation, trigonometry and Pythagoras' theorem.
ACTION OR OUTCOME	A measured plan with a shadow estimate.
BENEFICIARIES	Everyone who will use the structure.
EVIDENCE	The measurements, the plan and the shadow calculation.

Personal Project connection · MYP 5

PERSONAL INTEREST	Cycling, hiking or photography.
LEARNING GOAL	To measure a height or distance indirectly and report its precision.
PRODUCT OR OUTCOME	A measured profile of a local route or building.
MATHEMATICS	Similarity, Pythagoras' theorem and trigonometry.
PROCESS	Measure an angle and a distance, compute the height, then check the sense of the answer.
EVIDENCE	The measurements, the calculation and the stated precision.

CHAPTER 8 · PART III · SHAPE, SPACE AND PROOF

08

Three-dimensional solids and measurement

A solid is described by its faces, edges and vertices. Its net gives the surface area, its cross-section gives the volume, and its density connects the two to the material it is made from.

ENGAGE

A drinks manufacturer must choose between a cylindrical can and a rectangular carton of the same 330 ml capacity. Which uses less material? The answer is a comparison of surface areas, and the difference is not small.

Framing the inquiry

Key concept: Form
Related concepts: Measurement · Models
Global context: Globalisation and sustainability
Global context exploration: Human impact on the environment

Statement of inquiry: The form of a three-dimensional solid, and the measurement of its surface area and volume, allow the material it requires to be modelled, and those models show how design choices change the resources consumed.

Questions to guide us

- **Factual:** What are the formulas for the surface area and volume of prisms, cylinders, pyramids, cones and spheres?
- **Conceptual:** Why does the volume of a cone equal one third of the cylinder that contains it, and how does the precision of a measurement limit the precision of a volume?
- **Debatable:** Is a design that minimises material always the better environmental choice?

ATL · SELF-MANAGEMENT. In order for a student to manage a multi-step measurement task, students must select the correct formula, set out the substitution so each step can be checked, and review the units and the precision of the final result.

CHAPTER 8 · INVESTIGATION 8.1

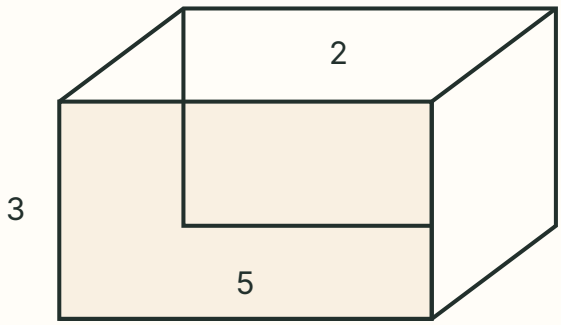
Solids, nets and views

INQUIRY QUESTION

How can a three-dimensional solid be represented on a flat surface, and what does each representation preserve?

YOUR TASK

Build the net of a cuboid, a cylinder and a square-based pyramid from card. Fold each one and record the number of faces, edges and vertices, then check Euler's relation for the two polyhedra.



a cuboid and its net

solid	faces	edges	vertices	V − E + F
cuboid	6	12	8	2
square-based pyramid	5	8	5	2
cylinder	3	2	0	not a polyhedron

CHAPTER 8 · INVESTIGATION 8.1

Solids, nets and views

What do you notice?

- Both polyhedra satisfied Euler's relation.
- The cylinder has no vertices or edges in the polyhedral sense, so the relation does not apply.
- The net of the cylinder is two circles and a rectangle whose length is the circumference.
- The net of the pyramid is one square and four triangles.

> **Observation.** A net flattens a solid without changing the shapes or areas of its faces, and Euler's relation checks the counts for any polyhedron.

MATHEMATICAL CONCLUSION

A solid can be represented by its net, by its views and by a sketch, and the net is the representation that preserves the areas needed for surface area.

Your own words

CHAPTER 8 · 8.1

Solids, nets and views

A solid is described by its faces, edges and vertices. Euler's relation for a polyhedron is $V - E + F = 2$, which holds for every solid without a hole.

A net is a flat arrangement of the faces that folds into the solid. It preserves area and shape, which makes it the natural representation for surface area.

Front, side and plan views each show two dimensions and hide the third. Together they determine the solid, and each alone does not.

A solid can also be represented by a sketch in oblique or isometric projection. The projection is a drawing convention, not a mathematical property, and it distorts some measurements.

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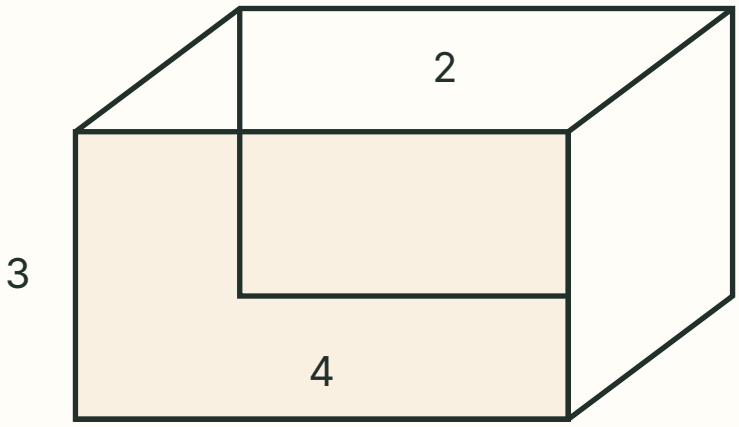
In your own words

CHAPTER 8 · 8.1 · WORKED EXAMPLE 1

Solids, nets and views

WORKED EXAMPLE 8.1.1

Problem. A cuboid has 6 faces, 12 edges and 8 vertices. Check Euler's relation.



a cuboid: 6 faces, 12 edges, 8 vertices

Mathematical reasoning. Euler's relation connects the counts of vertices, edges and faces for any polyhedron.

1. $V = 8$, $E = 12$ and $F = 6$.
2. $V - E + F = 8 - 12 + 6$.
3. $= 2$.

Final answer. the relation holds

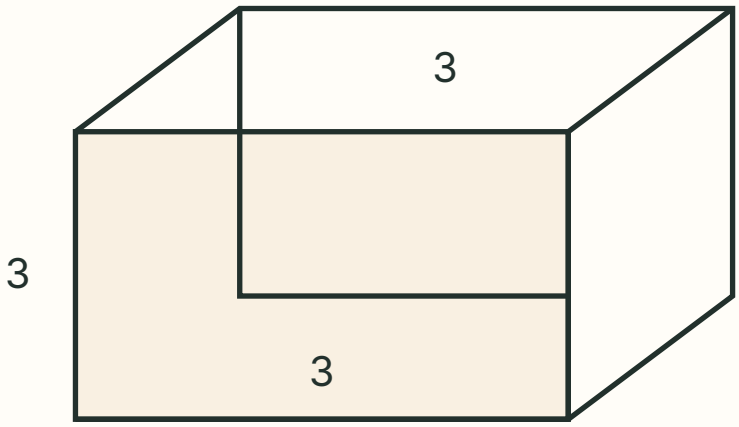
Interpretation. The relation is a check on a drawing: if it fails, the drawing is wrong.

CHAPTER 8 · 8.1 · WORKED EXAMPLE 2

Solids, nets and views

WORKED EXAMPLE 8.1.2

Problem. Describe the net of a cube of side 3 cm and find its area.



six square faces

Mathematical reasoning. The net of a cube is six squares, and its area is the surface area of the cube.

1. A cube has 6 identical square faces.
2. Each face has area $3 \times 3 = 9$ cm^2.
3. Total area $= 6 \times 9 = 54$ cm^2.

Final answer. six squares of area 9 cm² each, total 54 cm²

Interpretation. The net is the representation that turns a three-dimensional question into a flat one.

CHAPTER 8 · INVESTIGATION 8.2

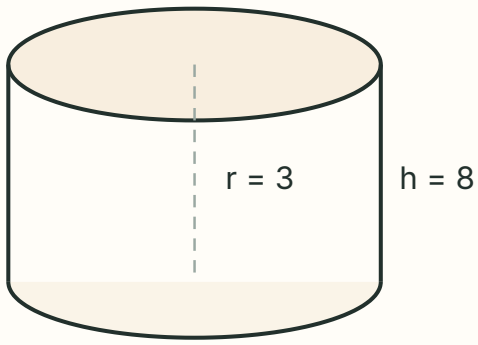
Surface area of prisms and cylinders

INQUIRY QUESTION

How can the surface area of a solid be found from its net?

YOUR TASK

Make the net of a cylinder from card, measure the rectangle and the two circles, and compute the total area. Then repeat for a cylinder of twice the radius and the same height, and compare the two surface areas.



the net of a cylinder

cylinder	radius	height	curved area	ends	total
A	3 cm	8 cm	48π	18π	66π cm^2
B	6 cm	8 cm	96π	72π	168π cm^2
ratio B/A	2	1	2	4	2.55

CHAPTER 8 · INVESTIGATION 8.2

Surface area of prisms and cylinders

What do you notice?

- Doubling the radius doubled the curved area but quadrupled the area of the ends.
- The total area grew by a factor of about 2.55, not by 4.
- The curved area depends on the radius linearly, while the ends depend on its square.
- For the same volume, a wider and shorter cylinder uses less material than a narrow tall one.

Observation. The curved area scales with the radius and the height, while the ends scale with the square of the radius.

MATHEMATICAL CONCLUSION

The surface area of a cylinder is $2\pi r^2 + 2\pi rh$, read directly from a net of two circles and a rectangle of length $2\pi r$ and height h.

Your own words

CHAPTER 8 · 8.2

Surface area of prisms and cylinders

The surface area of a prism is the sum of the areas of its faces: two congruent ends plus the lateral faces, which unfold into a rectangle.

For a cylinder, the two ends are circles of area πr^2 each and the curved surface unrolls into a rectangle of length $2\pi r$ and height h, so the curved area is $2\pi rh$.

The total surface area of a cylinder is therefore $2\pi r^2 + 2\pi rh$, and each term corresponds to a part of the net.

Reading the formula from the net is more reliable than memorising it, because the net shows where each term comes from.

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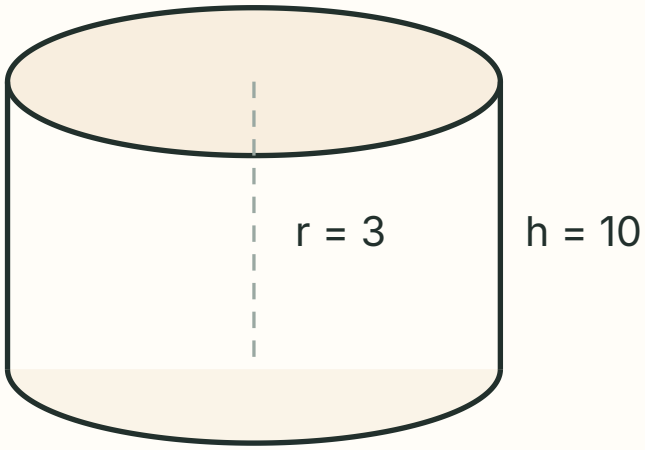
In your own words

CHAPTER 8 · 8.2 · WORKED EXAMPLE 1

Surface area of prisms and cylinders

WORKED EXAMPLE 8.2.1

Problem. Find the surface area of a cylinder of radius 3 cm and height 10 cm.



two circles and a rectangle

Mathematical reasoning. The net is two circles and a rectangle whose length is the circumference.

1. Area of the two ends = $2 \times \pi \times 3^2 = 18\pi$ cm^2.
2. Curved area = $2\pi \times 3 \times 10 = 60\pi$ cm^2.
3. Total = $18\pi + 60\pi = 78\pi$ cm^2.
4. = 245.0 cm^2 to one decimal place.

Final answer. 78π cm², or about 245.0 cm²

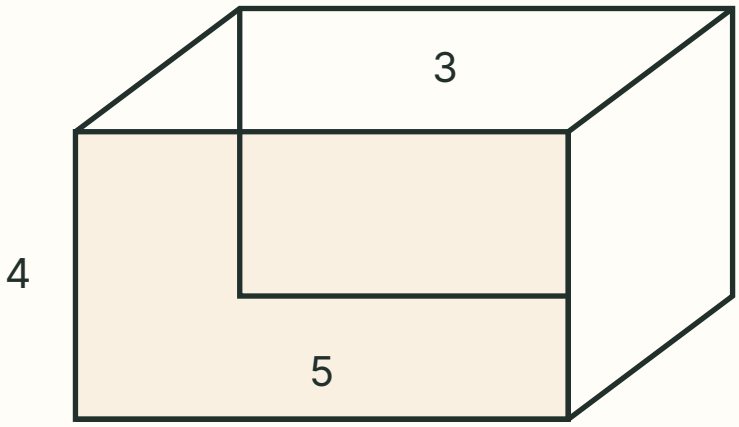
Interpretation. The exact answer is left in terms of π, and the decimal is the reporting form.

CHAPTER 8 · 8.2 · WORKED EXAMPLE 2

Surface area of prisms and cylinders

WORKED EXAMPLE 8.2.2

Problem. A rectangular prism measures 5 cm by 4 cm by 3 cm. Find its surface area.



three pairs of identical faces

Mathematical reasoning. A cuboid has three pairs of identical rectangular faces, so the total is twice the sum of the three distinct areas.

1. Face areas: $5 \times 4 = 20$, $5 \times 3 = 15$ and $4 \times 3 = 12$ cm^2.
2. Each appears twice.
3. Surface area $= 2 \times (20 + 15 + 12)$.
4. $= 2 \times 47 = 94$ cm^2.

Final answer. 94 cm²

Interpretation. Counting the pairs prevents the commonest error of missing a face.

CHAPTER 8 · INVESTIGATION 8.3

Surface area of pyramids and cones

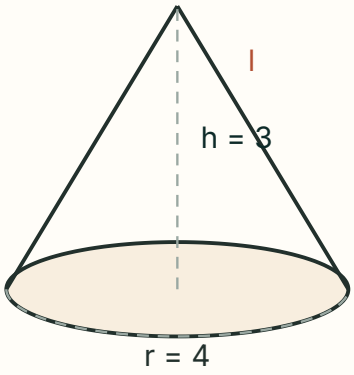
INQUIRY QUESTION

How does a slant height differ from a perpendicular height, and which one does each formula use?

YOUR TASK

Build a cone and a square-based pyramid from card. For each, measure the slant height and the perpendicular height, and check the relation between them using Pythagoras' theorem. Then compute the surface area from the net.

the slant height is the radius of the unrolled sector



solid	r or base	perpendicular height	slant height	check
cone	r = 4 cm	3 cm	5 cm	$3^2 + 4^2 = 5^2$
pyramid	base 6 cm	4 cm	5 cm	$4^2 + 3^2 = 5^2$

CHAPTER 8 · INVESTIGATION 8.3

Surface area of pyramids and cones

What do you notice?

- In both solids the slant height was the hypotenuse of a right-angled triangle inside the solid.
- Using the perpendicular height in the curved-area formula would have given the wrong answer.
- The cone's unrolled sector had radius 5 cm and arc length $2\pi \times 4$ cm.
- The pyramid's four faces were congruent because the base was square.

> **Observation.** The slant height is the height of a face, and it is related to the perpendicular height by Pythagoras' theorem.

MATHEMATICAL CONCLUSION

The curved surface area of a cone is πrl and the lateral area of a pyramid is $\frac{1}{2} \times$ perimeter $\times$ slant height, and in both cases the slant height is the height of a face rather than the height of the solid.

Your own words

CHAPTER 8 · 8.3

Surface area of pyramids and cones

A pyramid has a polygonal base and triangular faces that meet at an apex. The perpendicular height is measured from the apex to the base at right angles; the slant height is measured along a face.

The area of each triangular face is ½ × base × slant height, because the slant height is the height of the triangle in that face.

For a cone, the curved surface unrolls into a sector of a circle of radius l, the slant height, and its area is $\pi r l$. The base is a circle of area πr^2.

Confusing the slant height with the perpendicular height is the standard error. The relation between them is Pythagoras' theorem applied to the right-angled triangle inside the solid.

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In your own words

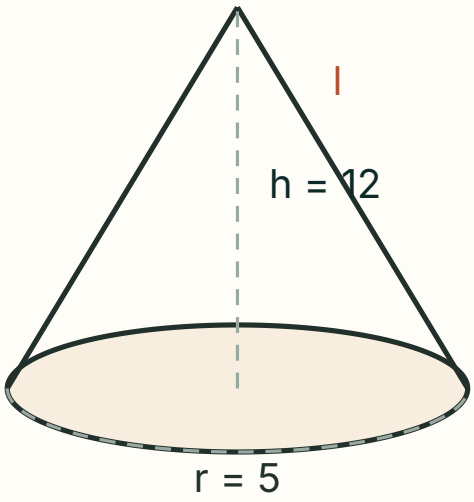
CHAPTER 8 · 8.3 · WORKED EXAMPLE 1

Surface area of pyramids and cones

WORKED EXAMPLE 8.3.1

Problem. A cone has radius 5 cm and slant height 13 cm. Find its curved surface area and total surface area.

r, h and the slant height l



Mathematical reasoning. The curved surface unrolls into a sector of radius l, so its area is πrl, and the base is a circle.

1. Curved area = $\pi rl = \pi \times 5 \times 13 = 65\pi$ cm^2.
2. Base area = $\pi r^2 = 25\pi$ cm^2.
3. Total = $65\pi + 25\pi = 90\pi$ cm^2.
4. = 282.7 cm^2 to one decimal place.

Final answer. 90π cm², or about 282.7 cm²

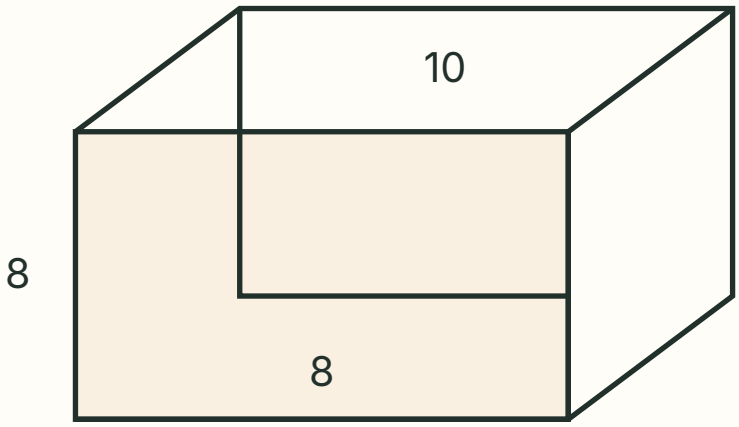
Interpretation. The slant height, not the perpendicular height, appears in the curved-area formula because the sector has radius l.

CHAPTER 8 · 8.3 · WORKED EXAMPLE 2

Surface area of pyramids and cones

WORKED EXAMPLE 8.3.2

Problem. A square-based pyramid has base side 8 cm and slant height 10 cm. Find its surface area.



a square base and four triangular faces

Mathematical reasoning. The base is a square and the four faces are congruent triangles whose height is the slant height.

1. Base area = $8 \times 8 = 64$ cm^2.
2. Area of one triangular face = $\frac{1}{2} \times 8 \times 10 = 40$ cm^2.
3. Four faces = 160 cm^2.
4. Total = $64 + 160 = 224$ cm^2.

Final answer. 224 cm²

Interpretation. Each face is a triangle, and the slant height is the height of that triangle.

CHAPTER 8 · INVESTIGATION 8.4

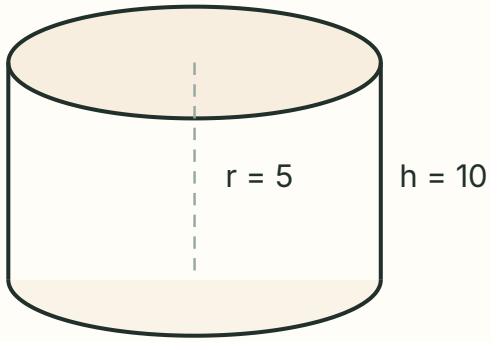
Volume of prisms and cylinders

INQUIRY QUESTION

How can the volume of any prism be found from a single cross-section?

YOUR TASK

Fill a cylinder with water and pour it into a rectangular tank to find its volume. Then compute the volume from $\pi r^2 h$ and compare. Repeat with a triangular prism made from card.



measuring a cylinder by displacement

solid	measured volume	computed volume	difference
cylinder r = 5, h = 10	785 cm^3	250π = 785.4 cm^3	0.4 cm^3
triangular prism	60 cm^3	½ × 3 × 4 × 10 = 60 cm^3	0 cm^3
cuboid 4 × 5 × 6	120 cm^3	120 cm^3	0 cm^3

CHAPTER 8 · INVESTIGATION 8.4

Volume of prisms and cylinders

What do you notice?

- The measured and computed volumes agreed to within the precision of the measuring cylinder.
- The prism rule worked for all three solids.
- Only the cross-section area changed between the three calculations.
- The agreement is a check on both the measurement and the formula.

Observation. The volume of a prism is the area of its cross-section times its length, whatever the shape of the cross-section.

MATHEMATICAL CONCLUSION

A prism has a constant cross-section and its volume is the area of that cross-section multiplied by the perpendicular length, which covers cylinders as well as polyhedra.

Your own words

CHAPTER 8 · 8.4

Volume of prisms and cylinders

A prism has a constant cross-section parallel to its base. Its volume is the area of that cross-section multiplied by the length perpendicular to it.

This one rule covers cuboids, triangular prisms, cylinders and any other prism, however irregular the cross-section. The cross-section replaces the formula.

A cylinder is a prism with a circular cross-section, so its volume is $\pi r^2 h$.

Volume is measured in cubic units, and the units must be consistent before multiplying. A common error is mixing metres and centimetres.

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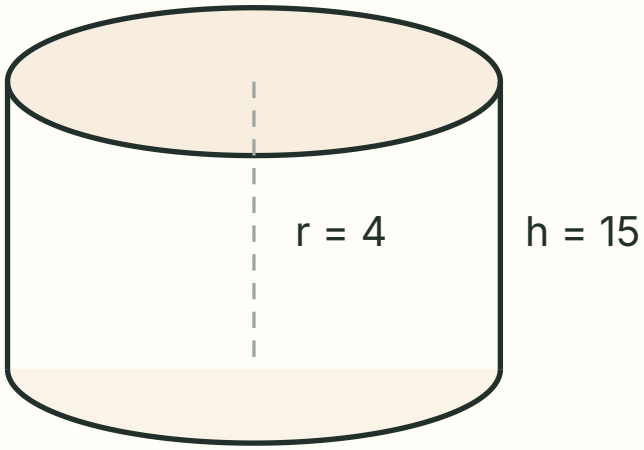
In your own words

CHAPTER 8 · 8.4 · WORKED EXAMPLE 1

Volume of prisms and cylinders

WORKED EXAMPLE 8.4.1

Problem. Find the volume of a cylinder of radius 4 cm and height 15 cm.



a circular cross-section

Mathematical reasoning. A cylinder is a prism with a circular cross-section, so the volume is the cross-section area times the height.

1. Cross-section area = $\pi \times 4^2 = 16\pi$ cm^2.
2. Volume = $16\pi \times 15 = 240\pi$ cm^3.
3. = 754.0 cm^3 to one decimal place.

Final answer. 240π cm^3, or about 754.0 cm^3

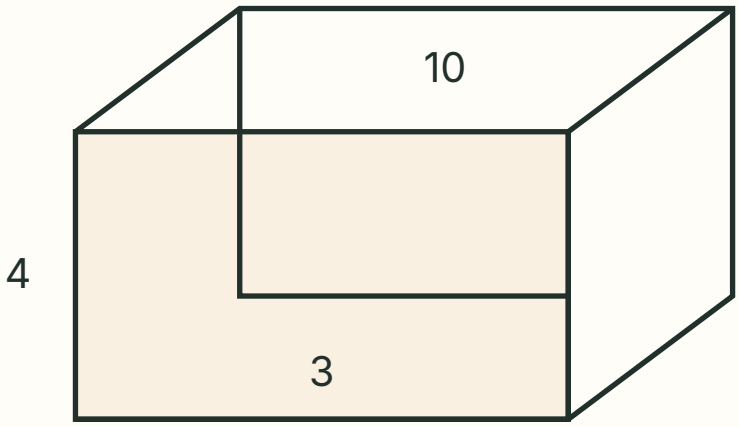
Interpretation. The formula is the prism rule applied to a circular cross-section.

CHAPTER 8 · 8.4 · WORKED EXAMPLE 2

Volume of prisms and cylinders

WORKED EXAMPLE 8.4.2

Problem. A triangular prism has a right-angled triangular cross-section with legs 3 cm and 4 cm, and length 10 cm. Find its volume.



a triangular cross-section

Mathematical reasoning. The volume is the area of the triangular cross-section times the length of the prism.

1. Cross-section area = ½ × 3 × 4 = 6 cm^2.
2. Volume = 6 × 10 = 60 cm^3.

Final answer. 60 cm³

Interpretation. The shape of the cross-section does not matter, only its area.

CHAPTER 8 · INVESTIGATION 8.5

Pyramids, cones, spheres and composite solids

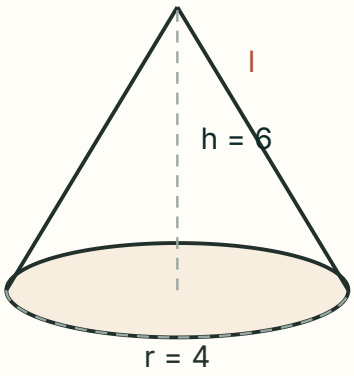
INQUIRY QUESTION

Why is the volume of a cone one third of the cylinder that contains it?

YOUR TASK

Fill a cone with water and pour it into a cylinder of the same base and height. Record how many fills are needed, then repeat with a pyramid and a prism of the same base and height.

three cone fills fill the cylinder



pair	container volume	part volume	number of fills
cylinder and cone	96π cm^3	32π cm^3	3
prism and pyramid	72 cm^3	24 cm^3	3
sphere and cylinder	250π cm^3	$500\pi/3$ cm^3	1.5 spheres

CHAPTER 8 · INVESTIGATION 8.5

Pyramids, cones, spheres and composite solids

What do you notice?

- Exactly three cone fills filled the cylinder, and three pyramid fills filled the prism.
- The relationship held for a circular base and for a square base.
- The sphere filled the cylinder one and a half times, which matches the ratio $4/3$ to 2.
- The relationship is about the shape, not the size.

Observation. A cone or pyramid has exactly one third of the volume of the prism or cylinder with the same base and height.

MATHEMATICAL CONCLUSION

The volume of a pyramid or cone is $\frac{1}{3}$ × base area × perpendicular height, which is one third of the containing prism or cylinder.

Your own words

CHAPTER 8 · 8.5

Pyramids, cones, spheres and composite solids

A pyramid or cone with the same base and height as a prism or cylinder has exactly one third of its volume. This can be shown by filling a cone three times into the matching cylinder.

The volume of a pyramid or cone is therefore $\frac{1}{3}$ × base area × perpendicular height, and for a cone $\frac{1}{3}\pi r^2 h$.

A sphere of radius r has volume $\frac{4}{3}\pi r^3$ and surface area $4\pi r^2$. Both formulas involve the radius cubed and squared respectively, as dimensions require.

A composite solid is handled by splitting it into parts whose volumes can be found, then adding or subtracting. The split is a modelling choice, and it must be stated.

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In your own words

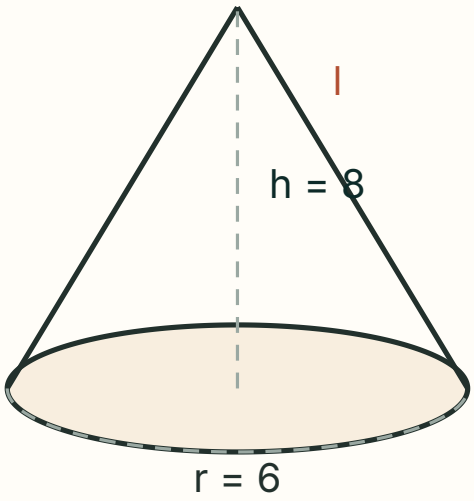
CHAPTER 8 · 8.5 · WORKED EXAMPLE 1

Pyramids, cones, spheres and composite solids

WORKED EXAMPLE 8.5.1

Problem. Find the volume of a cone of radius 6 cm and height 8 cm.

a cone inside its cylinder



Mathematical reasoning. The cone is one third of the cylinder with the same base and height.

1. Volume of the matching cylinder = $\pi \times 6^2 \times 8 = 288\pi$ cm^3.
2. Cone volume = $\frac{1}{3} \times 288\pi = 96\pi$ cm^3.
3. = 301.6 cm^3 to one decimal place.

Final answer. 96π cm³, or about 301.6 cm³

Interpretation. The factor of $\frac{1}{3}$ is the difference between a cone and a cylinder of the same base and height.

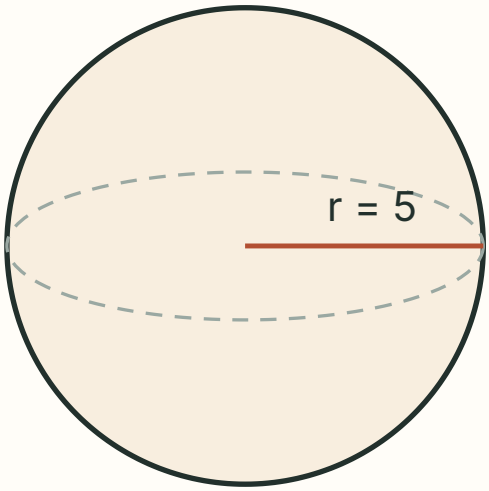
CHAPTER 8 · 8.5 · WORKED EXAMPLE 2

Pyramids, cones, spheres and composite solids

WORKED EXAMPLE 8.5.2

Problem. A sphere of radius 5 cm is placed in a cylindrical tin of radius 5 cm and height 10 cm. Find the volume of the space around it.

a sphere inside a cylinder



Mathematical reasoning. A composite volume is found by subtracting the part that is not wanted from the containing solid.

1. Tin volume = $\pi \times 5^2 \times 10 = 250\pi$ cm^3.
2. Sphere volume = $\frac{4}{3}\pi \times 5^3 = 500\pi/3$ cm^3.
3. Space = $250\pi - 500\pi/3 = 250\pi/3$ cm^3.
4. = 261.8 cm^3 to one decimal place.

Final answer. $250\pi/3$ cm^3, or about 261.8 cm^3

Interpretation. The subtraction is legitimate because the sphere sits entirely inside the tin.

CHAPTER 8 · INVESTIGATION 8.6

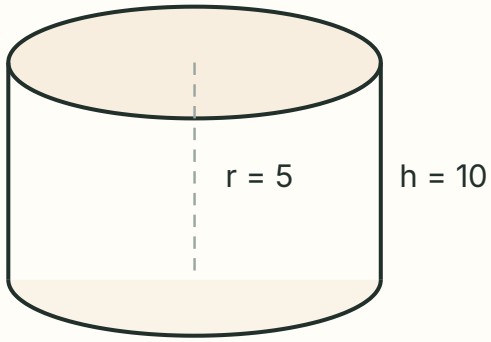
Measurement, density and precision

INQUIRY QUESTION

How does the precision of a measurement limit the precision of a volume?

YOUR TASK

Find the density of an irregular object by measuring its mass and its volume by displacement. Then estimate how the density would change if the volume measurement had a 5% error.



volume by displacement

measurement	value	precision
mass	156.0 g	0.1 g
initial water level	200 ml	1 ml
final water level	258 ml	1 ml
volume	58 ml	2 ml, about 3%
density	2.69 g/cm^3	about 3%

CHAPTER 8 · INVESTIGATION 8.6

Measurement, density and precision

What do you notice?

- The volume came from a difference of two readings, so the errors add.
- A 2 ml uncertainty in a 58 ml volume is about 3%.
- The density inherits that uncertainty, so 2.7 g/cm^3 is the honest report.
- The value is close to that of aluminium, which makes the result plausible.

Observation. A volume found by displacement carries the uncertainty of two readings, and the density inherits it.

MATHEMATICAL CONCLUSION

Density is mass per unit volume, and because volume is a product of measurements, a small percentage error in each measurement multiplies into the volume and then into the density.

Your own words

CHAPTER 8 · 8.6

Measurement, density and precision

Density is mass per unit volume, so density = mass ÷ volume and volume = mass ÷ density. The units are usually g/cm^3 or kg/m^3.

Rearranging between mass, volume and density is the same algebraic skill as rearranging any formula, and it is used constantly in materials work.

A volume is a product of three lengths, so a small error in each length is multiplied. A 1% error in a radius gives about a 3% error in a volume.

This is why reported volumes should carry no more significant figures than the measurements support. Extra digits are invented, not measured.

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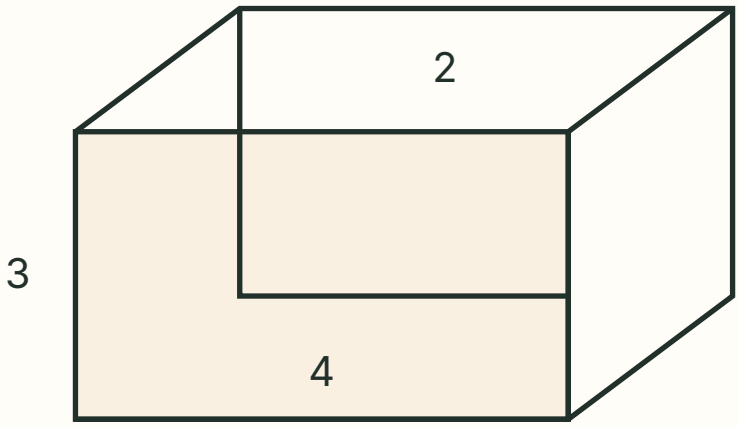
In your own words

CHAPTER 8 · 8.6 · WORKED EXAMPLE 1

Measurement, density and precision

WORKED EXAMPLE 8.6.1

Problem. A metal block measures 4.0 cm by 3.0 cm by 2.0 cm and has a mass of 190 g. Find its density.



a block of metal

Mathematical reasoning. Density is mass divided by volume, and the volume comes from the three measurements.

1. Volume = $4.0 \times 3.0 \times 2.0 = 24$ cm^3.
2. Density = $190 \div 24 = 7.92$ g/cm^3.
3. The measurements have two significant figures, so report 7.9 g/cm^3.

Final answer. 7.9 g/cm³

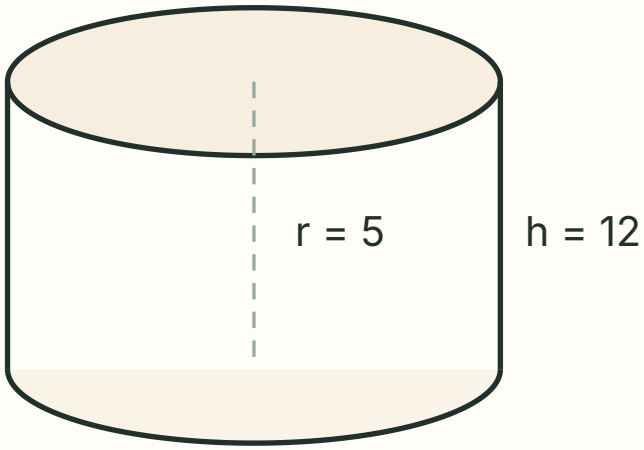
Interpretation. The density suggests the block is made of steel or iron, which is a check on the sense of the answer.

CHAPTER 8 · 8.6 · WORKED EXAMPLE 2

Measurement, density and precision

WORKED EXAMPLE 8.6.2

Problem. A cylinder has radius 5.0 cm and height 12.0 cm. Find its volume and the mass if the material has density 2.7 g/cm^3.



an aluminium cylinder

Mathematical reasoning. The volume is computed first and then converted to a mass using the density.

1. Volume = $\pi \times 5.0^2 \times 12.0 = 300\pi$ cm^3 = 942 cm^3.
2. Mass = $942 \times 2.7 = 2543$ g.
3. The measurements have two significant figures, so report 2.5 kg.

Final answer. about 2.5 kg

Interpretation. The mass follows from the volume, so it inherits the volume's precision.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding

8.A1 FLUENCY
Find the surface area of a cube of side 4 cm.

8.A2 FLUENCY
Find the volume of a cuboid 5 cm by 4 cm by 3 cm.

8.A3 FLUENCY
Find the volume of a cylinder of radius 3 cm and height 7 cm.

8.A4 FLUENCY
Find the curved surface area of a cylinder of radius 2 cm and height 9 cm.

8.A5 FLUENCY
Find the volume of a cone of radius 3 cm and height 4 cm.

8.A6 FLUENCY
Find the surface area of a sphere of radius 6 cm.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding (continued)

8.A7 FLUENCY

Find the volume of a sphere of radius 3 cm.

8.A8 FLUENCY

A square-based pyramid has base side 6 cm and height 4 cm. Find its volume.

8.A9 FLUENCY

Find the volume of a triangular prism with cross-section area 12 cm^2 and length 8 cm.

8.A10 FLUENCY

A block of mass 240 g has volume 30 cm^3. Find its density.

8.A11 FLUENCY

Find the curved surface area of a cone of radius 5 cm and slant height 13 cm.

8.A12 REASONING

Explain the difference between volume and surface area.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns

8.B1 REASONING

Investigate: for a fixed volume, does a tall narrow cylinder or a short wide one use less material?

8.B2 REASONING

Test the claim: doubling the radius of a cylinder doubles its volume.

8.B3 REASONING

Investigate whether a cube of side 2 cm has twice the surface area of a cube of side 1 cm.

8.B4 REASONING

Generalise: what happens to the volume of a sphere when the radius doubles?

8.B5 REASONING

Investigate the ratio of the volume of a cone to the cylinder with the same base and height.

8.B6 REASONING

Test the claim: every solid has a net that can be drawn without overlap.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns (continued)

8.B7 REASONING

Investigate how the surface area of a cuboid of fixed volume changes with its shape.

8.B8 REASONING

Generalise a rule connecting the percentage error in a length to the percentage error in a volume.

8.B9 REASONING

Investigate whether two objects of the same volume must have the same surface area.

8.B10 REASONING

Test the claim: the slant height of a cone is always longer than its perpendicular height.

8.B11 FLUENCY

Investigate the pattern of volumes of cubes of side 1, 2, 3, 4.

CRITERION C Communicating

Criterion C practice · Communicating

8.C1 REASONING

Explain in words what a net of a solid is and what it preserves.

8.C2 REASONING

Explain why the curved surface area of a cylinder is $2\pi rh$.

8.C3 REASONING

Describe the difference between a slant height and a perpendicular height.

8.C4 REASONING

Explain why the volume of a cone is one third of the matching cylinder.

8.C5 REASONING

Describe how you would find the volume of a composite solid.

8.C6 REASONING

Explain why density units are mass per unit volume.

CRITERION C Communicating

Criterion C practice · Communicating (continued)

8.C7 REASONING

Describe two representations of a cylinder and what each is best for.

8.C8 REASONING

Explain why a volume cannot be reported to more precision than its measurements support.

8.C9 REASONING

Describe how you would find the volume of an irregular object.

8.C10 REASONING

Explain why the surface area of a sphere is $4\pi r^2$ rather than $2\pi r^2$.

8.C11 REASONING

Describe the units you would use for the density of a liquid and why.

8.C12 REASONING

Explain how a design change that reduces surface area reduces material use.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts

8.D1 APPLICATION

A cylindrical tank of radius 0.5 m and height 1.2 m is full of water. Find the mass of the water, given a density of 1000 kg/m^3.

8.D2 APPLICATION

A rectangular box 20 cm by 15 cm by 10 cm is packed with cubes of side 5 cm. How many fit?

8.D3 APPLICATION

A cone-shaped cup has radius 3 cm and height 8 cm. How many cups fill a 1-litre jug?

8.D4 APPLICATION

A spherical ball of radius 4 cm is dropped into a cylinder of radius 5 cm containing water. By how much does the water level rise?

8.D5 APPLICATION

A storage container is a cuboid 1.2 m by 0.8 m by 0.7 m. Find how many 30 cm cubes fit, allowing for packing gaps.

8.D6 REASONING

Explain whether a spherical tank or a cylindrical tank of the same capacity uses less material.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts (continued)

8.D7 APPLICATION

A metal pipe is a cylinder of outer radius 5 cm and inner radius 4 cm, length 2 m. Find the volume of metal.

8.D8 APPLICATION

A grain silo is a cylinder of radius 3 m and height 8 m with a conical top of height 2 m. Find the total volume.

8.D9 APPLICATION

A block of wood measures 10.0 cm by 10.0 cm by 10.0 cm and has mass 620 g. Find its density and comment on its precision.

8.D10 REASONING

Explain whether a 1% error in the radius of a cylinder matters more or less than a 1% error in its height.

8.D11 APPLICATION

A conical tent has base radius 2.5 m and height 3 m. Find the canvas needed for the curved surface, allowing 10% extra for seams.

8.D12 REASONING

Explain why a designer choosing between containers of equal capacity should compare surface areas.

CHAPTER 8 · APPLICATION AND ASSESSMENT

Applying chapter 8

These tasks use nets, surface area, volume, density and precision. State the units of every answer and the precision you can justify.

A1 · The two containers

A drinks company must hold 330 ml in either a cylinder of radius 3.3 cm or a cuboid of square cross-section 4.5 cm. Find the height each container needs, the surface area of each, and recommend one with a justification.

Criteria A and D · state which container uses less material and comment on the limits of the comparison.

A2 · The irregular block

An irregular metal block has a mass of 512 g. It is lowered into a measuring cylinder and the water level rises from 180 ml to 240 ml. Find the density of the metal, state the precision of your answer, and identify a likely material.

Criteria A and C · show the displacement calculation and justify the precision.

CHAPTER 8 · REVIEW

What you should now be able to do

Key ideas

- A prism has a constant cross-section and volume = cross-section area × length.
- A cylinder is a prism with a circular cross-section: $V = \pi r^2 h$, curved area = $2\pi rh$.
- A pyramid or cone has volume $\frac{1}{3}$ × base area × perpendicular height.
- The curved surface area of a cone is πrl, where l is the slant height.
- A sphere has volume $\frac{4}{3}\pi r^3$ and surface area $4\pi r^2$.
- A composite solid is split into parts whose volumes are added or subtracted.
- Density = mass ÷ volume.
- A percentage error in a length multiplies by three in a volume.

Vocabulary

face · edge · vertex · net · cross-section · prism · cylinder · pyramid · cone · sphere · slant height · perpendicular height · composite solid · density · displacement

CHAPTER 8 · REVIEW

Mixed practice · consolidate and extend

Do not use a calculator unless a question asks for one. Show a method where one is requested.

8.R1 Find the volume of a cylinder of radius 5 cm and height 12 cm.

8.R2 Find the surface area of a cube of side 7 cm.

8.R3 Find the volume of a cone of radius 6 cm and height 9 cm.

8.R4 A mass of 350 g has volume 50 cm^3. Find the density.

8.R5 Find the curved surface area of a cone of radius 4 cm and slant height 10 cm.

8.R6 Explain why the slant height is used in the curved-area formula for a cone.

CHAPTER 8 · REVIEW

Returning to the statement of inquiry

> **STATEMENT OF INQUIRY**
>
> The form of a three-dimensional solid, and the measurement of its surface area and volume, allow the material it requires to be modelled, and those models show how design choices change the resources consumed.

Reflect on the chapter

Answer in your own words, using evidence from this chapter.

- Write the statement of inquiry for this chapter in your own words.
- Which mathematical ideas from this chapter did you use to understand it? Name at least two.
- How did the key concept of form and the related concepts of measurement and models appear in your work?
- How is the global context of globalisation and sustainability, and its exploration of human impact on the environment, visible in the investigations and examples you completed?
- Use evidence from an investigation or a worked example to justify your answer.
- Has your understanding of the statement of inquiry changed since the start of the chapter? Explain how.
- What are the limitations of a surface-area comparison when choosing a container, and what would you still like to ask?

CHAPTER 8 · PROJECTS, SERVICE AND ACTION

Acting on what you have learned

Every object made is material taken from somewhere. Use surface area and volume to see how much, and use precision to say how sure you are.

Service as Action

COMMUNITY NEED	Packaging waste in the community is high because containers are chosen without regard to the material they use.
MATHEMATICAL CONNECTION	Surface area, volume and density of materials.
PROPOSED ACTION	Compare the material used by two containers of equal capacity and publish the result with a recommendation.
EXPECTED IMPACT	A measurable reduction in packaging material for one product.
STUDENT REFLECTION	Record the comparison, the recommendation and what was actually adopted.

Community Project

PROJECT IDEA	A packaging audit for the school canteen.
NEED	The canteen does not know how much packaging it uses.
RESEARCH QUESTION	Which single change would reduce the packaging material the most?
MATHEMATICS	Surface area, volume and density.
ACTION OR OUTCOME	A ranked list of packaging changes with estimated material savings.
BENEFICIARIES	The canteen and everyone who uses it.
EVIDENCE	The measurements, the comparisons and the ranked list.

Personal Project connection · MYP 5

PERSONAL INTEREST	Cooking, recycling or model making.
LEARNING GOAL	To measure a real container and decide whether its design wastes material.
PRODUCT OR OUTCOME	A measured comparison of two containers at home.
MATHEMATICS	Surface area, volume and density.
PROCESS	Measure, compute the surface area and volume, then compare.
EVIDENCE	The measurements, the comparison and the conclusion.

CHAPTER 9 · PART IV · DATA AND EVIDENCE

09

Bivariate data, correlation and causation

Bivariate data records two measurements of the same subject. A scatter plot shows the association, a line of best fit models it, and the gap between association and cause is where most claims fail.

ENGAGE

A newspaper reports that children who eat breakfast get better exam grades. The data are real and the correlation is real. Does the headline follow from the data?

Framing the inquiry

Key concept: Relationships
Related concepts: Models · Justification
Global context: Fairness and development
Global context exploration: Inequality, difference and inclusion

Statement of inquiry: The relationship between two variables can be modelled by a line of best fit, and the justification required to treat that relationship as causal rather than merely correlated is what separates a mathematical claim from an unsupported one.

Questions to guide us

- **Factual:** What is bivariate data, what does a scatter plot show, and how is a line of best fit drawn?
- **Conceptual:** Why does a strong correlation not establish causation, and what would be needed to justify a causal claim?
- **Debatable:** Should a policy decision be based on a correlation that has no established mechanism?

ATL · RESEARCH. In order for a student to evaluate a media claim based on data, students must identify the source and the sample, check whether the measure supports the conclusion, and present the finding with its uncertainty stated.

CHAPTER 9 · INVESTIGATION 9.1

Two variables at once

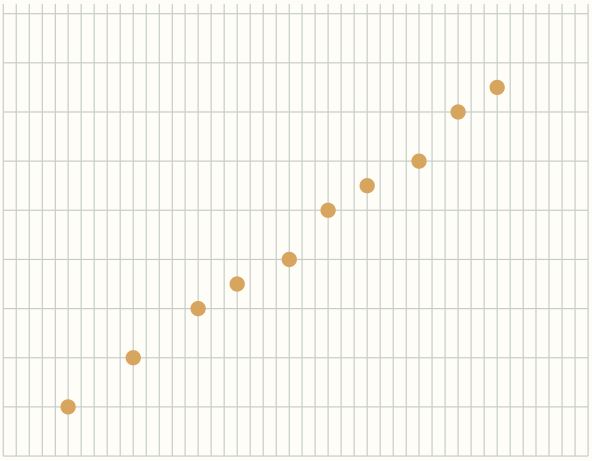
INQUIRY QUESTION

How can two measurements of the same subject be recorded and displayed together?

YOUR TASK

Collect paired data from your class on two variables of your choice (for example height and hand span, or practice minutes and a game score). Draw a scatter plot, describe the association, and identify any point that does not follow the pattern.

height against hand span



student	height (cm)	hand span (cm)	comment
1	150	17	follows the pattern
2	155	18	follows the pattern
3	162	19	follows the pattern
4	168	20.5	follows the pattern
5	178	22.5	follows the pattern
6	145	16	the smallest of both measures

CHAPTER 9 · INVESTIGATION 9.1

Two variables at once

What do you notice?

- The points rise from left to right, so the association is positive.
- The points lie close to a straight line, so the association is strong.
- No point was far enough from the pattern to call an outlier.
- The plot keeps the pairing, so each point is one person.

> **Observation.** A scatter plot keeps the pairing of the two measurements and shows the direction, strength and any exceptions of the association.

MATHEMATICAL CONCLUSION

Bivariate data is displayed as a scatter plot with one point per subject, and that plot shows the direction and strength of the association and any point that departs from it.

Your own words

CHAPTER 9 · 9.1

Two variables at once

Bivariate data records two measurements of the same subject: height and weight, study time and score, age and income. Each subject contributes one pair of values.

A scatter plot places one variable on each axis and plots one point per subject. It is the only display that keeps the pairing intact.

A scatter plot shows three things at once: the direction of the association, its strength, and any points that do not follow the pattern.

A point that lies far from the pattern is an outlier. It should be investigated, not deleted, because it may be the most informative point in the data set.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

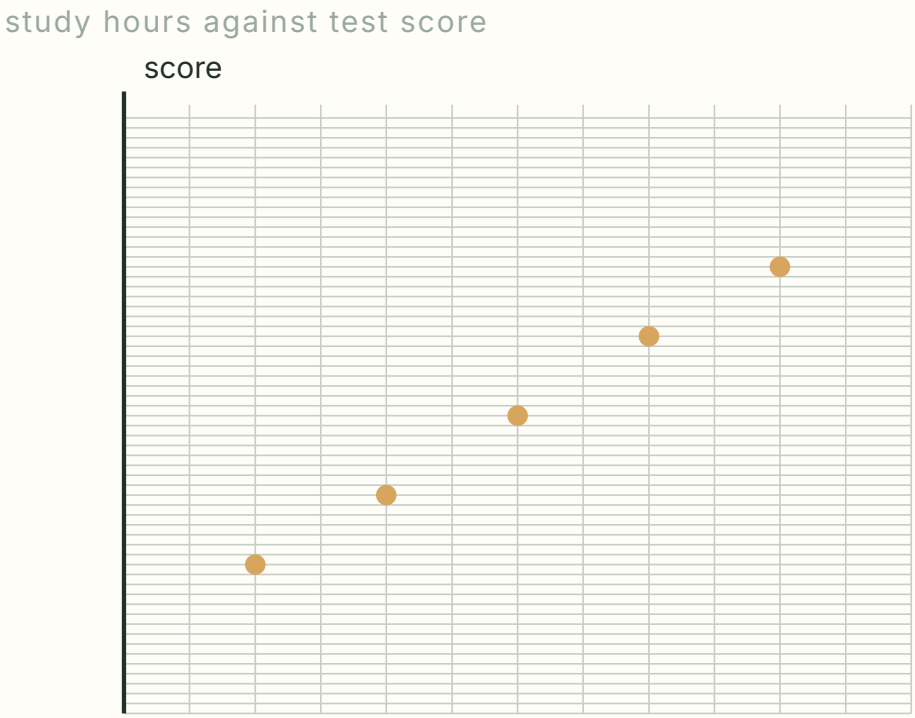
In your own words

CHAPTER 9 · 9.1 · WORKED EXAMPLE 1

Two variables at once

WORKED EXAMPLE 9.1.1

Problem. Five students record study hours and test scores: (2, 55), (4, 62), (6, 70), (8, 78), (10, 85). Draw a scatter plot and describe the association.



Mathematical reasoning. A scatter plot of paired values shows the direction and strength of the association in one picture.

1. Label the x-axis with study hours and the y-axis with score.
2. Plot one point per student.
3. The points rise from left to right, so the association is positive.
4. They lie close to a straight line, so the association is strong.

Final answer. a positive, strong, approximately linear association

Interpretation. The plot shows the pattern, but it does not explain it.

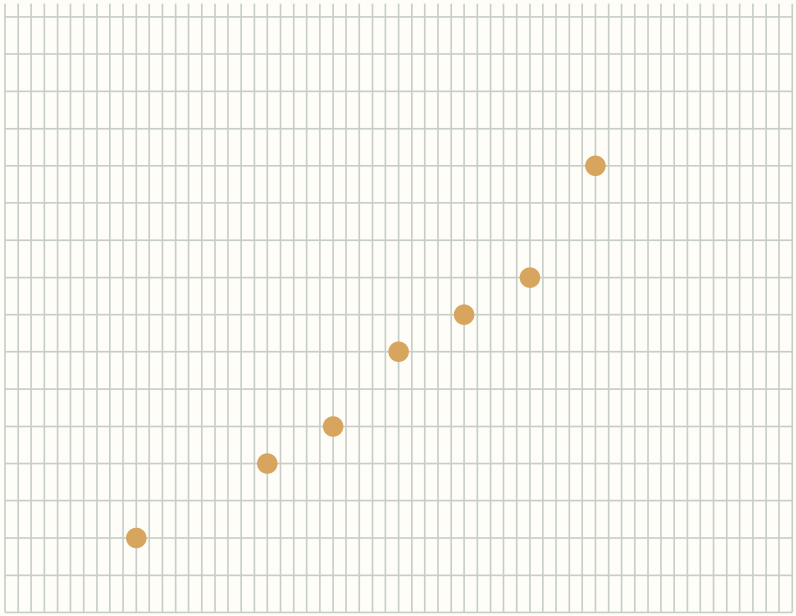
CHAPTER 9 · 9.1 · WORKED EXAMPLE 2

Two variables at once

WORKED EXAMPLE 9.1.2

Problem. A data set of height and shoe size contains one point far above the pattern. What should be done with it?

one point above the pattern



Mathematical reasoning. An outlier may be a recording error or a genuine extreme case, and both are informative.

1. Check the point against the original record.
2. If it is a recording error, correct or remove it and say so.
3. If it is genuine, keep it and note its effect on the line of best fit.
4. Never delete a point simply because it is inconvenient.

Final answer. investigate the point, then either correct it or keep it and report its effect

Interpretation. Silently removing an outlier changes the conclusion without saying so.

CHAPTER 9 · INVESTIGATION 9.2

Measuring linear association

INQUIRY QUESTION

How can the strength and direction of an association be measured?

YOUR TASK

Compute the correlation coefficient for a small data set by hand: find the mean of each variable, compute the products of the deviations, and compare with the product of the standard deviations. Then check your value against a calculator.

x	y	$x - \bar{x}$	$y - \bar{y}$	product
1	2	−2	−2.4	4.8
2	3	−1	−1.4	1.4
3	6	0	1.6	0
4	7	1	2.6	2.6
5	8	2	3.6	7.2

CHAPTER 9 · INVESTIGATION 9.2

Measuring linear association

What do you notice?

- The products of the deviations were all positive, which indicates a positive correlation.
- Their sum was 16, and dividing by the product of the standard deviations gave about 0.99.
- A calculator gave the same value to two decimal places.
- The sign of the sum of the products is the sign of the correlation.

Observation. The correlation coefficient is built from the products of the deviations, so its sign records the direction and its size records the strength.

MATHEMATICAL CONCLUSION

The correlation coefficient r lies between −1 and 1; its sign gives the direction of the linear association and its magnitude gives the strength, and it says nothing about cause.

Your own words

CHAPTER 9 · 9.2

Measuring linear association

The direction of an association is positive when both variables rise together and negative when one rises as the other falls.

The strength is measured by how closely the points follow a straight line. The correlation coefficient r takes values from −1 to 1, with the sign giving the direction and the size giving the strength.

A value of r near 0 means no linear association, though a strong curved association is still possible. The coefficient measures linear association only.

Correlation is not causation. Two variables can be strongly correlated because both depend on a third variable, or by coincidence.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 9 · 9.2 · WORKED EXAMPLE 1

Measuring linear association

WORKED EXAMPLE 9.2.1

Problem. Four data sets give $r = 0.95$, $r = -0.82$, $r = 0.12$ and $r = -0.05$. Describe each association.

r	direction	strength	meaning
0.95	positive	very strong	points close to a rising line
−0.82	negative	strong	points close to a falling line
0.12	positive	very weak	almost no linear pattern
−0.05	negative	essentially none	no linear pattern

The sign gives the direction and the size the strength.

Mathematical reasoning. The sign of r gives the direction and its absolute value gives the strength of the linear association.

1. $r = 0.95$: positive and very strong.
2. $r = -0.82$: negative and strong.
3. $r = 0.12$: positive but very weak.
4. $r = -0.05$: essentially no linear association.

Final answer. 0.95 very strong positive; −0.82 strong negative; 0.12 very weak positive; −0.05 none

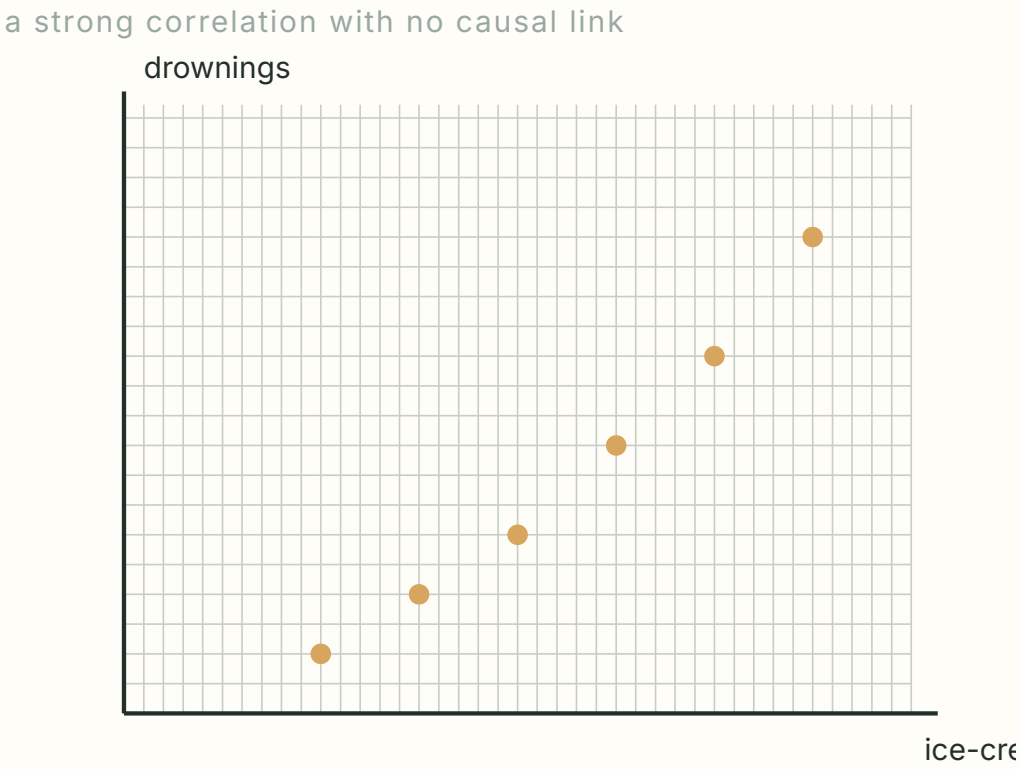
Interpretation. A weak correlation is not a small correlation with the sign removed; it is a different kind of result.

CHAPTER 9 · 9.2 · WORKED EXAMPLE 2

Measuring linear association

WORKED EXAMPLE 9.2.2

Problem. A study finds r = 0.88 between ice-cream sales and drowning deaths. What can and cannot be concluded?



Mathematical reasoning. A high correlation measures association only; causation requires a mechanism or a controlled study.

1. The correlation is strong and positive.
2. Both variables rise in summer, so a third variable explains both.
3. There is no mechanism by which ice-cream sales cause drowning.
4. So the correlation cannot support a causal claim.

Final answer. a strong positive association exists, but no causal conclusion is justified

Interpretation. The third variable — warm weather — is the explanation for both.

CHAPTER 9 · INVESTIGATION 9.3

The line of best fit

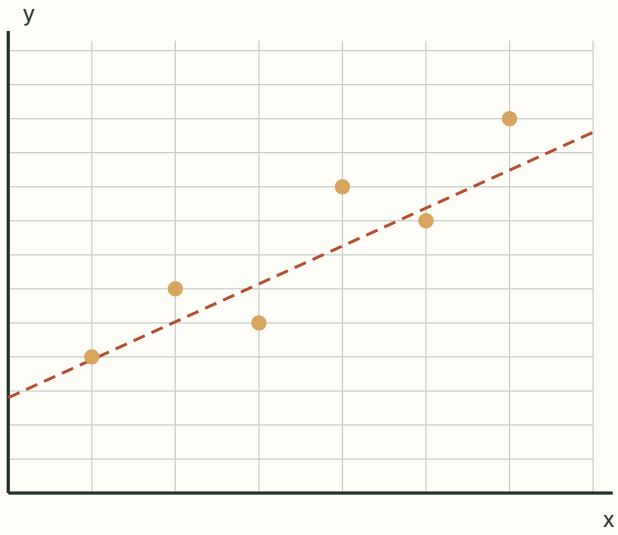
INQUIRY QUESTION

How can a line be drawn through a scatter so that it represents the whole data set?

YOUR TASK

Draw a line of best fit for the data (1, 4), (2, 6), (3, 5), (4, 9), (5, 8), (6, 11). Compute the residuals for every point, and adjust your line so that they balance.

a line with balanced residuals



x	y	predicted	residual
1	4	3.9	0.1
2	6	5.1	0.9
3	5	6.3	−1.3
4	9	7.5	1.5
5	8	8.7	−0.7
6	11	9.9	1.1

CHAPTER 9 · INVESTIGATION 9.3

The line of best fit

What do you notice?

- The residuals were both positive and negative.
- Their sum was about 1.6, close to zero, so the line was reasonably balanced.
- The largest residual was 1.5, at $x = 4$.
- Moving the line up would make the positive residuals larger and the negative ones smaller.

Observation. A line of best fit is positioned so that the residuals are small and their sum is close to zero.

MATHEMATICAL CONCLUSION

A line of best fit passes through the middle of a scatter so that the residuals are small, balanced and patternless, and it is a model of the trend rather than a record of the data.

Your own words

CHAPTER 9 · 9.3

The line of best fit

A line of best fit is a straight line drawn through the middle of a scatter so that the points are balanced on either side and the gaps are as small as possible.

The residuals are the vertical gaps between the points and the line. A good fit has residuals that are small, both positive and negative, and show no pattern.

The line is drawn by eye through the mean point, then adjusted so that the residuals balance. The exact position depends on the method, but any reasonable line gives similar predictions.

The line is a model, not the data. It summarises a trend, and every point that departs from it is information the model does not carry — which is why such models are checked against their residuals before they are used.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

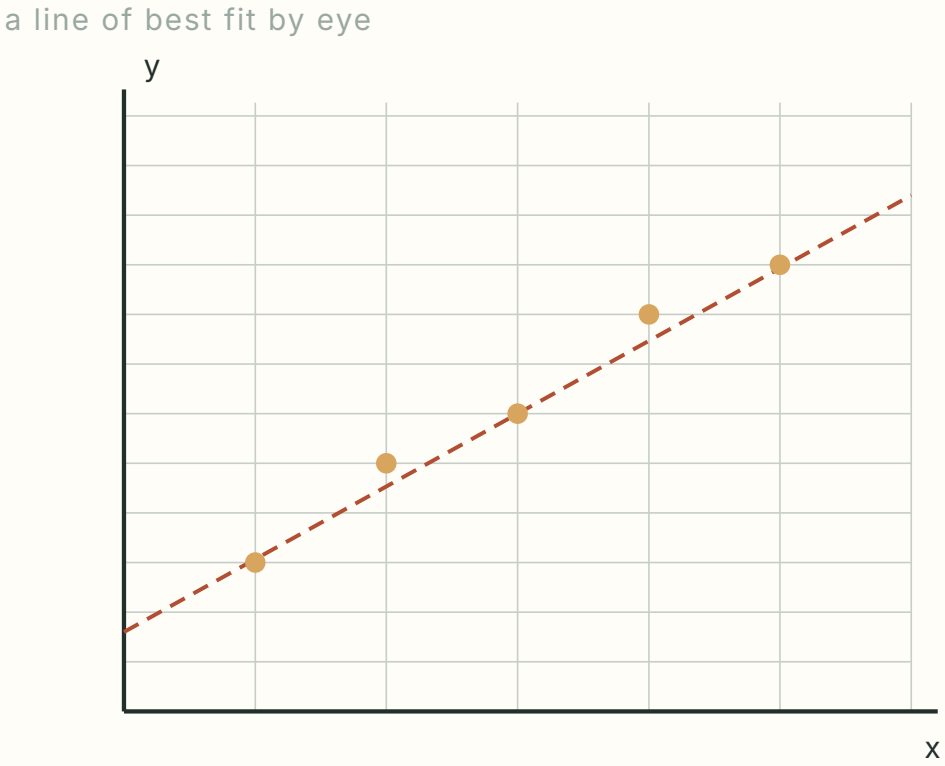
In your own words

CHAPTER 9 · 9.3 · WORKED EXAMPLE 1

The line of best fit

WORKED EXAMPLE 9.3.1

Problem. Draw a line of best fit for the data (1, 3), (2, 5), (3, 6), (4, 8), (5, 9) and find its equation.



Mathematical reasoning. A line through the mean point with a gradient estimated from two well-placed points on the line gives a reasonable fit.

1. The mean point is (3, 6).
2. Draw a line through (3, 6) that balances the points.
3. Two points on the line are (0, 1.6) and (6, 10.4).
4. Gradient = (10.4 − 1.6)/6 = 1.47, so $y = 1.47x + 1.6$.

Final answer. $y \approx 1.47x + 1.6$

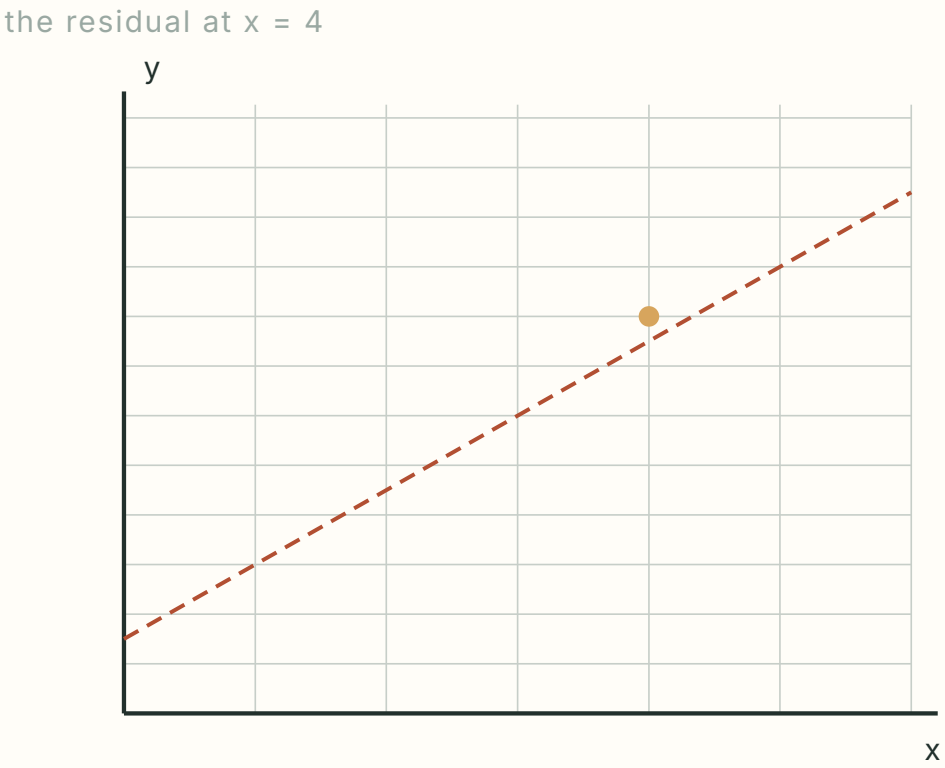
Interpretation. Another student's line will differ slightly, and both are acceptable if the residuals balance.

CHAPTER 9 · 9.3 · WORKED EXAMPLE 2

The line of best fit

WORKED EXAMPLE 9.3.2

Problem. For the line y = 1.5x + 1.5 and the point (4, 8), find the residual.



Mathematical reasoning. A residual is the vertical distance from the point to the line.

1. Predicted y at x = 4 is 1.5 × 4 + 1.5 = 7.5.
2. Actual y is 8.
3. Residual = 8 − 7.5 = 0.5.

Final answer. 0.5

Interpretation. A positive residual means the point lies above the line.

CHAPTER 9 · INVESTIGATION 9.4

Correlation, cause and consequence

INQUIRY QUESTION

When does a correlation justify a causal claim?

YOUR TASK

Find three real claims based on correlation: one where a mechanism is plausible, one where a common cause is likely, and one where the direction may be reversed. For each, state which explanation the data can and cannot distinguish.

claim	correlation	competing explanations	what the data can support
exercise and mood	positive	exercise improves mood; mood drives exercise; a third factor	an association only
shoe size and reading ability in children	positive	age causes both	an association, explained by age
smoking and lung disease	positive	smoking causes disease; disease changes behaviour	a causal claim, supported by controlled studies

CHAPTER 9 · INVESTIGATION 9.4

Correlation, cause and consequence

What do you notice?

- Two of the three claims had an obvious common cause.
- Only the third had a mechanism and controlled-study support.
- In every case the correlation itself was the same kind of evidence.
- The difference was in the justification, not in the number.

Observation. A correlation can be explained by A causing B, B causing A, or a common cause, and the data alone cannot distinguish them.

MATHEMATICAL CONCLUSION

A causal claim requires a mechanism and evidence from a controlled study; a correlation, however strong, supports only an association.

Your own words

CHAPTER 9 · 9.4

Correlation, cause and consequence

A causal claim says that changing one variable produces a change in the other. A correlation only says that the two variables move together.

Three explanations compete whenever a correlation is found: A causes B, B causes A, or a third variable C causes both. A correlation alone cannot distinguish them.

A mechanism is a plausible account of how the cause produces the effect. Without one, a causal claim is a hypothesis rather than a conclusion.

A controlled study, in which the suspected cause is varied while other factors are held constant, is the standard way to test a causal claim. Correlation is not a substitute for it.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 9 · 9.4 · WORKED EXAMPLE 1

Correlation, cause and consequence

WORKED EXAMPLE 9.4.1

Problem. Data show a strong correlation between the number of firefighters at a fire and the damage caused. Should the number of firefighters be reduced to limit damage?

explanation	assessment
firefighters cause damage	no mechanism
damage causes more firefighters	plausible — big fires need more crew
a third variable causes both	yes — the size of the fire

Three explanations compete whenever a correlation is found.

Mathematical reasoning. The third explanation is the correct one, and acting on the correlation would be a mistake.

1. The correlation is strong and positive.
2. Firefighters do not cause damage; there is no mechanism.
3. The size of the fire causes both the number of firefighters and the damage.
4. So reducing the crew would not reduce the damage.

Final answer. no — the size of the fire is the common cause

Interpretation. This is a case where acting on a correlation would make things worse.

CHAPTER 9 · 9.4 · WORKED EXAMPLE 2

Correlation, cause and consequence

WORKED EXAMPLE 9.4.2

Problem. A trial gives one group a new study method and another group the old method, with students assigned at random. The first group scores higher. What can be concluded?

feature	present?	what it supports
random assignment	yes	the groups are comparable
a varied factor	yes	the study method was changed
held-constant factors	yes	the trial controls them
a mechanism	plausible	the method changes practice time

A controlled study supports a causal claim.

Mathematical reasoning. Random assignment makes the groups comparable, so a difference in outcome can be attributed to the varied factor.

1. The groups were assigned at random, so they are comparable.
2. Only the study method differed.
3. The outcome differed, so the method is a plausible cause.
4. The conclusion is a supported causal claim, not merely a correlation.

Final answer. the study method is a supported cause of the higher scores

Interpretation. The support comes from the design of the study, not from the size of the correlation.

CHAPTER 9 · INVESTIGATION 9.5

Reading claims critically

INQUIRY QUESTION

How can a statistical claim be checked before it is believed?

YOUR TASK

Take three claims from news or advertising. For each, complete the four checks — source, sample, measure, conclusion — and decide what the data actually support.

claim	source	sample	measure	what the data support
'9 out of 10 dentists recommend'	a manufacturer	not stated	a percentage without a base rate	very little without the sample details
'test scores rose 5%'	a school	all students	a percentage of an unknown base	an increase, if the base is stated
'most people support the change'	a poll	1 000 respondents	a percentage with a margin	an estimate with a stated uncertainty

CHAPTER 9 · INVESTIGATION 9.5

Reading claims critically

What do you notice?

- Only the third claim stated its sample and its uncertainty.
- The first claim could not be checked at all.
- The second claim needed the base rate before it meant anything.
- The four checks were enough to rank the three claims.

Observation. A statistical claim can be checked by examining its source, its sample, its measure and its conclusion separately.

MATHEMATICAL CONCLUSION

A claim is only as strong as the sample that supports it and the measure that expresses it, and the four checks of source, sample, measure and conclusion reveal what the data actually support.

Your own words

CHAPTER 9 · 9.5

Reading claims critically

A statistical claim rests on four things: the source, the sample, the measure and the conclusion. Each can be checked independently.

The sample must represent the population the claim is about. A sample that is self-selected, small, or drawn from one group cannot support a claim about everyone.

The measure must be the one the conclusion needs. An average hides the spread, and a percentage without a base rate is easy to misread.

A claim that cannot be checked is not necessarily false, but it is not evidence either. Reporting the uncertainty is part of making the claim honestly.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 9 · 9.5 · WORKED EXAMPLE 1

Reading claims critically

WORKED EXAMPLE 9.5.1

Problem. A website reports that 90% of users recommend a product, based on 200 replies out of 50 000 customers. What is wrong with the claim?

check	finding
source	the seller's own website
sample	200 of 50 000, self-selected
measure	a percentage without the base rate
conclusion	not supported by the sample

Four checks, four problems.

Mathematical reasoning. A self-selected sample from a biased source cannot support a claim about the whole population.

1. The respondents chose to reply, so they are not representative.
2. 200 of 50 000 is 0.4% of the customers.
3. The source has an interest in the result.
4. So the claim is not supported by the evidence given.

Final answer. the sample is self-selected, small and drawn from a biased source

Interpretation. The percentage is arithmetically correct and still does not support the conclusion.

CHAPTER 9 · 9.5 · WORKED EXAMPLE 2

Reading claims critically

WORKED EXAMPLE 9.5.2

Problem. A hospital reports a higher death rate than a neighbouring hospital. What should be checked before concluding that its care is worse?

check	question to ask
case mix	are the patients equally ill?
base rate	how many patients in each hospital?
measure	is the same definition of death used?
time period	do the two periods match?

Comparing rates requires comparable populations.

Mathematical reasoning. A rate comparison is only meaningful when the populations and the definitions are comparable.

1. Check the case mix: a hospital treating more serious cases will have a higher rate.
2. Check the base rate: small hospitals have unstable rates.
3. Check the definitions and the time period.
4. Only then is a comparison of the two rates meaningful.

Final answer. case mix, base rate, definitions and time period must all be comparable

Interpretation. The same arithmetic on different populations gives different answers.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding

9.A1 REASONING
State what a scatter plot displays.

9.A2 FLUENCY
Describe the association when $r = -0.9$.

9.A3 FLUENCY
Find the residual when the predicted value is 12 and the actual value is 15.

9.A4 FLUENCY
State the range of possible values of the correlation coefficient.

9.A5 FLUENCY
A line of best fit is $y = 2x + 3$. Predict y when $x = 7$.

9.A6 FLUENCY
Describe the association when the points lie close to a rising line.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding (continued)

9.A7 FLUENCY

A line of best fit passes through (3, 6) and (7, 14). Find its gradient.

9.A8 REASONING

State the difference between interpolation and extrapolation on a scatter plot.

9.A9 FLUENCY

A data set has mean $x = 4$ and mean $y = 10$. Where does the line of best fit pass?

9.A10 FLUENCY

Find the predicted value from $y = 0.5x + 4$ when $x = 12$.

9.A11 REASONING

Describe what an outlier is in a scatter plot.

9.A12 REASONING

State the units of a correlation coefficient.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns

9.B1 REASONING

Investigate: can a strong correlation exist between two variables with no causal link? Give an example.

9.B2 REASONING

Test the claim: a correlation of 0.1 means there is no relationship at all.

9.B3 REASONING

Investigate whether reversing the two axes of a scatter plot changes the correlation coefficient.

9.B4 REASONING

Generalise: what does the sum of the residuals of a line of best fit equal, approximately?

9.B5 REASONING

Investigate whether an outlier always changes the line of best fit a great deal.

9.B6 REASONING

Test the claim: two variables that are correlated must have a linear relationship.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns (continued)

9.B7 REASONING

Investigate the pattern of residuals for a good line of best fit.

9.B8 REASONING

Generalise: what extra evidence is needed to turn a correlation into a causal claim?

9.B9 REASONING

Investigate whether a larger sample always gives a stronger correlation.

9.B10 REASONING

Test the claim: a percentage claim is meaningful without its base rate.

9.B11 REASONING

Investigate the effect of a single outlier on the mean of a small data set.

CRITERION C Communicating

Criterion C practice · Communicating

9.C1 REASONING
Explain in words what bivariate data are.

9.C2 REASONING
Explain the difference between correlation and causation.

9.C3 REASONING
Describe what a residual measures and what a pattern of residuals means.

9.C4 REASONING
Explain why a line of best fit passes through the mean point.

9.C5 REASONING
Describe the three explanations that compete when a correlation is found.

9.C6 REASONING
Explain why extrapolation is riskier than interpolation.

CRITERION C Communicating

Criterion C practice · Communicating (continued)

9.C7 REASONING
Describe how you would check a claim based on a survey.

9.C8 REASONING
Explain why a self-selected sample cannot support a claim about a whole population.

9.C9 REASONING
Describe what a base rate is and why it matters.

9.C10 REASONING
Explain why a strong correlation with no mechanism is not evidence of cause.

9.C11 REASONING
Describe two representations of the same bivariate data set and what each shows.

9.C12 REASONING
Explain why reporting the uncertainty of an estimate is part of an honest claim.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts

9.D1 APPLICATION

A line of best fit for study hours against score is $y = 4.2x + 38$. Predict the score for 6 hours of study and comment on the prediction.

9.D2 REASONING

A data set of 12 students gives a correlation of 0.91 between practice minutes and game score. A parent claims practice causes the score. Evaluate the claim.

9.D3 REASONING

A newspaper reports that school breakfast programmes raise test scores, citing a correlation. What further evidence would be needed?

9.D4 REASONING

Two neighbourhoods have average incomes differing by 30%. Explain why this does not by itself show unfairness.

9.D5 REASONING

A poll of 1 200 people finds 54% support with a margin of 3%. Explain what the result means.

9.D6 REASONING

A scatter plot of height against age for children aged 5 to 15 is strongly positive. A claim is made that growing taller makes children older. Evaluate.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts (continued)

9.D7 REASONING

A company reports that 95% of customers are satisfied, from 40 replies out of 8 000. Explain why the claim is weak.

9.D8 REASONING

A line of best fit predicts a plant height of 42 cm at week 10, but the collected data stop at week 6. Explain the status of the prediction.

9.D9 REASONING

A study finds a correlation of 0.6 between screen time and sleep quality in 1 500 teenagers. Explain what the data support and what they do not.

9.D10 REASONING

A hospital has a higher death rate than a neighbour. List four checks before concluding that its care is worse.

9.D11 REASONING

A school claims that a new reading scheme raised scores, comparing this year's class with last year's. Explain what is missing.

9.D12 REASONING

Explain why an average income figure can hide inequality within a community.

CHAPTER 9 · APPLICATION AND ASSESSMENT

Applying chapter 9

These tasks use scatter plots, correlation, lines of best fit, interpolation and extrapolation. State whether your prediction is interpolation or extrapolation, and what it assumes.

A1 · The breakfast claim

A school finds a correlation of 0.72 between eating breakfast and test scores among 90 students. The head teacher proposes making breakfast compulsory. Analyse the claim: state what the correlation supports, identify the competing explanations, and describe the study that would be needed to justify the policy.

Criteria A and D · justify your analysis with the three competing explanations.

A2 · The village water use

Data for ten households give household size and weekly water use in litres. Draw a scatter plot, draw a line of best fit, find its equation, predict the use for a household of 7, and state whether the prediction is interpolation or extrapolation.

Criteria A and C · label the axes with units and state the status of your prediction.

CHAPTER 9 · REVIEW

What you should now be able to do

Key ideas

- Bivariate data records two measurements of the same subject.
- A scatter plot shows direction, strength and outliers.
- The correlation coefficient r lies between −1 and 1 and measures linear association only.
- A line of best fit balances the residuals and passes through the mean point.
- A residual is the vertical gap between a point and the line.
- Interpolation is inside the range of the data; extrapolation is outside it.
- A causal claim needs a mechanism and a controlled study.
- A claim is checked through its source, sample, measure and conclusion.

Vocabulary

bivariate data · scatter plot · association · correlation coefficient · line of best fit · residual · outlier · interpolation · extrapolation · causation · common cause · base rate · sample · population

CHAPTER 9 · REVIEW

Mixed practice · consolidate and extend

Do not use a calculator unless a question asks for one. Show a method where one is requested.

9.R1 Describe the association when r = 0.85.

9.R2 Find the residual when the predicted value is 20 and the actual is 17.

9.R3 A line of best fit is y = 3x + 2. Predict y when x = 9.

9.R4 Explain why correlation does not imply causation.

9.R5 A line passes through (2, 5) and (6, 13). Find its equation.

9.R6 List the four checks for a statistical claim.

CHAPTER 9 · REVIEW

Returning to the statement of inquiry

> **STATEMENT OF INQUIRY**
>
> The relationship between two variables can be modelled by a line of best fit, and the justification required to treat that relationship as causal rather than merely correlated is what separates a mathematical claim from an unsupported one.

Reflect on the chapter

Answer in your own words, using evidence from this chapter.

- Write the statement of inquiry for this chapter in your own words.
- Which mathematical ideas from this chapter did you use to understand it? Name at least two.
- How did the key concept of relationships and the related concepts of models and justification appear in your work?
- How is the global context of fairness and development, and its exploration of inequality, difference and inclusion, visible in the investigations and examples you completed?
- Use evidence from an investigation or a worked example to justify your answer.
- Has your understanding of the statement of inquiry changed since the start of the chapter? Explain how.
- What are the limitations of a correlation for making a decision about people, and what would you still like to ask?

CHAPTER 9 · PROJECTS, SERVICE AND ACTION

Acting on what you have learned

Claims about people are made from data every day. Use correlation carefully, and require a mechanism before accepting a cause.

Service as Action

COMMUNITY NEED	Community decisions are made from claims that nobody has checked.
MATHEMATICAL CONNECTION	Correlation, causation and the critical reading of claims.
PROPOSED ACTION	Check one public claim about the community and publish the four checks with the verdict.
EXPECTED IMPACT	A decision is made from a claim that has been examined rather than repeated.
STUDENT REFLECTION	Record the claim, the checks and what you found.

Community Project

PROJECT IDEA	A claims audit for the school community.
NEED	Posters and announcements make statistical claims without sources.
RESEARCH QUESTION	Which claims made in the school are supported by checkable evidence?
MATHEMATICS	Sampling, correlation, base rates and margin of error.
ACTION OR OUTCOME	A short published audit of three claims.
BENEFICIARIES	Everyone who reads the school's announcements.
EVIDENCE	The audit sheet with the four checks for each claim.

Personal Project connection · MYP 5

PERSONAL INTEREST	Sport, gaming or health.
LEARNING GOAL	To check a claim about something you care about before acting on it.
PRODUCT OR OUTCOME	A one-page evaluation of a personal claim.
MATHEMATICS	Correlation, sampling and base rates.
PROCESS	Find the claim, apply the four checks, then state a verdict.
EVIDENCE	The evaluation and the verdict.

CHAPTER 10 · PART IV · DATA AND EVIDENCE

10

Probability, sampling and evidence

Probability is a number between 0 and 1 that measures uncertainty. It is built from a model of the outcomes, and the model must be stated before the number means anything.

ENGAGE

A medical test for a rare condition is 99% accurate. A patient tests positive. How likely is it that the patient actually has the condition? The answer surprises most people, and the reason is a base rate.

Framing the inquiry

Key concept: Logic
Related concepts: Quantity · Models
Global context: Identities and relationships
Global context exploration: Beliefs and values

Statement of inquiry: The logic of probability, and the models built from it, allow a quantity of uncertainty to be assigned to an event, and that quantity is what allows a decision to be justified rather than guessed.

Questions to guide us

- **Factual:** How is the probability of an event calculated, and what are the addition and multiplication rules?
- **Conceptual:** Why does the probability of a combined event depend on whether the events are independent, and how does a base rate change the interpretation of a test result?
- **Debatable:** Should a decision about a person be made on a probability alone?

ATL · COMMUNICATION. In order for a student to explain a probabilistic result to an audience, students must state the model used, express the result as a fraction, a decimal and a percentage, and say plainly what the number does and does not mean.

CHAPTER 10 · INVESTIGATION 10.1

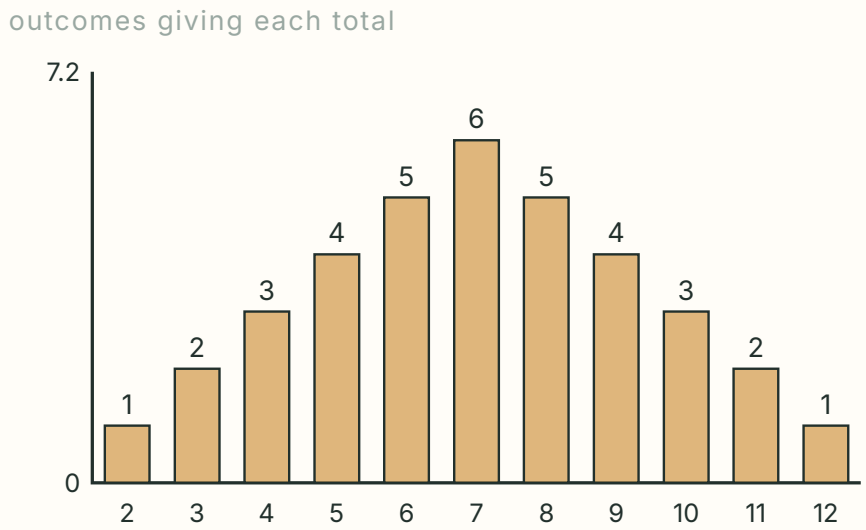
Outcomes, events and probability

INQUIRY QUESTION

How can the likelihood of an event be assigned a number?

YOUR TASK

Roll two dice 60 times and record the total each time. Compute the experimental probability of each total, then compare with the theoretical probabilities from the sample space of 36 outcomes.



total	theoretical P	experimental P (60 rolls)	difference
7	6/36 = 0.167	11/60 = 0.183	0.016
2	1/36 = 0.028	1/60 = 0.017	0.011
6	5/36 = 0.139	9/60 = 0.150	0.011
12	1/36 = 0.028	0/60 = 0	0.028

CHAPTER 10 · INVESTIGATION 10.1

Outcomes, events and probability

What do you notice?

- The experimental probabilities were close to the theoretical ones.
- No total appeared exactly as often as the theory predicts.
- The most likely total, 7, was also the most frequent in the experiment.
- The differences would shrink with more rolls.

> **Observation.** Experimental probabilities approach theoretical probabilities as the number of trials increases, but they do not match exactly in a small sample.

MATHEMATICAL CONCLUSION

P(event) = favourable outcomes ÷ total equally likely outcomes, and the experimental probability estimates it with an error that shrinks as the number of trials grows.

Your own words

CHAPTER 10 · 10.1

Outcomes, events and probability

The sample space is the set of all possible outcomes of an experiment. An event is a subset of the sample space, and the probability of an event measures how much of the sample space it covers.

When all outcomes are equally likely, P(event) = number of favourable outcomes ÷ total number of outcomes. This is the definition, and it depends on the assumption of equal likelihood.

Probabilities lie between 0 and 1, and the probabilities of all outcomes sum to 1. The complement rule P(not A) = 1 − P(A) follows directly.

The model is a choice. A fair die has six equally likely outcomes; a loaded one does not, and no calculation can detect that. The model must be stated and justified.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 10 · 10.1 · WORKED EXAMPLE 1

Outcomes, events and probability

WORKED EXAMPLE 10.1.1

Problem. Two fair dice are thrown. Find the probability that the total is 7.

first die	second die values giving 7
1	6
2	5
3	4
4	3
5	2
6	1

Six of the thirty-six outcomes give a total of 7.

Mathematical reasoning. There are 36 equally likely outcomes, and six of them give a total of 7.

1. The sample space has $6 \times 6 = 36$ equally likely outcomes.
2. The pairs giving 7 are (1,6), (2,5), (3,4), (4,3), (5,2), (6,1): six of them.
3. P(total is 7) = 6/36 = 1/6.

Final answer. 1/6

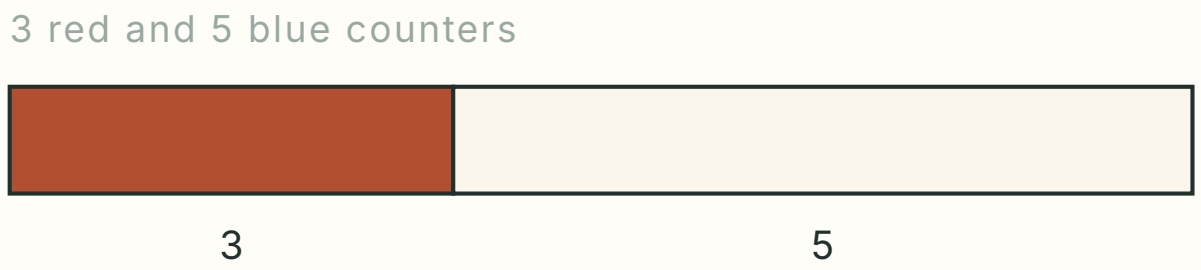
Interpretation. The assumption of equal likelihood is what makes the count of outcomes enough.

CHAPTER 10 · 10.1 · WORKED EXAMPLE 2

Outcomes, events and probability

WORKED EXAMPLE 10.1.2

Problem. A bag holds 3 red and 5 blue counters. Find P(red), P(not red) and P(red or blue).



Mathematical reasoning. The sample space has eight equally likely outcomes, and the events are subsets of it.

1. Total counters = 8.
2. P(red) = 3/8.
3. P(not red) = 1 − 3/8 = 5/8.
4. P(red or blue) = 1, because every counter is one or the other.

Final answer. P(red) = 3/8, P(not red) = 5/8, P(red or blue) = 1

Interpretation. The complement rule is faster than counting the blue counters, and it gives the same answer.

CHAPTER 10 · INVESTIGATION 10.2

Compound events: tables, trees and Venn diagrams

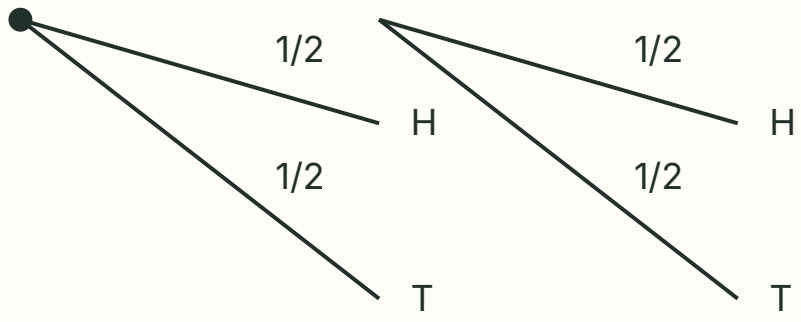
INQUIRY QUESTION

How can the probability of a combination of events be found?

YOUR TASK

Toss two fair coins 80 times. Record whether the results are the same or different, then compute the experimental probabilities. Compare with the theoretical probabilities from a tree diagram, and repeat with a spinner divided into three equal sectors.

two tosses of a fair coin



outcome	theoretical P	experimental P (80)	difference
two heads	1/4 = 0.25	22/80 = 0.275	0.025
one of each	1/2 = 0.50	39/80 = 0.488	0.012
two tails	1/4 = 0.25	19/80 = 0.238	0.012
the same	1/2 = 0.50	41/80 = 0.513	0.013

CHAPTER 10 · INVESTIGATION 10.2

Compound events: tables, trees and Venn diagrams

What do you notice?

- The experimental probabilities were close to the theoretical ones.
- 'One of each' was the most likely outcome, as the theory predicts.
- The two orderings, head then tail and tail then head, are separate outcomes.
- Combining them gives P(the same) = 1/2.

Observation. The probabilities of combined outcomes follow from multiplying along the branches of a tree and adding across the branches that belong to the event.

MATHEMATICAL CONCLUSION

For independent events $P(A \text{ and } B) = P(A) \times P(B)$, for mutually exclusive events $P(A \text{ or } B) = P(A) + P(B)$, and the general rule $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$ covers every case.

Your own words

CHAPTER 10 · 10.2

Compound events: tables, trees and Venn diagrams

Two events are independent when the outcome of one does not affect the other. For independent events, P(A and B) = P(A) × P(B).

Two events are mutually exclusive when they cannot both happen. For mutually exclusive events, P(A or B) = P(A) + P(B).

The general addition rule is P(A or B) = P(A) + P(B) − P(A and B), and the subtraction removes the outcomes counted twice.

Sample-space diagrams, tree diagrams and Venn diagrams are three representations of the same logic. The tree makes independence visible; the Venn diagram makes the overlap visible.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

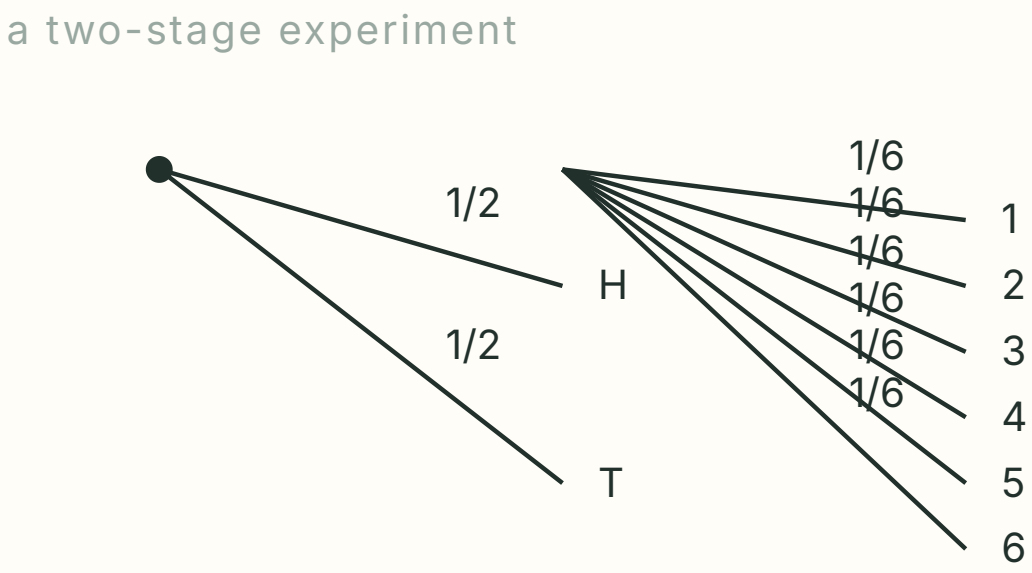
In your own words

CHAPTER 10 · 10.2 · WORKED EXAMPLE 1

Compound events: tables, trees and Venn diagrams

WORKED EXAMPLE 10.2.1

Problem. A fair coin is tossed and a fair die is rolled. Find P(head and 6).



Mathematical reasoning. The two events are independent, so their probabilities multiply.

1. P(head) = 1/2 and P(6) = 1/6.
2. The events are independent, so multiply.
3. P(head and 6) = 1/2 × 1/6 = 1/12.

Final answer. 1/12

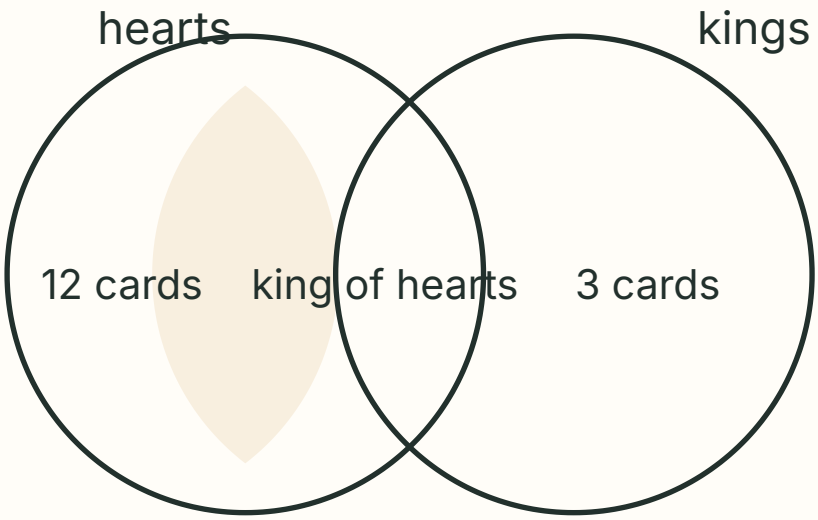
Interpretation. The tree diagram shows twelve equally likely combined outcomes, one of which is head and 6.

CHAPTER 10 · 10.2 · WORKED EXAMPLE 2

Compound events: tables, trees and Venn diagrams

WORKED EXAMPLE 10.2.2

Problem. A card is drawn from a standard pack. Find P(heart or king).



Mathematical reasoning. The general addition rule removes the card that is both a heart and a king, which would otherwise be counted twice.

1. P(heart) = 13/52 and P(king) = 4/52.
2. P(heart and king) = 1/52.
3. P(heart or king) = 13/52 + 4/52 − 1/52 = 16/52.
4. = 4/13.

Final answer. 4/13

Interpretation. The Venn diagram shows the overlap that the subtraction removes.

CHAPTER 10 · INVESTIGATION 10.3

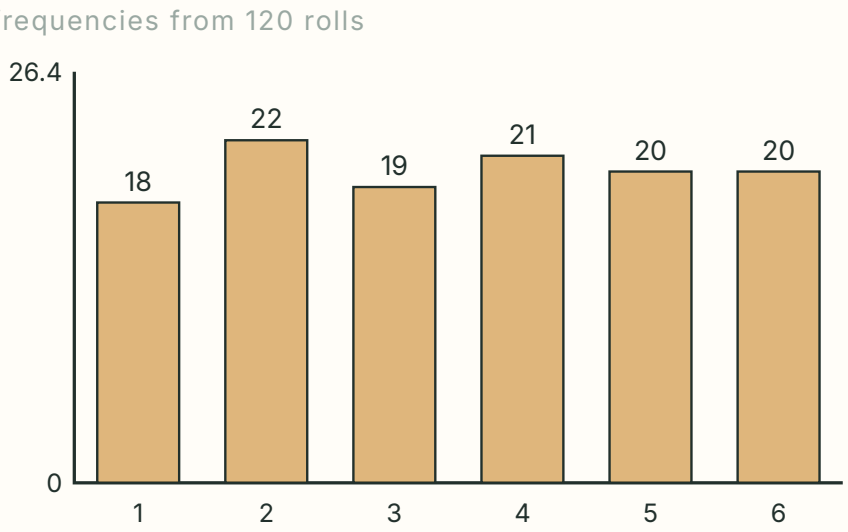
From theory to experiment

INQUIRY QUESTION

How does an experimental probability compare with a theoretical one?

YOUR TASK

Roll a die 120 times and record the frequency of each face. Compute the experimental probability of each face, compare with 1/6, and decide whether the die behaves fairly. Then pool your results with two other groups.



face	theoretical P	experimental P	difference
1	1/6 = 0.167	18/120 = 0.150	0.017
2	1/6 = 0.167	22/120 = 0.183	0.016
3	1/6 = 0.167	19/120 = 0.158	0.009
4	1/6 = 0.167	21/120 = 0.175	0.008
5	1/6 = 0.167	20/120 = 0.167	0.000
6	1/6 = 0.167	20/120 = 0.167	0.000

CHAPTER 10 · INVESTIGATION 10.3

From theory to experiment

What do you notice?

- All six frequencies were close to 20, which is 1/6 of 120.
- The largest difference was 0.017, which is small.
- Pooling three groups gave 360 rolls and even closer agreement.
- The evidence supports a fair die, but does not prove it.

Observation. Experimental probabilities from a large number of trials agree closely with the theoretical value, and pooling trials improves the agreement.

MATHEMATICAL CONCLUSION

An experimental probability estimates the theoretical probability, and a large difference between them is evidence that the model is wrong rather than that the calculation failed.

Your own words

CHAPTER 10 · 10.3

From theory to experiment

An experimental probability is a relative frequency: the number of times an event occurred divided by the number of trials. It is an estimate, not a definition.

The theoretical probability comes from a model of the outcomes. When the model is correct, the experimental probability approaches it as the number of trials grows.

A large difference between the two is evidence that the model is wrong, not that the mathematics failed. A die that never shows 6 in 60 rolls is probably loaded.

A small difference is expected. The question is always whether the difference is small enough to be explained by chance, which is the beginning of formal statistical testing.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

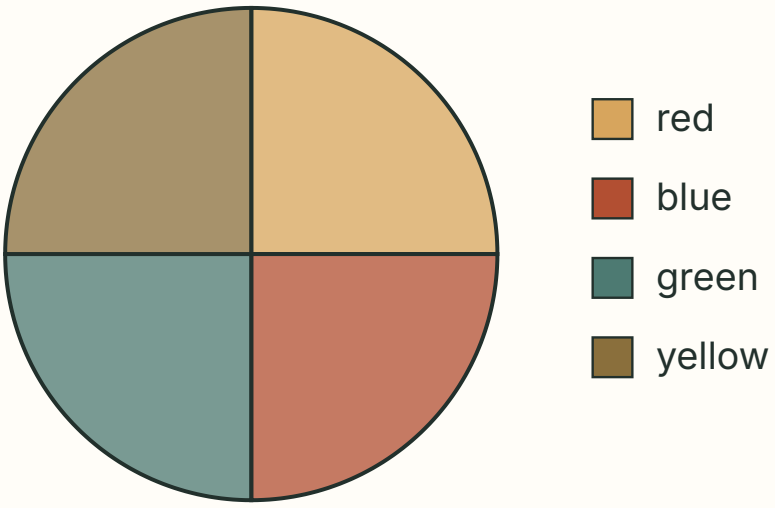
CHAPTER 10 · 10.3 · WORKED EXAMPLE 1

From theory to experiment

WORKED EXAMPLE 10.3.1

Problem. A spinner is divided into four equal sectors. In 200 spins it lands on red 46 times. Estimate P(red) and compare with the theoretical value.

a four-sector spinner



Mathematical reasoning. The experimental probability is a relative frequency and estimates the theoretical value.

1. Experimental P(red) = 46/200 = 0.23.
2. Theoretical P(red) = 1/4 = 0.25.
3. The difference is 0.02, which is small.
4. So the model of four equal sectors is plausible.

Final answer. 0.23 experimentally, against 0.25 theoretically

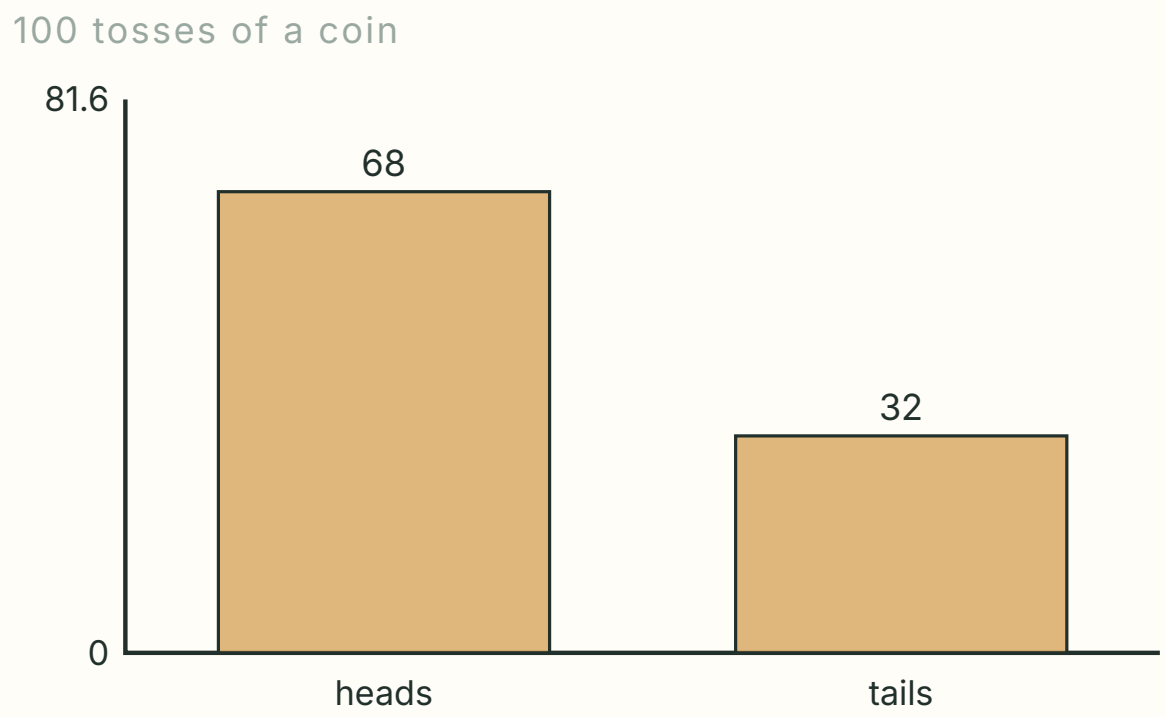
Interpretation. The small difference is what we expect from a fair spinner, and it does not overturn the model.

CHAPTER 10 · 10.3 · WORKED EXAMPLE 2

From theory to experiment

WORKED EXAMPLE 10.3.2

Problem. A coin is tossed 100 times and gives 68 heads. Is the coin fair?



Mathematical reasoning. A large departure from the theoretical value is evidence against the model.

1. Experimental P(head) = 68/100 = 0.68.
2. Theoretical P(head) for a fair coin = 0.5.
3. The difference of 0.18 is large for 100 tosses.
4. So there is evidence that the coin is not fair.

Final answer. the coin is probably not fair

Interpretation. The conclusion is about the model, and it is stated as evidence rather than proof.

CHAPTER 10 · INVESTIGATION 10.4

Populations, samples and bias

INQUIRY QUESTION

How can a sample be chosen so that it represents a population?

YOUR TASK

Estimate the proportion of students in the school who walk to school, using three samples: the first 20 people you meet, 20 people chosen at random from a list, and everyone in one class. Compare the three estimates.

sampling method	sample size	estimate of the proportion	biased?
first 20 met	20	0.75	yes — convenience
random from a list	20	0.45	no — random selection
one whole class	30	0.40	yes — one group only
random from a list	100	0.47	no, and less variable

CHAPTER 10 · INVESTIGATION 10.4

Populations, samples and bias

What do you notice?

- The convenience sample gave the highest estimate, because it met people arriving on foot.
- The class sample was not representative of the whole school.
- The two random samples agreed with each other.
- Doubling the random sample changed the estimate only slightly.

Observation. A random sample gives an estimate that does not depend on how the sample was reached, and a larger random sample reduces the variability.

MATHEMATICAL CONCLUSION

A sample represents a population only when selection is random, and bias from a non-random selection is not reduced by increasing the sample size.

Your own words

CHAPTER 10 · 10.4

Populations, samples and bias

A population is the whole group a claim is about, and a sample is the part that is measured. A parameter describes the population and a statistic describes the sample.

A sample is representative when every member of the population has a known, non-zero chance of being selected. Convenience and self-selected samples fail this test.

Bias is a systematic error in the selection. It cannot be reduced by taking a larger sample, because a bigger biased sample is still biased.

A random sample removes selection bias, and a larger sample reduces the variability of the estimate. The two effects are different and both matter.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 10 · 10.4 · WORKED EXAMPLE 1

Populations, samples and bias

WORKED EXAMPLE 10.4.1

Problem. A school of 1 200 students is surveyed by asking the first 40 students who arrive. Explain why the sample is biased.

check	finding
population	all 1 200 students
sample	the first 40 arrivals
selection	not random — early arrivals differ
effect	a systematic error that size cannot fix

A convenience sample is biased however large it is.

Mathematical reasoning. Students who arrive early differ systematically from those who arrive late, so the sample is not representative.

1. The population is all 1 200 students.
2. The sample is the first 40 arrivals, chosen by convenience.
3. Early arrivals are likely to differ from the rest in ways that matter.
4. So the estimate is biased, and taking 400 instead of 40 would not fix it.

Final answer. the sample is a convenience sample and is biased

Interpretation. Bias is about how the sample was chosen, not how big it is.

CHAPTER 10 · 10.4 · WORKED EXAMPLE 2

Populations, samples and bias

WORKED EXAMPLE 10.4.2

Problem. Explain the difference between a population parameter and a sample statistic.

term	describes	example
parameter	the population	the mean height of all students
statistic	the sample	the mean height of the 40 measured

A statistic estimates a parameter.

Mathematical reasoning. The parameter is fixed but unknown, and the statistic varies from sample to sample.

1. The parameter describes the whole population and cannot usually be computed.
2. The statistic is computed from the sample and varies between samples.
3. The statistic is used as an estimate of the parameter.
4. The quality of the estimate depends on the sampling method.

Final answer. a parameter describes the population; a statistic describes the sample

Interpretation. Confusing the two is what allows a biased sample to be reported as a fact about everyone.

CHAPTER 10 · INVESTIGATION 10.5

Risk, base rates and decisions

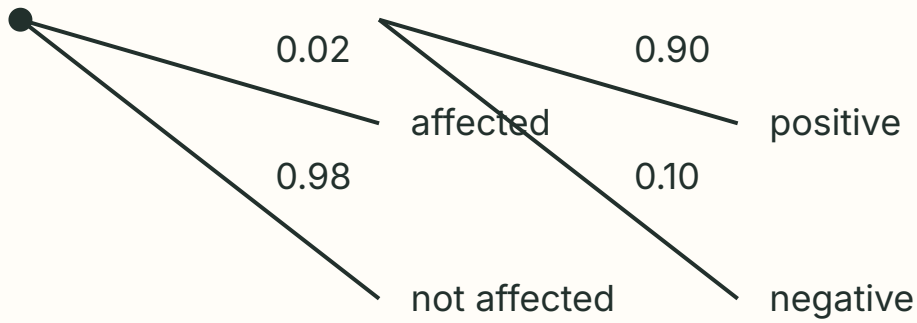
INQUIRY QUESTION

How should a probability be interpreted when a decision must be made?

YOUR TASK

Model a screening test for a condition with a base rate you choose. Build a table for 10 000 people, compute the true and false positives, and find the posterior probability. Then change the base rate and repeat.

a 2% base rate with a 90% accurate test



base rate	affected in 10 000	true positives	false positives	P(affected given positive)
1%	100	90	990	0.083
2%	200	180	980	0.155
10%	1 000	900	900	0.5
50%	5 000	4 500	500	0.9

CHAPTER 10 · INVESTIGATION 10.5

Risk, base rates and decisions

What do you notice?

- The posterior probability rose steadily with the base rate.
- The number of false positives fell as the base rate rose, even though the test's accuracy never changed.
- At a 1% base rate, more than nine in ten positive results were false.
- At a 50% base rate, nine in ten positive results were correct.

Observation. The probability that a positive result is correct depends far more on the base rate than on the accuracy of the test.

MATHEMATICAL CONCLUSION

A probability must be interpreted against its base rate, because a rare event produces mostly false positives even with an accurate test — and a decision must weigh the consequences as well as the number.

Your own words

CHAPTER 10 · 10.5

Risk, base rates and decisions

The base rate is the frequency of an event in the whole population, before any test or indicator is applied. It is the starting point for interpreting a test result.

A test with high accuracy can still give mostly false positives when the condition is rare, because the number of healthy people is far larger than the number of affected people.

The posterior probability is found by comparing the true positives with the total positives, and it depends on the base rate as much as on the accuracy.

A decision based on a probability must also weigh the consequences. The same probability can justify different actions when the costs of being wrong differ.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

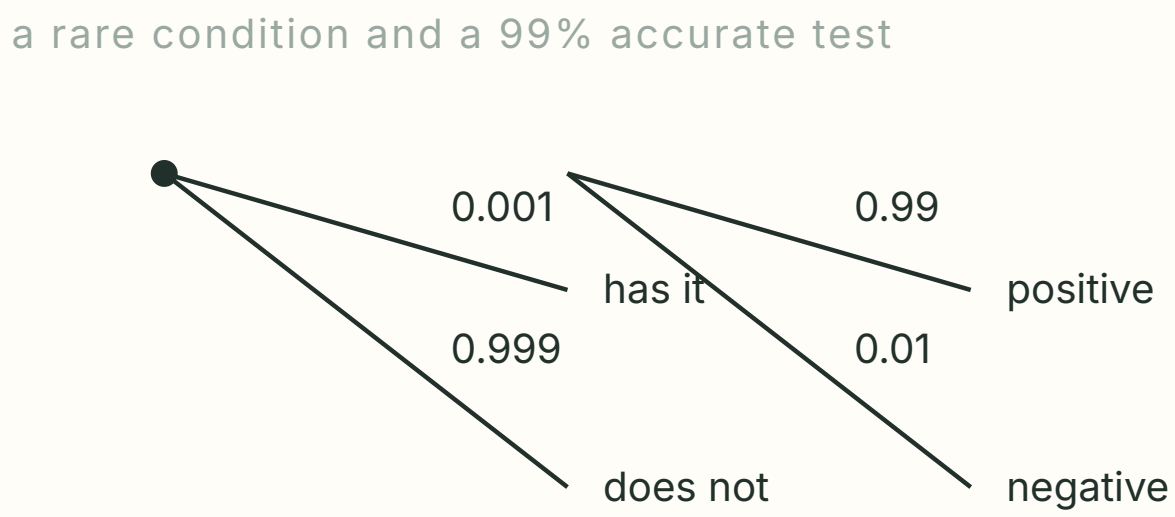
In your own words

CHAPTER 10 · 10.5 · WORKED EXAMPLE 1

Risk, base rates and decisions

WORKED EXAMPLE 10.5.1

Problem. A condition affects 1 in 1 000 people. A test is 99% accurate. A patient tests positive. Find the probability that the patient has the condition.



Mathematical reasoning. The posterior probability compares the true positives with all positives, and it is dominated by the base rate.

1. Of 100 000 people, 100 have the condition and 99 900 do not.
2. True positives ≈ 0.99 × 100 = 99.
3. False positives ≈ 0.01 × 99 900 = 999.
4. P(condition given positive) = 99/(99 + 999) ≈ 0.09.

Final answer. about 9%

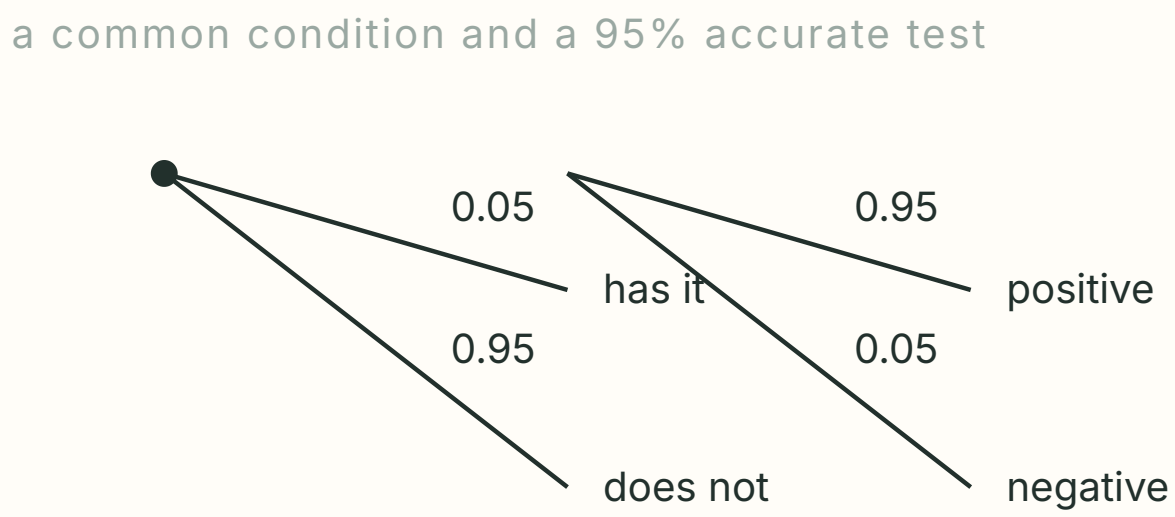
Interpretation. A 99% accurate test still gives mostly false positives when the condition is rare, because the base rate is tiny.

CHAPTER 10 · 10.5 · WORKED EXAMPLE 2

Risk, base rates and decisions

WORKED EXAMPLE 10.5.2

Problem. A screening programme for a condition affecting 1 in 20 people uses a test that is 95% accurate. Find the probability that a positive result is correct.



Mathematical reasoning. The same calculation with a different base rate gives a very different answer.

1. Of 1 000 people, 50 have the condition and 950 do not.
2. True positives ≈ 0.95 × 50 = 47.5.
3. False positives ≈ 0.05 × 950 = 47.5.
4. P(condition given positive) = 47.5/95 ≈ 0.5.

Final answer. about 50%

Interpretation. The test is only slightly less accurate, but the base rate is fifty times higher, so the result is far more informative.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding

10.A1 FLUENCY
A fair die is rolled. Find P(even).

10.A2 FLUENCY
A bag holds 4 red and 6 green counters. Find P(red).

10.A3 FLUENCY
Two fair coins are tossed. Find P(two heads).

10.A4 FLUENCY
Find P(not 6) when a fair die is rolled.

10.A5 FLUENCY
A card is drawn from a pack. Find P(king).

10.A6 FLUENCY
A spinner has 8 equal sectors, 3 of them gold. Find P(gold).

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding (continued)

10.A7 FLUENCY

Two events are independent with P(A) = 0.4 and P(B) = 0.5. Find P(A and B).

10.A8 FLUENCY

Events A and B are mutually exclusive with P(A) = 0.3 and P(B) = 0.45. Find P(A or B).

10.A9 FLUENCY

P(A) = 0.6, P(B) = 0.5 and P(A and B) = 0.3. Find P(A or B).

10.A10 FLUENCY

A fair coin is tossed three times. Find P(three heads).

10.A11 FLUENCY

A die is rolled twice. Find P(6 then 6).

10.A12 REASONING

Explain why the probabilities of all outcomes in a sample space sum to 1.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns

10.B1 REASONING

Investigate: in 100 rolls of a die, how many sixes would you expect, and would 20 sixes be surprising?

10.B2 REASONING

Test the claim: if a coin gives five heads in a row, the next toss is more likely to be tails.

10.B3 REASONING

Investigate whether two dice can be used to model a spinner with 12 equal sectors.

10.B4 REASONING

Generalise: when does P(A and B) = P(A) × P(B)?

10.B5 REASONING

Investigate the pattern of P(at least one six) when two dice are rolled.

10.B6 REASONING

Test the claim: mutually exclusive events cannot be independent.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns (continued)

10.B7 REASONING

Investigate whether P(A or B) can ever exceed 1.

10.B8 REASONING

Generalise a rule for the expected number of successes in n trials.

10.B9 REASONING

Investigate how the experimental probability changes as the number of trials rises from 10 to 1 000.

10.B10 REASONING

Test the claim: a larger sample always gives a less biased estimate.

10.B11 REASONING

Investigate the effect of a base rate on the number of false positives in 1 000 tests with 95% accuracy.

CRITERION C Communicating

Criterion C practice · Communicating

10.C1 REASONING

Explain in words what a sample space is.

10.C2 REASONING

Explain the difference between independent and mutually exclusive events.

10.C3 REASONING

Describe when to use the multiplication rule and when to use the addition rule.

10.C4 REASONING

Explain the general addition rule and why it subtracts an overlap.

10.C5 REASONING

Describe the difference between experimental and theoretical probability.

10.C6 REASONING

Explain why a large difference between experiment and theory suggests the model is wrong.

CRITERION C Communicating

Criterion C practice · Communicating (continued)

10.C7 REASONING
Describe two representations of a two-stage probability experiment.

10.C8 REASONING
Explain what bias is and why a larger sample does not remove it.

10.C9 REASONING
Describe the difference between a population and a sample.

10.C10 REASONING
Explain what a base rate is and why it changes the interpretation of a test result.

10.C11 REASONING
Describe how to find the probability that a positive test result is correct.

10.C12 REASONING
Explain why a probability alone is not enough to make a decision.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts

10.D1 APPLICATION

A factory finds 3 defective items in a sample of 250. Estimate the number of defective items in a batch of 20 000.

10.D2 APPLICATION

A test for a condition affecting 1 in 100 people is 98% accurate. Find the probability that a positive result is correct.

10.D3 REASONING

A school of 900 students is surveyed by asking every tenth student on a list. Explain whether the sample is representative.

10.D4 REASONING

A weather forecast gives a 70% chance of rain. Explain what this means and what it does not mean.

10.D5 REASONING

A lottery ticket has a 1 in 14 000 000 chance of winning. Explain why buying two tickets doubles the chance but does not make winning likely.

10.D6 REASONING

A doctor says a test is 99% accurate and the patient is positive. Explain why the patient may still be unlikely to have the condition.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts (continued)

10.D7 REASONING
A survey of 50 people at a football match finds 80% support a new stadium. Explain why this does not represent the town.

10.D8 REASONING
Two vaccines are reported as 90% and 95% effective. Explain what information is missing before they can be compared.

10.D9 APPLICATION
A quality check finds 5 faults in 400 items. Estimate the number of faults in 10 000 items and state the uncertainty.

10.D10 REASONING
Explain whether a decision to close a school should be based on a projection of a declining roll.

10.D11 APPLICATION
A screening test for a condition affecting 5 in 100 people is 90% accurate. Find the probability that a negative result is correct.

10.D12 REASONING
Explain why a rare but serious risk can justify a different decision from a common but minor one.

CHAPTER 10 · APPLICATION AND ASSESSMENT

Applying chapter 10

These tasks use probability models, sample spaces, compound events, experimental probability, sampling and base rates. State the model behind every number you report.

A1 · The screening decision

A condition affects 2 in 100 people. A test is 95% accurate. Find the probability that a person who tests positive actually has the condition, using a table for 10 000 people. Then explain what the result means for a decision about treatment.

Criteria A and D · justify the calculation with the base rate and comment on the consequences.

A2 · The biased survey

A company surveys 60 customers who visited its website and reports that 85% are satisfied. Evaluate the claim using the four checks, and describe the sampling method that would support a claim about all customers.

Criteria A and C · apply the checks of source, sample, measure and conclusion.

CHAPTER 10 · REVIEW

What you should now be able to do

Key ideas

- P(event) = favourable outcomes ÷ total equally likely outcomes.
- Probabilities lie between 0 and 1 and sum to 1 over the sample space.
- P(not A) = 1 − P(A).
- For independent events, P(A and B) = P(A) × P(B).
- P(A or B) = P(A) + P(B) − P(A and B).
- An experimental probability estimates the theoretical probability.
- A representative sample requires random selection; bias is not fixed by size.
- A posterior probability depends on the base rate as much as on the accuracy.

Vocabulary

sample space · outcome · event · equally likely · complement · independent · mutually exclusive · tree diagram · Venn diagram · experimental probability · theoretical probability · population · sample · bias · base rate · posterior probability

CHAPTER 10 · REVIEW

Mixed practice · consolidate and extend

Do not use a calculator unless a question asks for one. Show a method where one is requested.

10.R1 Find P(not a multiple of 3) when a fair die is rolled.

10.R2 Two independent events have probabilities 0.3 and 0.4. Find P(both).

10.R3 P(A) = 0.5, P(B) = 0.6, P(A and B) = 0.2. Find P(A or B).

10.R4 In 200 spins a spinner lands on red 52 times. Estimate P(red).

10.R5 Explain why a convenience sample is biased.

10.R6 A test is 99% accurate for a condition affecting 1 in 50 people. Explain why a positive result is not certain.

CHAPTER 10 · REVIEW

Returning to the statement of inquiry

> **STATEMENT OF INQUIRY**
>
> The logic of probability, and the models built from it, allow a quantity of uncertainty to be assigned to an event, and that quantity is what allows a decision to be justified rather than guessed.

Reflect on the chapter

Answer in your own words, using evidence from this chapter.

- Write the statement of inquiry for this chapter in your own words.
- Which mathematical ideas from this chapter did you use to understand it? Name at least two.
- How did the key concept of logic and the related concepts of quantity and models appear in your work?
- How is the global context of identities and relationships, and its exploration of beliefs and values, visible in the investigations and examples you completed?
- Use evidence from an investigation or a worked example to justify your answer.
- Has your understanding of the statement of inquiry changed since the start of the chapter? Explain how.
- What are the limitations of a probability for making a decision about people, and what would you still like to ask?

CHAPTER 10 · PROJECTS, SERVICE AND ACTION

Acting on what you have learned

Uncertainty is part of every decision about people. Use probability to size the uncertainty, and say what the number does not tell you.

Service as Action

COMMUNITY NEED	Community members make health and safety decisions from claims they cannot check.
MATHEMATICAL CONNECTION	Probability, base rates and the interpretation of risk.
PROPOSED ACTION	Publish a plain-language explanation of one probability claim that affects the community, with its base rate.
EXPECTED IMPACT	A decision is made with the uncertainty stated rather than hidden.
STUDENT REFLECTION	Record the claim, the base rate and what you explained.

Community Project

PROJECT IDEA	A probability explainer for the school community.
NEED	Screening, vaccination and safety claims are circulated without their base rates.
RESEARCH QUESTION	What does one commonly quoted probability actually mean for a person?
MATHEMATICS	Base rates, conditional probability and expected value.
ACTION OR OUTCOME	A one-page explainer with a worked example.
BENEFICIARIES	Students, families and staff.
EVIDENCE	The explainer and the worked calculation.

Personal Project connection · MYP 5

PERSONAL INTEREST	Sport, games or travel planning.
LEARNING GOAL	To interpret a probability in a decision you face.
PRODUCT OR OUTCOME	A decision analysis with the probabilities stated.
MATHEMATICS	Probability, expected value and base rates.
PROCESS	List the outcomes, assign probabilities, then weigh the consequences.
EVIDENCE	The analysis and the decision with its justification.

CHAPTER 1 · CRITERION A · KNOWING AND UNDERSTANDING

1.A1 **0.875 stops; 0.8·3· recurs; 0.275 stops**
Denominators of only 2s and 5s terminate: $8 = 2^3$ and $40 = 2^3 \times 5$.

1.A2 **3/11**
Let x = 0.2727…; 100x − x = 27, so 99x = 27.

1.A3 **The remainder in the division can only be 0–6, so one must repeat.**
Only seven remainders are possible, so the decimal cannot go on without repeating.

1.A4 **37/99 m**
0.3737… = 37/99.

1.A5 **No — the blocks grow, so the decimal never repeats.**
A rational decimal either stops or repeats a fixed block.

1.A6 **3.1429 (rounded); the exact value is 22/7**
22/7 recurs, so any finite decimal is a rounded claim.

1.A7 **Rational: 0.25, 7/3, −4, 0.·1·; irrational: √5, π**
Recurring and terminating decimals are rational; π and √5 are not fractions.

1.A8 **5/12**
1/3 = 4/12 and 1/2 = 6/12, so 5/12 lies between.

1.A9 **Yes — 3/5 = 0.6 exactly**
The denominator 5 gives a terminating decimal.

1.A10 **0.·9· = 1 exactly; 0.9999 is a rounded value below 1.**
A recurring decimal is exact, while a finite decimal is a claim about rounding.

1.A11 **1/22**
Let x = 0.04545…; then 100x = 4.5454… and 10x = 0.4545…, so 90x = 4.0909… − 0.4545… = 3.6363… = 40/11 and x = 1/22.

1.A12 **Because n = n/1**
The definition only requires a quotient of two integers.

CHAPTER 1 · CRITERION B · INVESTIGATING PATTERNS

1.B1 **Assume $\sqrt{5} = a/b$ in simplest form; then 5 divides a^2, so 5 divides a, so 5 divides b — a contradiction.**
A prime dividing a square divides the number itself.

1.B2 **Irrational**
If $\sqrt{(2/3)}$ were rational, its square 2/3 would have a rational square root, which would make 6 rational after clearing denominators.

1.B3 **$\sqrt{2} \times \sqrt{2} = 2$**
One counterexample refutes the claim.

1.B4 **The error falls by roughly a factor of 10 each time, but never reaches zero.**
Each added decimal place improves the approximation tenfold.

1.B5 **True — take the average of the two numbers and round it to a nearby fraction, or use a decimal truncation of the left-hand number.**
The rationals are dense on the number line.

1.B6 **$2\sqrt{3} = \sqrt{12}$ and $3\sqrt{2} = \sqrt{18}$, so $2\sqrt{3} < 3\sqrt{2}$.**
Squaring both sides preserves the order for positive numbers.

1.B7 **$\sqrt{n}$ is irrational**
The contradiction proof works for any non-square integer n.

1.B8 **$\sqrt{49} = 7$, $\sqrt{64} = 8$; the rule is $\sqrt{(n^2)} = n$**
Each term is the square root of a perfect square.

1.B9 **$\sqrt{2}$ and $-\sqrt{2}$, whose sum is 0**
Irrational numbers are not closed under addition.

1.B10 **True for $x \neq 0$ — otherwise $1/x = p/q$ would make $x = q/p$ rational.**
The rationals are closed under division by a non-zero rational.

1.B11 **Infinitely many — for example $1 + \sqrt{2}/n$ for any positive integer n.**
Adding an irrational to a rational gives an irrational.

CHAPTER 1 · CRITERION C · COMMUNICATING

1.C1 **$-7 \le x \le 3$, so $[-7, 3]$**
$|x + 2| \le 5$ means $-5 \le x + 2 \le 5$.

1.C2 **The distance between a and b on the number line**
Absolute value removes the direction and keeps the size of the gap.

1.C3 **$\{x : x \neq 3\}$; the single number 3 is excluded**
Union gathers both sides and the endpoint is in neither.

1.C4 **$7\sqrt{3}$**
$\sqrt{75} = 5\sqrt{3}$ and $\sqrt{12} = 2\sqrt{3}$, and both are multiples of $\sqrt{3}$.

1.C5 **$\sqrt{5} + 1$**
$(\sqrt{5} + 1)(\sqrt{5} - 1) = 4$, so the product clears the surd.

1.C6 **They are not multiples of the same surd.**
Only like surds combine, in the same way as like terms.

1.C7 **[3.55, 3.65) m**
Rounding to one decimal place includes the lower bound and excludes the upper.

1.C8 **3.60 m claims precision to the nearest centimetre; 3.6 m claims only the nearest 10 cm.**
Trailing zeros record the precision of the instrument.

1.C9 **{6, 12}**
The intersection holds only the numbers with both properties.

1.C10 **A counterexample disproves a universal claim; a contradiction proves one by ruling out the alternative.**
The two tools work in opposite directions.

1.C11 **$(-\infty, -2) \cup (5, \infty)$**
The complement keeps everything outside the interval, excluding both endpoints.

1.C12 **[3500, 4500) kg**
One significant figure rounds to the nearest thousand, including the lower bound.

CHAPTER 1 · CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS

1.D1 **30 m² exactly ($\sqrt{50} \times \sqrt{18} = \sqrt{900} = 30$)**
Multiplying surds multiplies the numbers under the root.

1.D2 **[12.65, 12.75) m, width 0.1 m**
Rounding to one decimal place gives an interval of width 0.1.

1.D3 **$0.4\dot{3} = 43/99 \approx 0.4343$, so it is larger by $1/99 - 0.43 \approx 0.0043$ km²**
Convert the recurring decimal to a fraction before comparing.

1.D4 **10 times larger**
The powers of 10 differ by 1, so the quantities differ by a factor of 10.

1.D5 **[24.5, 25.5) m**
The record fixes the length only to within half a metre either side.

1.D6 **$\sqrt{27} \approx 5.196$ cm, which is greater than 5.1, so it will not fit**
Compare the exact surd with the rounded measurement, and state the uncertainty.

1.D7 **61/99 ha ≈ 0.6162 ha ≈ 6162 m²**
Convert the recurring decimal to a fraction, then multiply.

1.D8 **The error can be up to 0.5 m per measurement, and errors add up over several measurements.**
Rounding errors accumulate, so the total can be far from the true amount.

1.D9 **$25000\sqrt{8}$ cm = $50000\sqrt{2}$ cm ≈ 70 710 cm ≈ 707 m**
Multiply by the scale factor, then convert units and round.

1.D10 **$1.4\dot{9} = 148/99 \approx 1.4949$, so it is larger by $1/1980$ km²**
Write both as fractions over a common denominator before subtracting.

1.D11 **$2\sqrt{3} \times 10^3 \approx 3464$ litres, so the first holds more by $1000(2\sqrt{3} - 3)$ litres**
Compare the exact surd form with the integer, then subtract.

1.D12 **The surd keeps the value exact, while a rounded decimal introduces an error that grows through the calculation.**
Rounding before calculating compounds the error.

CHAPTER 2 · CRITERION A · KNOWING AND UNDERSTANDING

2.A1 **$20a^7$**
Multiply the coefficients and add the indices.

2.A2 **1/8**
A negative index gives the reciprocal.

2.A3 **1, so that $a^m \div a^m = a^0$ continues to work**
The exponent laws require $a^0 = 1$.

2.A4 **$8x^6y^3$**
The power law multiplies every index by 3.

2.A5 **4**
A negative index inverts the fraction before squaring.

2.A6 **$3x^3y^2$**
Divide coefficients and subtract matching indices.

2.A7 **10^8**
$5 - (-3) = 8$.

2.A8 **Squaring is not distributive over addition; $(a + b)^2 = a^2 + 2ab + b^2$.**
The exponent applies to the whole bracket, not to each term separately.

2.A9 **3^3**
$2 + 5 - 4 = 3$.

2.A10 **5×2^n**
Repeated doubling is a power.

2.A11 **$1 + 1/4 = 5/4$**
$x^0 = 1$ and $x^{-1} = 1/x$.

2.A12 **$2x^3 = 2 \times x^3$; $(2x)^3 = 8x^3$**
The bracket changes which parts the index applies to.

CHAPTER 2 · CRITERION B · INVESTIGATING PATTERNS

2.B1 **Yes: $2^{-1} \times 3^{-1} = 1/6$ and $(6)^{-1} = 1/6$.**
The law holds for every integer index.

2.B2 **x^4, for any real x**
$\sqrt{(x^8)} = (x^8)\text{^}(1/2) = x^4$, and an even power is never negative.

2.B3 **Each step divides by 2; the rule is $2^{-n} = 1/2^n$.**
Extending the pattern backwards forces the negative index rule.

2.B4 **False: 4^(−1/2) = 1/2, not −2.**
A negative index gives a reciprocal, not a sign change.

2.B5 **None — the square root is not a real number for negative x.**
An even root of a negative number is not real.

2.B6 **4 + 9 = 13**
Take the cube root first, then square: $2^2 + 3^2$.

2.B7 **Yes: both give x^(ab).**
Multiplication of indices is commutative, so the order does not matter.

2.B8 **x^((m + n)/nm)**
Adding fractional indices requires a common denominator.

2.B9 **0^0 is undefined, so the law is stated for non-zero bases.**
The division that gives a^0 is not available when a = 0.

2.B10 **$2^4 \div 2^2 = 4$, but 2^(4÷2) = 4 — try $3^5 \div 3^2 = 27$ while 3^(5÷2) ≈ 15.6.**
Division subtracts indices; it never divides them.

2.B11 **n^2, so the nth term is the square of n**
Each term is a perfect square.

CHAPTER 2 · CRITERION C · COMMUNICATING

2.C1 **4.7×10^{-8} m**
Move the decimal point 8 places to the right.

2.C2 **It counts places the decimal point moves left, which happens for quantities smaller than 1.**
A small quantity has many leading zeros after the point.

2.C3 **1.0×10^{-2}**
$2.5 \times 4 = 10$ and $6 - 9 = -3$, so $10 \times 10^{-3} = 1.0 \times 10^{-2}$.

2.C4 **4×10^3 claims one significant figure; 4.0×10^3 claims two.**
The number of significant digits in the coefficient records precision.

2.C5 **3×10^{14}**
$12 - (-2) = 14$.

2.C6 **0.000012 m**
The exponent −5 moves the point five places left.

2.C7 **It shows the order of magnitude and the precision at once.**
The coefficient carries the significant digits and the exponent carries the scale.

2.C8 **3.4×10^4**
Rewrite 4×10^3 as 0.4×10^4, then add the coefficients.

2.C9 **The exponents differ by 6, so the quantities differ by a factor of 10^6.**
The exponent fixes the position of the decimal point.

2.C10 **3.00×10^8 m/s**
Round to three significant figures, keeping the trailing zeros.

2.C11 **10^{-7} kg, so it is between 10^{-7} and 10^{-6} kg**
The order of magnitude is the nearest power of 10.

2.C12 **Both are written as $a \times 10^n$, so comparing the exponents gives the comparison directly.**
The exponents differ by about 55, so the Earth is about 10^{55} times heavier.

CHAPTER 2 · CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS

2.D1 **about 3.9×10^6 s**
$180 \times 6 \times 3600 \approx 3.888 \times 10^6$, so about 4×10^6.

2.D2 **6×10^7 litres**
$3 \times 10^5 \times 2 \times 10^2 = 6 \times 10^7$.

2.D3 **about 1.6×10^8 bottles**
$4.5 \times 10^5 \times 365 \approx 1.64 \times 10^8$.

2.D4 **If the calculator answer is not within about a factor of 10 of the estimate, the calculation is wrong.**
The estimate fixes the scale, which is where most errors live.

2.D5 **about 6×10^3 hours**
$2.4 \times 10^8 \div 4 \times 10^4 = 6 \times 10^3$.

2.D6 **about 3×10^3 hours if the current were 1 A — the sense is wrong, so the units need converting: 2 A = 2000 mA gives about 1.5 hours.**
Converting to the same unit before dividing is what makes the answer sensible.

2.D7 **about 5.5×10^2 days**
$8.2 \times 10^9 \div 1.5 \times 10^7 \approx 5.5 \times 10^2$.

2.D8 **It claims precision that was never measured, so the extra digits are invented.**
Precision must match the instrument.

2.D9 **about 1600 m^2, within roughly 1400 m^2 and 1800 m^2**
Using 41.5 and 37.5 gives the low bound and 42.5 and 38.5 the high one.

2.D10 **400 doses**
$2 \times 10^{-1} \div 5 \times 10^{-4} = 0.4 \times 10^3 = 400$.

2.D11 **about 8.6×10^9, of order 10^{10}**
$3.6 \times 2.4 \approx 8.6$ and $4 + 5 = 9$.

2.D12 **For a mental check, rounding first keeps the arithmetic simple and the exponent exact.**
Rounding first is a checking strategy, not a reporting strategy.

CHAPTER 3 · CRITERION A · KNOWING AND UNDERSTANDING

3.A1 **$f(3) = 5$ and $f(-1) = -11$**
Substitute each value and evaluate.

3.A2 **$10x - 15$**
Distribute the 5 over both terms.

3.A3 **$6(2x + 3)$**
Take out the highest common factor.

3.A4 **$x^2 + 2x - 8$**
The coefficient of x is 4 − 2 and the constant is −8.

3.A5 **$3n + 2$**
The common difference is 3 and the first term is 5.

3.A6 **all real numbers except $x = 2$**
Division by zero is undefined, so $x = 2$ is excluded.

3.A7 **$5x - 4$**
Expand both brackets, then collect like terms.

3.A8 **$(x + 3)(x - 3)$**
A difference of two squares.

3.A9 **4**
16 − 12 = 4.

3.A10 **1, 7, 17**
Substitute $n = 1, 2, 3$.

3.A11 **$x = (y - 9)/4$**
Subtract 9, then divide by 4.

3.A12 **The term-to-term rule links consecutive terms; the position-to-term rule gives any term from its position.**
Only the position-to-term rule can jump directly to a distant term.

CHAPTER 3 · CRITERION B · INVESTIGATING PATTERNS

3.B1 **Neither — the first differences are 2, 4, 8, 16.**
The differences themselves grow, so the rule is not linear or quadratic.

3.B2 **True — a constant second difference is the signature of an n^2 term.**
The second difference equals twice the coefficient of n^2.

3.B3 **n^2; the next term is 36**
The terms are the perfect squares.

3.B4 **n^3**
The terms are the perfect cubes.

3.B5 **No — they differ by n for every n.**
Setting them equal gives $n = 0$, which is not a position in the sequence.

3.B6 **True — among any three consecutive integers one is divisible by 3 and at least one is even.**
The argument covers every case, not just the ones tested.

3.B7 **$n(n + 1)/2$**
Each term adds the next integer, which sums to $n(n + 1)/2$.

3.B8 **−3, −1, 1, 3 has a constant difference of 2 and a negative first term, so the nth term is $2n − 5$.**
A constant difference fixes the form but not the sign of every term.

3.B9 **Only when $x = 0$ or $y = 0$, since $(x + y)^2 = x^2 + 2xy + y^2$.**
The extra $2xy$ term disappears only when one factor is zero.

3.B10 **$n(2n − 1)$**
The first factor is n and the second rises by 2 from 1.

3.B11 **No — the differences are 5, 7, 9 and the second difference is constant at 2.**
A constant second difference means the rule is quadratic.

CHAPTER 3 · CRITERION C · COMMUNICATING

3.C1	**Expanding distributes the 2 over both terms inside the bracket.** Distribution preserves the value for every x.
3.C2	**An expression is a recipe with no equals sign; an equation states that two expressions are equal.** Only equations have solutions.
3.C3	**Double the input and add 5; f(0) is the value of the function at input 0.** Function notation names the rule and the input together.
3.C4	**Each table row is a pair (input, output) that the formula produces.** The table is a finite sample of the rule.
3.C5	**The middle term 6x is missing: $(x + 3)^2 = x^2 + 6x + 9$.** The bracket is multiplied by itself, so every term meets every term.
3.C6	**Nothing about the relationship; it changes which quantity is the subject.** Rearranging preserves the relationship and changes the perspective.
3.C7	**Time must be non-negative, and in practice it is limited to the growing season.** A domain is a modelling choice, not a mathematical fact.
3.C8	**A table shows values; a graph shows the shape and symmetry.** Different representations answer different questions.
3.C9	**A product equals zero only when a factor is zero, so factorising finds the solutions.** The expanded form hides the zeros.
3.C10	**A conjecture is a claim believed to be true; a generalisation is a claim shown to hold for every case.** The difference is the justification, not the wording.
3.C11	**$f(x) = (x - 4)^2$, and the subtraction happens first because of the brackets.** Brackets fix the order in which operations are applied.
3.C12	**A square is never negative, so the range is $y \geq 0$.** The range is limited by the operation, not by the domain.

CHAPTER 3 · CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS

3.D1 **$C = 40 + 55h$; 287.50 dollars**
Substitute $h = 4.5$: $40 + 247.50$.

3.D2 **375 minutes**
$0.04m = 15$, so $m = 375$.

3.D3 **7.50 dollars**
$1.5 \times (14 - 9) = 7.50$.

3.D4 **$x^2 + 2x - 3$; 45 m^2**
$36 + 12 - 3 = 45$.

3.D5 **$T = 80 + 6n$; cost per person $= 80/n + 6$**
Dividing the total by n gives the per-person cost, which falls as n rises.

3.D6 **$V = 50 + 12t$; $t = 25$ minutes**
$350 - 50 = 300$ and $300 \div 12 = 25$.

3.D7 **Area $= (2x + 1)^2$, perimeter $= 8x + 4$; $x = 13/4$**
$8x + 4 = 30$ gives $x = 26/8 = 13/4$.

3.D8 **20 kg**
$0.8w = 16$, so $w = 20$.

3.D9 **Yes — the fixed charge is recovered once the number of items exceeds 10.**
$20 + 2n < 4n$ gives $n > 10$.

3.D10 **45 tiles**
The terms are 5, 7, 9, 11, 13, which sum to 45.

3.D11 **$I = 0.05Pt$, so $P = I/(0.05t)$; $P = 1000$ dollars**
$100 \div (0.05 \times 2) = 1000$.

3.D12 **Plans usually include a monthly allowance, so the fixed part changes once the allowance is used up.**
A model is valid only over the range where its assumptions hold.

CHAPTER 4 · CRITERION A · KNOWING AND UNDERSTANDING

4.A1 **3**
Change in y is 12 and change in x is 4.

4.A2 **y = 4x − 2**
Use y = mx + c.

4.A3 **0**
The change in y is zero.

4.A4 **y = 2x + 5**
The gradient is 8 ÷ 4 = 2 and the intercept is 5.

4.A5 **Yes — the quotient is 3 throughout**
Direct proportion needs a constant quotient.

4.A6 **8**
The intercept is the constant term.

4.A7 **y = −5x + 11**
1 = −10 + c gives c = 11.

4.A8 **gradient 3, y = 3x + 4**
Each step of 2 in x gives 6 in y.

4.A9 **(2, 3)**
x + 1 = 5 − x gives x = 2.

4.A10 **y = 0.5x + 5**
8 = 3 + c gives c = 5.

4.A11 **−2**
−10 ÷ 5 = −2.

4.A12 **The change in x is zero, so the quotient is undefined.**
Division by zero is not defined.

CHAPTER 4 · CRITERION B · INVESTIGATING PATTERNS

4.B1 **No — they have the same gradient, so they are parallel.**
Lines with equal gradients and different intercepts never meet.

4.B2 **False — $y = 2x$ and $y = 3x$ both pass through the origin and meet there, but $y = 2x + 1$ and $y = 2x + 1$ are the same line. A counterexample is $y = x$ and $y = x + 1$, which do not share an intercept at all.**
Sharing an intercept means sharing a point, so the lines do meet there.

4.B3 **The first two have the same gradient 3 and are parallel; the third has gradient −3.**
Only the gradient controls steepness.

4.B4 **$(d - b)/(c - a)$**
The gradient is the change in y over the change in x.

4.B5 **Yes — the gradient is 2 between each consecutive pair.**
A constant gradient between consecutive points means collinearity.

4.B6 **False — $y = -2x + 5$ has gradient -2 and intercept 5.**
The gradient and the intercept are independent.

4.B7 **$k = 1.5$**
$y = 12$ at $x = 8$, so $8k = 12$.

4.B8 **$-1/m$**
The product of the gradients is -1.

4.B9 **3**
The common difference is the gradient.

4.B10 **True — equal gradients give parallel lines, which never meet.**
Distinct meeting lines must have different gradients.

4.B11 **Yes — both have gradient −3.**
Equal gradients mean the lines are parallel or identical.

CHAPTER 4 · CRITERION C · COMMUNICATING

4.C1	**The rate of change of the vertical quantity per unit of the horizontal quantity.** The gradient is a rate, so it carries units.
4.C2	**Interpolation predicts inside the range of the data; extrapolation predicts outside it.** Only the first is supported by the evidence.
4.C3	**Because $y = kx$ gives $y = 0$ when $x = 0$.** The origin corresponds to both quantities being zero.
4.C4	**Use two points for the gradient, or use a point and the gradient directly.** The choice depends on what information is given.
4.C5	**The gap between the model's prediction and the measured value.** Small balanced residuals indicate a good fit.
4.C6	**It fixes both the steepness and the position, which determine the line uniquely.** Two numbers are enough to fix a straight line.
4.C7	**There is a fixed charge payable even for zero distance.** The intercept is the value at zero input.
4.C8	**Equal gradients mean equal rates of change, so the lines can never meet.** The gradient is the same everywhere on a line.
4.C9	**Compute the gradient between consecutive pairs and check they are equal.** A constant gradient between consecutive points means collinearity.
4.C10	**Direct proportion is the special case whose line passes through the origin.** A linear relationship may have a non-zero intercept.
4.C11	**The gradient and intercept describe that community's situation, not a universal rule.** A model inherits the range and the population it was fitted to.
4.C12	**Plot both lines and find the crossing point, which is the break-even quantity.** The crossing point is the decision boundary.

CHAPTER 4 · CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS

4.D1 **C = 120 + 7n; 435 dollars**
120 + 7 × 45 = 435.

4.D2 **21.60 dollars; 10 km**
20 − 4 = 16 and 16 ÷ 1.6 = 10.

4.D3 **500 minutes**
10 + 0.05m = 25 + 0.02m gives 0.03m = 15.

4.D4 **20 minutes**
360 − 60 = 300 and 300 ÷ 15 = 20.

4.D5 **P = 0.35m − 50; about 143 kg**
0.35m = 50 gives m ≈ 142.9.

4.D6 **No — the fixed fee gives a non-zero intercept, so the line does not pass through the origin.**
Direct proportion requires the line to pass through the origin.

4.D7 **18 km/h; 22.5 km**
12 ÷ (40/60) = 18 and 18 × 1.25 = 22.5.

4.D8 **400 pages**
47 − 15 = 32 and 32 ÷ 0.08 = 400.

4.D9 **allowance = (household size ÷ total size) × total water; fairness depends on whether size is the agreed basis.**
A proportional model is only fair if the basis is agreed.

4.D10 **520 now; the assumption is that the growth rate stays at 20 a year**
620 − 5 × 20 = 520.

4.D11 **20 kg; the first service is cheaper above it**
8 + 0.5m = 3 + 0.75m gives 5 = 0.25m.

4.D12 **Plants stop growing, so the constant-rate assumption fails.**
A model is valid only where its assumptions hold.

CHAPTER 5 · CRITERION A · KNOWING AND UNDERSTANDING

5.A1 **x = 5**
5x = 25, so x = 5.

5.A2 **x = 5**
x + 4 = 9, so x = 5.

5.A3 **x = 7**
2x + 1 = 15, so 2x = 14.

5.A4 **x = −5**
Divide both sides by −4.

5.A5 **x = 3**
−2x = −6, so x = 3.

5.A6 **x = 20**
x/4 = 5.

5.A7 **x = 7**
Collect the x terms on the left.

5.A8 **x = 5**
6x − 2 = 4x + 8, so 2x = 10.

5.A9 **x = 10**
2x = 20.

5.A10 **x = 8**
Multiply by 10: 2(x − 2) = x + 4.

5.A11 **x = 5**
x = 5, and 23 = 23 confirms it.

5.A12 **Because x might be zero, and division by zero is undefined.**
The permitted steps require a non-zero divisor.

CHAPTER 5 · CRITERION B · INVESTIGATING PATTERNS

5.B1 **No — subtracting 2x gives 3 = 5, which is false.**
The variable cancels and leaves a false statement.

5.B2 **False — 2x + 3 = 2x + 3 is true for every x, so it has infinitely many solutions.**
The variable can cancel to a true statement.

5.B3 **No solution — the second equation is not a multiple of the first in the constants.**
Doubling the first gives 2x + 2y = 8, not 9.

5.B4 **Always — the second equation is a multiple of the first.**
The two equations state the same condition.

5.B5 **It is one condition written twice, so there are infinitely many solutions.**
The equations are identical.

5.B6 **True — different gradients mean different rates of change, so the lines meet exactly once.**
The gradient fixes the steepness, and equal steepness with different intercepts gives parallel lines.

5.B7 **y = (12 − 2x)/3, and every x gives a y, so there are infinitely many pairs.**
One equation in two unknowns has infinitely many solutions.

5.B8 **Compare the ratios of the coefficients: equal ratios with unequal constants means no solution; equal ratios with equal constants means infinitely many.**
The ratios reveal the geometric relationship of the lines.

5.B9 **No — it has infinitely many, because the second equation is double the first.**
The equations are equivalent.

5.B10 **False — three lines need not pass through one point.**
Each equation adds a condition that may be incompatible.

5.B11 **Each has infinitely many solutions, one for each choice of x.**
A single equation in two unknowns leaves one degree of freedom.

CHAPTER 5 · CRITERION C · COMMUNICATING

5.C1 **The pair of values that makes both equations true at the same time.**
The solution is the intersection of the two conditions.

5.C2 **An equation usually has one solution; an inequality has a set of solutions.**
The answer to an inequality is a region or interval.

5.C3 **Because the order of the numbers reverses when the scale is reflected through zero.**
The inequality is preserved only if the direction is reversed.

5.C4 **Substitution when one variable is easy to isolate; elimination when the coefficients can be made to match.**
The choice depends on the structure of the equations.

5.C5 **The two conditions cannot both hold, which happens when the lines are parallel.**
Incompatible conditions have no common solution.

5.C6 **Because infinitely many pairs can satisfy all the inequalities at once.**
The solution set is the intersection of the half-planes.

5.C7 **Substitute the pair into both original equations and confirm both are true.**
A check in one equation is not enough.

5.C8 **The feasible region is the set of allowed choices; a solution is one choice made from it.**
The mathematics narrows the choices; the decision is separate.

5.C9 **A budget is an upper bound, not an exact amount, so it is naturally written with ≤.**
The limit allows any amount up to the maximum.

5.C10 **One solution, no solution, or infinitely many, recognised from the ratios of the coefficients.**
Elimination produces a solvable equation, a false statement or a true one.

5.C11 **Because the equations state the same condition, so any pair satisfying one satisfies both.**
The two equations are equivalent.

5.C12 **By solving the two boundary equations of the edges that meet there.**
A vertex is the intersection of two boundary lines.

CHAPTER 5 · CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS

5.D1 **8 bags**
With 3 bags of fertiliser at 18 dollars, 32 dollars remain for 8 bags of seed.

5.D2 **120 pencils**
20 notebooks cost 24 dollars, leaving 36 dollars for 120 pencils.

5.D3 **45 boxes**
10 crates weigh 250 kg, leaving 550 kg for 45 boxes at 12.2 kg each — 45 boxes weigh 540 kg, which fits.

5.D4 **$b \geq 15$ and $w + b \leq 60$, so $w \leq 45$; for example $w = 30$ and $b = 15$**
Any pair inside the region works, and the plan is a choice.

5.D5 **$2r + 3b \leq 200$ with $b \geq 20$, so $2r \leq 140$ and $r \leq 70$**
The bean minimum is a fixed cost that reduces what is left for rice.

5.D6 **No — the constraints describe what is possible, not what is best.**
Choosing among feasible options needs an additional criterion.

5.D7 **10 km; the first is cheaper below it**
$3 + 0.5d = 5 + 0.3d$ gives $0.2d = 2$.

5.D8 **Yes — $8 \times 30 = 240$ and the second pipe supplies the remaining 260 litres in about 22 minutes, so both together fill it inside 30 minutes**
$240 + 12 \times 22 = 504$, so about 21.7 minutes of the second pipe is enough.

5.D9 **Widths w with $w(20 - w) \geq 96$, so $8 \leq w \leq 12$**
The perimeter gives length $20 - w$, and the area condition restricts w.

5.D10 **4 kg**
$4a + 6b \leq 48$ with $a + b = 10$ gives $40 + 2b \leq 48$, so $b \leq 4$.

5.D11 **47 students**
$500 - 120 = 380$ and $380 \div 8 = 47.5$, so 47 students.

5.D12 **A resource has a limit rather than an exact requirement, so $\leq$ is the honest relation.**
Inequalities describe feasibility; equations describe exact balances.

CHAPTER 6 · CRITERION A · KNOWING AND UNDERSTANDING

6.A1 **(1, 9)**
Add the vector to the point.

6.A2 **(−4, 3)**
Reflection in the y-axis changes the sign of x.

6.A3 **(4, −3)**
Reflection in the x-axis changes the sign of y.

6.A4 **(−2, −5)**
A half turn changes both signs.

6.A5 **(1, −3)**
The rule is (x, y) → (y, −x).

6.A6 **Lengths, angles and area**
A translation is an isometry, so all lengths and angles are invariant.

6.A7 **(2, 6)**
Reflection in y = x interchanges the coordinates.

6.A8 **12 cm**
Multiply every length by the scale factor.

6.A9 **4**
A square fits onto itself four times in a full turn.

6.A10 **(−1, 1), (−4, 1), (−1, 5)**
Change the sign of each x-coordinate.

6.A11 **(−2, −3)**
The rule is (x, y) → (−y, x).

6.A12 **A translation moves every point by the same vector; a rotation turns every point about a fixed centre.**
Only the rotation has a centre and an angle.

CHAPTER 6 · CRITERION B · INVESTIGATING PATTERNS

6.B1 **Yes — reflecting first gives (−1, 3), translating first gives (3, 3).**
Transformations generally do not commute.

6.B2 **True — the composition is a rotation of 180° about the intersection of the lines.**
Two perpendicular mirrors produce a half turn.

6.B3 **Yes — rotations about the same centre add their angles.**
The composition of rotations is a rotation.

6.B4 **The identity — the shape returns to its original position.**
A reflection is its own inverse.

6.B5 **No — only isosceles and equilateral triangles do; a scalene triangle has none.**
A line of symmetry requires two matching halves.

6.B6 **False — the side must be included for ASA; otherwise the triangles can differ.**
ASA requires the side between the two angles.

6.B7 **Three independent measurements, such as SSS, SAS or ASA.**
Each sufficient condition determines a unique triangle.

6.B8 **n = 3, 4 and 6 only**
The interior angle $(n - 2) \times 180°/n$ must divide 360° exactly.

6.B9 **Yes — its three angles sum to 180°, so two of each fit around a point.**
The angle sum of a triangle guarantees a tessellation.

6.B10 **False — the letter S has rotational symmetry of order 2 but no line of symmetry.**
The two kinds of symmetry are independent.

6.B11 **6**
It fits onto itself six times in a full turn.

CHAPTER 6 · CRITERION C · COMMUNICATING

6.C1	**One can be mapped onto the other by an isometry, so they have the same form and size.** Congruence is a claim about a transformation, not about appearance.
6.C2	**A clockwise order of vertices becomes anticlockwise in the image.** Reflection turns the shape over.
6.C3	**The centre, the angle and the direction.** Without all three the rotation is not determined.
6.C4	**Three sides determine a unique triangle, so no other shape is possible.** The measurements fix the shape completely.
6.C5	**In SAS the angle is between the two sides; in SSA it is not, and two triangles may result.** Only the included angle determines a unique triangle.
6.C6	**Small differences are invisible in a drawing, so measurements are needed.** A justification must be checkable by another person.
6.C7	**A coordinate rule and a diagram: the rule is exact and the diagram shows the structure.** Different representations answer different questions.
6.C8	**Line symmetry is about a reflection; rotational symmetry is about a rotation.** The two are independent properties.
6.C9	**Because the interior angle must divide 360°, which happens only for n = 3, 4 and 6.** The angle condition is a proof, not a search.
6.C10	**Angles and shape are preserved; lengths and area are multiplied by k and k^2.** An enlargement is not an isometry.
6.C11	**Check that two angles match and that the side between them matches.** The side must be the included one.
6.C12	**Gaps and overlaps would break the surface, so the angles must sum to exactly 360°.** The condition is about the angles at each meeting point.

CHAPTER 6 · CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS

6.D1 **A rotation of 120°; three copies give 360°**
Three rotations of 120° complete a full turn.

6.D2 **108 tiles**
12 by 9 tiles.

6.D3 **90 cm**
Two squares minus the shared edge: 8 sides of 15 cm.

6.D4 **36 cm; the interior angle is 120° and 3 × 120° = 360°**
The angle condition guarantees the tessellation.

6.D5 **360° ÷ 50° = 7.2, so the 50° angles cannot tile a point alone; a combination of angles is needed.**
Tessellation needs the angles at each point to sum to exactly 360°.

6.D6 **A reflection is an isometry, so the logo and its image are congruent.**
The invariance of lengths and angles is the justification.

6.D7 **360°; four copies**
Four rotations of 90° complete a full turn.

6.D8 **They are congruent if the water surface is perfectly flat, because the reflection is an isometry.**
Real water distorts the image, so the claim depends on the surface.

6.D9 **Find pairs whose angles sum to 360° at a point, such as 120° and 240° from two hexagons and a square.**
The angle sum at each vertex must be exactly 360°.

6.D10 **Measure two sides and the included angle, or all three sides once, and apply SAS or SSS.**
A sufficient condition is shorter than a full measurement.

6.D11 **Each line reflects the kite onto itself, so the design has two mirror lines.**
A line of symmetry is a reflection that maps the shape to itself.

6.D12 **Lengths multiply by the scale factor and areas by its square, while angles are unchanged.**
An enlargement preserves shape but not size.

CHAPTER 7 · CRITERION A · KNOWING AND UNDERSTANDING

7.A1 **12 cm**
Multiply by the scale factor.

7.A2 **10 cm**
$\sqrt{(36 + 64)} = 10$.

7.A3 **30°**
The angle whose sine is 0.5.

7.A4 **12 cm**
$\sqrt{(169 - 25)} = 12$.

7.A5 **300 cm²**
Area scales by $5^2 = 25$, and $12 \times 25 = 300$.

7.A6 **opposite ÷ adjacent**
The tangent ratio.

7.A7 **0.28**
$7 \div 25$.

7.A8 **Yes — two equal angles force the third, so AA applies.**
Two equal angles are sufficient for similarity of triangles.

7.A9 **1.7 km**
$3.4 \times 50\,000 = 170\,000$ cm = 1.7 km.

7.A10 **$\sqrt{5}$ cm**
$\sqrt{(1 + 4)} = \sqrt{5}$.

7.A11 **60 m**
$\tan 45° = 1$, so the height equals the distance.

7.A12 **Lengths multiply by k and areas by k^2.**
Area is a product of two lengths.

CHAPTER 7 · CRITERION B · INVESTIGATING PATTERNS

7.B1 **Yes — they are similar, so the ratios are identical.**
Similarity forces proportional sides.

7.B2 **True — this is the SSS similarity condition.**
Proportional sides determine the shape.

7.B3 **7-24-25, since 49 + 576 = 625**
The pattern follows odd starting values: 3, 5, 7 with the other two from $n^2 - 1$ and $n^2 + 1$ over 2.

7.B4 **Right-angled, because $3^2 + 4^2 = 5^2$**
The converse of Pythagoras' theorem.

7.B5 **No — 25 + 36 = 61, which is not 100.**
The sum of squares must equal the square of the longest side.

7.B6 **True — all three share the same angles.**
Each smaller triangle has a right angle and shares an acute angle with the original.

7.B7 **The area ratio is the square of the length ratio.**
Area is a product of two lengths.

7.B8 **height = distance × tan(angle of elevation)**
The tangent ratio relates the opposite side to the adjacent side.

7.B9 **Yes — they are alternate angles between parallel horizontals.**
The two horizontals are parallel, so the alternate angles are equal.

7.B10 **False — the ratios depend only on the angles, so they are unchanged.**
Similarity preserves angles and scales sides equally.

7.B11 **0, 0.5, 0.707, 0.866, 1**
Sine rises from 0 to 1 as the angle goes from 0° to 90°.

CHAPTER 7 · CRITERION C · COMMUNICATING

7.C1	**Corresponding angles are equal and corresponding sides are in a constant ratio.** Similarity preserves shape but not size.
7.C2	**Because such triangles are similar, so their sides are proportional.** The ratios are quotients of proportional lengths.
7.C3	**Elevation is measured upward from the horizontal; depression is measured downward.** The two are equal by alternate angles.
7.C4	**The calculation gives the height above the observer's eye, not above the ground.** The right-angled triangle starts at eye level.
7.C5	**By rearranging four copies of the triangle in a square, or by the similar-triangles argument on the altitude to the hypotenuse.** Both compare areas that must be equal.
7.C6	**SSS compares side ratios; AA compares angles.** Either is sufficient for triangles.
7.C7	**Measure the angle of elevation and the horizontal distance, apply the tangent ratio, then add the eye height.** The two measurements define a right-angled triangle.
7.C8	**It relies on assumptions about level ground, verticality and the accuracy of the angle.** Each assumption adds uncertainty.
7.C9	**It shows shape and proportion but not the true size or the accuracy of the measurement.** A drawing is a representation with its own limits.
7.C10	**It tests whether an angle is a right angle without measuring the angle.** The side lengths determine the angle.
7.C11	**Measure 3 m along one wall and 4 m along the other; if the diagonal is 5 m the corner is square.** A 3-4-5 triangle is right-angled, which is the practical form of the theorem.
7.C12	**The same angular error covers a larger distance when the object is far away.** Height error ≈ distance^2 × angular error for small angles.

CHAPTER 7 · CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS

7.D1 **12 m**
The ratio is 2/2.5 = 0.8, so the height is 15 × 0.8.

7.D2 **15.6 m**
25 × tan 32° = 25 × 0.6249.

7.D3 **√32 ≈ 5.66 m**
√(36 − 4) = √32.

7.D4 **1.5 km; 18 minutes**
7.5 × 20 000 cm = 1.5 km, and 1.5 ÷ 5 = 0.3 h.

7.D5 **30 cm; area factor 9**
Scale factor 3, so the area factor is 9.

7.D6 **No — the method needs a shadow, so it cannot be used without direct sunlight.**
The method depends on parallel light rays.

7.D7 **about 18.4°**
tan θ = 40/120, so θ = $\tan^{-1}(1/3)$.

7.D8 **about 31°**
cos θ = 3.6/4.2, so θ ≈ 31°.

7.D9 **Yes — the two angles and the baseline determine two similar right-angled triangles, so the height follows from the difference in their horizontal distances.**
Two angles and the included side determine the geometry.

7.D10 **24 m at 1 : 200; 6 m at 1 : 50**
12 × 200 cm = 24 m and 12 × 50 cm = 6 m.

7.D11 **about 36.9°**
sin θ = 6/10.

7.D12 **Because the result inherits the uncertainty of the angle and the distance, so extra digits would be invented.**
Indirect measurement compounds errors.

CHAPTER 8 · CRITERION A · KNOWING AND UNDERSTANDING

8.A1 **96 cm²**
$6 \times 4^2 = 96$.

8.A2 **60 cm³**
$5 \times 4 \times 3$.

8.A3 **63π cm³**
$\pi \times 9 \times 7$.

8.A4 **36π cm²**
$2\pi \times 2 \times 9$.

8.A5 **12π cm³**
$\frac{1}{3} \times \pi \times 9 \times 4$.

8.A6 **144π cm²**
$4\pi r^2$.

8.A7 **36π cm³**
$\frac{4}{3}\pi \times 27$.

8.A8 **48 cm³**
$\frac{1}{3} \times 36 \times 4$.

8.A9 **96 cm³**
Area × length.

8.A10 **8 g/cm³**
$240 \div 30$.

8.A11 **65π cm²**
πrl.

8.A12 **Volume measures the space inside and surface area the material of the boundary.**
They have different units and answer different questions.

CHAPTER 8 · CRITERION B · INVESTIGATING PATTERNS

8.B1 **The short wide one — surface area grows with the radius more slowly than with the height for a fixed volume.**
For fixed V, A = 2V/h + 2π(V/(πh))^(2/3) falls as h rises then rises, with a minimum when h = 2r.

8.B2 **False — the volume depends on r^2, so it quadruples.**
Volume is proportional to the square of the radius.

8.B3 **No — it has four times the surface area and eight times the volume.**
Area scales as k^2 and volume as k^3.

8.B4 **It becomes eight times as large**
Volume is proportional to r^3.

8.B5 **1 : 3**
The cone is one third of the cylinder.

8.B6 **False for some solids, and the net of a sphere cannot be drawn without distortion.**
A sphere's surface cannot be flattened without stretching.

8.B7 **It is smallest when the cuboid is closest to a cube.**
For a fixed volume, area is minimised when the dimensions are equal.

8.B8 **The volume error is about three times the length error.**
Volume is a product of three lengths.

8.B9 **No — a sphere has the smallest surface area for a given volume.**
Shape affects area even when volume is fixed.

8.B10 **True — the slant height is the hypotenuse of the right-angled triangle inside the cone.**
The hypotenuse is the longest side.

8.B11 **1, 8, 27, 64 — the cubes of the side length**
Volume scales as the cube of the linear dimension.

CHAPTER 8 · CRITERION C · COMMUNICATING

8.C1 **A flat arrangement of the faces that folds into the solid, preserving the shapes and areas of the faces.**
The net turns a surface-area question into a flat one.

8.C2 **Because the curved surface unrolls into a rectangle of length $2\pi r$ and height h.**
The circumference becomes the length of the rectangle.

8.C3 **The slant height is measured along a face; the perpendicular height is measured at right angles to the base.**
Only the slant height appears in a face-area formula.

8.C4 **Because three cone fills exactly fill the cylinder of the same base and height.**
The relationship is geometric and holds for any size.

8.C5 **Split it into parts whose volumes can be computed, then add or subtract.**
The split must be stated so it can be checked.

8.C6 **Because density is defined as mass divided by volume.**
The units follow from the definition.

8.C7 **A net for surface area and a cross-section for volume.**
Each representation answers a different question.

8.C8 **Because the volume is a product of the measurements and inherits their uncertainty.**
Extra digits would be invented.

8.C9 **By displacement of water in a measuring cylinder.**
The displaced volume equals the object's volume.

8.C10 **Because the sphere's curved surface is equivalent to four circles of radius r.**
The formula can be checked against the containing cylinder.

8.C11 **g/cm^3 for small quantities and kg/m^3 for large ones.**
The units must match the mass and volume measurements.

8.C12 **Material is proportional to surface area for a thin-walled container.**
The surface area is the amount of material in the wall.

CHAPTER 8 · CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS

8.D1 **942 kg**
$V = \pi \times 0.25 \times 1.2 = 0.942\ m^3$, so mass = 942 kg.

8.D2 **24 cubes**
$4 \times 3 \times 2$.

8.D3 **about 13 cups**
Cup volume = $24\pi \approx 75.4\ cm^3$, and $1000 \div 75.4 \approx 13.3$.

8.D4 **about 3.4 cm**
Sphere volume = $256\pi/3 \approx 268\ cm^3$, and the cylinder's base area is $25\pi \approx 78.5\ cm^2$, so the rise is $268 \div 78.5 \approx 3.4$ cm.

8.D5 **16 cubes**
Along the sides only 4, 2 and 2 cubes of 30 cm fit, so $4 \times 2 \times 2 = 16$, even though the volume ratio is 24.9.

8.D6 **The sphere, because it has the smallest surface area for a given volume.**
Shape affects area even when volume is fixed.

8.D7 **about 5655 cm^3**
$\pi(5^2 - 4^2) \times 200 = 1800\pi \approx 5655\ cm^3$.

8.D8 **78π m^3, or about 245.0 m^3**
Cylinder $\pi \times 3^2 \times 8 = 72\pi$ and cone $\frac{1}{3} \times \pi \times 3^2 \times 2 = 6\pi$, giving a total of $78\pi\ m^3$.

8.D9 **0.62 g/cm^3; about three significant figures**
$620 \div 1000$, and the measurements support three figures.

8.D10 **The same — both are lengths, and both scale the volume linearly.**
Volume is proportional to each length separately.

8.D11 **about 33.8 m^2**
$l = \sqrt{(2.5^2 + 3^2)} \approx 3.91$ m, curved area = $\pi \times 2.5 \times 3.91 \approx 30.7\ m^2$, and 10% extra gives $30.7 \times 1.1 \approx 33.8\ m^2$.

8.D12 **Because material use is proportional to surface area for a thin wall.**
Equal capacity fixes the volume, so the area decides the material.

CHAPTER 9 · CRITERION A · KNOWING AND UNDERSTANDING

9.A1 **One point per subject, showing the pairing of two measurements**
A scatter plot keeps the pairing intact.

9.A2 **Strong and negative**
The sign gives the direction and the magnitude the strength.

9.A3 **3**
Residual = actual − predicted.

9.A4 **−1 to 1**
The coefficient is bounded by −1 and 1.

9.A5 **17**
Substitute $x = 7$.

9.A6 **Strong and positive**
Closeness to a line gives the strength and the direction of the slope gives the sign.

9.A7 **2**
$(14 - 6)/(7 - 3) = 2$.

9.A8 **Interpolation predicts inside the range of the data; extrapolation predicts outside it.**
Only interpolation is supported by the data.

9.A9 **Through the point (4, 10)**
A line of best fit passes through the mean point.

9.A10 **10**
$0.5 \times 12 + 4$.

9.A11 **A point that lies far from the pattern of the other points**
An outlier departs from the general trend.

9.A12 **It has no units**
The coefficient is a dimensionless number.

CHAPTER 9 · CRITERION B · INVESTIGATING PATTERNS

9.B1 **Yes — ice-cream sales and drowning deaths, both driven by warm weather.**
A common cause produces correlation without causation.

9.B2 **False — there may be a strong curved relationship that the coefficient does not measure.**
The coefficient measures linear association only.

9.B3 **No — the coefficient is the same in both directions.**
Correlation is symmetric in the two variables.

9.B4 **Zero**
A line of best fit balances the residuals.

9.B5 **No — an outlier far from the other points in x has a large effect, while one that follows the trend has little.**
The effect depends on the leverage of the point.

9.B6 **False — the relationship may be curved while the linear correlation is weak.**
The coefficient measures linear association only.

9.B7 **They are small, both positive and negative, and show no pattern.**
A pattern in the residuals means the model is wrong.

9.B8 **A mechanism and evidence from a controlled study**
Correlation alone cannot distinguish the three explanations.

9.B9 **No — sample size affects the reliability of the estimate, not its value.**
A large sample can show a weak correlation precisely.

9.B10 **False — 90% of 10 is very different from 90% of 10 000.**
A percentage needs the size of the group it came from.

9.B11 **One extreme value can move the mean a long way, while the median moves much less.**
The mean is not resistant to outliers.

CHAPTER 9 · CRITERION C · COMMUNICATING

9.C1	**Two measurements recorded for the same subject, forming one pair per subject.** The pairing is what distinguishes bivariate from univariate data.
9.C2	**Correlation is an observed association; causation is a claim that changing one variable produces the change in the other.** Only a controlled study with a mechanism supports a causal claim.
9.C3	**The vertical gap between a point and the line; a pattern means the linear model is wrong.** Residuals diagnose the model.
9.C4	**Because the residuals must balance, and balancing them forces the line through the centre of the data.** The mean point is the balance point of the scatter.
9.C5	**A causes B, B causes A, or a third variable causes both.** A correlation alone cannot distinguish them.
9.C6	**Because the trend that held inside the data may not continue outside it.** The model is only supported where there are data.
9.C7	**Check the source, the sample, the measure and the conclusion.** Each can be examined independently.
9.C8	**Because the people who chose to respond differ systematically from those who did not.** The sample is not representative.
9.C9	**The frequency of an event in the whole population, needed to interpret a test result or a percentage.** Without the base rate a percentage cannot be interpreted.
9.C10	**Because the correlation may come from a common cause or from coincidence.** A mechanism is part of a causal claim.
9.C11	**A table shows the exact pairs; a scatter plot shows the pattern.** Different representations answer different questions.
9.C12	**Because the estimate is a range, and reporting a single value hides that.** Uncertainty is information about the claim.

CHAPTER 9 · CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS

9.D1	**63.2** $4.2 \times 6 + 38 = 63.2$, and this is interpolation if 6 hours is inside the collected range.
9.D2	**The correlation supports an association, but a causal claim needs a controlled study and a mechanism.** Correlation is not causation.
9.D3	**A controlled study in which the programme is varied and other factors held constant.** Only a controlled study supports a causal claim.
9.D4	**Because the averages hide the spread, the sample sizes and the definitions used.** A comparison of averages needs comparable populations.
9.D5	**The true support is estimated to lie between 51% and 57%.** The margin of error gives the uncertainty of the estimate.
9.D6	**The claim reverses the direction: age causes height, not the other way round.** The direction of a causal claim is part of what the data must support.
9.D7	**The sample is self-selected and tiny, and the source has an interest.** The claim fails the source and sample checks.
9.D8	**It is an extrapolation and assumes the growth rate continues unchanged.** Extrapolation is an assumption, not a result.
9.D9	**They support a moderate negative association across a large sample, but not a causal claim in either direction.** Sample size supports reliability, not causation.
9.D10	**Case mix, base rate, definitions and time period** A rate comparison needs comparable populations.
9.D11	**A control group and random assignment, so that other changes are ruled out.** A before-and-after comparison cannot rule out other causes.
9.D12	**Because the average says nothing about the spread of the distribution.** Two communities with the same average can have very different distributions.

CHAPTER 10 · CRITERION A · KNOWING AND UNDERSTANDING

10.A1 **1/2**
Three of the six outcomes are even.

10.A2 **2/5**
4 out of 10.

10.A3 **1/4**
One of the four equally likely outcomes.

10.A4 **5/6**
$1 - 1/6$.

10.A5 **1/13**
Four kings in 52 cards.

10.A6 **3/8**
Three of the eight sectors.

10.A7 **0.2**
Multiply the probabilities.

10.A8 **0.75**
Add the probabilities.

10.A9 **0.8**
$0.6 + 0.5 - 0.3$.

10.A10 **1/8**
$(1/2)^3$.

10.A11 **1/36**
The two rolls are independent.

10.A12 **Because the outcomes cover the whole sample space exactly once.**
One of the outcomes must occur.

CHAPTER 10 · CRITERION B · INVESTIGATING PATTERNS

10.B1 **About 16.7; 20 sixes is more than expected but not impossible**
The expected value is a mean, not a prediction of one outcome.

10.B2 **False — the tosses are independent, so P(tails) is still 1/2.**
Independence means the past does not affect the future.

10.B3 **Yes — the 36 outcomes of two dice map onto 12 sectors if the dice are fair.**
A model needs equally likely outcomes, which two fair dice provide.

10.B4 **When A and B are independent**
Multiplication is the rule for independent events only.

10.B5 **$1 - (5/6)^2 = 11/36 \approx 0.306$**
The complement rule is easier than counting the favourable outcomes.

10.B6 **True — if they cannot both happen, knowing one occurred tells you the other did not.**
Independence requires the outcome of one to leave the other unchanged.

10.B7 **No — the general addition rule subtracts the overlap, so the result is at most 1.**
The rule removes the double counting.

10.B8 **$n \times P(\text{success})$**
Expectation is linear, so it scales with the number of trials.

10.B9 **It settles closer to the theoretical value**
The law of large numbers describes the convergence.

10.B10 **False — bias comes from the selection method, and a larger biased sample stays biased.**
Size affects variability, not bias.

10.B11 **At a 1% base rate there are about 50 false positives; at 10% about 86.**
The false-positive count depends on the number of unaffected people.

CHAPTER 10 · CRITERION C · COMMUNICATING

10.C1 **The set of all possible outcomes of an experiment.**
An event is a subset of the sample space.

10.C2 **Independent events do not affect each other; mutually exclusive events cannot both happen.**
The two ideas describe different relationships.

10.C3 **Multiplication for 'and' with independent events; addition for 'or' with mutually exclusive events.**
The wording of the question decides the rule.

10.C4 **P(A or B) = P(A) + P(B) − P(A and B), because the overlap is counted twice.**
The subtraction removes the double counting.

10.C5 **Experimental is a measured relative frequency; theoretical comes from a model of the outcomes.**
The experiment estimates the theory.

10.C6 **Because the theory is a consequence of the model, and a large departure means the assumptions fail.**
The evidence is against the model, not the arithmetic.

10.C7 **A tree diagram and a sample-space table.**
Both show the combined outcomes, in different arrangements.

10.C8 **A systematic error in the selection, which scales with the sample rather than shrinking.**
Bias is about the method, not the size.

10.C9 **The population is the whole group a claim is about; the sample is the part measured.**
A statistic from a sample estimates a parameter of the population.

10.C10 **The frequency of the event in the whole population, which sets how many false positives occur.**
A rare event produces mostly false positives.

10.C11 **Compare the true positives with all positives, using the base rate to count each.**
The posterior probability depends on both the accuracy and the base rate.

10.C12 **Because the consequences of being wrong also matter, and they differ between situations.**
A decision weighs the number and the cost of an error.

CHAPTER 10 · CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS

10.D1 **240 items**
3/250 × 20 000.

10.D2 **about 33%**
Of 10 000 people, 98 true positives and about 198 false positives, so 98/296 ≈ 0.33.

10.D3 **Yes, if the list is not ordered by a relevant characteristic**
Systematic selection from an unordered list approximates random selection.

10.D4 **It means rain is expected on about 7 of 10 similar days; it does not mean it will rain for 70% of the day.**
A probability is a long-run frequency, not a duration.

10.D5 **The chance becomes 2 in 14 000 000, which is still about 0.000014%.**
Doubling a very small probability leaves it very small.

10.D6 **Because the base rate of the condition is very low, so most positives are false.**
The base rate dominates the posterior probability.

10.D7 **The sample is self-selected and drawn from one group with an interest.**
Attendance at a match is not a random selection from the town.

10.D8 **The base rate of the condition, the size of the trials and the definition of effectiveness.**
An effectiveness figure is meaningless without the trial details.

10.D9 **about 125 faults, with an uncertainty of roughly ±20 from the sample size**
The estimate inherits the sampling variability.

10.D10 **It should also weigh the base rate, the uncertainty of the projection and the consequences for the community.**
A probability is one input to a decision, not the decision.

10.D11 **about 99.4%**
Of 1 000 people, 50 are affected and 950 are not. False negatives = 0.1 × 50 = 5 and true negatives = 0.9 × 950 = 855, so P(correct negative) = 855/860 ≈ 0.994.

10.D12 **Because the expected cost is the probability multiplied by the consequence, and a small probability of a large loss can exceed a large probability of a small one.**
Expected value combines probability and consequence.

Formula bank (1 of 6)

Key facts and formulas, chapter by chapter. Use them to check work, not to replace reasoning.

THE REAL NUMBER SYSTEM

Rational number

any number that can be written as a/b with integers a, b and $b \neq 0$; its decimal either terminates or recurs.

Irrational number

a number that cannot be written as a fraction; its decimal neither terminates nor recurs.

Surd

an exact irrational root, such as $\sqrt{2}$ or $3\sqrt{5}$.

Absolute value

$|x|$ is the distance of x from 0 on a number line.

Interval notation

[a, b] includes both ends; (a, b) excludes both; [a, b) includes a only.

Set notation

$A \cup B$ is the union, $A \cap B$ the intersection, A' the complement.

Formula bank (2 of 6)

Key facts and formulas, chapter by chapter. Use them to check work, not to replace reasoning.

POWERS, ROOTS AND NOTATION

Product law

$a^m \times a^n = a^{m+n}$.

Quotient law

$a^m \div a^n = a^{m-n}$ for $a \neq 0$.

Power law

$(a^m)^n = a^{mn}$.

Zero and negative indices

$a^0 = 1$ for $a \neq 0$; $a^{-n} = 1/a^n$.

Rational index

a^(1/n) is the nth root of a; a^(m/n) = $(\sqrt[n]{a})^m$.

Scientific notation

$a \times 10^n$ with $1 \leq a < 10$.

Significant figures

the digits that carry measured meaning, counted from the first non-zero digit.

Formula bank (3 of 6)

Key facts and formulas, chapter by chapter. Use them to check work, not to replace reasoning.

ALGEBRA AND FUNCTIONS

Function

a rule that gives exactly one output for each input in its domain.

Domain and range

the set of permitted inputs and the set of resulting outputs.

Expanding

$a(b + c) = ab + ac$.

Difference of squares

$a^2 - b^2 = (a - b)(a + b)$.

Perfect square

$a^2 + 2ab + b^2 = (a + b)^2$.

Rearranging a formula

apply the same inverse operation to both sides to isolate the wanted letter.

nth term

a linear sequence has term = $mn + c$; a quadratic sequence has constant second difference.

Formula bank (4 of 6)

Key facts and formulas, chapter by chapter. Use them to check work, not to replace reasoning.

LINEAR MODELS AND SYSTEMS

Straight line

$y = mx + c$, where m is the gradient and c the y-intercept.

Gradient

$m = (y_2 - y_1) \div (x_2 - x_1)$, the change in y per unit change in x.

Direct proportion

$y = kx$, a straight line through the origin.

Simultaneous equations

solve by substitution or elimination; check in both equations.

Inequality

multiplying or dividing by a negative number reverses the inequality sign.

Formula bank (5 of 6)

Key facts and formulas, chapter by chapter. Use them to check work, not to replace reasoning.

SHAPE, SPACE AND TRIGONOMETRY

Congruence

SSS, SAS, ASA, RHS for triangles.

Similarity

corresponding angles equal and corresponding sides in the same ratio.

Pythagoras

$a^2 + b^2 = c^2$ for the hypotenuse c of a right-angled triangle.

Trigonometric ratios

$\sin\theta$ = opposite/hypotenuse, $\cos\theta$ = adjacent/hypotenuse, $\tan\theta$ = opposite/adjacent.

Angles of elevation and depression

measured from the horizontal, and equal by alternate angles.

Prism

volume = area of cross-section × length; surface area is the sum of the face areas.

Cylinder

volume = $\pi r^2 h$; curved surface area = $2\pi rh$; total surface area = $2\pi rh + 2\pi r^2$.

Pyramid

volume = $\frac{1}{3}$ × base area × perpendicular height.

Cone

volume = $\frac{1}{3}\pi r^2 h$; curved surface area = πrl, where l is the slant height.

Sphere

volume = $\frac{4}{3}\pi r^3$; surface area = $4\pi r^2$.

Density

density = mass ÷ volume.

Formula bank (6 of 6)

Key facts and formulas, chapter by chapter. Use them to check work, not to replace reasoning.

DATA, PROBABILITY AND EVIDENCE

Bivariate data

paired measurements of two variables for the same subject.

Correlation

the strength and direction of a linear association, measured by r with $-1 \le r \le 1$.

Line of best fit

a line drawn through the middle of a scatter so that the residuals are balanced.

Interpolation and extrapolation

predicting inside or outside the range of the collected data.

Causation

correlation alone never establishes cause; a mechanism or a controlled study is required.

Probability of an event

P(event) = favourable outcomes ÷ total equally likely outcomes.

Compound events

P(A and B) = P(A) × P(B) for independent events; P(A or B) = P(A) + P(B) − P(A and B).

Sample and population

a sample estimates a population parameter; bias makes the estimate unreliable.

Base rate

the underlying frequency of an event in the whole population, needed to interpret a test result.

Glossary (1 of 2)

('Absolute value', 'The distance of a number from zero on a number line, written |x|.', 1) ch. 1
The distance of a number from zero on a number line, written |x|.

('Base rate', 'The frequency of an event in the whole population, before any test result is known.', 10) ch. 10
The frequency of an event in the whole population, before any test result is known.

('Bias', 'A systematic error that makes a sample unrepresentative of the population.', 10) ch. 10
A systematic error that makes a sample unrepresentative of the population.

('Bivariate data', 'Paired measurements of two variables recorded for the same subject.', 9) ch. 9
Paired measurements of two variables recorded for the same subject.

('Causation', 'A relationship in which changing one variable produces the change in the other.', 9) ch. 9
A relationship in which changing one variable produces the change in the other.

('Complement (probability)', 'The set of outcomes that are not in the event; P(not A) = 1 − P(A).', 10) ch. 10
The set of outcomes that are not in the event; P(not A) = 1 − P(A).

('Composite solid', 'A solid made by joining two or more simpler solids.', 8) ch. 8
A solid made by joining two or more simpler solids.

('Congruent', 'Identical in shape and size; for triangles, SSS, SAS, ASA or RHS proves it.', 6) ch. 6
Identical in shape and size; for triangles, SSS, SAS, ASA or RHS proves it.

('Conjecture', 'A mathematical claim believed to be true but not yet justified.', 1) ch. 1
A mathematical claim believed to be true but not yet justified.

('Continuous data', 'Numerical data that can take any value in an interval.', 9) ch. 9
Numerical data that can take any value in an interval.

('Correlation', 'The strength and direction of a linear association between two variables.', 9) ch. 9
The strength and direction of a linear association between two variables.

('Correlation coefficient', 'A number r between −1 and 1 that measures linear association.', 9) ch. 9
A number r between −1 and 1 that measures linear association.

('Cross-section', 'The shape exposed when a solid is cut parallel to a face.', 8) ch. 8
The shape exposed when a solid is cut parallel to a face.

('Density', 'Mass per unit volume, in g/cm³ or kg/m³.', 8) ch. 8
Mass per unit volume, in g/cm^3 or kg/m^3.

('Direct proportion', 'A relationship y = kx whose graph is a straight line through the origin.', 4) ch. 4
A relationship y = kx whose graph is a straight line through the origin.

('Discrete data', 'Numerical data that counts separate values.', 9) ch. 9
Numerical data that counts separate values.

('Domain', 'The set of permitted inputs of a function.', 3) ch. 3
The set of permitted inputs of a function.

('Elimination', 'Solving simultaneous equations by adding or subtracting to remove one variable.', 5) ch. 5
Solving simultaneous equations by adding or subtracting to remove one variable.

('Function', 'A rule that gives exactly one output for each permitted input.', 3) ch. 3
A rule that gives exactly one output for each permitted input.

('Gradient', 'The change in y per unit change in x; the steepness of a line.', 4) ch. 4
The change in y per unit change in x; the steepness of a line.

('Interpolation', 'Predicting a value inside the range of the collected data.', 9) ch. 9
Predicting a value inside the range of the collected data.

('Interval', 'A continuous set of numbers between two endpoints, written with square or round brackets.', 1) ch. 1
A continuous set of numbers between two endpoints, written with square or round brackets.

('Irrational number', 'A number that cannot be written as a fraction; its decimal neither terminates nor recurs.', 1) ch. 1
A number that cannot be written as a fraction; its decimal neither terminates nor recurs.

('Line of best fit', 'A line drawn through the middle of a scatter so that the residuals balance.', 9) ch. 9
A line drawn through the middle of a scatter so that the residuals balance.

Glossary (2 of 2)

('Mutually exclusive', 'Events that cannot both happen, so P(A or B) = P(A) + P(B).', 10) ch. 10
Events that cannot both happen, so P(A or B) = P(A) + P(B).

('nth term', 'A rule that gives the term in any position of a sequence.', 3) ch. 3
A rule that gives the term in any position of a sequence.

('Population', 'The whole group a claim is about.', 10) ch. 10
The whole group a claim is about.

("Pythagoras' theorem", 'In a right-angled triangle, $a^2 + b^2 = c^2$ for the hypotenuse c.', 7) ch. 7
In a right-angled triangle, $a^2 + b^2 = c^2$ for the hypotenuse c.

('Range (of a function)', 'The set of outputs a function can produce.', 3) ch. 3
The set of outputs a function can produce.

('Rational number', 'A number that can be written as a fraction of two integers.', 1) ch. 1
A number that can be written as a fraction of two integers.

('Sample', 'The part of a population that is actually measured.', 10) ch. 10
The part of a population that is actually measured.

('Scatter plot', 'A graph of paired data used to look for association.', 9) ch. 9
A graph of paired data used to look for association.

('Scientific notation', 'A number written as $a \times 10^n$ with $1 \le a < 10$.', 2) ch. 2
A number written as $a \times 10^n$ with $1 \le a < 10$.

('Significant figures', 'The digits of a measurement that carry meaning, counted from the first non-zero digit.', 2) ch. 2
The digits of a measurement that carry meaning, counted from the first non-zero digit.

('Similar', 'Same shape, possibly different size: corresponding angles equal, sides in ratio.', 7) ch. 7
Same shape, possibly different size: corresponding angles equal, sides in ratio.

('Simultaneous equations', 'Two or more equations that must be true at the same time.', 5) ch. 5
Two or more equations that must be true at the same time.

('Slant height', 'The distance from the apex of a cone or pyramid to the edge of its base.', 8) ch. 8
The distance from the apex of a cone or pyramid to the edge of its base.

('Substitution', 'Replacing a variable by a value or by an equal expression.', 3) ch. 3
Replacing a variable by a value or by an equal expression.

('Surd', 'An exact irrational root left in root form, such as √2.', 1) ch. 1
An exact irrational root left in root form, such as √2.

('Tessellation', 'A covering of the plane by shapes that meet edge to edge with no gaps.', 6) ch. 6
A covering of the plane by shapes that meet edge to edge with no gaps.

('Trigonometric ratio', 'Sine, cosine or tangent: a ratio of two sides of a right-angled triangle.', 7) ch. 7
Sine, cosine or tangent: a ratio of two sides of a right-angled triangle.

('Union and intersection', 'A ∪ B contains what is in either set; A ∩ B only what is in both.', 1) ch. 1
A ∪ B contains what is in either set; A ∩ B only what is in both.

('y-intercept', 'The value of y where a line crosses the y-axis.', 4) ch. 4
The value of y where a line crosses the y-axis.

REFERENCES

References and acknowledgements

Official IB sources

- IB Middle Years Programme curriculum — https://www.ibo.org/programmes/middle-years-programme/curriculum/ (page states last updated 19 February 2026). Used for: at least 50 hours per subject per year, global contexts, conceptual understanding, approaches to learning and service as action.
- IB MYP mathematics subject page — https://www.ibo.org/programmes/middle-years-programme/curriculum/mathematics/ Used for: the four branches of MYP mathematics, standard and extended mathematics, and the inquiry and application focus.
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- IB MYP general regulations (2026 assessment procedures) — https://www.ibo.org/globalassets/new-structure/become-an-ib-school/pdfs/myp-2026-general-regulations.pdf Used for: the required use of MYP objectives and assessment criteria, minimum teaching hours, and Year 5 eAssessment eligibility.

Principal content reference

- Pre-rewrite MYP 3 chapter and subtopic audit (second-edition/audit/chapter-audits/myp3-audit.md with the chapter and subtopic registers). Used as the principal content reference for chapter scope, teaching order and the key concepts, related concepts, global contexts and inquiry questions of each chapter.

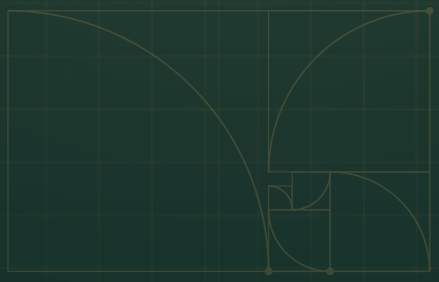
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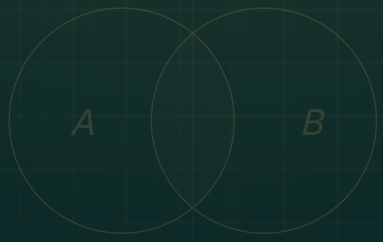
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Year 3 reasoning is organised, not just correct: Criterion C tasks ask you to structure an argument, and Criterion B tasks ask you to justify a generalisation rather than to list cases. Every chapter closes with a reflection that returns to the statement of inquiry.



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